CALCULUS

CALCULUS

Third Edition

Michael Spivak

CAMBRIDGE
UNIVERSITY PRESS

CAMBRIDGE UNIVERSITY PRESS
Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore, São Paulo,
Delhi, Dubai, Tokyo

Cambridge University Press
The Edinburgh Building, Cambridge CB2 8RU, UK

Published in the United States of America by Publish or Perish Inc., Houston

Published outside North America by Cambridge University Press

www.cambridge.org
Information on this title: www.cambridge.org/9780521867443

© Michael Spivak 1967, 1980, 1994

This publication is in copyright. Subject to statutory exception
and to the provisions of relevant collective licensing agreements,
no reproduction of any part may take place without
the written permission of Publish or Perish Inc.

First published 1967, second edition 1980, third edition 1994
Fourth printing 2009

Printed in the United Kingdom at the University Press, Cambridge

A catalog record for this publication is available from the British Library

ISBN 978-0-521-86744-3 hardback

Cambridge University Press has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication, and does not guarantee that any content on such websites is, or will remain, accurate or appropriate.

Dedicated to the Memory of Y. P.

PREFACE

I hold every man a debtor
to his profession,
from the which as men of course
doe seeke to receive countenance and profit,
so ought they of duty to endeavour
themselves by way of amends,
to be a help and
ornament therunto.

FRANCIS BACON

PREFACE TO THE FIRST EDITION

Every aspect of this book was influenced by the desire to present calculus not merely as a prelude to but as the first real encounter with mathematics. Since the foundations of analysis provided the arena in which modern modes of mathematical thinking developed, calculus ought to be the place in which to expect, rather than avoid, the strengthening of insight with logic. In addition to developing the students' intuition about the beautiful concepts of analysis, it is surely equally important to persuade them that precision and rigor are neither deterrents to intuition, nor ends in themselves, but the natural medium in which to formulate and think about mathematical questions.

This goal implies a view of mathematics which, in a sense, the entire book attempts to defend. No matter how well particular topics may be developed, the goals of this book will be realized only if it succeeds as a whole. For this reason, it would be of little value merely to list the topics covered, or to mention pedagogical practices and other innovations. Even the cursory glance customarily bestowed on new calculus texts will probably tell more than any such extended advertisement, and teachers with strong feelings about particular aspects of calculus will know just where to look to see if this book fulfills their requirements.

A few features do require explicit comment, however. Of the twenty-nine chapters in the book, two (starred) chapters are optional, and the three chapters comprising Part V have been included only for the benefit of those students who might want to examine on their own a construction of the real numbers. Moreover, the appendices to Chapters 3 and 11 also contain optional material.

The order of the remaining chapters is intentionally quite inflexible, since the purpose of the book is to present calculus as the evolution of one idea, not as a collection of "topics." Since the most exciting concepts of calculus do not appear until Part III, it should be pointed out that Parts I and II will probably require less time than their length suggests—although the entire book covers a one-year course, the chapters are not meant to be covered at any uniform rate. A rather natural dividing point does occur between Parts II and III, so it is possible to reach differentiation and integration even more quickly by treating Part II very briefly, perhaps returning later for a more detailed treatment. This arrangement corresponds to the traditional organization of most calculus courses, but I feel that it will only diminish the value of the book for students who have seen a small amount of calculus previously, and for bright students with a reasonable background.

The problems have been designed with this particular audience in mind. They range from straightforward, but not overly simple, exercises which develop basic techniques and test understanding of concepts, to problems of considerable difficulty and, I hope, of comparable interest. There are about 625 problems in all. Those which emphasize manipulations usually contain many examples, numbered

with small Roman numerals, while small letters are used to label interrelated parts in other problems. Some indication of relative difficulty is provided by a system of starring and double starring, but there are so many criteria for judging difficulty, and so many hints have been provided, especially for harder problems, that this guide is not completely reliable. Many problems are so difficult, especially if the hints are not consulted, that the best of students will probably have to attempt only those which especially interest them; from the less difficult problems it should be easy to select a portion which will keep a good class busy, but not frustrated. The answer section contains solutions to about half the examples from an assortment of problems that should provided a good test of technical competence. A separate answer book contains the solutions of the other parts of these problems, and of all the other problems as well. Finally, there is a Suggested Reading list, to which the problems often refer, and a glossary of symbols.

I am grateful for the opportunity to mention the many people to whom I owe my thanks. Jane Bjorkgren performed prodigious feats of typing that compensated for my fitful production of the manuscript. Richard Serkey helped collect the material which provides historical sidelights in the problems, and Richard Weiss supplied the answers appearing in the back of the book. I am especially grateful to my friends Michael Freeman, Jay Goldman, Anthony Phillips, and Robert Wells for the care with which they read, and the relentlessness with which they criticized, a preliminary version of the book. Needles to say, they are not responsible for the deficiencies which remain, especially since I sometimes rejected suggestions which would have made the book appear suitable for a larger group of students. I must express my admiration for the editors and staff of W. A. Benjamin, Inc., who were always eager to increase the appeal of the book, while recognizing the audience for which it was intended.

The inadequacies which preliminary editions always involve were gallantly endured by a rugged group of freshmen in the honors mathematics course at Brandeis University during the academic year 1965–1966. About half of this course was devoted to algebra and topology, while the other half covered calculus, with the preliminary edition as the text. It is almost obligatory in such circumstances to report that the preliminary version was a gratifying success. This is always safe—after all, the class is unlikely to rise up in a body and protest publicly—but the students themselves, it seems to me, deserve the right to assign credit for the thoroughness with which they absorbed an impressive amount of mathematics. I am content to hope that some other students will be able to use the book to such good purpose, and with such enthusiasm.

MICHAEL SPIVAK
Waltham, Massachusetts
February 1967

PREFACE TO THE SECOND EDITION

I have often been told that the title of this book should really be something like "An Introduction to Analysis," because the book is usually used in courses where the students have already learned the mechanical aspects of calculus—such courses are standard in Europe, and they are becoming more common in the United States. After thirteen years it seems too late to change the title, but other changes, in addition to the correction of numerous misprints and mistakes, seemed called for. There are now separate Appendices for many topics that were previously slighted: polar coordinates, uniform continuity, parameterized curves, Riemann sums, and the use of integrals for evaluating lengths, volumes and surface areas. A few topics, like manipulations with power series, have been discussed more thoroughly in the text, and there are also more problems on these topics, while other topics, like Newton's method and the trapezoid rule and Simpson's rule, have been developed in the problems. There are in all about 160 new problems, many of which are intermediate in difficulty between the few routine problems at the beginning of each chapter and the more difficult ones that occur later.

Most of the new problems are the work of Ted Shifrin. Frederick Gordon pointed out several serious mistakes in the original problems, and supplied some non-trivial corrections, as well as the neat proof of Theorem 12-2, which took two Lemmas and two pages in the first edition. Joseph Lipman also told me of this proof, together with the similar trick for the proof of the last theorem in the Appendix to Chapter 11, which went unproved in the first edition. Roy O. Davies told me the trick for Problem 11-66, which previously was proved only in Problem 20-8 [21-8 in the third edition], and Marina Ratner suggested several interesting problems, especially ones on uniform continuity and infinite series. To all these people go my thanks, and the hope that in the process of fashioning the new edition their contributions weren't too badly botched.

MICHAEL SPIVAK

PREFACE TO THE THIRD EDITION

The most significant change in this third edition is the inclusion of a new (starred) Chapter 17 on planetary motion, in which calculus is employed for a substantial physics problem.

In preparation for this, the old Appendix to Chapter 4 has been replaced by three Appendices: the first two cover vectors and conic sections, while polar coordinates are now deferred until the third Appendix, which also discusses the polar coordinate equations of the conic sections. Moreover, the Appendix to Chapter 12 has been extended to treat vector operations on vector-valued curves.

Another large change is merely a rearrangement of old material: "The Cosmopolitan Integral," previously a second Appendix to Chapter 13, is now an Appendix to the chapter on "Integration in Elementary Terms" (previously Chapter 18, now Chapter 19); moreover, those problems from that chapter which used the material from that Appendix now appear as problems in the newly placed Appendix.

A few other changes and renumbering of Problems result from corrections, and elimination of incorrect problems.

I was both startled and somewhat dismayed when I realized that after allowing 13 years to elapse between the first and second editions of the book, I have allowed another 14 years to elapse before this third edition. During this time I seem to have accumulated a not-so-short list of corrections, but no longer have the original communications, and therefore cannot properly thank the various individuals involved (who by now have probably lost interest anyway). I have had time to make only a few changes to the Suggested Reading, which after all these years probably requires a complete revision; this will have to wait until the next edition, which I hope to make in a more timely fashion.

MICHAEL SPIVAK

CONTENTS

PREFACE **vi**

PART I **Prologue**

1 Basic Properties of Numbers 3
2 Numbers of Various Sorts 21

PART II **Foundations**

3 Functions 39
Appendix. Ordered Pairs 54
4 Graphs 56
Appendix 1. Vectors 75
Appendix 2. The Conic Sections 80
Appendix 3. Polar Coordinates 84
5 Limits 90
6 Continuous Functions 113
7 Three Hard Theorems 120
8 Least Upper Bounds 131
Appendix. Uniform Continuity 142

PART III **Derivatives and Integrals**

9 Derivatives 147
10 Differentiation 166
11 Significance of the Derivative 185
Appendix. Convexity and Concavity 216
12 Inverse Functions 227
Appendix. Parametric Representation of Curves 241
13 Integrals 250
Appendix. Riemann Sums 279
14 The Fundamental Theorem of Calculus 282

15 The Trigonometric Functions 300
*16 π is Irrational 321
*17 Planetary Motion 327
18 The Logarithm and Exponential Functions 336
19 Integration in Elementary Terms 359
Appendix. The Cosmopolitan Integral 397

PART IV Infinite Sequences and Infinite Series

20 Approximation by Polynomial Functions 405
*21 *e* is Transcendental 435
22 Infinite Sequences 445
23 Infinite Series 464
24 Uniform Convergence and Power Series 491
25 Complex Numbers 517
26 Complex Functions 532
27 Complex Power Series 546

PART V Epilogue

28 Fields 571
29 Construction of the Real Numbers 578
30 Uniqueness of the Real Numbers 591
Suggested Reading 599
Answers (to selected problems) 609
Glossary of Symbols 655
Index 659

PART 1
PROLOGUE

To be conscious that
you are ignorant is a great step
to knowledge.

BENJAMIN DISRAELI

CHAPTER 1 BASIC PROPERTIES OF NUMBERS

The title of this chapter expresses in a few words the mathematical knowledge required to read this book. In fact, this short chapter is simply an explanation of what is meant by the "basic properties of numbers," all of which—addition and multiplication, subtraction and division, solutions of equations and inequalities, factoring and other algebraic manipulations—are already familiar to us. Nevertheless, this chapter is not a review. Despite the familiarity of the subject, the survey we are about to undertake will probably seem quite novel; it does not aim to present an extended review of old material, but to condense this knowledge into a few simple and obvious properties of numbers. Some may even seem too obvious to mention, but a surprising number of diverse and important facts turn out to be consequences of the ones we shall emphasize.

Of the twelve properties which we shall study in this chapter, the first nine are concerned with the fundamental operations of addition and multiplication. For the moment we consider only addition: this operation is performed on a pair of numbers—the sum $a + b$ exists for any two given numbers a and b (which may possibly be the same number, of course). It might seem reasonable to regard addition as an operation which can be performed on several numbers at once, and consider the sum $a_1 + \cdots + a_n$ of n numbers $a_1, \ldots, a_n$ as a basic concept. It is more convenient, however, to consider addition of pairs of numbers only, and to define other sums in terms of sums of this type. For the sum of three numbers a, b, and c, this may be done in two different ways. One can first add b and c, obtaining $b + c$, and then add a to this number, obtaining $a + (b + c)$; or one can first add a and b, and then add the sum $a + b$ to c, obtaining $(a + b) + c$. Of course, the two compound sums obtained are equal, and this fact is the very first property we shall list:

(P1) If a, b, and c are any numbers, then

$$a + (b + c) = (a + b) + c.$$

The statement of this property clearly renders a separate concept of the sum of three numbers superfluous; we simply agree that $a + b + c$ denotes the number $a+(b+c) = (a+b)+c$. Addition of four numbers requires similar, though slightly more involved, considerations. The symbol $a + b + c + d$ is defined to mean

$$\begin{array}{lll} & (1) & ((a + b) + c) + d, \\ \text{or} & (2) & (a + (b + c)) + d, \\ \text{or} & (3) & a + ((b + c) + d), \\ \text{or} & (4) & a + (b + (c + d)), \\ \text{or} & (5) & (a + b) + (c + d). \end{array}$$

This definition is unambiguous since these numbers are all equal. Fortunately, *this* fact need not be listed separately, since it follows from the property P1 already listed. For example, we know from P1 that

$$(a + b) + c = a + (b + c),$$

and it follows immediately that (1) and (2) are equal. The equality of (2) and (3) is a direct consequence of P1, although this may not be apparent at first sight (one must let $b + c$ play the role of b in P1, and d the role of c). The equalities (3) = (4) = (5) are also simple to prove.

It is probably obvious that an appeal to P1 will also suffice to prove the equality of the 14 possible ways of summing five numbers, but it may not be so clear how we can reasonably arrange a proof that this is so without actually listing these 14 sums. Such a procedure is feasible, but would soon cease to be if we considered collections of six, seven, or more numbers; it would be totally inadequate to prove the equality of all possible sums of an arbitrary finite collection of numbers $a_1, \ldots, a_n$. This fact may be taken for granted, but for those who would like to worry about the proof (and it is worth worrying about once) a reasonable approach is outlined in Problem 24. Henceforth, we shall usually make a tacit appeal to the results of this problem and write sums $a_1 + \cdots + a_n$ with a blithe disregard for the arrangement of parentheses.

The number 0 has one property so important that we list it next:

(P2) If a is any number, then

$$a + 0 = 0 + a = a.$$

An important role is also played by 0 in the third property of our list:

(P3) For every number a, there is a number $-a$ such that

$$a + (-a) = (-a) + a = 0.$$

Property P2 ought to represent a distinguishing characteristic of the number 0, and it is comforting to note that we are already in a position to prove this. Indeed, if a number x satisfies

$$a + x = a$$

for any one number a, then $x = 0$ (and consequently this equation also holds for all numbers a). The proof of this assertion involves nothing more than subtracting a from both sides of the equation, in other words, adding $-a$ to both sides; as the following detailed proof shows, all three properties P1–P3 must be used to justify this operation.

$$\begin{array}{ll} \text{If} & a + x = a, \\ \text{then} & (-a) + (a + x) = (-a) + a = 0; \\ \text{hence} & ((-a) + a) + x = 0; \\ \text{hence} & 0 + x = 0; \\ \text{hence} & x = 0. \end{array}$$

As we have just hinted, it is convenient to regard subtraction as an operation derived from addition: we consider $a - b$ to be an abbreviation for $a + (-b)$. It is then possible to find the solution of certain simple equations by a series of steps (each justified by P1, P2, or P3) similar to the ones just presented for the equation $a + x = a$. For example:

$$\begin{array}{ll} \text{If} & x + 3 = 5, \\ \text{then} & (x + 3) + (-3) = 5 + (-3); \\ \text{hence} & x + (3 + (-3)) = 5 - 3 = 2; \\ \text{hence} & x + 0 = 2; \\ \text{hence} & x = 2. \end{array}$$

Naturally, such elaborate solutions are of interest only until you become convinced that they can always be supplied. In practice, it is usually just a waste of time to solve an equation by indicating so explicitly the reliance on properties P1, P2, and P3 (or any of the further properties we shall list).

Only one other property of addition remains to be listed. When considering the sums of three numbers a, b, and c, only two sums were mentioned: $(a + b) + c$ and $a + (b + c)$. Actually, several other arrangements are obtained if the order of a, b, and c is changed. That these sums are all equal depends on

(P4) If a and b are any numbers, then

$$a + b = b + a.$$

The statement of P4 is meant to emphasize that although the operation of addition of pairs of numbers might conceivably depend on the order of the two numbers, in fact it does not. It is helpful to remember that not all operations are so well behaved. For example, subtraction does not have this property: usually $a - b \neq b - a$. In passing we might ask just when $a - b$ does equal $b - a$, and it is amusing to discover how powerless we are if we rely only on properties P1–P4 to justify our manipulations. Algebra of the most elementary variety shows that $a - b = b - a$ only when $a = b$. Nevertheless, it is impossible to derive this fact from properties P1–P4; it is instructive to examine the elementary algebra carefully and determine which step(s) cannot be justified by P1–P4. We will indeed be able to justify all steps in detail when a few more properties are listed. Oddly enough, however, the crucial property involves multiplication.

The basic properties of multiplication are fortunately so similar to those for addition that little comment will be needed; both the meaning and the consequences should be clear. (As in elementary algebra, the product of a and b will be denoted by $a \cdot b$, or simply ab.)

(P5) If a, b, and c are any numbers, then

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

(P6) If a is any number, then

$$a \cdot 1 = 1 \cdot a = a.$$

Moreover, $1 \neq 0$.

(The assertion that $1 \neq 0$ may seem a strange fact to list, but we have to list it, because there is no way it could possibly be proved on the basis of the other properties listed—these properties would all hold if there were only one number, namely, 0.)

(P7) For every number $a \neq 0$, there is a number a^{-1} such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

(P8) If a and b are any numbers, then

$$a \cdot b = b \cdot a.$$

One detail which deserves emphasis is the appearance of the condition $a \neq 0$ in P7. This condition is quite necessary; since $0 \cdot b = 0$ for all numbers b, there is *no* number 0^{-1} satisfying $0 \cdot 0^{-1} = 1$. This restriction has an important consequence for division. Just as subtraction was defined in terms of addition, so division is defined in terms of multiplication: The symbol a/b means $a \cdot b^{-1}$. Since 0^{-1} is meaningless, $a/0$ is also meaningless—division by 0 is *always* undefined.

Property P7 has two important consequences. If $a \cdot b = a \cdot c$, it does not necessarily follow that $b = c$; for if $a = 0$, then both $a \cdot b$ and $a \cdot c$ are 0, no matter what b and c are. However, if $a \neq 0$, then $b = c$; this can be deduced from P7 as follows:

$$\begin{aligned}
&\text{If} && a \cdot b = a \cdot c \text{ and } a \neq 0,\\
&\text{then} && a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c);\\
&\text{hence} && (a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c;\\
&\text{hence} && 1 \cdot b = 1 \cdot c;\\
&\text{hence} && b = c.
\end{aligned}$$

It is also a consequence of P7 that if $a \cdot b = 0$, then either $a = 0$ or $b = 0$. In fact,

$$\begin{aligned}
&\text{if} && a \cdot b = 0 \text{ and } a \neq 0,\\
&\text{then} && a^{-1} \cdot (a \cdot b) = 0;\\
&\text{hence} && (a^{-1} \cdot a) \cdot b = 0;\\
&\text{hence} && 1 \cdot b = 0;\\
&\text{hence} && b = 0.
\end{aligned}$$

(It may happen that $a = 0$ and $b = 0$ are both true; this possibility is not excluded when we say "either $a = 0$ or $b = 0$"; in mathematics "or" is always used in the sense of "one or the other, or both.")

This latter consequence of P7 is constantly used in the solution of equations. Suppose, for example, that a number x is known to satisfy

$$(x - 1)(x - 2) = 0.$$

Then it follows that either $x - 1 = 0$ or $x - 2 = 0$; hence $x = 1$ or $x = 2$.

On the basis of the eight properties listed so far it is still possible to prove very little. Listing the next property, which combines the operations of addition and multiplication, will alter this situation drastically.

(P9) If a, b, and c are any numbers, then

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

(Notice that the equation $(b + c) \cdot a = b \cdot a + c \cdot a$ is also true, by P8.)

As an example of the usefulness of P9 we will now determine just when $a - b = b - a$:

If	$a - b = b - a,$
then	$(a - b) + b = (b - a) + b = b + (b - a);$
hence	$a = b + b - a;$
hence	$a + a = (b + b - a) + a = b + b.$
Consequently	$a \cdot (1 + 1) = b \cdot (1 + 1),$
and therefore	$a = b.$

A second use of P9 is the justification of the assertion $a \cdot 0 = 0$ which we have already made, and even used in a proof on page 6 (can you find where?). This fact was not listed as one of the basic properties, even though no proof was offered when it was first mentioned. With P1–P8 alone a proof was not possible, since the number 0 appears only in P2 and P3, which concern addition, while the assertion in question involves multiplication. With P9 the proof is simple, though perhaps not obvious: We have

$$\begin{aligned} a \cdot 0 + a \cdot 0 &= a \cdot (0 + 0) \\ &= a \cdot 0; \end{aligned}$$

as we have already noted, this immediately implies (by adding $-(a \cdot 0)$ to both sides) that $a \cdot 0 = 0$.

A series of further consequences of P9 may help explain the somewhat mysterious rule that the product of two negative numbers is positive. To begin with, we will establish the more easily acceptable assertion that $(-a) \cdot b = -(a \cdot b)$. To prove this, note that

$$\begin{aligned} (-a) \cdot b + a \cdot b &= [(-a) + a] \cdot b \\ &= 0 \cdot b \\ &= 0. \end{aligned}$$

It follows immediately (by adding $-(a \cdot b)$ to both sides) that $(-a) \cdot b = -(a \cdot b)$. Now note that

$$\begin{aligned} (-a) \cdot (-b) + [-(a \cdot b)] &= (-a) \cdot (-b) + (-a) \cdot b \\ &= (-a) \cdot [(-b) + b] \\ &= (-a) \cdot 0 \\ &= 0. \end{aligned}$$

Consequently, adding $(a \cdot b)$ to both sides, we obtain

$$(-a) \cdot (-b) = a \cdot b.$$

The fact that the product of two negative numbers is positive is thus a consequence of P1–P9. In other words, *if we want P1 to P9 to be true, the rule for the product of two negative numbers is forced upon us.*

The various consequences of P9 examined so far, although interesting and important, do not really indicate the significance of P9; after all, we could have listed each of these properties separately. Actually, P9 is the justification for almost all algebraic manipulations. For example, although we have shown how to solve the equation

$$(x - 1)(x - 2) = 0,$$

we can hardly expect to be presented with an equation in this form. We are more likely to be confronted with the equation

$$x^2 - 3x + 2 = 0.$$

The "factorization" $x^2 - 3x + 2 = (x - 1)(x - 2)$ is really a triple use of P9:

$$\begin{aligned}(x - 1) \cdot (x - 2) &= x \cdot (x - 2) + (-1) \cdot (x - 2) \\ &= x \cdot x + x \cdot (-2) + (-1) \cdot x + (-1) \cdot (-2) \\ &= x^2 + x[(-2) + (-1)] + 2 \\ &= x^2 - 3x + 2.\end{aligned}$$

A final illustration of the importance of P9 is the fact that this property is actually used every time one multiplies arabic numerals. For example, the calculation

$$\begin{array}{r} 13 \\ \times 24 \\ \hline 52 \\ 26 \\ \hline 312 \end{array}$$

is a concise arrangement for the following equations:

$$\begin{aligned}13 \cdot 24 &= 13 \cdot (2 \cdot 10 + 4) \\ &= 13 \cdot 2 \cdot 10 + 13 \cdot 4 \\ &= 26 \cdot 10 + 52.\end{aligned}$$

(Note that moving 26 to the left in the above calculation is the same as writing $26 \cdot 10$.) The multiplication $13 \cdot 4 = 52$ uses P9 also:

$$\begin{aligned}13 \cdot 4 &= (1 \cdot 10 + 3) \cdot 4 \\ &= 1 \cdot 10 \cdot 4 + 3 \cdot 4 \\ &= 4 \cdot 10 + 12 \\ &= 4 \cdot 10 + 1 \cdot 10 + 2 \\ &= (4 + 1) \cdot 10 + 2 \\ &= 5 \cdot 10 + 2 \\ &= 52.\end{aligned}$$

The properties P1–P9 have descriptive names which are not essential to remember, but which are often convenient for reference. We will take this opportunity to list properties P1–P9 together and indicate the names by which they are commonly designated.

(P1)	(Associative law for addition)	$a + (b + c) = (a + b) + c.$
(P2)	(Existence of an additive identity)	$a + 0 = 0 + a = a.$
(P3)	(Existence of additive inverses)	$a + (-a) = (-a) + a = 0.$
(P4)	(Commutative law for addition)	$a + b = b + a.$
(P5)	(Associative law for multiplication)	$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$
(P6)	(Existence of a multiplicative identity)	$a \cdot 1 = 1 \cdot a = a; \quad 1 \neq 0.$
(P7)	(Existence of multiplicative inverses)	$a \cdot a^{-1} = a^{-1} \cdot a = 1,$ for $a \neq 0.$
(P8)	(Commutative law for multiplication)	$a \cdot b = b \cdot a.$
(P9)	(Distributive law)	$a \cdot (b + c) = a \cdot b + a \cdot c.$

The three basic properties of numbers which remain to be listed are concerned with inequalities. Although inequalities occur rarely in elementary mathematics, they play a prominent role in calculus. The two notions of inequality, $a < b$ (a is less than b) and $a > b$ (a is greater than b), are intimately related: $a < b$ means the same as $b > a$ (thus $1 < 3$ and $3 > 1$ are merely two ways of writing the same assertion). The numbers a satisfying $a > 0$ are called **positive**, while those numbers a satisfying $a < 0$ are called **negative**. While positivity can thus be defined in terms of $<$, it is possible to reverse the procedure: $a < b$ can be defined to mean that $b - a$ is positive. In fact, it is convenient to consider the collection of all positive numbers, denoted by P, as the basic concept, and state all properties in terms of P:

(P10) (Trichotomy law) For every number a, one and only one of the following holds:

(i) $a = 0$,
(ii) a is in the collection P,
(iii) $-a$ is in the collection P.

(P11) (Closure under addition) If a and b are in P, then $a + b$ is in P.

(P12) (Closure under multiplication) If a and b are in P, then $a \cdot b$ is in P.

These three properties should be complemented with the following definitions:

$$\begin{aligned} a > b \quad &\text{if} \quad a - b \text{ is in } P; \\ a < b \quad &\text{if} \quad b > a; \\ a \geq b \quad &\text{if} \quad a > b \text{ or } a = b; \\ a \leq b \quad &\text{if} \quad a < b \text{ or } a = b.* \end{aligned}$$

Note, in particular, that $a > 0$ if and only if a is in P.

All the familiar facts about inequalities, however elementary they may seem, are consequences of P10–P12. For example, if a and b are any two numbers, then precisely one of the following holds:

(i) $a - b = 0$,
(ii) $a - b$ is in the collection P,
(iii) $-(a - b) = b - a$ is in the collection P.

Using the definitions just made, it follows that precisely one of the following holds:

(i) $a = b$,
(ii) $a > b$,
(iii) $b > a$.

A slightly more interesting fact results from the following manipulations. If $a < b$, so that $b - a$ is in P, then surely $(b + c) - (a + c)$ is in P; thus, if $a < b$, then $a + c < b + c$. Similarly, suppose $a < b$ and $b < c$. Then

$$\begin{aligned} &b - a \text{ is in } P, \\ \text{and} \quad &c - b \text{ is in } P, \\ \text{so} \quad &c - a = (c - b) + (b - a) \text{ is in } P. \end{aligned}$$

This shows that if $a < b$ and $b < c$, then $a < c$. (The two inequalities $a < b$ and $b < c$ are usually written in the abbreviated form $a < b < c$, which has the third inequality $a < c$ almost built in.)

The following assertion is somewhat less obvious: If $a < 0$ and $b < 0$, then $ab > 0$. The only difficulty presented by the proof is the unraveling of definitions. The symbol $a < 0$ means, by definition, $0 > a$, which means $0 - a = -a$ is in P. Similarly $-b$ is in P, and consequently, by P12, $(-a)(-b) = ab$ is in P. Thus $ab > 0$.

The fact that $ab > 0$ if $a > 0$, $b > 0$ and also if $a < 0$, $b < 0$ has one special consequence: $a^2 > 0$ if $a \neq 0$. Thus squares of nonzero numbers are always positive, and in particular we have proved a result which might have seemed sufficiently elementary to be included in our list of properties: $1 > 0$ (since $1 = 1^2$).

*There is one slightly perplexing feature of the symbols $\geq$ and $\leq$. The statements

$$1 + 1 \leq 3$$
$$1 + 1 \leq 2$$

are both true, even though we know that $\leq$ could be replaced by $<$ in the first, and by $=$ in the second. This sort of thing is bound to occur when $\leq$ is used with specific numbers; the usefulness of the symbol is revealed by a statement like Theorem 1—here equality holds for some values of a and b, while inequality holds for other values.

The fact that $-a > 0$ if $a < 0$ is the basis of a concept which will play an extremely important role in this book. For any number a, we define the **absolute value** $|a|$ of a as follows:

$$|a| = \begin{cases} a, & a \geq 0 \\ -a, & a \leq 0. \end{cases}$$

Note that $|a|$ is always positive, except when $a = 0$. For example, we have $|-3| = 3$, $|7| = 7$, $|1 + \sqrt{2} - \sqrt{3}| = 1 + \sqrt{2} - \sqrt{3}$, and $|1 + \sqrt{2} - \sqrt{10}| = \sqrt{10} - \sqrt{2} - 1$. In general, the most straightforward approach to any problem involving absolute values requires treating several cases separately, since absolute values are defined by cases to begin with. This approach may be used to prove the following very important fact about absolute values.

THEOREM 1 For all numbers a and b, we have

$$|a + b| \leq |a| + |b|.$$

PROOF We will consider 4 cases:

$$\begin{array}{lll} (1) & a \geq 0, & b \geq 0; \\ (2) & a \geq 0, & b \leq 0; \\ (3) & a \leq 0, & b \geq 0; \\ (4) & a \leq 0, & b \leq 0. \end{array}$$

In case (1) we also have $a + b \geq 0$, and the theorem is obvious; in fact,

$$|a + b| = a + b = |a| + |b|,$$

so that in this case equality holds.

In case (4) we have $a + b \leq 0$, and again equality holds:

$$|a + b| = -(a + b) = -a + (-b) = |a| + |b|.$$

In case (2), when $a \geq 0$ and $b \leq 0$, we must prove that

$$|a + b| \leq a - b.$$

This case may therefore be divided into two subcases. If $a + b \geq 0$, then we must prove that

$$\begin{array}{rc} & a + b \leq a - b, \\ \text{i.e.,} & b \leq -b, \end{array}$$

which is certainly true since $b \leq 0$ and hence $-b \geq 0$. On the other hand, if $a + b \leq 0$, we must prove that

$$\begin{array}{rc} & -a - b \leq a - b, \\ \text{i.e.,} & -a \leq a, \end{array}$$

which is certainly true since $a \geq 0$ and hence $-a \leq 0$.

Finally, note that case (3) may be disposed of with no additional work, by applying case (2) with a and b interchanged. ∎

Although this method of treating absolute values (separate consideration of various cases) is sometimes the only approach available, there are often simpler methods which may be used. In fact, it is possible to give a much shorter proof of Theorem 1; this proof is motivated by the observation that

$$|a| = \sqrt{a^2}.$$

(Here, and throughout the book, $\sqrt{x}$ denotes the *positive* square root of x; this symbol is defined only when $x \geq 0$.) We may now observe that

$$\begin{aligned}(|a+b|)^2 = (a+b)^2 &= a^2 + 2ab + b^2 \\ &\leq a^2 + 2|a| \cdot |b| + b^2 \\ &= |a|^2 + 2|a| \cdot |b| + |b|^2 \\ &= (|a| + |b|)^2.\end{aligned}$$

From this we can conclude that $|a+b| \leq |a| + |b|$ because $x^2 < y^2$ implies $x < y$, provided that x and y are both nonnegative; a proof of *this* fact is left to the reader (Problem 5).

One final observation may be made about the theorem we have just proved: a close examination of either proof offered shows that

$$|a+b| = |a| + |b|$$

if a and b have the same sign (i.e., are both positive or both negative), or if one of the two is 0, while

$$|a+b| < |a| + |b|$$

if a and b are of opposite signs.

We will conclude this chapter with a subtle point, neglected until now, whose inclusion is required in a conscientious survey of the properties of numbers. After stating property P9, we proved that $a - b = b - a$ implies $a = b$. The proof began by establishing that

$$a \cdot (1+1) = b \cdot (1+1),$$

from which we concluded that $a = b$. This result is obtained from the equation $a \cdot (1+1) = b \cdot (1+1)$ by dividing both sides by $1+1$. Division by 0 should be avoided scrupulously, and it must therefore be admitted that the validity of the argument depends on knowing that $1+1 \neq 0$. Problem 25 is designed to convince you that this fact cannot possibly be proved from properties P1–P9 alone! Once P10, P11, and P12 are available, however, the proof is very simple: We have already seen that $1 > 0$; it follows that $1+1 > 0$, and in particular $1+1 \neq 0$.

This last demonstration has perhaps only strengthened your feeling that it is absurd to bother proving such obvious facts, but an honest assessment of our present situation will help justify serious consideration of such details. In this chapter we have assumed that numbers are familiar objects, and that P1–P12 are merely explicit statements of obvious, well-known properties of numbers. It would be difficult, however, to justify this assumption. Although one learns how to "work with" numbers in school, just what numbers *are*, remains rather vague. A great deal of this book is devoted to elucidating the concept of numbers, and by the end

of the book we will have become quite well acquainted with them. But it will be necessary to work with numbers throughout the book. It is therefore reasonable to admit frankly that we do not yet thoroughly understand numbers; we may still say that, in whatever way numbers are finally defined, they should certainly have properties P1–P12.

Most of this chapter has been an attempt to present convincing evidence that P1–P12 are indeed basic properties which we should assume in order to deduce other familiar properties of numbers. Some of the problems (which indicate the derivation of other facts about numbers from P1–P12) are offered as further evidence. It is still a crucial question whether P1–P12 actually account for *all* properties of numbers. As a matter of fact, we shall soon see that they do *not*. In the next chapter the deficiencies of properties P1–P12 will become quite clear, but the proper means for correcting these deficiencies is not so easily discovered. The crucial additional basic property of numbers which we are seeking is profound and subtle, quite unlike P1–P12. The discovery of this crucial property will require all the work of Part II of this book. In the remainder of Part I we will begin to see why some additional property is required; in order to investigate this we will have to consider a little more carefully what we mean by "numbers."

PROBLEMS

1. Prove the following:

(i) If $ax = a$ for some number $a \neq 0$, then $x = 1$.
(ii) $x^2 - y^2 = (x - y)(x + y)$.
(iii) If $x^2 = y^2$, then $x = y$ or $x = -y$.
(iv) $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.
(v) $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1})$.
(vi) $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$. (There is a particularly easy way to do this, using (iv), and it will show you how to find a factorization for $x^n + y^n$ whenever n is odd.)

2. What is wrong with the following "proof"? Let $x = y$. Then

$$\begin{aligned} x^2 &= xy, \\ x^2 - y^2 &= xy - y^2, \\ (x + y)(x - y) &= y(x - y), \\ x + y &= y, \\ 2y &= y, \\ 2 &= 1. \end{aligned}$$

3. Prove the following:

(i) $\dfrac{a}{b} = \dfrac{ac}{bc}$, if $b, c \neq 0$.

(ii) $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{ad + bc}{bd}$, if $b, d \neq 0$.

(iii) $(ab)^{-1} = a^{-1}b^{-1}$, if $a, b \neq 0$. (To do this you must remember the defining property of $(ab)^{-1}$.)

(iv) $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{ac}{db}$, if $b, d \neq 0$.

(v) $\dfrac{a}{b} \Big/ \dfrac{c}{d} = \dfrac{ad}{bc}$, if $b, c, d \neq 0$.

(vi) If $b, d \neq 0$, then $\dfrac{a}{b} = \dfrac{c}{d}$ if and only if $ad = bc$. Also determine when $\dfrac{a}{b} = \dfrac{b}{a}$.

4. Find all numbers x for which

(i) $4 - x < 3 - 2x$.
(ii) $5 - x^2 < 8$.
(iii) $5 - x^2 < -2$.
(iv) $(x - 1)(x - 3) > 0$. (When is a product of two numbers positive?)
(v) $x^2 - 2x + 2 > 0$.
(vi) $x^2 + x + 1 > 2$.
(vii) $x^2 - x + 10 > 16$.
(viii) $x^2 + x + 1 > 0$.
(ix) $(x - \pi)(x + 5)(x - 3) > 0$.
(x) $(x - \sqrt[3]{2})(x - \sqrt{2}) > 0$.
(xi) $2^x < 8$.
(xii) $x + 3^x < 4$.
(xiii) $\dfrac{1}{x} + \dfrac{1}{1 - x} > 0$.
(xiv) $\dfrac{x - 1}{x + 1} > 0$.

5. Prove the following:

(i) If $a < b$ and $c < d$, then $a + c < b + d$.
(ii) If $a < b$, then $-b < -a$.
(iii) If $a < b$ and $c > d$, then $a - c < b - d$.
(iv) If $a < b$ and $c > 0$, then $ac < bc$.
(v) If $a < b$ and $c < 0$, then $ac > bc$.
(vi) If $a > 1$, then $a^2 > a$.
(vii) If $0 < a < 1$, then $a^2 < a$.
(viii) If $0 \leq a < b$ and $0 \leq c < d$, then $ac < bd$.
(ix) If $0 \leq a < b$, then $a^2 < b^2$. (Use (viii).)
(x) If $a, b \geq 0$ and $a^2 < b^2$, then $a < b$. (Use (ix), backwards.)

6. (a) Prove that if $0 \leq x < y$, then $x^n < y^n$, $n = 1, 2, 3, \ldots$.
(b) Prove that if $x < y$ and n is odd, then $x^n < y^n$.
(c) Prove that if $x^n = y^n$ and n is odd, then $x = y$.
(d) Prove that if $x^n = y^n$ and n is even, then $x = y$ or $x = -y$.

7. Prove that if $0 < a < b$, then

$$a < \sqrt{ab} < \frac{a+b}{2} < b.$$

Notice that the inequality $\sqrt{ab} \leq (a+b)/2$ holds for all $a, b \geq 0$. A generalization of this fact occurs in Problem 2-22.

***8.** Although the basic properties of inequalities were stated in terms of the collection P of all positive numbers, and $<$ was defined in terms of P, this procedure can be reversed. Suppose that P10–P12 are replaced by

(P′10) For any numbers a and b one, and only one, of the following holds:
(i) $a = b$,
(ii) $a < b$,
(iii) $b < a$.

(P′11) For any numbers a, b, and c, if $a < b$ and $b < c$, then $a < c$.

(P′12) For any numbers a, b, and c, if $a < b$, then $a + c < b + c$.

(P′13) For any numbers a, b, and c, if $a < b$ and $0 < c$, then $ac < bc$.

Show that P10–P12 can then be deduced as theorems.

9. Express each of the following with at least one less pair of absolute value signs.

(i) $|\sqrt{2} + \sqrt{3} - \sqrt{5} + \sqrt{7}|$.
(ii) $|(|a+b| - |a| - |b|)|$.
(iii) $|(|a+b| + |c| - |a+b+c|)|$.
(iv) $|x^2 - 2xy + y^2|$.
(v) $|(|\sqrt{2} + \sqrt{3}| - |\sqrt{5} - \sqrt{7}|)|$.

10. Express each of the following without absolute value signs, treating various cases separately when necessary.

(i) $|a+b| - |b|$.
(ii) $|(|x| - 1)|$.
(iii) $|x| - |x^2|$.
(iv) $a - |(a - |a|)|$.

11. Find all numbers x for which

(i) $|x - 3| = 8$.
(ii) $|x - 3| < 8$.
(iii) $|x + 4| < 2$.
(iv) $|x - 1| + |x - 2| > 1$.
(v) $|x - 1| + |x + 1| < 2$.

(vi) $|x-1|+|x+1|<1$.
(vii) $|x-1|\cdot|x+1|=0$.
(viii) $|x-1|\cdot|x+2|=3$.

12. Prove the following:

(i) $|xy|=|x|\cdot|y|$.
(ii) $\left|\frac{1}{x}\right|=\frac{1}{|x|}$, if $x\neq 0$. (The best way to do this is to remember what $|x|^{-1}$ is.)
(iii) $\frac{|x|}{|y|}=\left|\frac{x}{y}\right|$, if $y\neq 0$.
(iv) $|x-y|\leq|x|+|y|$. (Give a very short proof.)
(v) $|x|-|y|\leq|x-y|$. (A very short proof is possible, if you write things in the right way.)
(vi) $|(|x|-|y|)|\leq|x-y|$. (Why does this follow immediately from (v)?)
(vii) $|x+y+z|\leq|x|+|y|+|z|$. Indicate when equality holds, and prove your statement.

13. The maximum of two numbers x and y is denoted by $\max(x,y)$. Thus $\max(-1,3)=\max(3,3)=3$ and $\max(-1,-4)=\max(-4,-1)=-1$. The minimum of x and y is denoted by $\min(x,y)$. Prove that

$$\max(x,y)=\frac{x+y+|y-x|}{2},$$
$$\min(x,y)=\frac{x+y-|y-x|}{2}.$$

Derive a formula for $\max(x,y,z)$ and $\min(x,y,z)$, using, for example

$$\max(x,y,z)=\max(x,\max(y,z)).$$

14. (a) Prove that $|a|=|-a|$. (The trick is not to become confused by too many cases. First prove the statement for $a\geq 0$. Why is it then obvious for $a\leq 0$?)
(b) Prove that $-b\leq a\leq b$ if and only if $|a|\leq b$. In particular, it follows that $-|a|\leq a\leq|a|$.
(c) Use this fact to give a new proof that $|a+b|\leq|a|+|b|$.

***15.** Prove that if x and y are not both 0, then

$$x^2+xy+y^2>0,$$
$$x^4+x^3y+x^2y^2+xy^3+y^4>0.$$

Hint: Use Problem 1.

***16.** (a) Show that

$(x+y)^2=x^2+y^2$ only when $x=0$ or $y=0$,
$(x+y)^3=x^3+y^3$ only when $x=0$ or $y=0$ or $x=-y$.

(b) Using the fact that

$$x^2 + 2xy + y^2 = (x + y)^2 \geq 0,$$

show that $4x^2 + 6xy + 4y^2 > 0$ unless x and y are both 0.

(c) Use part (b) to find out when $(x + y)^4 = x^4 + y^4$.

(d) Find out when $(x+y)^5 = x^5+y^5$. Hint: From the assumption $(x+y)^5 = x^5+y^5$ you should be able to derive the equation $x^3+2x^2y+2xy^2+y^3 = 0$, if $xy \neq 0$. This implies that $(x + y)^3 = x^2y + xy^2 = xy(x + y)$.

You should now be able to make a good guess as to when $(x+y)^n = x^n+y^n$; the proof is contained in Problem 11-57.

17. (a) Find the smallest possible value of $2x^2 - 3x + 4$. Hint: "Complete the square," i.e., write $2x^2 - 3x + 4 = 2(x - 3/4)^2 + ?$

(b) Find the smallest possible value of $x^2 - 3x + 2y^2 + 4y + 2$.

(c) Find the smallest possible value of $x^2 + 4xy + 5y^2 - 4x - 6y + 7$.

18. (a) Suppose that $b^2 - 4c \geq 0$. Show that the numbers

$$\frac{-b + \sqrt{b^2 - 4c}}{2}, \qquad \frac{-b - \sqrt{b^2 - 4c}}{2}$$

both satisfy the equation $x^2 + bx + c = 0$.

(b) Suppose that $b^2 - 4c < 0$. Show that there are no numbers x satisfying $x^2 + bx + c = 0$; in fact, $x^2 + bx + c > 0$ for all x. Hint: Complete the square.

(c) Use this fact to give another proof that if x and y are not both 0, then $x^2 + xy + y^2 > 0$.

(d) For which numbers α is it true that $x^2 + \alpha xy + y^2 > 0$ whenever x and y are not both 0?

(e) Find the smallest possible value of $x^2 + bx + c$ and of $ax^2 + bx + c$, for $a > 0$.

19. The fact that $a^2 \geq 0$ for all numbers a, elementary as it may seem, is nevertheless the fundamental idea upon which most important inequalities are ultimately based. The great-granddaddy of all inequalities is the *Schwarz inequality*:

$$x_1y_1 + x_2y_2 \leq \sqrt{{x_1}^2 + {x_2}^2}\sqrt{{y_1}^2 + {y_2}^2}.$$

(A more general form occurs in Problem 2-21.) The three proofs of the Schwarz inequality outlined below have only one thing in common—their reliance on the fact that $a^2 \geq 0$ for all a.

(a) Prove that if $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$ for some number λ, then equality holds in the Schwarz inequality. Prove the same thing if $y_1 = y_2 = 0$. Now suppose that y_1 and y_2 are not both 0, and that there is no number

λ such that $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$. Then

$$\begin{aligned} 0 &< (\lambda y_1 - x_1)^2 + (\lambda y_2 - x_2)^2 \\ &= \lambda^2({y_1}^2 + {y_2}^2) - 2\lambda(x_1 y_1 + x_2 y_2) + ({x_1}^2 + {x_2}^2). \end{aligned}$$

Using Problem 18, complete the proof of the Schwarz inequality.

(b) Prove the Schwarz inequality by using $2xy \leq x^2 + y^2$ (how is this derived?) with

$$x = \frac{x_i}{\sqrt{{x_1}^2 + {x_2}^2}}, \qquad y = \frac{y_i}{\sqrt{{y_1}^2 + {y_2}^2}},$$

first for $i = 1$ and then for $i = 2$.

(c) Prove the Schwarz inequality by first proving that

$$({x_1}^2 + {x_2}^2)({y_1}^2 + {y_2}^2) = (x_1 y_1 + x_2 y_2)^2 + (x_1 y_2 - x_2 y_1)^2.$$

(d) Deduce, from each of these three proofs, that equality holds only when $y_1 = y_2 = 0$ or when there is a number λ such that $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$.

In our later work, three facts about inequalities will be crucial. Although proofs will be supplied at the appropriate point in the text, a personal assault on these problems is infinitely more enlightening than a perusal of a completely worked-out proof. The statements of these propositions involve some weird numbers, but their basic message is very simple: if x is close enough to x_0, and y is close enough to y_0, then $x + y$ will be close to $x_0 + y_0$, and xy will be close to $x_0 y_0$, and $1/y$ will be close to $1/y_0$. The symbol "ε" which appears in these propositions is the fifth letter of the Greek alphabet ("epsilon"), and could just as well be replaced by a less intimidating Roman letter; however, tradition has made the use of ε almost sacrosanct in the contexts to which these theorems apply.

20. Prove that if

$$|x - x_0| < \frac{\varepsilon}{2} \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2},$$

then

$$\begin{aligned} |(x + y) - (x_0 + y_0)| &< \varepsilon, \\ |(x - y) - (x_0 - y_0)| &< \varepsilon. \end{aligned}$$

***21.** Prove that if

$$|x - x_0| < \min\left(\frac{\varepsilon}{2(|y_0| + 1)}, 1\right) \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then $|xy - x_0 y_0| < \varepsilon$.

(The notation "min" was defined in Problem 13, but the formula provided by that problem is irrelevant at the moment; the first inequality in the hypothesis just means that

$$|x - x_0| < \frac{\varepsilon}{2(|y_0| + 1)} \quad \text{and} \quad |x - x_0| < 1;$$

at one point in the argument you will need the first inequality, and at another point you will need the second. One more word of advice: since the hypotheses only provide information about $x - x_0$ and $y - y_0$, it is almost a foregone conclusion that the proof will depend upon writing $xy - x_0y_0$ in a way that involves $x - x_0$ and $y - y_0$.)

***22.** Prove that if $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon|y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| < \varepsilon.$$

***23.** Replace the question marks in the following statement by expressions involving ε, x_0, and y_0 so that the conclusion will be true:

If $y_0 \neq 0$ and

$$|y - y_0| < ? \quad \text{and} \quad |x - x_0| < ?$$

then $y \neq 0$ and

$$\left|\frac{x}{y} - \frac{x_0}{y_0}\right| < \varepsilon.$$

This problem is trivial in the sense that its solution follows from Problems 21 and 22 with almost no work at all (notice that $x/y = x \cdot 1/y$). The crucial point is not to become confused; decide which of the two problems should be used first, and don't panic if your answer looks unlikely.

***24.** This problem shows that the actual placement of parentheses in a sum is irrelevant. The proofs involve "mathematical induction"; if you are not familiar with such proofs, but still want to tackle this problem, it can be saved until after Chapter 2, where proofs by induction are explained.

Let us agree, for definiteness, that $a_1 + \cdots + a_n$ will denote

$$a_1 + (a_2 + (a_3 + \cdots + (a_{n-2} + (a_{n-1} + a_n))) \cdots).$$

Thus $a_1 + a_2 + a_3$ denotes $a_1 + (a_2 + a_3)$, and $a_1 + a_2 + a_3 + a_4$ denotes $a_1 + (a_2 + (a_3 + a_4))$, etc.

(a) Prove that

$$(a_1 + \cdots + a_k) + a_{k+1} = a_1 + \cdots + a_{k+1}.$$

Hint: Use induction on k.

(b) Prove that if $n \geq k$, then

$$(a_1 + \cdots + a_k) + (a_{k+1} + \cdots + a_n) = a_1 + \cdots + a_n.$$

Hint: Use part (a) to give a proof by induction on k.

(c) Let $s(a_1, \ldots, a_k)$ be some sum formed from $a_1, \ldots, a_k$. Show that

$$s(a_1, \ldots, a_k) = a_1 + \cdots + a_k.$$

Hint: There must be two sums $s'(a_1, \ldots, a_l)$ and $s''(a_{l+1}, \ldots, a_k)$ such that

$$s(a_1, \ldots, a_k) = s'(a_1, \ldots, a_l) + s''(a_{l+1}, \ldots, a_k).$$

25. Suppose that we interpret "number" to mean either 0 or 1, and $+$ and $\cdot$ to be the operations defined by the following two tables.

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Check that properties P1–P9 all hold, even though $1 + 1 = 0$.

CHAPTER 2 NUMBERS OF VARIOUS SORTS

In Chapter 1 we used the word "number" very loosely, despite our concern with the basic properties of numbers. It will now be necessary to distinguish carefully various kinds of numbers.

The simplest numbers are the "counting numbers"

$$1, 2, 3, \ldots .$$

The fundamental significance of this collection of numbers is emphasized by its symbol **N** (for **natural numbers**). A brief glance at P1–P12 will show that our basic properties of "numbers" do not apply to **N**—for example, P2 and P3 do not make sense for **N**. From this point of view the system **N** has many deficiencies. Nevertheless, **N** is sufficiently important to deserve several comments before we consider larger collections of numbers.

The most basic property of **N** is the principle of "mathematical induction." Suppose $P(x)$ means that the property P holds for the number x. Then the principle of mathematical induction states that $P(x)$ is true for all natural numbers x provided that

(1) $P(1)$ is true.

(2) Whenever $P(k)$ is true, $P(k+1)$ is true.

Note that condition (2) merely asserts the truth of $P(k+1)$ under the assumption that $P(k)$ is true; this suffices to ensure the truth of $P(x)$ for all x, if condition (1) also holds. In fact, if $P(1)$ is true, then it follows that $P(2)$ is true (by using (2) in the special case $k = 1$). Now, since $P(2)$ is true it follows that $P(3)$ is true (using (2) in the special case $k = 2$). It is clear that each number will eventually be reached by a series of steps of this sort, so that $P(k)$ is true for all numbers k.

A favorite illustration of the reasoning behind mathematical induction envisions an infinite line of people,

$$\text{person number 1, person number 2, person number 3, } \ldots .$$

If each person has been instructed to tell any secret he hears to the person behind him (the one with the next largest number) and a secret is told to person number 1, then clearly every person will eventually learn the secret. If $P(x)$ is the assertion that person number x will learn the secret, then the instructions given (to tell all secrets learned to the next person) assures that condition (2) is true, and telling the secret to person number 1 makes (1) true. The following example is a less facetious use of mathematical induction. There is a useful and striking formula which expresses the sum of the first n numbers in a simple way:

$$1 + \cdots + n = \frac{n(n+1)}{2}.$$

To prove this formula, note first that it is clearly true for $n = 1$. Now *assume* that for some natural number k we have

$$1 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned} 1 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + k + 1 \\ &= \frac{k(k+1) + 2k + 2}{2} \\ &= \frac{k^2 + 3k + 2}{2} \\ &= \frac{(k+1)(k+2)}{2}, \end{aligned}$$

so the formula is also true for $k + 1$. By the principle of induction this proves the formula for all natural numbers n. This particular example illustrates a phenomenon that frequently occurs, especially in connection with formulas like the one just proved. Although the proof by induction is often quite straightforward, the method by which the formula was discovered remains a mystery. Problems 5 and 6 indicate how some formulas of this type may be derived.

The principle of mathematical induction may be formulated in an equivalent way without speaking of "properties" of a number, a term which is sufficiently vague to be eschewed in a mathematical discussion. A more precise formulation states that if A is any collection (or "set"—a synonymous mathematical term) of natural numbers and

(1) 1 is in A,

(2) $k + 1$ is in A whenever k is in A,

then A is the set of all natural numbers. It should be clear that this formulation adequately replaces the less formal one given previously—we just consider the set A of natural numbers x which satisfy $P(x)$. For example, suppose A is the set of natural numbers n for which it is true that

$$1 + \cdots + n = \frac{n(n+1)}{2}.$$

Our previous proof of this formula showed that A contains 1, and that $k + 1$ is in A, if k is. It follows that A is the set of all natural numbers, i.e., that the formula holds for all natural numbers n.

There is yet another rigorous formulation of the principle of mathematical induction, which looks quite different. If A is any collection of natural numbers, it

is tempting to say that A must have a smallest member. Actually, this statement can fail to be true in a rather subtle way. A particularly important set of natural numbers is the collection A that contains no natural numbers at all, the "empty collection" or "null set,"* denoted by $\emptyset$. The null set $\emptyset$ is a collection of natural numbers that has no smallest member—in fact, it has no members at all. This is the only possible exception, however; if A is a nonnull set of natural numbers, then A has a least member. This "intuitively obvious" statement, known as the "well-ordering principle," can be proved from the principle of induction as follows. Suppose that the set A has no least member. Let B be the set of natural numbers n such that $1, \ldots, n$ are all *not* in A. Clearly 1 is in B (because if 1 were in A, then A would have 1 as smallest member). Moreover, if $1, \ldots, k$ are not in A, surely $k+1$ is not in A (otherwise $k+1$ would be the smallest member of A), so $1, \ldots, k+1$ are all not in A. This shows that if k is in B, then $k+1$ is in B. It follows that every number n is in B, i.e., the numbers $1, \ldots, n$ are *not* in A for any natural number n. Thus $A = \emptyset$, which completes the proof.

It is also possible to prove the principle of induction from the well-ordering principle (Problem 10). Either principle may be considered as a basic assumption about the natural numbers.

There is still another form of induction which should be mentioned. It sometimes happens that in order to prove $P(k+1)$ we must assume not only $P(k)$, but also $P(l)$ for all natural numbers $l \leq k$. In this case we rely on the "principle of complete induction": If A is a set of natural numbers and

(1) 1 is in A,

(2) $k+1$ is in A if $1, \ldots, k$ are in A,

then A is the set of all natural numbers.

Although the principle of complete induction may appear much stronger than the ordinary principle of induction, it is actually a consequence of that principle. The proof of this fact is left to the reader, with a hint (Problem 11). Applications will be found in Problems 7, 17, 20 and 22.

Closely related to proofs by induction are "recursive definitions." For example, the number $n!$ (read "n factorial") is defined as the product of all the natural numbers less than or equal to n:

$$n! = 1 \cdot 2 \cdot \cdots \cdot (n-1) \cdot n.$$

This can be expressed more precisely as follows:

(1) $1! = 1$

(2) $n! = n \cdot (n-1)!$.

This form of the definition exhibits the relationship between $n!$ and $(n-1)!$ in an

*Although it may not strike you as a collection, in the ordinary sense of the word, the null set arises quite naturally in many contexts. We frequently consider the set A, consisting of all x satisfying some property P; often we have no guarantee that P is satisfied by *any* number, so that A might be $\emptyset$—in fact often one proves that P is always false by showing that $A = \emptyset$.

explicit way that is ideally suited for proofs by induction. Problem 23 reviews a definition already familiar to you, which may be expressed more succinctly as a recursive definition; as this problem shows, the recursive definition is really necessary for a rigorous proof of some of the basic properties of the definition.

One definition which may not be familiar involves some convenient notation which we will constantly be using. Instead of writing

$$a_1 + \cdots + a_n,$$

we will usually employ the Greek letter Σ (capital sigma, for "sum") and write

$$\sum_{i=1}^{n} a_i.$$

In other words, $\sum_{i=1}^{n} a_i$ denotes the sum of the numbers obtained by letting $i = 1, 2, \ldots, n$. Thus

$$\sum_{i=1}^{n} i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Notice that the letter i really has nothing to do with the number denoted by $\sum_{i=1}^{n} i$, and can be replaced by any convenient symbol (except n, of course!):

$$\sum_{j=1}^{n} j = \frac{n(n+1)}{2},$$
$$\sum_{j=1}^{i} j = \frac{i(i+1)}{2},$$
$$\sum_{n=1}^{j} n = \frac{j(j+1)}{2}.$$

To define $\sum_{i=1}^{n} a_i$ precisely really requires a recursive definition:

$$(1) \quad \sum_{i=1}^{1} a_i = a_1,$$
$$(2) \quad \sum_{i=1}^{n} a_i = \sum_{i=1}^{n-1} a_i + a_n.$$

But only purveyors of mathematical austerity would insist too strongly on such precision. In practice, all sorts of modifications of this symbolism are used, and no one ever considers it necessary to add any words of explanation. The symbol

$$\sum_{\substack{i=1\\ i\neq 4}}^{n} a_i,$$

for example, is an obvious way of writing

$$a_1 + a_2 + a_3 + a_5 + a_6 + \cdots + a_n,$$

or more precisely,

$$\sum_{i=1}^{3} a_i + \sum_{i=5}^{n} a_i.$$

The deficiencies of the natural numbers which we discovered at the beginning of this chapter may be partially remedied by extending this system to the set of **integers**

$$\ldots, -2, -1, 0, 1, 2, \ldots.$$

This set is denoted by **Z** (from German "Zahl," number). Of properties P1–P12, only P7 fails for **Z**.

A still larger system of numbers is obtained by taking quotients m/n of integers (with $n \neq 0$). These numbers are called **rational numbers**, and the set of all rational numbers is denoted by **Q** (for "quotients"). In this system of numbers all of P1–P12 are true. It is tempting to conclude that the "properties of numbers," which we studied in some detail in Chapter 1, refer to just one set of numbers, namely, **Q**. There is, however, a still larger collection of numbers to which properties P1–P12 apply—the set of all **real numbers**, denoted by **R**. The real numbers include not only the rational numbers, but other numbers as well (the **irrational numbers**) which can be represented by infinite decimals; π and $\sqrt{2}$ are both examples of irrational numbers. The proof that π is irrational is not easy—we shall devote all of Chapter 16 of Part III to a proof of this fact. The irrationality of $\sqrt{2}$, on the other hand, is quite simple, and was known to the Greeks. (Since the Pythagorean theorem shows that an isosceles right triangle, with sides of length 1, has a hypotenuse of length $\sqrt{2}$, it is not surprising that the Greeks should have investigated this question.) The proof depends on a few observations about the natural numbers. Every natural number n can be written either in the form $2k$ for some integer k, or else in the form $2k + 1$ for some integer k (this "obvious" fact has a simple proof by induction (Problem 8)). Those natural numbers of the form $2k$ are called **even**; those of the form $2k + 1$ are called **odd**. Note that even numbers have even squares, and odd numbers have odd squares:

$$(2k)^2 = 4k^2 = 2 \cdot (2k^2),$$
$$(2k + 1)^2 = 4k^2 + 4k + 1 = 2 \cdot (2k^2 + 2k) + 1.$$

In particular it follows that the converse must also hold: if n^2 is even, then n is even; if n^2 is odd, then n is odd. The proof that $\sqrt{2}$ is irrational is now quite simple. Suppose that $\sqrt{2}$ were rational; that is, suppose there were natural numbers p

and q such that

$$\left(\frac{p}{q}\right)^2 = 2.$$

We can assume that p and q have no common divisor (since all common divisors could be divided out to begin with). Now we have

$$p^2 = 2q^2.$$

This shows that p^2 is even, and consequently p must be even; that is, $p = 2k$ for some natural number k. Then

$$p^2 = 4k^2 = 2q^2,$$

so

$$2k^2 = q^2.$$

This shows that q^2 is even, and consequently that q is even. Thus both p and q are even, contradicting the fact that p and q have no common divisor. This contradiction completes the proof.

It is important to understand precisely what this proof shows. We have demonstrated that there is no rational number x such that $x^2 = 2$. This assertion is often expressed more briefly by saying that $\sqrt{2}$ is irrational. Note, however, that the use of the symbol $\sqrt{2}$ implies the existence of *some* number (necessarily irrational) whose square is 2. We have not proved that such a number exists and we can assert confidently that, at present, a proof is *impossible* for us. Any proof at this stage would have to be based on P1–P12 (the only properties of $\mathbf{R}$ we have mentioned); since P1–P12 are also true for $\mathbf{Q}$ the exact same argument would show that there is a rational number whose square is 2, and this we know is false. (Note that the reverse argument will not work—our proof that there is no rational number whose square is 2 cannot be used to show that there is no real number whose square is 2, because our proof used not only P1–P12 but also a special property of $\mathbf{Q}$, the fact that every number in $\mathbf{Q}$ can be written p/q for integers p and q.)

This particular deficiency in our list of properties of the real numbers could, of course, be corrected by adding a new property which asserts the existence of square roots of positive numbers. Resorting to such a measure is, however, neither aesthetically pleasing nor mathematically satisfactory; we would still not know that every number has an nth root if n is odd, and that every positive number has an nth root if n is even. Even if we assumed this, we could not prove the existence of a number x satisfying $x^5 + x + 1 = 0$ (even though there does happen to be one), since we do not know how to write the solution of the equation in terms of nth roots (in fact, it is known that the solution cannot be written in this form). And, of course, we certainly do not wish to assume that all equations have solutions, since this is false (no real number x satisfies $x^2 + 1 = 0$, for example). In fact, this direction of investigation is not a fruitful one. The most useful hints about the property distinguishing $\mathbf{R}$ from $\mathbf{Q}$, the most compelling evidence for the necessity of elucidating this property, do not come from the study of numbers alone. In order to study the properties of the real numbers in a more profound way, we

must study more than the real numbers. At this point we must begin with the foundations of calculus, in particular the fundamental concept on which calculus is based—functions.

PROBLEMS

1. Prove the following formulas by induction.

(i) $1^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$.

(ii) $1^3 + \cdots + n^3 = (1 + \cdots + n)^2$.

2. Find a formula for

(i) $\displaystyle\sum_{i=1}^{n}(2i-1) = 1 + 3 + 5 + \cdots + (2n-1)$.

(ii) $\displaystyle\sum_{i=1}^{n}(2i-1)^2 = 1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2$.

Hint: What do these expressions have to do with $1 + 2 + 3 + \cdots + 2n$ and $1^2 + 2^2 + 3^2 + \cdots + (2n)^2$?

3. If $0 \le k \le n$, the "binomial coefficient" $\binom{n}{k}$ is defined by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}, \quad \text{if } k \neq 0, n$$

$\binom{n}{0} = \binom{n}{n} = 1$. (This becomes a special case of the first formula if we define $0! = 1$.)

(a) Prove that

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

(The proof does not require an induction argument.)

This relation gives rise to the following configuration, known as "Pascal's triangle"—a number not on one of the sides is the sum of the two numbers above it; the binomial coefficient $\binom{n}{k}$ is the $(k+1)$st number in the $(n+1)$st row.

$$\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1 \\
 & & & & & \cdots & & & & &
\end{array}$$

(b) Notice that all the numbers in Pascal's triangle are natural numbers. Use part (a) to prove by induction that $\binom{n}{k}$ is always a natural number. (Your entire proof by induction will, in a sense, be summed up in a glance by Pascal's triangle.)

(c) Give another proof that $\binom{n}{k}$ is a natural number by showing that $\binom{n}{k}$ is the number of sets of exactly k integers each chosen from $1, \ldots, n$.

(d) Prove the "binomial theorem": If a and b are any numbers and n is a natural number, then

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + b^n$$
$$= \sum_{j=0}^{n}\binom{n}{j}a^{n-j}b^j.$$

(e) Prove that

(i) $$\sum_{j=0}^{n}\binom{n}{j} = \binom{n}{0} + \cdots + \binom{n}{n} = 2^n.$$

(ii) $$\sum_{j=0}^{n}(-1)^j\binom{n}{j} = \binom{n}{0} - \binom{n}{1} + \cdots \pm \binom{n}{n} = 0.$$

(iii) $$\sum_{l \text{ odd}}\binom{n}{l} = \binom{n}{1} + \binom{n}{3} + \cdots = 2^{n-1}.$$

(iv) $$\sum_{l \text{ even}}\binom{n}{l} = \binom{n}{0} + \binom{n}{2} + \cdots = 2^{n-1}.$$

4. (a) Prove that

$$\sum_{k=0}^{l}\binom{n}{k}\binom{m}{l-k} = \binom{n+m}{l}.$$

Hint: Apply the binomial theorem to $(1+x)^n(1+x)^m$.

(b) Prove that

$$\sum_{k=0}^{n}\binom{n}{k}^2 = \binom{2n}{n}.$$

5. (a) Prove by induction on n that

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$

if $r \neq 1$ (if $r = 1$, evaluating the sum certainly presents no problem).

(b) Derive this result by setting $S = 1 + r + \cdots + r^n$, multiplying this equation by r, and solving the two equations for S.

6. The formula for $1^2 + \cdots + n^2$ may be derived as follows. We begin with the formula

$$(k+1)^3 - k^3 = 3k^2 + 3k + 1.$$

Writing this formula for $k = 1, \ldots, n$ and adding, we obtain

$$\begin{aligned}
2^3 - 1^3 &= 3 \cdot 1^2 + 3 \cdot 1 + 1 \\
3^3 - 2^3 &= 3 \cdot 2^2 + 3 \cdot 2 + 1 \\
&\vdots \\
(n+1)^3 - n^3 &= 3 \cdot n^2 + 3 \cdot n + 1 \\
\hline
(n+1)^3 - 1 \quad &= 3[1^2 + \cdots + n^2] + 3[1 + \cdots + n] + n.
\end{aligned}$$

Thus we can find $\sum_{k=1}^{n} k^2$ if we already know $\sum_{k=1}^{n} k$ (which could have been found in a similar way). Use this method to find

(i) $1^3 + \cdots + n^3$.

(ii) $1^4 + \cdots + n^4$.

(iii) $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \cdots + \dfrac{1}{n(n+1)}$.

(iv) $\dfrac{3}{1^2 \cdot 2^2} + \dfrac{5}{2^2 \cdot 3^2} + \cdots + \dfrac{2n+1}{n^2(n+1)^2}$.

***7.** Use the method of Problem 6 to show that $\sum_{i=1}^{n} k^p$ can always be written in the form

$$\frac{n^{p+1}}{p+1} + An^p + Bn^{p-1} + Cn^{p-2} + \cdots .$$

(The first 10 such expressions are

$$\sum_{k=1}^{n} k = \tfrac{1}{2}n^2 + \tfrac{1}{2}n$$

$$\sum_{k=1}^{n} k^2 = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n$$

$$\sum_{k=1}^{n} k^3 = \tfrac{1}{4}n^4 + \tfrac{1}{2}n^3 + \tfrac{1}{4}n^2$$

$$\sum_{k=1}^{n} k^4 = \tfrac{1}{5}n^5 + \tfrac{1}{2}n^4 + \tfrac{1}{3}n^3 - \tfrac{1}{30}n$$

$$\sum_{k=1}^{n} k^5 = \tfrac{1}{6}n^6 + \tfrac{1}{2}n^5 + \tfrac{5}{12}n^4 - \tfrac{1}{12}n^2$$

$$\sum_{k=1}^{n} k^6 = \tfrac{1}{7}n^7 + \tfrac{1}{2}n^6 + \tfrac{1}{2}n^5 - \tfrac{1}{6}n^3 + \tfrac{1}{42}n$$

$$\sum_{k=1}^{n} k^7 = \tfrac{1}{8}n^8 + \tfrac{1}{2}n^7 + \tfrac{7}{12}n^6 - \tfrac{7}{24}n^4 + \tfrac{1}{12}n^2$$

$$\sum_{k=1}^{n} k^8 = \tfrac{1}{9}n^9 + \tfrac{1}{2}n^8 + \tfrac{2}{3}n^7 - \tfrac{7}{15}n^5 + \tfrac{2}{9}n^3 - \tfrac{1}{30}n$$

$$\sum_{k=1}^{n} k^9 = \tfrac{1}{10}n^{10} + \tfrac{1}{2}n^9 + \tfrac{3}{4}n^8 - \tfrac{7}{10}n^6 + \tfrac{1}{2}n^4 - \tfrac{3}{20}n^2$$

$$\sum_{k=1}^{n} k^{10} = \tfrac{1}{11}n^{11} + \tfrac{1}{2}n^{10} + \tfrac{5}{6}n^9 - 1n^7 + 1n^5 - \tfrac{1}{2}n^3 + \tfrac{5}{66}n.$$

Notice that the coefficients in the second column are always $\frac{1}{2}$, and that after the third column the powers of n with nonzero coefficients decrease by 2 until n^2 or n is reached. The coefficients in all but the first two columns seem to be rather haphazard, but there actually is some sort of pattern; finding it may be regarded as a super-perspicacity test. See Problem 27-17 for the complete story.)

8. Prove that every natural number is either even or odd.

9. Prove that if a set A of natural numbers contains n_0 and contains $k + 1$ whenever it contains k, then A contains all natural numbers $\geq n_0$.

10. Prove the principle of mathematical induction from the well-ordering principle.

11. Prove the principle of complete induction from the ordinary principle of induction. Hint: If A contains 1 and A contains $n + 1$ whenever it contains $1, \ldots, n$, consider the set B of all k such that $1, \ldots, k$ are all in A.

12. (a) If a is rational and b is irrational, is $a + b$ necessarily irrational? What if a and b are both irrational?

(b) If a is rational and b is irrational, is ab necessarily irrational? (Careful!)
(c) Is there a number a such that a^2 is irrational, but a^4 is rational?
(d) Are there two irrational numbers whose sum and product are both rational?

13. (a) Prove that $\sqrt{3}$, $\sqrt{5}$, and $\sqrt{6}$ are irrational. Hint: To treat $\sqrt{3}$, for example, use the fact that every integer is of the form $3n$ or $3n+1$ or $3n+2$. Why doesn't this proof work for $\sqrt{4}$?
(b) Prove that $\sqrt[3]{2}$ and $\sqrt[3]{3}$ are irrational.

14. Prove that

(a) $\sqrt{2}+\sqrt{6}$ is irrational.
(b) $\sqrt{2}+\sqrt{3}$ is irrational.

15. (a) Prove that if $x = p+\sqrt{q}$ where p and q are rational, and m is a natural number, then $x^m = a+b\sqrt{q}$ for some rational a and b.
(b) Prove also that $(p-\sqrt{q}\,)^m = a-b\sqrt{q}$.

16. (a) Prove that if m and n are natural numbers and $m^2/n^2 < 2$, then $(m+2n)^2/(m+n)^2 > 2$; show, moreover, that

$$\frac{(m+2n)^2}{(m+n)^2} - 2 < 2 - \frac{m^2}{n^2}.$$

(b) Prove the same results with all inequality signs reversed.
(c) Prove that if $m/n < \sqrt{2}$, then there is another rational number m'/n' with $m/n < m'/n' < \sqrt{2}$.

***17.** It seems likely that $\sqrt{n}$ is irrational whenever the natural number n is not the square of another natural number. Although the method of Problem 13 may actually be used to treat any particular case, it is not clear in advance that it will always work, and a proof for the general case requires some extra information. A natural number p is called a **prime number** if it is impossible to write $p = ab$ for natural numbers a and b unless one of these is p, and the other 1; for convenience we also agree that 1 is *not* a prime number. The first few prime numbers are 2, 3, 5, 7, 11, 13, 17, 19. If $n > 1$ is not a prime, then $n = ab$, with a and b both $< n$; if either a or b is not a prime it can be factored similarly; continuing in this way proves that we can write n as a product of primes. For example, $28 = 4 \cdot 7 = 2 \cdot 2\dot{7}$.

(a) Turn this argument into a rigorous proof by complete induction. (To be sure, any reasonable mathematician would accept the informal argument, but this is partly because it would be obvious to her how to state it rigorously.)

A fundamental theorem about integers, which we will not prove here, states that this factorization is unique, except for the order of the factors. Thus, for example, 28 can never be written as a product of primes one of which

is 3, nor can it be written in a way that involves 2 only once (now you should appreciate why 1 is not allowed as a prime).

(b) Using this fact, prove that $\sqrt{n}$ is irrational unless $n = m^2$ for some natural number m.

(c) Prove more generally that $\sqrt[k]{n}$ is irrational unless $n = m^k$.

(d) No discussion of prime numbers should fail to allude to Euclid's beautiful proof that there are infinitely many of them. Prove that there cannot be only finitely many prime numbers $p_1, p_2, p_3, \ldots, p_n$ by considering $p_1 \cdot p_2 \cdot \ldots \cdot p_n + 1$.

***18.** (a) Prove that if x satisfies

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

for some integers $a_{n-1}, \ldots, a_0$, then x is irrational unless x is an integer. (Why is this a generalization of Problem 17?)

(b) Prove that $\sqrt{6} - \sqrt{2} - \sqrt{3}$ is irrational.

(c) Prove that $\sqrt{2} + \sqrt[3]{2}$ is irrational. Hint: Start by working out the first 6 powers of this number.

19. Prove Bernoulli's inequality: If $h > -1$, then

$$(1 + h)^n \geq 1 + nh.$$

Why is this trivial if $h > 0$?

20. The Fibonacci sequence $a_1, a_2, a_3, \ldots$ is defined as follows:

$$\begin{aligned} a_1 &= 1, \\ a_2 &= 1, \\ a_n &= a_{n-1} + a_{n-2} \qquad \text{for } n \geq 3. \end{aligned}$$

This sequence, which begins 1, 1, 2, 3, 5, 8, ..., was discovered by Fibonacci (circa 1175–1250), in connection with a problem about rabbits. Fibonacci assumed that an initial pair of rabbits gave birth to one new pair of rabbits per month, and that after two months each new pair behaved similarly. The number a_n of pairs born in the nth month is $a_{n-1} + a_{n-2}$, because a pair of rabbits is born for each pair born the previous month, and moreover each pair born two months ago now gives birth to another pair. The number of interesting results about this sequence is truly amazing—there is even a Fibonacci Association which publishes a journal, *The Fibonacci Quarterly*. Prove that

$$a_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

One way of deriving this astonishing formula is presented in Problem 24-15.

21. The Schwarz inequality (Problem 1-19) actually has a more general form:

$$\sum_{i=1}^{n} x_i y_i \leq \sqrt{\sum_{i=1}^{n} {x_i}^2} \sqrt{\sum_{i=1}^{n} {y_i}^2}.$$

Give three proofs of this, analogous to the three proofs in Problem 1-19.

22. The result in Problem 1-7 has an important generalization: If $a_1, \ldots, a_n \geq 0$, then the "arithmetic mean"

$$A_n = \frac{a_1 + \cdots + a_n}{n}$$

and "geometric mean"

$$G_n = \sqrt[n]{a_1 \ldots a_n}$$

satisfy

$$G_n \leq A_n.$$

(a) Suppose that $a_1 < A_n$. Then some a_i satisfies $a_i > A_n$; for convenience, say $a_2 > A_n$. Let $\bar{a}_1 = A_n$ and let $\bar{a}_2 = a_1 + a_2 - \bar{a}_1$. Show that

$$\bar{a}_1 \bar{a}_2 \geq a_1 a_2.$$

Why does repeating this process enough times eventually prove that $G_n \leq A_n$? (This is another place where it is a good exercise to provide a formal proof by induction, as well as an informal reason.) When does equality hold in the formula $G_n \leq A_n$?

The reasoning in this proof is related to another interesting proof.

(b) Using the fact that $G_n \leq A_n$ when $n = 2$, prove, by induction on k, that $G_n \leq A_n$ for $n = 2^k$.

(c) For a general n, let $2^m > n$. Apply part (b) to the 2^m numbers

$$a_1, \ldots, a_n, \underbrace{A_n, \ldots, A_n}_{2^m - n \text{ times}}$$

to prove that $G_n \leq a_n$.

23. The following is a recursive definition of a^n:

$$\begin{aligned} a^1 &= a, \\ a^{n+1} &= a^n \cdot a. \end{aligned}$$

Prove, by induction, that

$$\begin{aligned} a^{n+m} &= a^n \cdot a^m, \\ (a^n)^m &= a^{nm}. \end{aligned}$$

(Don't try to be fancy: use either induction on n or induction on m, not both at once.)

24. Suppose we know properties P1 and P4 for the natural numbers, but that multiplication has never been mentioned. Then the following can be used as a recursive definition of multiplication:

$$1 \cdot b = b,$$
$$(a+1) \cdot b = a \cdot b + b.$$

Prove the following (in the order suggested!):

$a \cdot (b+c) = a \cdot b + a \cdot c$ (use induction on a),
$a \cdot 1 = a$,
$a \cdot b = b \cdot a$ (you just finished proving the case $b = 1$).

25. In this chapter we began with the natural numbers and gradually built up to the real numbers. A completely rigorous discussion of this process requires a little book in itself (see Part V). No one has ever figured out how to get to the real numbers without going through this process, but if we do accept the real numbers as given, then the natural numbers can be *defined* as the real numbers of the form 1, $1+1$, $1+1+1$, etc. The whole point of this problem is to show that there is a rigorous mathematical way of saying "etc."

(a) A set A of real numbers is called **inductive** if

(1) 1 is in A,
(2) $k+1$ is in A whenever k is in A.

Prove that

(i) **R** is inductive.
(ii) The set of positive real numbers is inductive.
(iii) The set of positive real numbers unequal to $\frac{1}{2}$ is inductive.
(iv) The set of positive real numbers unequal to 5 is not inductive.
(v) If A and B are inductive, then the set C of real numbers which are in both A and B is also inductive.

(b) A real number n will be called a **natural number** if n is in *every* inductive set.

(i) Prove that 1 is a natural number.
(ii) Prove that $k+1$ is a natural number if k is a natural number.

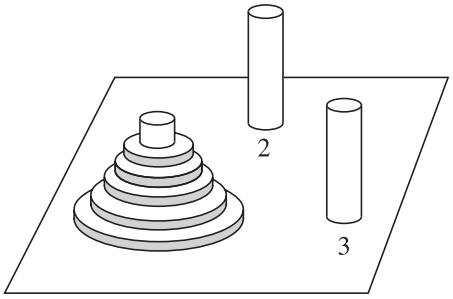


FIGURE 1

26. There is a puzzle consisting of three spindles, with n concentric rings of decreasing diameter stacked on the first (Figure 1). A ring at the top of a stack may be moved from one spindle to another spindle, provided that it is not placed on top of a smaller ring. For example, if the smallest ring is moved to spindle 2 and the next-smallest ring is moved to spindle 3, then the smallest ring may be moved to spindle 3 also, on top of the next-smallest. Prove that the entire stack of n rings can be moved onto spindle 3 in $2^n - 1$ moves, and that this cannot be done in fewer than $2^n - 1$ moves.

***27.** University B. once boasted 17 tenured professors of mathematics. Tradition prescribed that at their weekly luncheon meeting, faithfully attended by all 17, any members who had discovered an error in their published work should make an announcement of this fact, and promptly resign. Such an announcement had never actually been made, because no professor was aware of any errors in her or his work. This is not to say that no errors existed, however. In fact, over the years, in the work of every member of the department at least one error had been found, by some other member of the department. This error had been mentioned to all other members of the department, but the actual author of the error had been kept ignorant of the fact, to forestall any resignations.

One fateful year, the department was augmented by a visitor from another university, one Prof. X, who had come with hopes of being offered a permanent position at the end of the academic year. Naturally, he was apprised, by various members of the department, of the published errors which had been discovered. When the hoped-for appointment failed to materialize, Prof. X obtained his revenge at the last luncheon of the year. "I have enjoyed my visit here very much," he said, "but I feel that there is one thing that I have to tell you. At least one of you has published an incorrect result, which has been discovered by others in the department." What happened the next year?

****28.** After figuring out, or looking up, the answer to Problem 27, consider the following: Each member of the department already knew what Prof. X asserted, so how could his saying it change anything?

PART 2
FOUNDATIONS

The statement is so frequently made
that the differential calculus deals with
continuous magnitude, and yet
an explanation of this continuity is
nowhere given;
even the most rigorous expositions
of the differential calculus do not base
their proofs upon continuity but,
with more or less consciousness of the fact,
they either appeal to geometric notions
or those suggested by geometry,
or depend upon theorems which are never
established in a purely arithmetic manner.
Among these, for example,
belongs the above-mentioned theorem,
and a more careful investigation
convinced me that this theorem, or
or any one equivalent to it, can be regarded
in some way as a sufficient basis
for infinitesimal analysis.
It then only remained to discover its true
origin in the elements of arithmetic
and thus at the same time
to secure a real definition of
the essence of continuity
I succeeded Nov. 24, 1858, and
a few days afterward I communicated
the results
of my meditations to my dear friend
Durège with whom I had a long
and lively discussion.

RICHARD DEDEKIND

CHAPTER 3 FUNCTIONS

Undoubtedly the most important concept in all of mathematics is that of a function—in almost every branch of modern mathematics functions turn out to be the central objects of investigation. It will therefore probably not surprise you to learn that the concept of a function is one of great generality. Perhaps it will be a relief to learn that, for the present, we will be able to restrict our attention to functions of a very special kind; even this small class of functions will exhibit sufficient variety to engage our attention for quite some time. We will not even begin with a proper definition. For the moment a provisional definition will enable us to discuss functions at length, and will illustrate the intuitive notion of functions, as understood by mathematicians. Later, we will consider and discuss the advantages of the modern mathematical definition. Let us therefore begin with the following:

PROVISIONAL DEFINITION A function is a rule which assigns, to each of certain real numbers, some other real number.

The following examples of functions are meant to illustrate and amplify this definition, which, admittedly, requires some such clarification.

Example 1 The rule which assigns to each number the square of that number.

Example 2 The rule which assigns to each number y the number

$$\frac{y^3 + 3y + 5}{y^2 + 1}.$$

Example 3 The rule which assigns to each number $c \neq 1, -1$ the number

$$\frac{c^3 + 3c + 5}{c^2 - 1}.$$

Example 4 The rule which assigns to each number x satisfying $-17 \leq x \leq \pi/3$ the number x^2.

Example 5 The rule which assigns to each number a the number 0 if a is irrational, and the number 1 if a is rational.

Example 6 The rule which assigns

$$\text{to 2 the number 5,}$$
$$\text{to 17 the number } \frac{36}{\pi},$$

$$\text{to } \frac{\pi^2}{17} \text{ the number 28,}$$

$$\text{to } \frac{36}{\pi} \text{ the number 28,}$$

and to any $y \neq 2$, 17, $\pi^2/17$, or $36/\pi$, the number 16 if y is of the form $a + b\sqrt{2}$ for a, b in $\mathbf{Q}$.

Example 7 The rule which assigns to each number t the number $t^3 + x$. (This rule depends, of course, on what the number x is, so we are really describing infinitely many different functions, one for each number x.)

Example 8 The rule which assigns to each number z the number of 7's in the decimal expansion of z, if this number is finite, and $-\pi$ if there are infinitely many 7's in the decimal expansion of z.

One thing should be abundantly clear from these examples—a function is *any* rule that assigns numbers to certain other numbers, not just a rule which can be expressed by an algebraic formula, or even by one uniform condition which applies to every number; nor is it necessarily a rule which you, or anybody else, can actually apply in practice (no one knows, for example, what rule 8 associates to π). Moreover, the rule may neglect some numbers and it may not even be clear to which numbers the function applies (try to determine, for example, whether the function in Example 6 applies to π). The set of numbers to which a function *does* apply is called the *domain* of the function.

Before saying anything else about functions we badly need some notation. Since throughout this book we shall frequently be talking about functions (indeed we shall hardly ever talk about anything else) we need a convenient way of naming functions, and of referring to functions in general. The standard practice is to denote a function by a letter. For obvious reasons the letter "f" is a favorite, thereby making "g" and "h" other obvious candidates, but any letter (or any reasonable symbol, for that matter) will do, not excluding "x" and "y", although these letters are usually reserved for indicating numbers. If f is a function, then the number which f associates to a number x is denoted by $f(x)$—this symbol is read "f of x" and is often called the **value of f at x**. Naturally, if we denote a function by x, some other letter must be chosen to denote the number (a perfectly legitimate, though perverse, choice would be "f," leading to the symbol $x(f)$). Note that the symbol $f(x)$ makes sense only for x in the domain of f; for other x the symbol $f(x)$ is not defined.

If the functions defined in Examples 1–8 are denoted by f, g, h, r, s, θ, α_x, and y, then we can rewrite their definitions as follows:

(1) $f(x) = x^2$ for all x.

(2) $g(y) = \dfrac{y^3 + 3y + 5}{y^2 + 1}$ for all y.

(3) $h(c) = \dfrac{c^3 + 3c + 5}{c^2 - 1}$ for all $c \neq 1, -1$.

(4) $r(x) = x^2$ for all x such that $-17 \leq x \leq \pi/3$.

(5) $$s(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational.} \end{cases}$$

(6) $$\theta(x) = \begin{cases} 5, & x = 2 \\ \dfrac{36}{\pi}, & x = 17 \\ 28, & x = \dfrac{\pi^2}{17} \\ 28, & x = \dfrac{36}{\pi} \\ 16, & x \neq 2,\ 17,\ \dfrac{\pi^2}{17}, \text{ or } \dfrac{36}{\pi}, \text{ and } x = a + b\sqrt{2} \text{ for } a, b \text{ in } \mathbf{Q}. \end{cases}$$

(7) $\alpha_x(t) = t^3 + x$ for all numbers t.

(8) $$y(x) = \begin{cases} n, & \text{exactly } n \text{ 7's appear in the decimal expansion of } x \\ -\pi, & \text{infinitely many 7's appear in the decimal expansion of } x. \end{cases}$$

These definitions illustrate the common procedure adopted for defining a function f—indicating what $f(x)$ is for every number x in the domain of f. (Notice that this is exactly the same as indicating $f(a)$ for every number a, or $f(b)$ for every number b, etc.) In practice, certain abbreviations are tolerated. Definition (1) could be written simple

$$(1) \quad f(x) = x^2$$

the qualifying phrase "for all x" being understood. Of course, for definition (4) the only possible abbreviation is

$$(4) \quad r(x) = x^2, \qquad -17 \leq x \leq \pi/3.$$

It is usually understood that a definition such as

$$k(x) = \frac{1}{x} + \frac{1}{x-1}, \quad x \neq 0, 1$$

can be shortened to

$$k(x) = \frac{1}{x} + \frac{1}{x-1};$$

in other words, *unless the domain is explicitly restricted further, it is understood to consist of all numbers for which the definition makes any sense at all.*

You should have little difficulty checking the following assertions about the functions defined above:

$$f(x+1) = f(x) + 2x + 1;$$
$$g(x) = h(x) \text{ if } x^3 + 3x + 5 = 0;$$
$$r(x+1) = r(x) + 2x + 1 \text{ if } -17 \leq x \leq \frac{\pi}{3} - 1;$$

$$s(x+y) = s(x) \text{ if } y \text{ is rational};$$
$$\theta\left(\frac{\pi^2}{17}\right) = \theta\left(\frac{36}{\pi}\right);$$
$$\alpha_x(x) = x \cdot [f(x) + 1];$$
$$y\left(\frac{1}{3}\right) = 0, \; y\left(\frac{7}{9}\right) = -\pi.$$

If the expression $f(s(a))$ looks unreasonable to you, then you are forgetting that $s(a)$ is a number like any other number, so that $f(s(a))$ makes sense. As a matter of fact, $f(s(a)) = s(a)$ for all a. Why? Even more complicated expressions than $f(s(a))$ are, after a first exposure, no more difficult to unravel. The expression

$$f(r(s(\theta(\alpha_3(y(\tfrac{1}{3})))))),$$

formidable as it appears, may be evaluated quite easily with a little patience:

$$\begin{aligned} & f(r(s(\theta(\alpha_3(y(\tfrac{1}{3}))))))\\ &= f(r(s(\theta(\alpha_3(0))))) \\ &= f(r(s(\theta(3)))) \\ &= f(r(s(16))) \\ &= f(r(1)) \\ &= f(1) \\ &= 1. \end{aligned}$$

The first few problems at the end of this chapter give further practice manipulating this symbolism.

The function defined in (1) is a rather special example of an extremely important class of functions, the polynomial functions. A function f is a **polynomial function** if there are real numbers $a_0, \ldots, a_n$ such that

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0, \qquad \text{for all } x$$

(when $f(x)$ is written in this form it is usually tacitly assumed that $a_n \neq 0$). The highest power of x with a nonzero coefficient is called the **degree** of f; for example, the polynomial function f defined by $f(x) = 5x^6 + 137x^4 - \pi$ has degree 6.

The functions defined in (2) and (3) belong to a somewhat larger class of functions, the **rational functions**; these are the functions of the form p/q where p and q are polynomial functions (and q is not the function which is always 0). The rational functions are themselves quite special examples of an even larger class of functions, very thoroughly studied in calculus, which are simpler than many of the functions first mentioned in this chapter. The following are examples of this kind of function:

(9) $\quad f(x) = \dfrac{x + x^2 + x\sin^2 x}{x\sin x + x\sin^2 x}$

(10) $\quad f(x) = \sin(x^2).$

(11) $\quad f(x) = \sin(\sin(x^2)).$

$$(12)\quad f(x) = \sin^2(\sin(\sin^2(x \sin^2 x^2))) \cdot \sin\left(\frac{x + \sin(x \sin x)}{x + \sin x}\right).$$

By what criterion, you may feel impelled to ask, can such functions, especially a monstrosity like (12), be considered simple? The answer is that they can be built up from a few simple functions using a few simple means of combining functions. In order to construct the functions (9)–(12) we need to start with the "identity function" I, for which $I(x) = x$, and the "sine function" sin, whose value $\sin(x)$ at x is often written simple $\sin x$. The following are some of the important ways in which functions may be combined to produce new functions.

If f and g are any two functions, we can define a new function $f + g$, called the **sum** of f and g, by the equation

$$(f + g)(x) = f(x) + g(x).$$

Note that according to the conventions we have adopted, the domain of $f + g$ consists of all x for which "$f(x) + g(x)$" makes sense, i.e., the set of all x in both domain f and domain g. If A and B are any two sets, then $A \cap B$ (read "A intersect B" or "the intersection of A and B") denotes the set of x in both A and B; this notation allows us to write $\text{domain}(f + g) = \text{domain } f \cap \text{domain } g$.

In a similar vein, we define the **product** $f \cdot g$ and the **quotient** $\frac{f}{g}$ (or f/g) of f and g by

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

and

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Moreover, if g is a function and c is a number, we define a new function $c \cdot g$ by

$$(c \cdot g)(x) = c \cdot g(x).$$

This becomes a special case of the notation $f \cdot g$ if we agree that the symbol c should also represent the function f defined by $f(x) = c$; such a function, which has the same value for all numbers x, is called a **constant function**.

The domain of $f \cdot g$ is $\text{domain } f \cap \text{domain } g$, and the domain of $c \cdot g$ is simply the domain of g. On the other hand, the domain of f/g is rather complicated—it may be written $\text{domain } f \cap \text{domain } g \cap \{x : g(x) \neq 0\}$, the symbol $\{x : g(x) \neq 0\}$ denoting the set of numbers x such that $g(x) \neq 0$. In general, $\{x : \ldots\}$ denotes the set of all x such that " $\ldots$ " is true. Thus $\{x : x^3 + 3 < 11\}$ denotes the set of all numbers x such that $x^3 < 8$, and consequently $\{x : x^3 + 3 < 11\} = \{x : x < 2\}$. Either of these symbols could just as well have been written using y everywhere instead of x. Variations of this notation are common, but hardly require any discussion. Any one can guess that $\{x > 0 : x^3 < 8\}$ denotes the set of positive numbers whose cube is less than 8; it could be expressed more formally as $\{x : x > 0 \text{ and } x^3 < 8\}$. Incidentally, this set is equal to the set $\{x : 0 < x < 2\}$. One

variation is slightly less transparent, but very standard. The set $\{1, 3, 2, 4\}$, for example, contains just the four numbers 1, 2, 3, and 4; it can also be denoted by $\{x : x = 1 \text{ or } x = 3 \text{ or } x = 2 \text{ or } x = 4\}$.

Certain facts about the sum, product, and quotient of functions are obvious consequences of facts about sums, products, and quotients of numbers. For example, it is very easy to prove that

$$(f + g) + h = f + (g + h).$$

The proof is characteristic of almost every proof which demonstrates that two functions are equal—the two functions must be shown to have the same domain, and the same value at any number in the domain. For example, to prove that $(f+g)+h = f+(g+h)$, note that unraveling the definition of the two sides gives

$$\begin{aligned}[(f + g) + h](x) &= (f + g)(x) + h(x) \\ &= [f(x) + g(x)] + h(x)\end{aligned}$$

and

$$\begin{aligned}[f + (g + h)](x) &= f(x) + (g + h)(x) \\ &= f(x) + [g(x) + h(x)],\end{aligned}$$

and the equality of $[f(x) + g(x)] + h(x)$ and $f(x) + [g(x) + h(x)]$ is a fact about numbers. In this proof the equality of the two domains was not explicitly mentioned because this is obvious, as soon as we begin to write down these equations; the domain of $(f + g) + h$ and of $f + (g + h)$ is clearly domain $f \cap$ domain $g \cap$ domain h. We naturally write $f + g + h$ for $(f + g) + h = f + (g + h)$, precisely as we did for numbers.

It is just as easy to prove that $(f \cdot g) \cdot h = f \cdot (g \cdot h)$, and this function is denoted by $f \cdot g \cdot h$. The equations $f + g = g + f$ and $f \cdot g = g \cdot f$ should also present no difficulty.

Using the operations $+, \cdot, /$ we can now express the function f defined in (9) by

$$f = \frac{I + I \cdot I + I \cdot \sin \cdot \sin}{I \cdot \sin + I \cdot \sin \cdot \sin}.$$

It should be clear, however, that we cannot express function (10) this way. We require yet another way of combining functions. This combination, the composition of two functions, is by far the most important.

If f and g are any two functions, we define a new function $f \circ g$, the **composition** of f and g, by

$$(f \circ g)(x) = f(g(x));$$

the domain of $f \circ g$ is $\{x : x \text{ is in domain } g \text{ and } g(x) \text{ is in domain } f\}$. The symbol "$f \circ g$" is often read "$f$ circle g." Compared to the phrase "the composition of f and g" this has the advantage of brevity, of course, but there is another advantage of far greater import: there is much less chance of confusing $f \circ g$ with $g \circ f$, and

these must *not* be confused, since they are not usually equal; in fact, almost any f and g chosen at random will illustrate this point (try $f = I \cdot I$ and $g = \sin$, for example). Lest you become too apprehensive about the operation of composition, let us hasten to point out that composition *is* associative:

$$(f \circ g) \circ h = f \circ (g \circ h)$$

(and the proof is a triviality); this function is denoted by $f \circ g \circ h$. We can now write the functions (10), (11), (12) as

(10) $f = \sin \circ (I \cdot I)$,

(11) $f = \sin \circ \sin \circ (I \cdot I)$,

(12) $f = (\sin \cdot \sin) \circ \sin \circ (\sin \cdot \sin) \circ (I \cdot [(\sin \cdot \sin) \circ (I \cdot I)]) \cdot \sin \circ \left(\dfrac{I + \sin \circ (I \cdot \sin)}{I + \sin} \right)$.

One fact has probably already become clear. Although this method of writing functions reveals their "structure" very clearly, it is hardly short or convenient. The shortest name for the function f such that $f(x) = \sin(x^2)$ for all x unfortunately seems to be "the function f such that $f(x) = \sin(x^2)$ for all x." The need for abbreviating this clumsy description has been clear for two hundred years, but no reasonable abbreviation has received universal acclaim. At present the strongest contender for this honor is something like

$$x \to \sin(x^2)$$

(read "x goes to $\sin(x^2)$" or just "x arrow $\sin(x^2)$"), but it is hardly popular among writers of calculus textbooks. In this book we will tolerate a certain amount of ellipsis, and speak of "the function $f(x) = \sin(x^2)$." Even more popular is the quite drastic abbreviation: "the function $\sin(x^2)$." For the sake of precision we will never use this description, which, strictly speaking, confuses a number and a function, but it is so convenient that you will probably end up adopting it for personal use. As with any convention, utility is the motivating factor, and this criterion is reasonable so long as the slight logical deficiencies cause no confusion. On occasion, confusion *will* arise unless a more precise description is used. For example, "the function $x + t^3$" is an ambiguous phrase; it could mean either

$$x \to x + t^3, \text{ i.e., the function } f \text{ such that } f(x) = x + t^3 \text{ for all } x$$

or

$$t \to x + t^3, \text{ i.e., the function } f \text{ such that } f(t) = x + t^3 \text{ for all } t.$$

As we shall see, however, for many important concepts associated with functions, calculus has a notation which contains the "$x \to$" built in.

By now we have made a sufficiently extensive investigation of functions to warrant reconsidering our definition. We have defined a function as a "rule," but it is hardly clear what this means. If we ask "What happens if you break this rule?" it is not easy to say whether this question is merely facetious or actually profound.

A more substantial objection to the use of the word "rule" is that

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - 3(x + 1)$$

are certainly *different* rules, if by a rule we mean the actual instructions given for determining $f(x)$; nevertheless, we want

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - c(x + 1)$$

to define the same function. For this reason, a function is sometimes defined as an "association" between numbers; unfortunately the word "association" escapes the objections raised against "rule" only because it is even more vague.

There is, of course, a satisfactory way of defining functions, or we should never have gone to the trouble of criticizing our original definition. But a satisfactory definition can never be constructed by finding synonyms for English words which are troublesome. The definition which mathematicians have finally accepted for "function" is a beautiful example of the means by which intuitive ideas have been incorporated into rigorous mathematics. The correct question to ask about a function is not "What is a rule?" or "What is an association?" but "What does one have to know about a function in order to know all about it?" The answer to the last question is easy—for each number x one needs to know the number $f(x)$; we can imagine a table which would display all the information one could desire about the function $f(x) = x^2$:

x	$f(x)$
1	1
-1	1
2	4
-2	4
$\sqrt{2}$	2
$-\sqrt{2}$	2
π	π^2
$-\pi$	π^2

It is not even necessary to arrange the numbers in a table (which would actually be impossible if we wanted to list all of them). Instead of a two column array we can consider various pairs of numbers

$$(1, 1),\ (-1, 1),\ (2, 4),\ (-2, 4),\ (\pi, \pi^2),\ (\sqrt{2}, 2), \ldots$$

simply collected together into a set.* To find $f(1)$ we simply take the second number of the pair whose first member is 1; to find $f(\pi)$ we take the second number of the pair whose first member is π. We seem to be saying that a function might as well be defined as a collection of pairs of numbers. For example, if we were given the following collection (which contains just 5 pairs):

$$f = \{(1,7),\ (3,7),\ (5,3),\ (4,8),\ (8,4)\},$$

then $f(1) = 7$, $f(3) = 7$, $f(5) = 3$, $f(4) = 8$, $f(8) = 4$ and 1, 3, 4, 5, 8 are the only numbers in the domain of f. If we consider the collection

$$f = \{(1,7),\ (3,7),\ (2,5),\ (1,8),\ (8,4)\},$$

then $f(3) = 7$, $f(2) = 5$, $f(8) = 4$; but it is impossible to decide whether $f(1) = 7$ or $f(1) = 8$. In other words, a function cannot be defined to be any old collection of pairs of numbers; we must rule out the possibility which arose in this case. We are therefore led to the following definition.

DEFINITION

A **function** is a collection of pairs of numbers with the following property: if (a, b) and (a, c) are both in the collection, then $b = c$; in other words, the collection must not contain two different pairs with the same first element.

This is our first full-fledged definition, and illustrates the format we shall always use to define significant new concepts. These definitions are so important (at least as important as theorems) that it is essential to know when one is actually at hand, and to distinguish them from comments, motivating remarks, and casual explanations. They will be preceded by the word **DEFINITION**, contain the term being defined in boldface letters, and constitute a paragraph unto themselves.

There is one more definition (actually defining two things at once) which can now be made rigorously:

DEFINITION

If f is a function, the **domain** of f is the set of all a for which there is some b such that (a, b) is in f. If a is in the domain of f, it follows from the definition of a function that there is, in fact, a *unique* number b such that (a, b) is in f. This unique b is denoted by $\boldsymbol{f(a)}$.

With this definition we have reached our goal: the important thing about a function f is that a number $f(x)$ is determined for each number x in its domain. You may feel that we have also reached the point where an intuitive definition has been replaced by an abstraction with which the mind can hardly grapple. Two consolations may be offered. First, although a function has been defined as a

*The pairs occurring here are often called "ordered pairs," to emphasize that, for example, (2, 4) is not the same pair as (4, 2). It is only fair to warn that we are going to define functions in terms of ordered pairs, another undefined term. Ordered pairs *can* be defined, however, and an appendix to this chapter has been provided for skeptics.

collection of pairs, there is nothing to stop you from *thinking* of a function as a rule. Second, neither the intuitive nor the formal definition indicates the best way of thinking about functions. The best way is to draw pictures; but this requires a chapter all by itself.

PROBLEMS

1. Let $f(x) = 1/(1+x)$. What is

(i) $f(f(x))$ (for which x does this make sense?).

(ii) $f\left(\frac{1}{x}\right)$.

(iii) $f(cx)$.

(iv) $f(x+y)$.

(v) $f(x) + f(y)$.

(vi) For which numbers c is there a number x such that $f(cx) = f(x)$. Hint: There are a lot more than you might think at first glance.

(vii) For which numbers c is it true that $f(cx) = f(x)$ for two different numbers x?

2. Let $g(x) = x^2$, and let

$$h(x) = \begin{cases} 0, & x \text{ rational} \\ 1, & x \text{ irrational.} \end{cases}$$

(i) For which y is $h(y) \leq y$?

(ii) For which y is $h(y) \leq g(y)$?

(iii) What is $g(h(z)) - h(z)$?

(iv) For which w is $g(w) \leq w$?

(v) For which ε is $g(g(\varepsilon)) = g(\varepsilon)$?

3. Find the domain of the functions defined by the following formulas.

(i) $f(x) = \sqrt{1-x^2}$.

(ii) $f(x) = \sqrt{1-\sqrt{1-x^2}}$.

(iii) $f(x) = \dfrac{1}{x-1} + \dfrac{1}{x-2}$.

(iv) $f(x) = \sqrt{1-x^2} + \sqrt{x^2-1}$.

(v) $f(x) = \sqrt{1-x} + \sqrt{x-2}$.

4. Let $S(x) = x^2$, let $P(x) = 2^x$, and let $s(x) = \sin x$. Find each of the following. In each case you answer should be a *number*.

(i) $(S \circ P)(y)$.

(ii) $(S \circ s)(y)$.

(iii) $(S \circ P \circ s)(t) + (s \circ P)(t)$.

(iv) $s(t^3)$.

5. Express each of the following functions in terms of S, P, s, using only $+$, $\cdot$, and $\circ$ (for example, the answer to (i) is $P \circ s$). In each case your

answer should be a *function*.

(i) $f(x) = 2^{\sin x}$.
(ii) $f(x) = \sin 2^x$.
(iii) $f(x) = \sin x^2$.
(iv) $f(x) = \sin^2 x$ (remember that $\sin^2 x$ is an abbreviation for $(\sin x)^2$).
(v) $f(t) = 2^{2^t}$. (Note: a^{b^c} *always* means $a^{(b^c)}$; this convention is adopted because $(a^b)^c$ can be written more simply as a^{bc}.)
(vi) $f(u) = \sin(2^u + 2^{u^2})$.
(vii) $f(y) = \sin\big(\sin\big(\sin\big(2^{2^{2^{\sin y}}}\big)\big)\big)$.
(viii) $f(a) = 2^{\sin^2 a} + \sin(a^2) + 2^{\sin(a^2+\sin a)}$.

Polynomial functions, because they are simple, yet flexible, occupy a favored role in most investigations of functions. The following two problems illustrate their flexibility, and guide you through a derivation of their most important elementary properties.

6. (a) If $x_1, \ldots, x_n$ are distinct numbers, find a polynomial function f_i of degree $n-1$ which is 1 at x_i and 0 at x_j for $j \neq i$. Hint: the product of all $(x - x_i)$ for $j \neq i$, is 0 at x_j if $j \neq i$. (This product is usually denoted by

$$\prod_{\substack{j=1 \\ j\neq i}}^{n} (x - x_j),$$

the symbol Π (capital pi) playing the same role for products that Σ plays for sums.)

(b) Now find a polynomial function f of degree $n-1$ such that $f(x_i) = a_i$, where $a_1, \ldots, a_n$ are given numbers. (You should use the functions f_i from part (a). The formula you will obtain is called the "Lagrange interpolation formula.")

7. (a) Prove that for any polynomial function f, and any number a, there is a polynomial function g, and a number b, such that $f(x) = (x-a)g(x)+b$ for all x. (The idea is simply to divide $(x-a)$ into $f(x)$ by long division, until a constant remainder is left. For example, the calculation

$$\begin{array}{r|l} & x^2 + x - 2 \\ \hline x-1 & x^3 \qquad\quad -3x+1 \\ & x^3 - x^2 \\ \hline & \quad x^2 - 3x \\ & \quad x^2 - x \\ \hline & \qquad -2x+1 \\ & \qquad -2x+2 \\ \hline & \qquad\quad -1 \end{array}$$

shows that $x^3 - 3x + 1 = (x-1)(x^2 + x - 2) - 1$. A formal proof is possible by induction on the degree of f.)

(b) Prove that if $f(a) = 0$, then $f(x) = (x-a)g(x)$ for some polynomial function g. (The converse is obvious.)

(c) Prove that if f is a polynomial function of degree n, then f has at most n roots, i.e., there are at most n numbers a with $f(a) = 0$.

(d) Show that for each n there is a polynomial function of degree n with n roots. If n is even find a polynomial function of degree n with no roots, and if n is odd find one with only one root.

8. For which numbers a, b, c, and d will the function

$$f(x) = \frac{ax + b}{cx + d}$$

satisfy $f(f(x)) = x$ for all x?

9. (a) If A is any set of real numbers, define a function C_A as follows:

$$C_A(x) = \begin{cases} 1, & x \text{ in } A \\ 0, & x \text{ not in } A. \end{cases}$$

Find expressions for $C_{A\cap B}$ and $C_{A\cup B}$ and $C_{\mathbf{R}-A}$, in terms of C_A and C_B. (The symbol $A \cap B$ was defined in this chapter, but the other two may be new to you. They can be defined as follows:

$$A \cup B = \{x : x \text{ is in } A \text{ or } x \text{ is in } B\},$$
$$\mathbf{R} - A = \{x : x \text{ is in } \mathbf{R} \text{ but } x \text{ is not in } A\}.)$$

(b) Suppose f is a function such that $f(x) = 0$ or 1 for each x. Prove that there is a set A such that $f = C_A$.

(c) Show that $f = f^2$ if and only if $f = C_A$ for some set A.

10. (a) For which functions f is there a function g such that $f = g^2$? Hint: You can certainly answer this question if "function" is replaced by "number."

(b) For which functions f is there a function g such that $f = 1/g$?

*(c) For which functions b and c can we find a function x such that

$$(x(t))^2 + b(t)x(t) + c(t) = 0$$

for all numbers t?

*(d) What conditions must the functions a and b satisfy if there is to be a function x such that

$$a(t)x(t) + b(t) = 0$$

for all numbers t? How many such functions x will there be?

11. (a) Suppose that H is a function and y is a number such that $H(H(y)) = y$. What is

$$\underbrace{H(H(H(\cdots(H(y)\cdots)}_{80 \text{ times}})?$$

(b) Same question if 80 is replaced by 81.
(c) Same question if $H(H(y)) = H(y)$.
*(d) Find a function H such that $H(H(x)) = H(x)$ for all numbers x, and such that $H(1) = 36$, $H(2) = \pi/3$, $H(13) = 47$, $H(36) = 36$, $H(\pi/3) = \pi/3$, $H(47) = 47$. (Don't try to "solve" for $H(x)$; there are many functions H with $H(H(x)) = H(x)$. The extra conditions on H are supposed to suggest a way of finding a suitable H.)
*(e) Find a function H such that $H(H(x)) = H(x)$ for all x, and such that $H(1) = 7$, $H(17) = 18$.

12. A function f is **even** if $f(x) = f(-x)$ and **odd** if $f(x) = -f(-x)$. For example, f is even if $f(x) = x^2$ or $f(x) = |x|$ or $f(x) = \cos x$, while f is odd if $f(x) = x$ or $f(x) = \sin x$.

(a) Determine whether $f + g$ is even, odd, or not necessarily either, in the four cases obtained by choosing f even or odd, and g even or odd. (Your answers can most conveniently be displayed in a 2×2 table.)
(b) Do the same for $f \cdot g$.
(c) Do the same for $f \circ g$.
(d) Prove that every even function f can be written $f(x) = g(|x|)$, for infinitely many functions g.

***13.** (a) Prove that any function f with domain $\mathbf{R}$ can be written $f = E + O$, where E is even and O is odd.
(b) Prove that this way of writing f is unique. (If you try to do part (b) first, by "solving" for E and O you will probably find the solution to part (a).)

14. If f is any function, define a new function $|f|$ by $|f|(x) = |f(x)|$. If f and g are functions, define two new functions, $\max(f, g)$ and $\min(f, g)$, by

$$\max(f, g)(x) = \max(f(x), g(x)),$$
$$\min(f, g)(x) = \min(f(x), g(x)).$$

Find an expression for $\max(f, g)$ and $\min(f, g)$ in terms of $|\ |$.

15. (a) Show that $f = \max(f, 0) + \min(f, 0)$. This particular way of writing f is fairly useful; the functions $\max(f, 0)$ and $\min(f, 0)$ are called the **positive** and **negative parts** of f.
(b) A function f is called **nonnegative** if $f(x) \geq 0$ for all x. Prove that any function f can be written $f = g - h$, where g and h are nonnegative, in infinitely many ways. (The "standard way" is $g = \max(f, 0)$ and $h = -\min(f, 0)$.) Hint: Any *number* can certainly be written as the difference of two nonnegative *numbers* in infinitely many ways.

***16.** Suppose f satisfies $f(x + y) = f(x) + f(y)$ for all x and y.

(a) Prove that $f(x_1 + \cdots + x_n) = f(x_1) + \cdots + f(x_n)$.
(b) Prove that there is some number c such that $f(x) = cx$ for all *rational* numbers x (at this point we're not trying to say anything about $f(x)$ for irrational x). Hint: First figure out what c must be. Now prove that

$f(x) = cx$, first when x is a natural number, then when x is an integer, then when x is the reciprocal of an integer and, finally, for all rational x.

***17.** If $f(x) = 0$ for all x, then f satisfies $f(x+y) = f(x) + f(y)$ for all x and y, and also $f(x \cdot y) = f(x) \cdot f(y)$ for all x and y. Now suppose that f satisfies these two properties, but that $f(x)$ is not always 0. Prove that $f(x) = x$ for all x, as follows:

(a) Prove that $f(1) = 1$.
(b) Prove that $f(x) = x$ if x is rational.
(c) Prove that $f(x) > 0$ if $x > 0$. (This part is tricky, but if you have been paying attention to the philosophical remarks accompanying the problems in the last two chapters, you will know what to do.)
(d) Prove that $f(x) > f(y)$ if $x > y$.
(e) Prove that $f(x) = x$ for all x. Hint: Use the fact that between any two numbers there is a rational number.

***18.** Precisely what conditions must f, g, h, and k satisfy in order that $f(x)g(y) = h(x)k(y)$ for all x and y?

***19.** (a) Prove that there do *not* exist functions f and g with either of the following properties:

(i) $f(x) + g(y) = xy$ for all x and y.
(ii) $f(x) \cdot g(y) = x + y$ for all x and y.

Hint: Try to get some information about f or g by choosing particular values of x and y.

(b) Find functions f and g such that $f(x+y) = g(xy)$ for all x and y.

***20.** (a) Find a function f, other than a constant function, such that $|f(y) - f(x)| \leq |y - x|$.
(b) Suppose that $f(y) - f(x) \leq (y-x)^2$ for all x and y. (Why does this imply that $|f(y) - f(x)| \leq (y-x)^2$?) Prove that f is a constant function. Hint: Divide the interval from x to y into n equal pieces.

21. Prove or give a counterexample for each of the following assertions:

(a) $f \circ (g + h) = f \circ g + f \circ h$.
(b) $(g + h) \circ f = g \circ f + h \circ f$.
(c) $\dfrac{1}{f \circ g} = \dfrac{1}{f} \circ g$.
(d) $\dfrac{1}{f \circ g} = f \circ \left(\dfrac{1}{g}\right)$.

22. (a) Suppose $g = h \circ f$. Prove that if $f(x) = f(y)$, then $g(x) = g(y)$.
(b) Conversely, suppose that f and g are two functions such that $g(x) = g(y)$ whenever $f(x) = f(y)$. Prove that $g = h \circ f$ for some function h. Hint: Just try to define $h(z)$ when z is of the form $z = f(x)$ (these are the only z that matter) and use the hypotheses to show that your definition will not run into trouble.

23. Suppose that $f \circ g = I$, where $I(x) = x$. Prove that

(a) if $x \neq y$, then $g(x) \neq g(y)$;
(b) every number b can be written $b = f(a)$ for some number a.

***24.** (a) Suppose g is a function with the property that $g(x) \neq g(y)$ if $x \neq y$. Prove that there is a function f such that $f \circ g = I$.
(b) Suppose that f is a function such that every number b can be written $b = f(a)$ for some number a. Prove that there is a function g such that $f \circ g = I$.

***25.** Find a function f such that $g \circ f = I$ for some g, but such that there is no function h with $f \circ h = I$.

***26.** Suppose $f \circ g = I$ and $h \circ f = I$. Prove that $g = h$. Hint: Use the fact that composition is associative.

27. (a) Suppose $f(x) = x + 1$. Are there any functions g such that $f \circ g = g \circ f$?
(b) Suppose f is a constant function. For which functions g does $f \circ g = g \circ f$?
(c) Suppose that $f \circ g = g \circ f$ for *all* functions g. Show that f is the identity function, $f(x) = x$.

28. (a) Let F be the set of all functions whose domain is $\mathbf{R}$. Prove that, using $+$ and $\cdot$ as defined in this chapter, all of properties P1–P9 except P7 hold for F, provided 0 and 1 are interpreted as constant functions.
(b) Show that P7 does not hold.
*(c) Show that P10–P12 cannot hold. In other words, show that there is no collection P of functions in F, such that P10–P12 hold for P. (It is sufficient, and will simplify things, to consider only functions which are 0 except at two points x_0 and x_1.)
(d) Suppose we define $f < g$ to mean that $f(x) < g(x)$ for all x. Which of P′10–P′13 (in Problem 1-8) now hold?
(e) If $f < g$, is $h \circ f < h \circ g$? Is $f \circ h < g \circ h$?

APPENDIX. ORDERED PAIRS

Not only in the definition of a function, but in other parts of the book as well, it is necessary to use the notion of an ordered pair of objects. A definition has not yet been given, and we have never even stated explicitly what properties an ordered pair is supposed to have. The one property which we will require states formally that the ordered pair (a, b) should be determined by a and b, and the order in which they are given:

$$\text{if } (a, b) = (c, d), \text{ then } a = c \text{ and } b = d.$$

Ordered pairs may be treated most conveniently by simply introducing (a, b) as an undefined term and adopting the basic property as an axiom—since this property is the only significant fact about ordered pairs, there is not much point worrying about what an ordered pair "really" is. Those who find this treatment satisfactory need read no further.

The rest of this short appendix is for the benefit of those readers who will feel uncomfortable unless ordered pairs are somehow defined so that this basic property becomes a theorem. There is no point in restricting our attention to ordered pairs of numbers; it is just as reasonable, and just as important, to have available the notion of an ordered pair of any two mathematical objects. This means that our definition ought to involve only concepts common to all branches of mathematics. The one common concept which pervades all areas of mathematics is that of a set, and ordered pairs (like everything else in mathematics) can be defined in this context; an ordered pair will turn out to be a set of a rather special sort.

The set $\{a, b\}$, containing the two elements a and b, is an obvious first choice, but will not do as a definition for (a, b), because there is no way of determining from $\{a, b\}$ which of a or b is meant to be the first element. A more promising candidate is the rather startling set:

$$\{\{a\}, \{a, b\}\}.$$

This set has two members, both of which are *themselves* sets; one member is the set $\{a\}$, containing the single member a, the other is the set $\{a, b\}$. Shocking as it may seem, we are going to define (a, b) to be this set. The justification for this choice is given by the theorem immediately following the definition—the definition works, and there really isn't anything else worth saying.

DEFINITION

$$\boldsymbol{(a, b)} = \{\{a\}, \{a, b\}\}.$$

THEOREM 1 If $(a, b) = (c, d)$, then $a = c$ and $b = d$.

PROOF The hypothesis means that

$$\{\{a\},\ \{a, b\}\} = \{\{c\},\ \{c, d\}\}.$$

Now $\{\{a\},\ \{a, b\}\}$ contains just two members, $\{a\}$ and $\{a, b\}$; and a is the only common element of these two members of $\{\{a\},\ \{a, b\}\}$. Similarly, c is the unique

common member of both members of $\{\{c\},\ \{c,d\}\}$. Therefore $a = c$. We therefore have

$$\{\{a\},\ \{a,b\}\} = \{\{a\},\ \{a,d\}\},$$

and only the proof that $b = d$ remains. It is convenient to distinguish 2 cases.

Case 1. $b = a$. In this case, $\{a,b\} = \{a\}$, so the set $\{\{a\},\ \{a,b\}\}$ really has only one member, namely, $\{a\}$. The same must be true of $\{\{a\},\ \{a,d\}\}$, so $\{a,d\} = \{a\}$, which implies that $d = a = b$.

Case 2. $b \neq a$. In this case, b is in one member of $\{\{a\},\ \{a,b\}\}$ but not in the other. It must therefore be true that b is in one member of $\{\{a\},\ \{a,d\}\}$ but not in the other. This can happen only if b is in $\{a,d\}$, but b is not in $\{a\}$; thus $b = a$ or $b = d$, but $b \neq a$; so $b = d$. ∎

CHAPTER 4 GRAPHS

Mention the real numbers to a mathematician and the image of a straight line will probably form in her mind, quite involuntarily. And most likely she will neither banish nor too eagerly embrace this mental picture of the real numbers. "Geometric intuition" will allow her to interpret statements about numbers in terms of this picture, and may even suggest methods of proving them. Although the properties of the real numbers which were studied in Part I are not greatly illuminated by a geometric picture, such an interpretation will be a great aid in Part II.

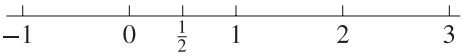


FIGURE 1

You are probably already familiar with the conventional method of considering the straight line as a picture of the real numbers, i.e., of associating to each real number a point on a line. To do this (Figure 1) we pick, arbitrarily, a point which we label 0, and a point to the right, which we label 1. The point twice as far to the right is labeled 2, the point the same distance from 0 to 1, but to the left of 0, is labeled -1, etc. With this arrangement, if $a < b$, then the point corresponding to a lies to the left of the point corresponding to b. We can also draw rational numbers, such as $\frac{1}{2}$, in the obvious way. It is usually taken for granted that the irrational numbers also somehow fit into this scheme, so that every real number can be drawn as a point on the line. We will not make too much fuss about justifying this assumption, since this method of "drawing" numbers is intended solely as a method of picturing certain abstract ideas, and our proofs will never rely on these pictures (although we will frequently use a picture to suggest or help explain a proof). Because this geometric picture plays such a prominent, albeit inessential role, geometric terminology is frequently employed when speaking of numbers—thus a number is sometimes called a *point*, and $\mathbf{R}$ is often called the *real line*.

The number $|a-b|$ has a simple interpretation in terms of this geometric picture: it is the distance between a and b, the length of the line segment which has a as one end point and b as the other. This means, to choose an example whose frequent occurrence justifies special consideration, that the set of numbers x which satisfy $|x - a| < \varepsilon$ may be pictured as the collection of points whose distance from a is less than ε. This set of points is the "interval" from $a - \varepsilon$ to $a + \varepsilon$, which may also be described as the points corresponding to numbers x with $a - \varepsilon < x < a + \varepsilon$ (Figure 2).

FIGURE 2

Sets of numbers which correspond to intervals arise so frequently that it is desirable to have special names for them. The set $\{x : a < x < b\}$ is denoted by (a, b) and called the **open interval** from a to b. This notation naturally creates some ambiguity, since (a, b) is also used to denote a pair of numbers, but in context it is always clear (or can easily be made clear) whether one is talking about a pair or an interval. Note that if $a \geq b$, then $(a, b) = \emptyset$, the set with no elements; in prac-

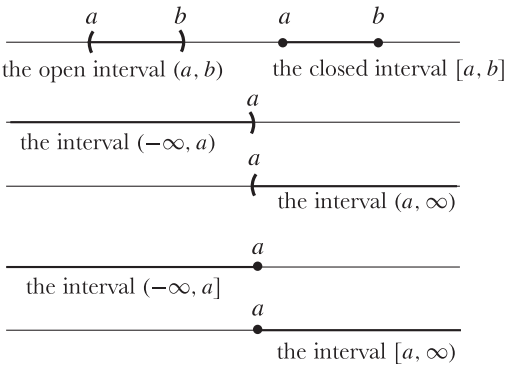


FIGURE 3

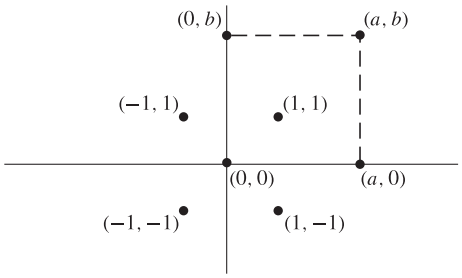


FIGURE 4

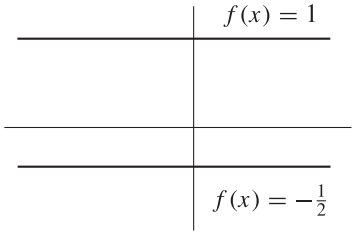


FIGURE 5

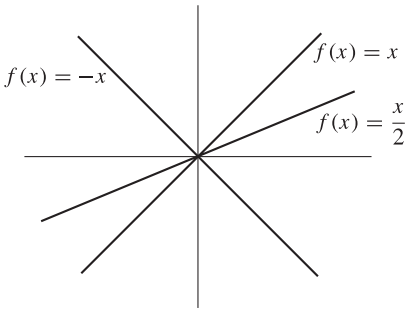


FIGURE 6

tice, however, it is almost always assumed (explicitly if one has been careful, and implicitly otherwise), that whenever an interval (a, b) is mentioned, the number a is less than b.

The set $\{x : a \leq x \leq b\}$ is denoted by $[a, b]$ and is called the **closed interval** from a to b. This symbol is usually reserved for the case $a < b$, but it is sometimes used for $a = b$, also. The usual pictures for the intervals (a, b) and $[a, b]$ are shown in Figure 3; since no reasonably accurate picture could ever indicate the difference between the two intervals, various conventions have been adopted. Figure 3 also shows certain "infinite" intervals. The set $\{x : x > a\}$ is denoted by (a, ∞), while the set $\{x : x \geq a\}$ is denoted by $[a, \infty)$; the sets $(-\infty, a)$ and $(-\infty, a]$ are defined similarly. At this point a standard warning must be issued: the symbols ∞ and $-\infty$, though usually read "infinity" and "minus infinity," are *purely* suggestive; there is no number "∞" which satisfies $\infty \geq a$ for all numbers a. While the symbols ∞ and $-\infty$ will appear in many contexts, it is always necessary to define these uses in ways that refer only to numbers. The set $\mathbf{R}$ of all real numbers is also considered to be an "interval," and is sometimes denoted by $(-\infty, \infty)$.

Of even greater interest to us than the method of drawing numbers is a method of drawing pairs of numbers. This procedure, probably also familiar to you, requires a "coordinate system," two straight lines intersecting at right angles. To distinguish these straight lines, we call one the *horizontal axis*, and one the *vertical axis*. (More prosaic terminology, such as the "first" and "second" axes, is probably preferable from a logical point of view, but most people hold their books, or at least their blackboards, in the same way, so that "horizontal" and "vertical" are more descriptive.) Each of the two axes could be labeled with real numbers, but we can also label points on the horizontal axis with pairs $(a, 0)$ and points on the vertical axis with pairs $(0, b)$, so that the intersection of the two axes, the "origin" of the coordinate system, is labeled $(0, 0)$. Any pair (a, b) can now be drawn as in Figure 4, lying at the vertex of the rectangle whose other three vertices are labeled $(0, 0)$, $(a, 0)$, and $(0, b)$. The numbers a and b are called the *first* and *second coordinates*, respectively, of the point determined in this way.

Our real concern, let us recall, is a method of drawing functions. Since a function is just a collection of pairs of numbers, we can draw a function by drawing each of the pairs in the function. The drawing obtained in this way is called the **graph** of the function. In other words, the graph of f contains all the points corresponding to pairs $(x, f(x))$. Since most functions contain infinitely many pairs, drawing the graph promises to be a laborious undertaking, but, in fact, many functions have graphs which are quite easy to draw.

Not surprisingly, the simplest functions of all, the constant functions $f(x) = c$, have the simplest graphs. It is easy to see that the graph of the function $f(x) = c$ is a straight line parallel to the horizontal axis, at distance c from it (Figure 5).

The functions $f(x) = cx$ also have particularly simple graphs—straight lines through $(0, 0)$, as in Figure 6. A proof of this fact is indicated in Figure 7:

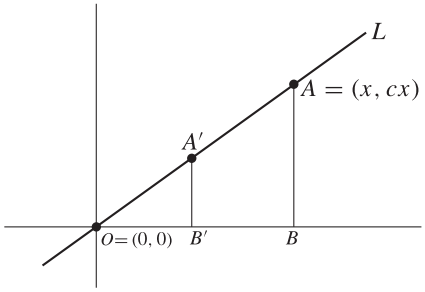


FIGURE 7

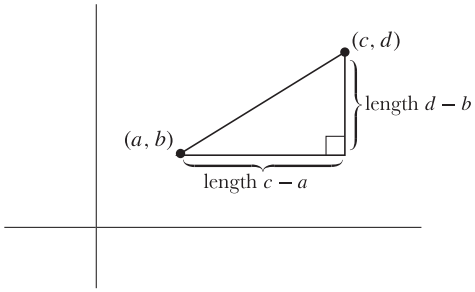


FIGURE 8

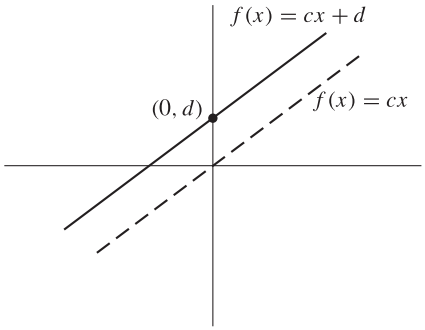


FIGURE 9

Let x be some number not equal to 0, and let L be the straight line which passes through the origin O, corresponding to $(0, 0)$, and through the point A, corresponding to (x, cx). A point A', with first coordinate y, will lie on L when the triangle $A'B'O$ is similar to the triangle ABO, thus when

$$\frac{A'B'}{OB'} = \frac{AB}{OB} = c;$$

this is precisely the condition that A' corresponds to the pair (y, cy), i.e., that A' lies on the graph of f. The argument has implicitly assumed that $c > 0$, but the other cases are treated easily enough. The number c, which measures the ratio of the sides of the triangles appearing in the proof, is called the *slope* of the straight line, and a line parallel to this line is also said to have slope c.

This demonstration has neither been labeled nor treated as a formal proof. Indeed, a rigorous demonstration would necessitate a digression which we are not at all prepared to follow. The rigorous proof of *any* statement connecting geometric and algebraic concepts would first require a real proof (or a precisely stated assumption) that the points on a straight line correspond in an exact way to the real numbers. Aside from this, it would be necessary to develop plane geometry as precisely as we intend to develop the properties of real numbers. Now the detailed development of plane geometry is a beautiful subject, but it is by no means a prerequisite for the study of calculus. We shall use geometric pictures only as an aid to intuition; for our purposes (and for most of mathematics) it is perfectly satisfactory to *define* the plane to be the set of all pairs of real numbers, and to *define* straight lines as certain collections of pairs, including, among others, the collections $\{(x, cx) : x \text{ a real number}\}$. To provide this artificially constructed geometry with all the structure of geometry studied in high school, one more definition is required. If (a, b) and (c, d) are two points in the plane, i.e., pairs of real numbers, we *define* the **distance** between (a, b) and (c, d) to be

$$\sqrt{(a - c)^2 + (b - d)^2}.$$

If the motivation for this definition is not clear, Figure 8 should serve as adequate explanation—with this definition the Pythagorean theorem has been built into our geometry.*

Reverting once more to our informal geometric picture, it is not hard to see (Figure 9) that the graph of the function $f(x) = cx + d$ is a straight line with slope c, passing through the point $(0, d)$. For this reason, the functions $f(x) = cx + d$ are called **linear functions**. Simple as they are, linear functions occur frequently, and you should feel comfortable working with them. The following is a typical problem whose solution should not cause any trouble. Given two distinct points (a, b) and (c, d), find the linear function f whose graph goes through (a, b) and (c, d). This amounts to saying that $f(a) = b$ and $f(c) = d$. If

*The fastidious reader might object to this definition on the grounds that nonnegative numbers are not yet known to have square roots. This objection is really unanswerable at the moment—the definition will just have to be accepted with reservations, until this little point is settled.

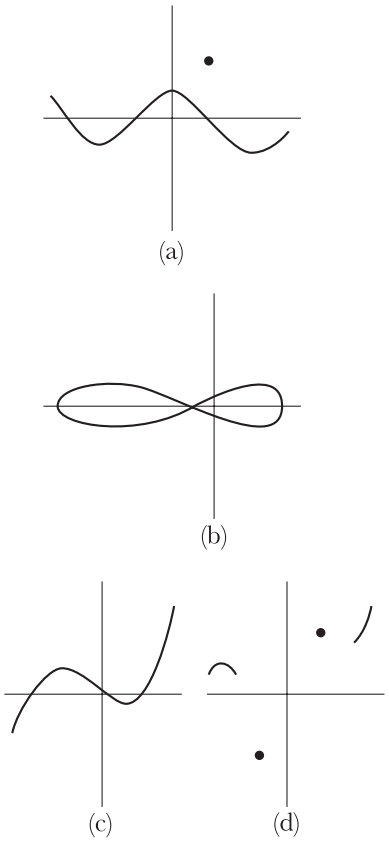


FIGURE 10

f is to be of the form $f(x) = \alpha x + \beta$, then we must have

$$\alpha a + \beta = b,$$
$$\alpha c + \beta = d;$$

therefore $\alpha = (d - b)/(c - a)$ and $\beta = b - [(d - b)/(c - a)]a$, so

$$f(x) = \frac{d-b}{c-a}x + b - \frac{d-b}{c-a}a = \frac{d-b}{c-a}(x - a) + b,$$

a formula most easily remembered by using the "point-slope form" (see Problem 6).

Of course, this solution is possible only if $a \neq c$; the graphs of linear functions account only for the straight lines which are not parallel to the vertical axis. The vertical straight lines are not the graph of *any* function at all; in fact, the graph of a function can never contain even two distinct points on the same vertical line. This conclusion is immediate from the definition of a function—two points on the same vertical line correspond to pairs of the form (a, b) and (a, c) and, by definition, a function cannot contain (a, b) and (a, c) if $b \neq c$. Conversely, if a set of points in the plane has the property that no two points lie on the same vertical line, then it is surely the graph of a function. Thus, the first two sets in Figure 10 are not graphs of functions and the last two are; notice that the fourth is the graph of a function whose domain is not all of $\mathbf{R}$, since some vertical lines have no points on them at all.

After the linear functions the simplest is perhaps the function $f(x) = x^2$. If we draw some of the pairs in f, i.e., some of the pairs of the form (x, x^2), we obtain a picture like Figure 11.

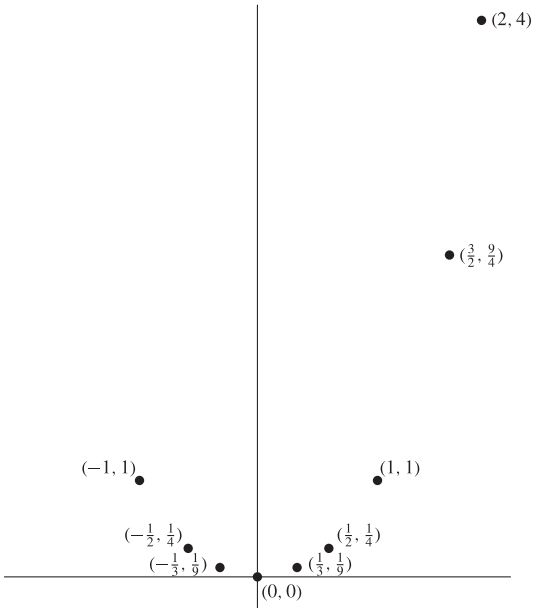


FIGURE 11

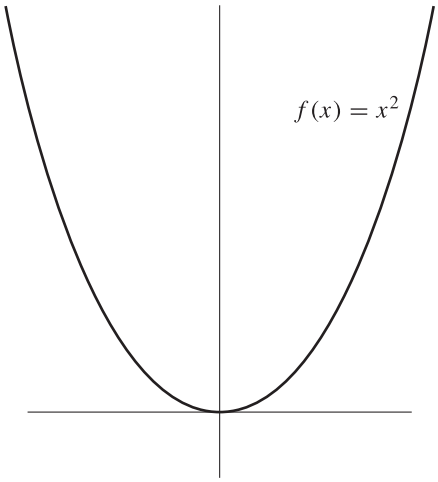


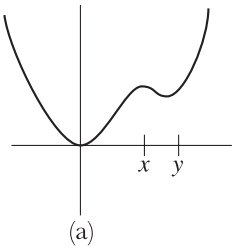
FIGURE 12

It is not hard to convince yourself that all the pairs (x, x^2) lie along a curve like the one shown in Figure 12; this curve is known as a **parabola**.

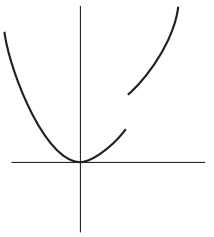
Since a graph is just a drawing on paper, made (in this case) with printer's ink, the question "Is this what the graph really looks like?" is hard to phrase in any sensible manner. No drawing is ever *really* correct since the line has thickness. Nevertheless, there are some questions which one *can* ask: for example, how can you be sure that the graph does not look like one of the drawings in Figure 13? It is easy to see, and even to prove, that the graph cannot look like (a); for if $0 < x < y$, then $x^2 < y^2$, so the graph should be higher at y than at x, which is not the case in (a) . It is also easy to see, simply by drawing a very accurate graph, first plotting many pairs (x, x^2), that the graph cannot have a large "jump" as in (b) or a "corner" as in (c). In order to prove these assertions, however, we first need to say, in a mathematical way, what it means for a function not to have a "jump" or "corner"; these ideas already involve some of the fundamental concepts of calculus. Eventually we will be able to define them rigorously, but meanwhile you may amuse yourself by attempting to define these concepts, and then examining your definitions critically. Later these definitions may be compared with the ones mathematicians have agreed upon. If they compare favorably, you are certainly to be congratulated!

The functions $f(x) = x^n$, for various natural numbers n, are sometimes called **power functions**. Their graphs are most easily compared as in Figure 14, by drawing several at once.

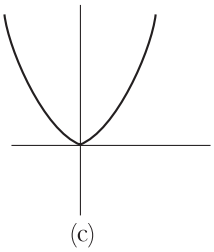
The power functions are only special cases of polynomial functions, introduced in the previous chapter. Two particular polynomial functions are graphed in



(a)



(b)



(c)

FIGURE 13

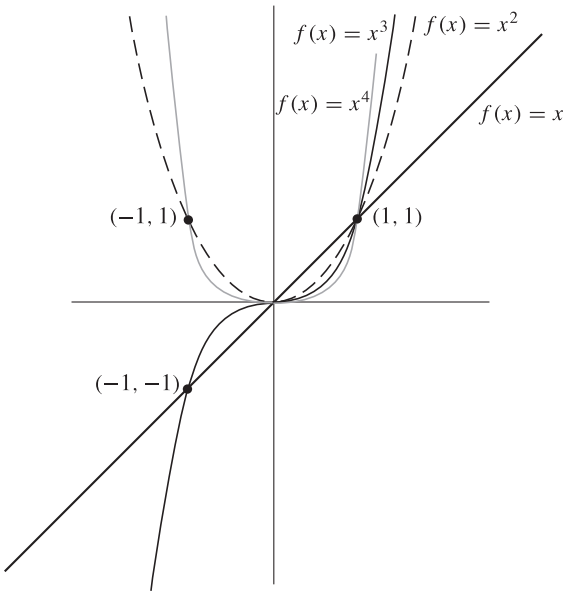


FIGURE 14

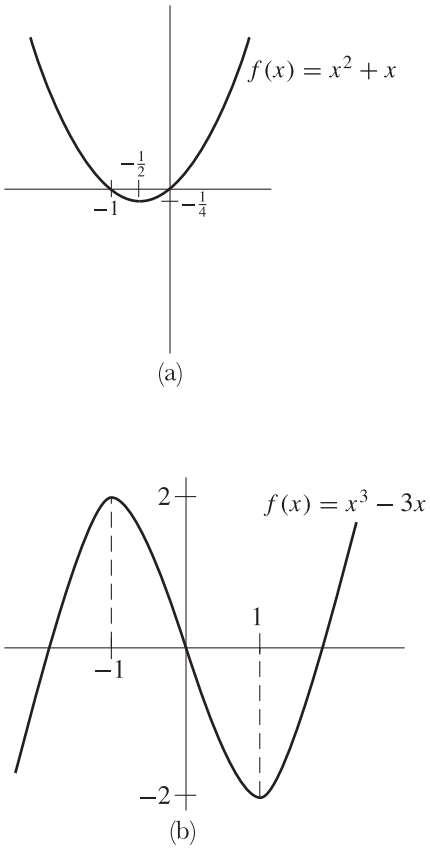


FIGURE 15

Figure 15, while Figure 16 is meant to give a general idea of the graph of the polynomial function

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

in the case $a_n > 0$.

In general, the graph of f will have at most $n-1$ "peaks" or "valleys" (a "peak" is a point like $(x, f(x))$ in Figure 16, while a "valley" is a point like $(y, f(y))$. The number of peaks and valleys may actually be much smaller (the power functions, for example, have at most one valley). Although these assertions are easy to make, we will not even contemplate giving proofs until Part III (once the powerful methods of Part III are available, the proofs will be very easy).

Figure 17 illustrates the graphs of several rational functions. The rational functions exhibit even greater variety than the polynomial functions, but their behavior will also be easy to analyze once we can use the derivative, the basic tool of Part III.

Many interesting graphs can be constructed by "piecing together" the graphs of functions already studied. The graph in Figure 18 is made up entirely of straight lines. The function f with this graph satisfies

$$f\left(\frac{1}{n}\right) = (-1)^{n+1},$$

$$f\left(\frac{-1}{n}\right) = (-1)^{n+1},$$

$$f(x) = 1, \qquad |x| \geq 1,$$

and is a linear function on each interval $[1/(n+1), 1/n]$ and $[-1/n, -1/(n+1)]$. (The number 0 is not in the domain of f.) Of course, one can write out an explicit formula for $f(x)$, when x is in $[1/(n+1), 1/n]$; this is a good exercise in the use of linear functions, and will also convince you that a picture is worth a thousand words.

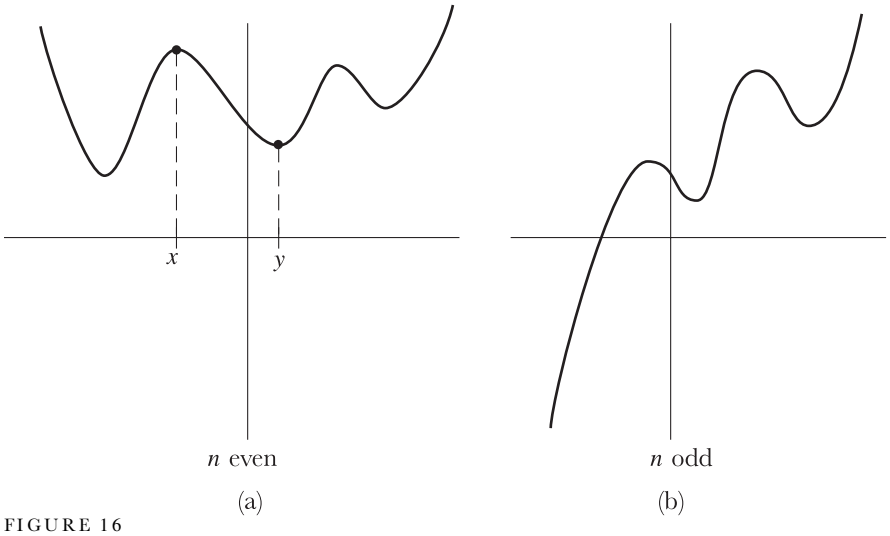


FIGURE 16

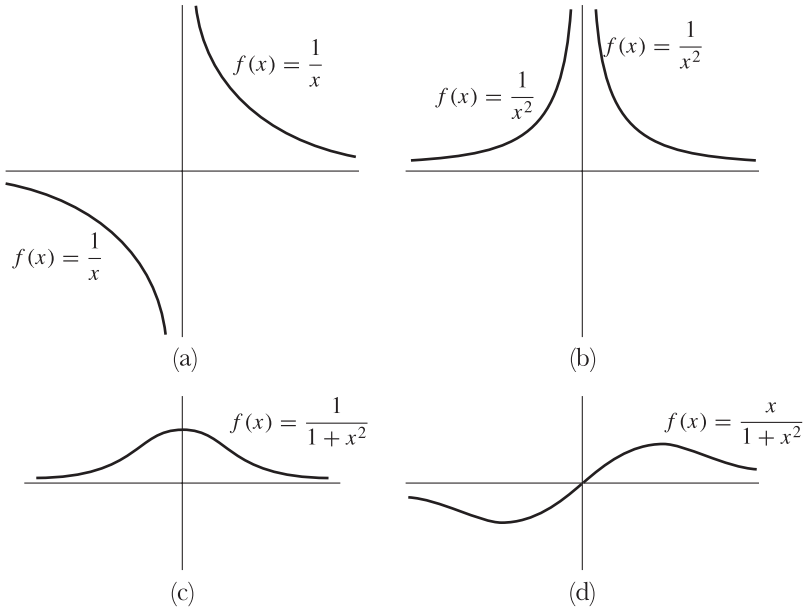


FIGURE 17

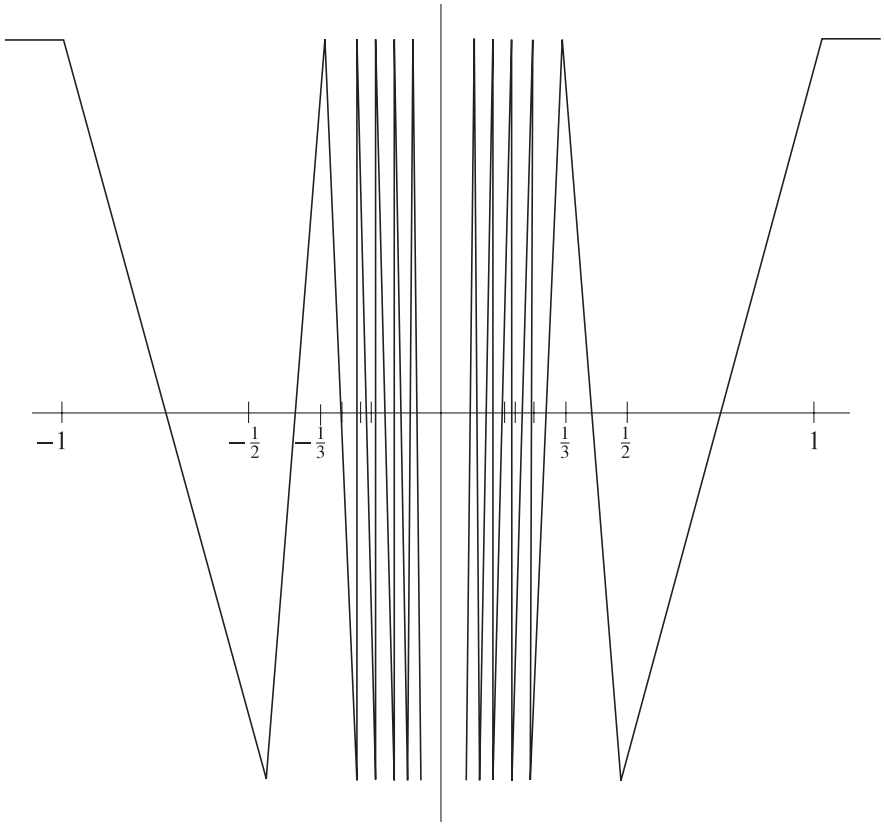


FIGURE 18

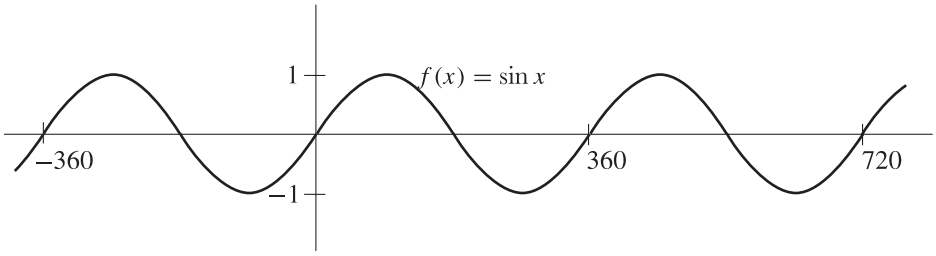


FIGURE 19

It is actually possible to define, in a much simpler way, a function which exhibits this same property of oscillating infinitely often near 0, by using the sine function. In Chapter 15 we will discuss this function in detail, and radian measure in particular; for the time being it will be easiest to use degree measurements for angles. The graph of the sine function is shown in Figure 19 (the scale on the horizontal axis has been altered so that the graph will be clearer; radian measure has, besides important mathematical properties, the additional advantage that such changes are unnecessary).

Now consider the function $f(x) = \sin 1/x$. The graph of f is shown in Figure 20. To draw this graph it helps to first observe that

$$f(x) = 0 \qquad \text{for } x = \frac{1}{180}, \frac{1}{360}, \frac{1}{540}, \ldots,$$

$$f(x) = 1 \qquad \text{for } x = \frac{1}{90}, \frac{1}{90+360}, \frac{1}{90+720}, \ldots,$$

$$f(x) = -1 \qquad \text{for } x = \frac{1}{270}, \frac{1}{270+360}, \frac{1}{270+720}, \ldots.$$

Notice that when x is large, so that $1/x$ is small, $f(x)$ is also small; when x is

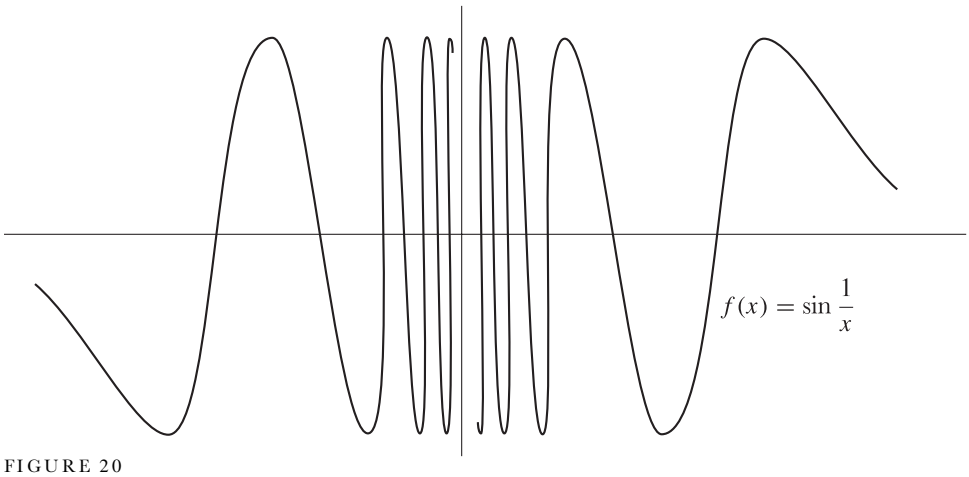


FIGURE 20

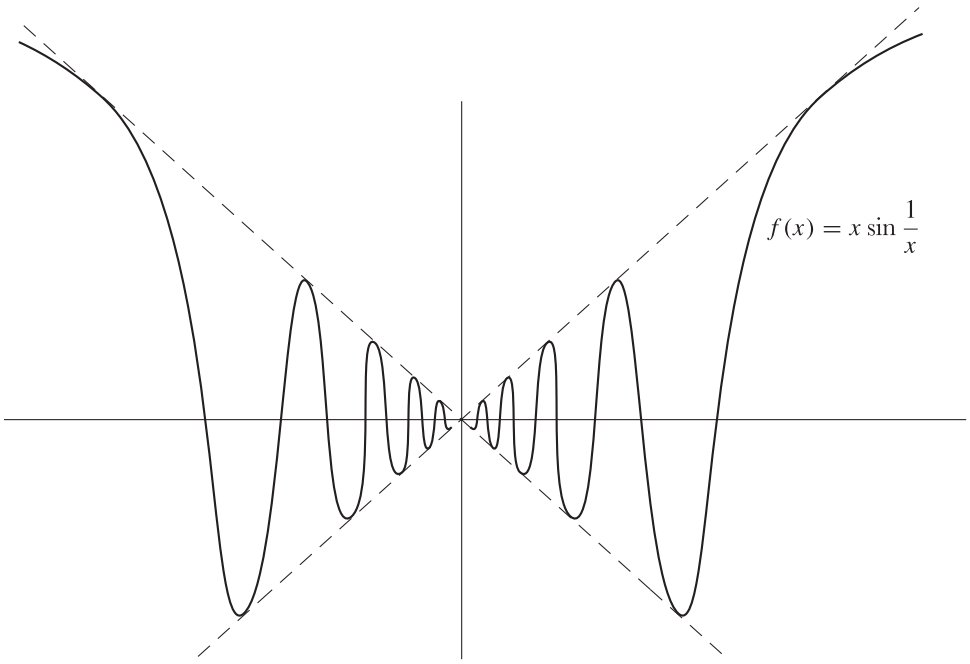


FIGURE 21

"large negative," that is, when $|x|$ is large for negative x, again $f(x)$ is close to 0, although $f(x) < 0$.

An interesting modification of this function is $f(x) = x \sin 1/x$. The graph of this function is sketched in Figure 21. Since $\sin 1/x$ oscillates infinitely often near 0 between 1 and -1, the function $f(x) = x \sin 1/x$ oscillates infinitely often between x and $-x$. The behavior of the graph for x large or large negative is harder to

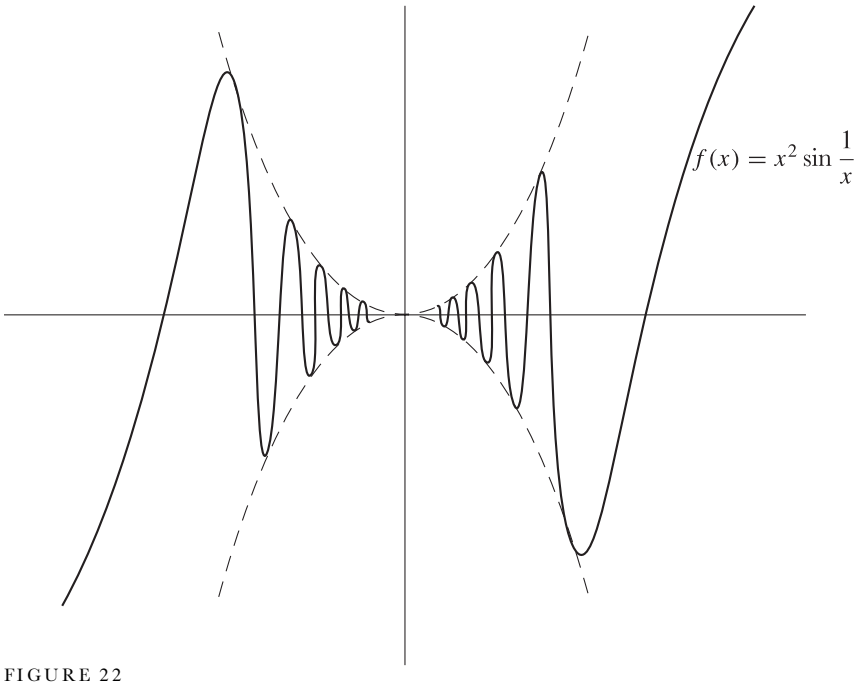


FIGURE 22

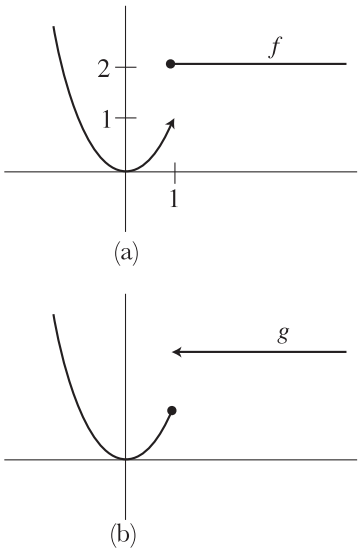


FIGURE 23

analyze. Since $\sin 1/x$ is getting close to 0, while x is getting larger and larger, there seems to be no telling what the product will do. It *is* possible to decide, but this is another question that is best deferred to Part III. The graph of $f(x) = x^2 \sin 1/x$ has also been illustrated (Figure 22).

For these infinitely oscillating functions, it is clear that the graph cannot hope to be really "accurate." The best we can do is to show part of it, and leave out the part near 0 (which is the interesting part). Actually, it is easy to find much simpler functions whose graphs cannot be "accurately" drawn. The graphs of

$$f(x) = \begin{cases} x^2, & x < 1 \\ 2, & x \geq 1 \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} x^2, & x \leq 1 \\ 2, & x > 1 \end{cases}$$

can only be distinguished by some convention similar to that used for open and closed intervals (Figure 23).

Out last example is a function whose graph is spectacularly nondrawable:

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational.} \end{cases}$$

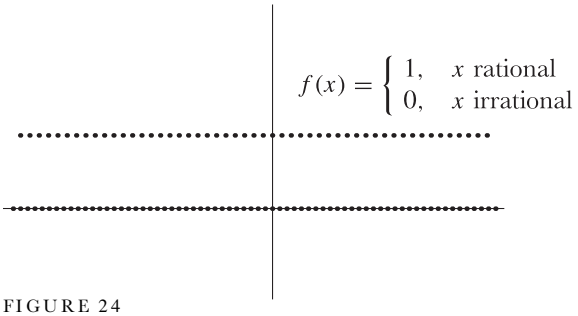


FIGURE 24

The graph of f must contain infinitely many points on the horizontal axis and also infinitely many points on a line parallel to the horizontal axis, but it must not contain either of these lines entirely. Figure 24 shows the usual textbook picture of the graph. To distinguish the two parts of the graph, the dots are placed closer together on the line corresponding to irrational x. (There is actually a mathematical reason behind this convention, but it depends on some sophisticated ideas, introduced in Problems 21-5 and 21-6.)

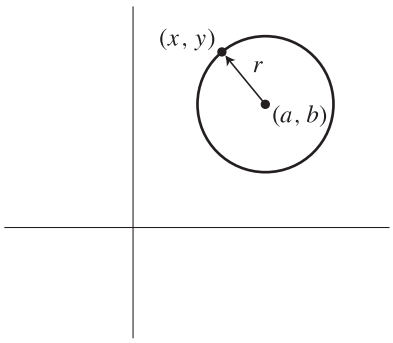


FIGURE 25

The peculiarities exhibited by some functions are so engrossing that it is easy to forget some of the simplest, and most important, subsets of the plane, which are not the graphs of functions. The most important example of all is the **circle**. A circle with center (a, b) and radius $r > 0$ contains, by definition, all the points (x, y) whose distance from (a, b) is equal to r. The circle thus consists (Figure 25) of all points (x, y) with

$$\sqrt{(x-a)^2 + (y-b)^2} = r$$

or

$$(x-a)^2 + (y-b)^2 = r^2.$$

The circle with center (0,0) and radius 1, often regarded as a sort of standard copy, is called the *unit circle*.

A close relative of the circle is the **ellipse**. This is defined as the set of points, the *sum* of whose distances from two "focus" points is a constant. (When the two foci are the same, we obtain a circle.) If, for convenience, the focus points are taken to be $(-c, 0)$ and $(c, 0)$, and the sum of the distances is taken to be $2a$ (the factor 2 simplifies some algebra), then (x, y) is on the ellipse if and only if

$$\sqrt{(x-(-c))^2+y^2}+\sqrt{(x-c)^2+y^2}=2a$$

or

$$\sqrt{(x+c)^2+y^2}=2a-\sqrt{(x-c)^2+y^2}$$

or

$$x^2+2cx+c^2+y^2=4a^2-4a\sqrt{(x-c)^2+y^2}+x^2-2cx+c^2+y^2$$

or

$$4(cx-a^2)=-4a\sqrt{(x-c)^2+y^2}$$

or

$$c^2x^2-2cxa^2+a^4=a^2(x^2-2cx+c^2+y^2)$$

or

$$(c^2-a^2)x^2-a^2y^2=a^2(c^2-a^2)$$

or

$$\frac{x^2}{a^2}+\frac{y^2}{a^2-c^2}=1.$$

This is usually written simply

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,$$

where $b=\sqrt{a^2-c^2}$ (since we must clearly choose $a > c$, it follows that $a^2-c^2>0$). A picture of an ellipse is shown in Figure 26. The ellipse intersects the horizontal axis when $y=0$, so that

$$\frac{x^2}{a^2}=1, \qquad x=\pm a,$$

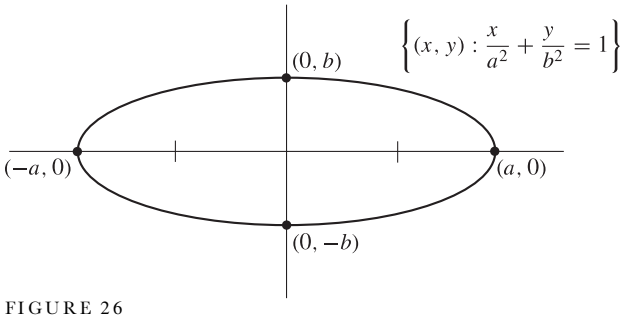


FIGURE 26

and it intersects the vertical axis when $x = 0$, so that

$$\frac{y^2}{b^2} = 1, \qquad y = \pm b.$$

The **hyperbola** is defined analogously, except that we require the *difference* of the two distances to be constant. Choosing the points $(-c, 0)$ and $(c, 0)$ once again, and the constant difference as $2a$, we obtain, as the condition that (x, y) be on the hyperbola,

$$\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = \pm 2a,$$

which may be simplified to

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

In this case, however, we must clearly choose $c > a$, so that $a^2 - c^2 < 0$. If $b = \sqrt{c^2 - a^2}$, then (x, y) is on the hyperbola if and only if

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The picture is shown in Figure 27. It contains two pieces, because the difference between the distances of (x, y) from $(-c, 0)$ and $(c, 0)$ may be taken in two different orders. The hyperbola intersects the horizontal axis when $y = 0$, so that $x = \pm a$, but it never intersects the vertical axis.

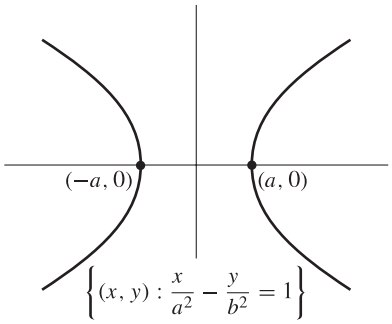


FIGURE 27

It is interesting to compare (Figure 28) the hyperbola with $a = b = \sqrt{2}$ and the graph of the function $f(x) = 1/x$. The drawings look quite similar, and the two sets are actually identical, except for a rotation through an angle of 45° (Problem 23).

Clearly no rotation of the plane will change circles or ellipses into the graphs of functions. Nevertheless, the study of these important geometric figures can often be reduced to the study of functions. Ellipses, for example, are made up of the

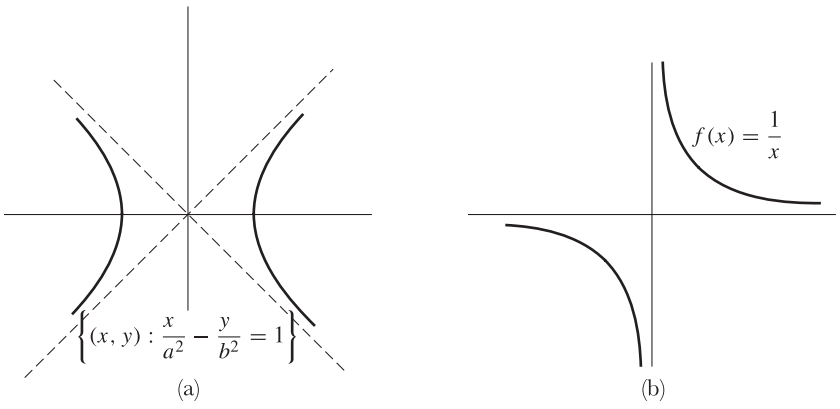


FIGURE 28

graphs of two functions,

$$f(x) = b\sqrt{1 - (x^2/a^2)}, \qquad -a \le x \le a$$

and

$$g(x) = -b\sqrt{1 - (x^2/a^2)}, \qquad -a \le x \le a.$$

Of course, there are many other pairs of functions with this same property. For example, we can take

$$f(x) = \begin{cases} b\sqrt{1 - (x^2/a^2)}, & 0 < x \le a \\ -b\sqrt{1 - (x^2/a^2)}, & -a \le x \le 0 \end{cases}$$

and

$$g(x) = \begin{cases} -b\sqrt{1 - (x^2/a^2)}, & 0 < x \le a \\ b\sqrt{1 - (x^2/a^2}, & -a \le x \le 0. \end{cases}$$

We could also choose

$$f(x) = \begin{cases} b\sqrt{1 - (x^2/a^2)}, & x \text{ rational}, \quad -a \le x \le a \\ -b\sqrt{1 - (x^2/a^2)}, & x \text{ irrational}, -a \le x \le a \end{cases}$$

and

$$g(x) = \begin{cases} -b\sqrt{1 - (x^2/a^2)}, & x \text{ rational}, \quad -a \le x \le a \\ b\sqrt{1 - (x^2/a^2)}, & x \text{ irrational}, -a \le x \le a. \end{cases}$$

But all these other pairs necessarily involve unreasonable functions which jump around. A proof, or even a precise statement of this fact, is too difficult at present. Although you have probably already begun to make a distinction between those functions with reasonable graphs, and those with unreasonable graphs, you may find it very difficult to state a reasonable definition of reasonable functions. A mathematical definition of this concept is by no means easy, and a great deal of this book may be viewed as successive attempts to impose more and more conditions that a "reasonable" function must satisfy. As we define some of these conditions, we will take time out to ask if we have really succeeded in isolating the functions which deserve to be called reasonable. The answer, unfortunately, will always be "no," or at best, a qualified "yes."

PROBLEMS

1. Indicate on a straight line the set of all x satisfying the following conditions. Also name each set, using the notation for intervals (in some cases you will also need the $\cup$ sign).

(i) $|x - 3| < 1$.
(ii) $|x - 3| \le 1$.
(iii) $|x - a| < \varepsilon$.
(iv) $|x^2 - 1| < \frac{1}{2}$.

(v) $\dfrac{1}{1+x^2} \geq \frac{1}{5}$.

(vi) $\dfrac{1}{1+x^2} \leq a$ (give an answer in terms of a, distinguishing various cases).

(vii) $x^2 + 1 \geq 2$.

(viii) $(x+1)(x-1)(x-2) > 0$.

2. There is a very useful way of describing the points of the closed interval $[a, b]$ (where we assume, as usual, that $a < b$).

(a) First consider the interval $[0, b]$, for $b > 0$. Prove that if x is in $[0, b]$, then $x = tb$ for some t with $0 \leq t \leq 1$. What is the significance of the number t? What is the mid-point of the interval $[0, b]$?

(b) Now prove that if x is in $[a, b]$, then $x = (1-t)a + tb$ for some t with $0 \leq t \leq 1$. Hint: This expression can also be written as $a + t(b-a)$. What is the midpoint of the interval $[a, b]$? What is the point $1/3$ of the way from a to b?

(c) Prove, conversely, that if $0 \leq t \leq 1$, then $(1-t)a + tb$ is in $[a, b]$.

(d) The points of the *open* interval (a, b) are those of the form $(1-t)a + tb$ for $0 < t < 1$.

3. Draw the set of all points (x, y) satisfying the following conditions. (In most cases your picture will be a sizable portion of a plane, not just a line or curve.)

(i) $x > y$.

(ii) $x + a > y + b$.

(iii) $y < x^2$.

(iv) $y \leq x^2$.

(v) $|x - y| < 1$.

(vi) $|x + y| < 1$.

(vii) $x + y$ is an integer.

(viii) $\dfrac{1}{x+y}$ is an integer.

(ix) $(x-1)^2 + (y-2)^2 < 1$.

(x) $x^2 < y < x^4$.

4. Draw the set of all points (x, y) satisfying the following conditions:

(i) $|x| + |y| = 1$.

(ii) $|x| - |y| = 1$.

(iii) $|x - 1| = |y - 1|$.

(iv) $|1 - x| = |y - 1|$.

(v) $x^2 + y^2 = 0$.

(vi) $xy = 0$.

(vii) $x^2 - 2x + y^2 = 4$.

(viii) $x^2 = y^2$.

5. Draw the set of all points (x, y) satisfying the following conditions:

(i) $x = y^2$.

(ii) $\dfrac{y^2}{a^2} - \dfrac{x^2}{b^2} = 1$.

(iii) $x = |y|$.

(iv) $x = \sin y$.

Hint: You already know the answers when x and y are interchanged.

6. (a) Show that the straight line through (a, b) with slope m is the graph of the function $f(x) = m(x - a) + b$. This formula, known as the "point-slope form" is far more convenient than the equivalent expression $f(x) = mx + (b - ma)$; it is immediately clear from the point-slope form that the slope is m, and that the value of f at a is b.

(b) For $a \neq c$, show that the straight line through (a, b) and (c, d) is the graph of the function

$$f(x) = \frac{d - b}{c - a}(x - a) + b.$$

(c) When are the graphs of $f(x) = mx + b$ and $g(x) = m'x + b'$ parallel straight lines?

7. (a) For any numbers A, B, and C, with A and B not both 0, show that the set of all (x, y) satisfying $Ax + By + C = 0$ is a straight line (possibly a vertical one). Hint: First decide when a vertical straight line is described.

(b) Show conversely that every straight line, including vertical ones, can be described as the set of all (x, y) satisfying $Ax + By + C = 0$.

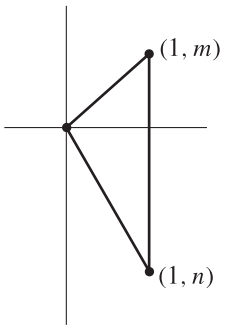


FIGURE 29

8. (a) Prove that the graphs of the functions

$$f(x) = mx + b,$$
$$g(x) = nx + c,$$

are perpendicular if $mn = -1$, by computing the squares of the lengths of the sides of the triangle in Figure 29. (Why is this special case, where the lines intersect at the origin, as good as the general case?)

(b) Prove that the two straight lines consisting of all (x, y) satisfying the conditions

$$Ax + By + C = 0,$$
$$A'x + B'y + C' = 0,$$

are perpendicular if and only if $AA' + BB' = 0$.

9. (a) Prove, using Problem 1-19, that

$$\sqrt{(x_1 + y_1)^2 + (x_2 + y_2)^2} \leq \sqrt{{x_1}^2 + {x_2}^2} + \sqrt{{y_1}^2 + {y_2}^2}.$$

(b) Prove that

$$\sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} \leq \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} + \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}.$$

Interpret this inequality geometrically (it is called the "triangle inequality"). When does strict inequality hold?

10. Sketch the graphs of the following functions, plotting enough points to get a good idea of the general appearance. (Part of the problem is to make a reasonable decision how many is "enough"; the queries posed below are meant to show that a little thought will often be more valuable than hundreds of individual points.)

(i) $f(x) = x + \dfrac{1}{x}$. (What happens for x near 0, and for large x? Where does the graph lie in relation to the graph of the identify function? Why does it suffice to consider only positive x at first?)

(ii) $f(x) = x - \dfrac{1}{x}$.

(iii) $f(x) = x^2 + \dfrac{1}{x^2}$.

(iv) $f(x) = x^2 - \dfrac{1}{x^2}$.

11. Describe the general features of the graph of f if

(i) f is even.
(ii) f is odd.
(iii) f is nonnegative.
(iv) $f(x) = f(x + a)$ for all x (a function with this property is called **periodic**, with **period** a.

12. Graph the functions $f(x) = \sqrt[m]{x}$ for $m = 1, 2, 3, 4$. (There is an easy way to do this, using Figure 14. Be sure to remember, however, that $\sqrt[m]{x}$ means the *positive* mth root of x when m is even; you should also note that there will be an important difference between the graphs when m is even and when m is odd.)

13. (a) Graph $f(x) = |x|$ and $f(x) = x^2$.
(b) Graph $f(x) = |\sin x|$ and $f(x) = \sin^2 x$. (There is an important difference between the graphs, which we cannot yet even describe rigorously. See if you can discover what it is; part (a) is meant to be a clue.)

14. Describe the graph of g in terms of the graph of f if

(i) $g(x) = f(x) + c$.
(ii) $g(x) = f(x + c)$. (It is easy to make a mistake here.)
(iii) $g(x) = cf(x)$.
(iv) $g(x) = f(cx)$. (Distinguish the cases $c = 0$, $c > 0$, $c < 0$.)
(v) $g(x) = f(1/x)$.
(vi) $g(x) = f(|x|)$.
(vii) $g(x) = |f(x)|$.
(viii) $g(x) = \max(f, 0)$.
(ix) $g(x) = \min(f, 0)$.
(x) $g(x) = \max(f, 1)$.

15. Draw the graph of $f(x) = ax^2 + bx + c$. Hint: Use the methods of Problem 1-18.

16. Suppose that A and C are not both 0. Show that the set of all (x, y) satisfying

$$Ax^2 + Bx + Cy^2 + Dy + E = 0$$

is either a parabola, an ellipse, or an hyperbola (or possibly $\emptyset$). Hint: The case $C = 0$ is essentially Problem 15, and the case $A = 0$ is just a minor variant. Now consider separately the cases where A and B are both positive or negative, and where one is positive while the other is negative.

17. The symbol $[x]$ denotes the largest integer which is $\leq x$. Thus, $[2.1] = [2] = 2$ and $[-0.9] = [-1] = -1$. Draw the graph of the following functions (they are all quite interesting, and several will reappear frequently in other problems).

(i) $f(x) = [x]$.
(ii) $f(x) = x - [x]$.
(iii) $f(x) = \sqrt{x - [x]}$.
(iv) $f(x) = [x] + \sqrt{x - [x]}$.
(v) $f(x) = \left[\dfrac{1}{x}\right]$.
(vi) $f(x) = \dfrac{1}{\left[\dfrac{1}{x}\right]}$.

18. Graph the following functions.

(i) $f(x) = \{x\}$, where $\{x\}$ is defined to be the distance from x to the nearest integer.
(ii) $f(x) = \{2x\}$.
(iii) $f(x) = \{x\} + \frac{1}{2}\{2x\}$.
(iv) $f(x) = \{4x\}$.
(v) $f(x) = \{x\} + \frac{1}{2}\{2x\} + \frac{1}{4}\{4x\}$.

Many functions may be described in terms of the decimal expansion of a number. Although we will not be in a position to describe infinite decimals rigorously until Chapter 23, your intuitive notion of infinite decimals should suffice to carry you through the following problem, and others which occur before Chapter 23. There is one ambiguity about infinite decimals which must be eliminated: Every decimal ending in a string of 9's is equal to another ending in a string of 0's (e.g., $1.23999\ldots = 1.24000\ldots$). We will always use the one ending in 9's.

***19.** Describe as best you can the graphs of the following functions (a complete picture is usually out of the question).

(i) $f(x) =$ the 1st number in the decimal expansion of x.
(ii) $f(x) =$ the 2nd number in the decimal expansion of x.
(iii) $f(x) =$ the number of 7's in the decimal expansion of x if this number is finite, and 0 otherwise.
(iv) $f(x) = 0$ if the number of 7's in the decimal expansion of x is finite, and 1 otherwise.
(v) $f(x) =$ the number obtained by replacing all digits in the decimal expansion of x which come after the first 7 (if any) by 0.
(vi) $f(x) = 0$ if 1 never appears in the decimal expansion of x, and n if 1 first appears in the nth place.

***20.** Let

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ \dfrac{1}{q}, & x = \dfrac{p}{q} \text{ rational in lowest terms.} \end{cases}$$

(A number p/q is in **lowest terms** if p and q are integers with no common factor, and $q > 0$). Draw the graph of f as well as you can (don't sprinkle points randomly on the paper; consider first the rational numbers with $q = 2$, then those with $q = 3$, etc.).

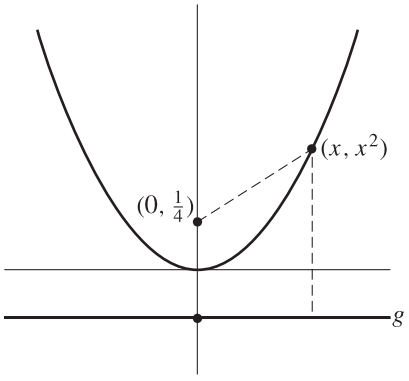


FIGURE 30

21. (a) The points on the graph of $f(x) = x^2$ are the ones of the form (x, x^2). Prove that each such point is equidistant from the point $(0, \frac{1}{4})$ and the graph of $g(x) = -\frac{1}{4}$. (See Figure 30.)
(b) Given a point $P = (\alpha, \beta)$ and a horizontal line L, the graph of $g(x) = \gamma$, show that the set of all points (x, y) equidistant from P and L is the graph of a function of the form $f(x) = ax^2 + bx + c$.

***22.** (a) Show that the square of the distance from (c, d) to (x, mx) is

$$x^2(m^2 + 1) + x(-2md - 2c) + d^2 + c^2.$$

Using Problem 1-18 to find the minimum of these numbers, show that the distance from (c, d) to the graph of $f(x) = mx$ is

$$|cm - d|/\sqrt{m^2 + 1}.$$

(b) Find the distance from (c, d) to the graph of $f(x) = mx + b$. (Reduce this case to part (a).)

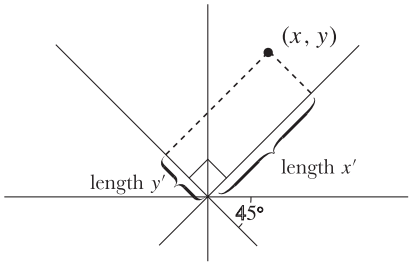


FIGURE 31

***23.** (a) Using Problem 22, show that the numbers x' and y' indicated in Figure 31 are given by

$$x' = \frac{1}{\sqrt{2}}\,x + \frac{1}{\sqrt{2}}\,y,$$
$$y' = -\frac{1}{\sqrt{2}}\,x + \frac{1}{\sqrt{2}}\,y.$$

(b) Show that the set of all (x, y) with $(x'/\sqrt{2})^2 - (y'/\sqrt{2})^2 = 1$ is the same as the set of all (x, y) with $xy = 1$.

APPENDIX 1. VECTORS

Suppose that v is a point in the plane; in other words, v is a pair of numbers

$$v = (v_1, v_2).$$

For convenience, we will use this convention that subscripts indicate the first and second pairs of a point that has been described by a single letter. Thus, if we mention the points w and z, it will be understood that w is the pair (w_1, w_2), while z is the pair (z_1, z_2).

Instead of the actual pair of numbers (v_1, v_2), we often picture v as an arrow from the origin O to this point (Figure 1), and we refer to these arrows as *vectors* in the plane. Of course, we've haven't really said anything new yet, we've simply introduced an alternate term for a point of the plane, and another mental picture. The real point of the new terminology is to emphasize that we are going to do some new things with points in the plane.

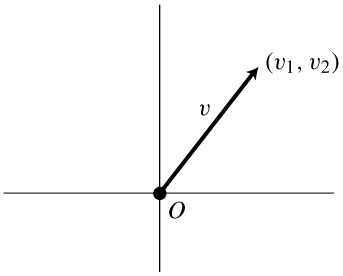


FIGURE 1

For example, suppose that we have two vectors (i.e., points) in the plane,

$$v = (v_1, v_2), \quad w = (w_1, w_2).$$

Then we can define a new vector (a new point of the plane) $v + w$ by the equation

$$v + w = (v_1 + w_1, v_2 + w_2). \tag{1}$$

Notice that all the letters on the right side of this equation are numbers, and the $+$ sign is just our usual addition of numbers. On the other hand, the $+$ sign on the left side is new: previously, the sum of two points in the plane wasn't defined, and we've simply used equation (1) as a *definition.*

A very fussy mathematician might want to use some new symbol for this newly defined operation, like

$$v \boldsymbol{+} w, \qquad \text{or perhaps} \qquad v \oplus w,$$

but there's really no need to insist on this; since $v + w$ hasn't been defined before, there's no possibility of confusion, so we might as well keep the notation simple.

Of course, any one can make new notation; for example, since it's our definition, we could just as well have defined $v + w$ as $(v_1 + w_1 \cdot w_2,\ v_2 + {w_1}^2)$, or by some other equally weird formula. The real question is, does our new construction have any particular significance?

Figure 2 shows two vectors v and w, as well as the point

$$(v_1 + w_1, v_2 + w_2),$$

which, for the moment, we have simply indicated in the usual way, without drawing an arrow. Note that it is easy to compute the slope of the line L between v and our new point: as indicated in Figure 2, this slope is just

$$\frac{(v_2 + w_2) - v_2}{(v_1 + w_1) - v_1} = \frac{w_2}{w_1},$$

and this, of course, is the slope of our vector w, from the origin O to (w_1, w_2). In other words, the line L is parallel to w.

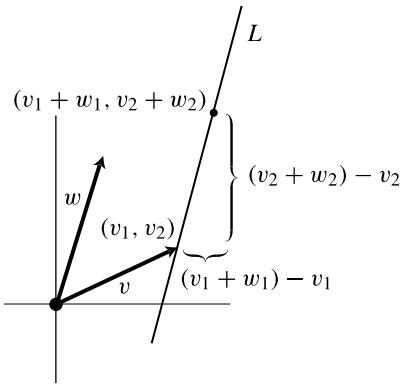


FIGURE 2

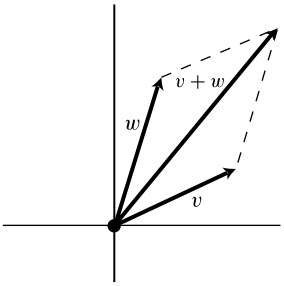


FIGURE 3

Similarly, the slope of the line M between (w_1, w_2) and our new point is

$$\frac{(v_2 + w_2) - w_2}{(v_1 + w_1) - v_2} = \frac{v_2}{v_1},$$

which is the slope of the vector v; so M is parallel to v. In short, the new point $v + w$ lies on the parallelogram having v and w as sides. When we draw $v + w$ as an arrow (Figure 3), it points along the diagonal of this parallelogram. In physics, vectors are used to symbolize forces, and the sum of two vectors represents the resultant force when two different forces are applied simultaneously to the same object.

Figure 4 shows another way of visualizing the sum $v+w$. If we use "w" to denote an arrow parallel to w, and having the same length, but starting at v instead of at the origin, then $v + w$ is the vector from O to the final endpoint; thus we get to $v + w$ by first following v, and then following w.

Many of the properties of $+$ for ordinary numbers also hold for this new $+$ for vectors. For example, the "commutative law"

$$v + w = w + v,$$

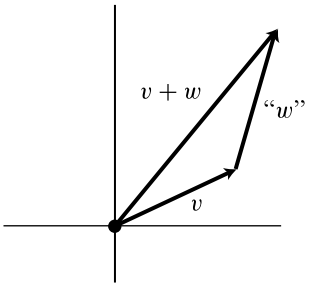


FIGURE 4

is obvious from the geometric picture, since the parallelogram spanned by v and w is the same as the parallelogram spanned by w and v. It is also easily checked analytically, since it states that

$$(v_1 + w_1, v_2 + w_2) = (w_1 + v_1, w_2 + v_2),$$

and thus simply depends on the commutative law for numbers:

$$v_1 + w_1 = w_1 + v_1,$$
$$v_2 + w_2 = w_2 + v_2.$$

Similarly, unraveling definitions, we find the "associative law"

$$[v + w] + z = v + [w + z].$$

Figure 5 indicates a method of finding $v + w + z$.

The origin $O = (0, 0)$ is an "additive identity,"

$$O + v = v + O = v,$$

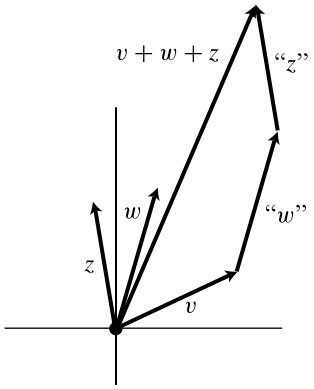


FIGURE 5

and if we define

$$-v = (-v_1, -v_2),$$

then we also have

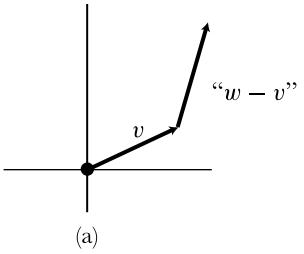
$$v + (-v) = -v + v = O.$$

Naturally we can also define

$$w - v = w + (-v),$$

exactly as with numbers; equivalently,

$$w - v = (w_1 - v_1, w_2 - v_2).$$



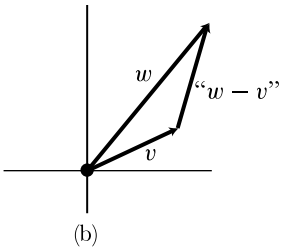


FIGURE 6

Just as with numbers, our definition of $w - v$ simply means that it satisfies

$$v + (w - v) = w.$$

Figure 6(a) shows v and an arrow "$w - v$" that is parallel to $w - v$ but that starts at the endpoint of v. As we established with Figure 4, the vector from the origin to the endpoint of this arrow is just $v + (w - v) = w$ (Figure 6(b)). In other words, we can picture $w - v$ geometrically as the arrow that goes from v to w (except that it must then be moved back to the origin).

There is also a way of multiplying a number by a vector: For a number a and a vector $v = (v_1, v_2)$, we define

$$a \cdot v = (av_1, av_2)$$

(We sometimes simply write av instead of $a \cdot v$; of course, it is then especially important to remember that v denotes a vector, rather than a number.) The vector $a \cdot v$ points in the same direction as v when $a > 0$ and in the opposite direction when $a < 0$ (Figure 7).

You can easily check the following formulas:

$$\begin{aligned} a \cdot (b \cdot v) &= (ab) \cdot v, \\ 1 \cdot v &= v, \\ 0 \cdot v &= O, \\ -1 \cdot v &= -v. \end{aligned}$$

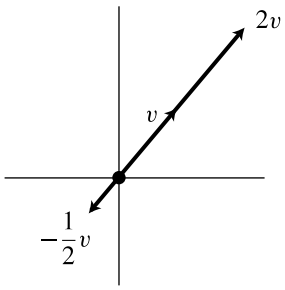


FIGURE 7

Notice that we have only defined a product of a number and a vector, we have not defined a way of 'multiplying' two vectors to get another vector.* However, there are various ways of 'multiplying' vectors to get numbers, which are explored in the following problems.

PROBLEMS

1. Given a point v of the plane, let $R_\theta(v)$ be the result of rotating v around the origin through an angle of θ (Figure 8). The aim of this problem is to obtain a formula for R_θ, with minimal calculation.

(a) Show that

$$\begin{aligned} R_\theta(1, 0) &= (\cos\theta, \sin\theta), \\ R_\theta(0, 1) &= (-\sin\theta, \cos\theta). \end{aligned}$$

(b) Explain why we have

$$\begin{aligned} R_\theta(v + w) &= R_\theta(v) + R_\theta(w), \\ R_\theta(a \cdot w) &= a \cdot R_\theta(w). \end{aligned}$$

(c) Now show that for any point (x, y) we have

$$R_\theta(x, y) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta).$$

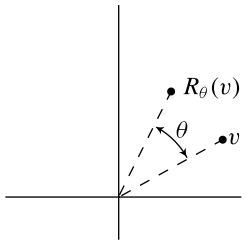


FIGURE 8

*If you jump to Chapter 25, you'll find that there is an important way of defining a product, but this is something very special for the plane—it doesn't work for vectors in 3-space, for example, even though the other constructions do.

(d) Use this result to give another solution to Problem 4-23.

2. Given v and w, we define the *number*

$$v \cdot w = v_1 w_1 + v_2 w_2;$$

this is often called the 'dot product' or 'scalar product' of v and w ('scalar' being a rather old-fashioned word for a number, as opposed to a vector).

(a) Given v, find a vector w such that $v \cdot w = 0$. Now describe the set of all such vectors w.

(b) Show that

$$v \cdot w = w \cdot v$$
$$v \cdot (w + z) = v \cdot w + v \cdot z$$

and that

$$a \cdot (v \cdot w) = (a \cdot v) \cdot w = v \cdot (a \cdot w).$$

Notice that the last of these equations involves *three* products: the dot product $\cdot$ of two vectors; the product $\cdot$ of a number and a vector; and the ordinary product $\cdot$ of two numbers.

(c) Show that $v \cdot v \geq 0$, and that $v \cdot v = 0$ only when $v = O$. Hence we can define the *norm* $\|v\|$ as

$$\|v\| = \sqrt{v \cdot v},$$

which will be 0 only for $v = O$. What is the geometric interpretation of the norm?

(d) Prove that

$$\|v + w\| \leq \|v\| + \|w\|,$$

and that equality holds if and only if $v = 0$ or $w = 0$ or $w = a \cdot v$ for some number $a > 0$.

(e) Show that

$$v \cdot w = \frac{\|v + w\|^2 - \|v - w\|^2}{4}.$$

3. (a) Let R_θ be rotation by an angle of θ (Problem 1). Show that

$$R_\theta(v) \cdot R_\theta(w) = v \cdot w.$$

(b) Let $e = (1, 0)$ be the vector of length 1 pointing along the first axis, and let $w = (\cos\theta, \sin\theta)$; this is a vector of length 1 that makes an angle of θ with the first axis (compare Problem 1). Calculate that

$$e \cdot w = \cos\theta.$$

Conclude that in general

$$v \cdot w = \|v\| \cdot \|w\| \cdot \cos\theta,$$

where θ is the angle between v and w.

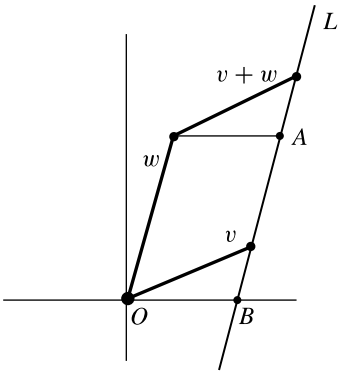


FIGURE 9

4. Given two vectors v and w, we'd expect to have a simple formula, involving the coordinates v_1, v_2, w_1, w_2, for the area of the parallelogram they span. Figure 9 indicates a strategy for finding such a formula: since the triangle with vertices w, A, $v + w$ is congruent to the triangle OBv, we can reduce the problem to an easier one where one side of the parallelogram lies along the horizontal axis:

(a) The line L passes through v and is parallel to w, so has slope w_2/w_1. Conclude that the point B has coordinate

$$\frac{v_1 w_2 - w_1 v_2}{w_2},$$

and that the parallelogram therefore has area

$$\det(v, w) = v_1 w_2 - w_1 v_2.$$

This formula, which defines the *determinant* det, certainly seems to be simple enough, but it can't really be true that $\det(v, w)$ always gives the area. After all, we clearly have

$$\det(w, v) = -\det(v, w),$$

so sometimes det will be negative! Indeed, it is easy to see that our "derivation" made all sorts of assumptions (that w_2 was positive, that B had a positive coordinate, etc.) Nevertheless, it seems likely that $\det(v, w)$ is $\pm$ the area; the next problem gives an independent proof.

5. (a) If v points along the positive horizontal axis, show that $\det(v, w)$ is the area of the parallelogram spanned by v and w for w above the horizontal axis ($w_2 > 0$), and the negative of the area for w below this axis.

(b) If R_θ is rotation by an angle of θ (Problem 1), show that

$$\det(R_\theta v, R_\theta w) = \det(v, w).$$

Conclude that $\det(v, w)$ is the area of the parallelogram spanned by v and w when the rotation from v to w is counterclockwise, and the negative of the area when it is clockwise.

6. Show that

$$\det(v, w + z) = \det(v, w) + \det(v, z)$$
$$\det(v + w, z) = \det(v, z) + \det(w, z)$$

and that

$$a \det(v, w) = \det(a \cdot v, w) = \det(v, a \cdot w).$$

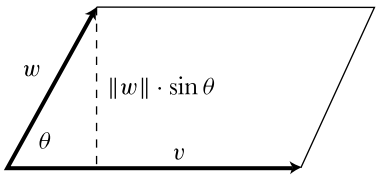


FIGURE 10

7. Using the method of Problem 3, show that

$$\det(v, w) = \|v\| \cdot \|w\| \cdot \sin\theta,$$

which is also obvious from the geometric interpretation (Figure 10).

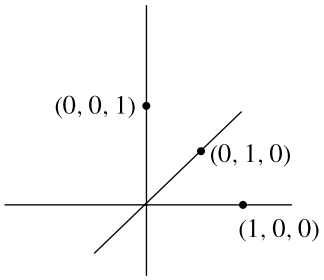


FIGURE 1

APPENDIX 2. THE CONIC SECTIONS

Although we will be concerned almost exclusively with figures in the plane, defined formally as the set of all pairs of real numbers, in this Appendix we want to consider three-dimensional space, which we can describe in terms of triples of real numbers, using a "three-dimensional coordinate system," consisting of three straight lines intersecting at right angles (Figure 1). Our *horizontal* and *vertical* axes now mutate to two axes in a horizontal plane, with the third axis perpendicular to both.

One of the simplest subsets of this three-dimensional space is the (infinite) *cone* illustrated in Figure 2; this cone may be produced by rotating a "generating line," of slope C say, around the third axis.

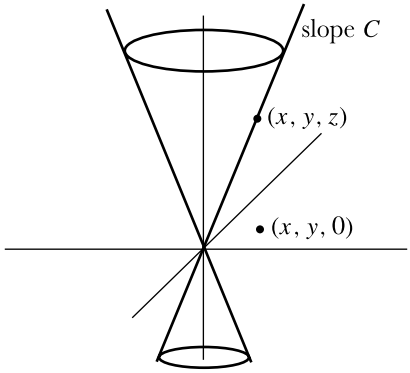


FIGURE 2

For any given first two coordinates x and y, the point $(x, y, 0)$ in the horizontal plane has distance $\sqrt{x^2 + y^2}$ from the origin, and thus

(1) (x, y, z) is in the cone if and only if $z = \pm C\sqrt{x^2 + y^2}$.

We can descend from these three-dimensional vistas to the more familiar two-dimensional one by asking what happens when we intersect this cone with some plane P (Figure 3).

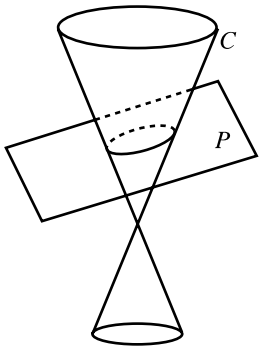


FIGURE 3

If the plane is parallel to the horizontal plane, there's certainly no mystery—the intersection is just a circle. Otherwise, the plane P intersects the horizontal plane

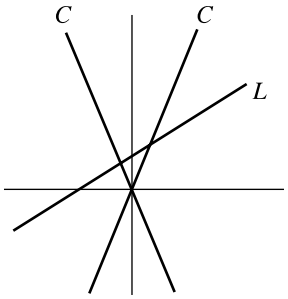


FIGURE 4

in a straight line. We can make things a lot simpler for ourselves if we rotate everything so that this intersection line points straight out from the plane of the paper, while the first axis is in the usual position that we are familiar with. The plane P is thus viewed "straight on," so that all we see (Figure 4) is its intersection L with the plane of the first and third axes; from this view-point the cone itself simply appears as two straight lines.

In the plane of the first and third axes, the line L can be described as the collection of all points of the form

$$(x, Mx + B),$$

where M is the slope of L. For an arbitrary point (x, y, z) it follows that

(2) (x, y, z) is in the plane P if and only if $z = Mx + B$.

Combining (1) and (2), we see that (x, y, z) is in the intersection of the cone and the plane if and only if

$$(*) \qquad Mx + B = \pm C\sqrt{x^2 + y^2}.$$

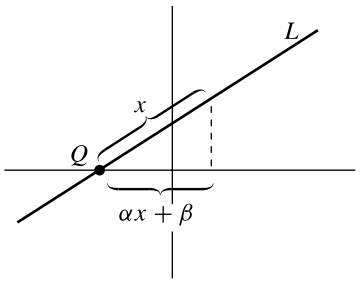


FIGURE 5

Now we have to choose coordinate axes in the plane P. We can choose L as the first axis, measuring distances from the intersection Q with the horizontal plane (Figure 5); for the second axis we just choose the line through Q parallel to our original second axis. If the first coordinate of a point in P with respect to these axes is x, then the first coordinate of this point with respect to the original axes can be written in the form

$$\alpha x + \beta$$

for some α and β. On the other hand, if the second coordinate of the point with respect to these axes is y, then y is also the second coordinate with respect to the original axes.

Consequently, $(*)$ says that the point lies on the intersection of the plane and the cone if and only if

$$M(\alpha x + \beta) + B = \pm C\sqrt{(\alpha x + \beta)^2 + y^2}.$$

Although this looks fairly complicated, after squaring we can write this as

$$\alpha^2 C^2 y^2 + \alpha^2 (M^2 - A^2)x^2 + Ex + F = 0$$

for some E and F that we won't bother writing out. Dividing by α^2 simplifies this to

$$C^2 y^2 + (C^2 - M^2)x^2 + Gx + H = 0.$$

Now Problem 4-16 indicates that this is either a parabola, an ellipse, or an hyperbola. In fact, looking a little more closely at the solution (and interchanging

the roles of x and y), we see that the values of G and H are irrelevant:

(1) If $M = \pm C$ we obtain a parabola;
(2) If $C^2 > M^2$ we obtain an ellipse;
(3) If $C^2 < M^2$ we obtain an hyperbola.

These analytic conditions are easy to interpret geometrically (Figure 6):

(1) If our plane is parallel to one of the generating lines of the cone we obtain a parabola;
(2) If our plane slopes less than the generating line of the cone (so that our intersection omits one half of the cone) we obtain an ellipse;
(3) If our plane slopes more than the generating line of the cone we obtain an hyperbola.

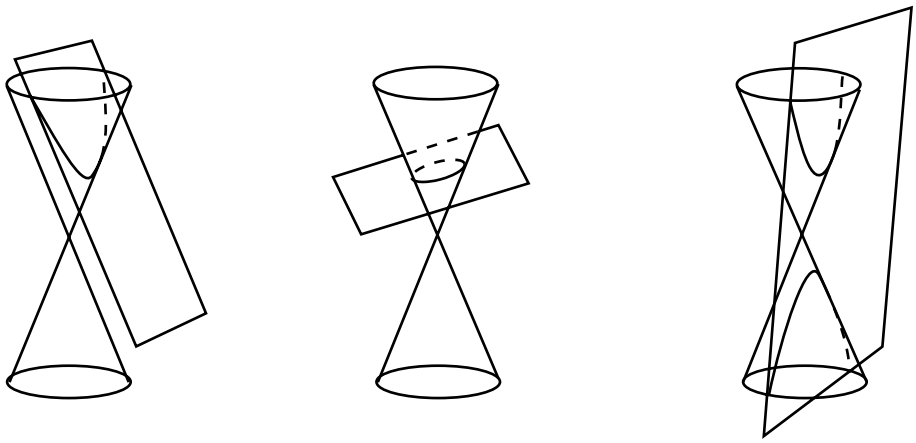


FIGURE 6

In fact, the very names of these "conic sections" are related to this description. The word *parabola* comes from a Greek root meaning 'alongside,' the same root that appears in parable, not to mention paradigm, paradox, paragon, paragraph, paralegal, parallax, parallel, even parachute. *Ellipse* comes from a Greek root meaning 'defect,' or omission, as in ellipsis (an omission, ... or the dots that indicate it). And *hyperbola* comes from a Greek root meaning 'throwing beyond,' or excess. With the currency of words like hyperactive, hypersensitive, and hyperventilate, not to mention hype, one can probably say, without risk of hyperbole, that this root is familiar to almost everyone.*

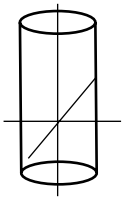


FIGURE 7

PROBLEMS

1. Consider a cylinder with a generator perpendicular to the horizontal plane (Figure 7); the only requirement for a point (x, y, z) to lie on this cylinder is

*Although the correspondence between these roots and the geometric picture correspond so beautifully, for the sake of dull accuracy it has to be reported that the Greeks originally applied the words to describe features of certain equations involving the conic sections.

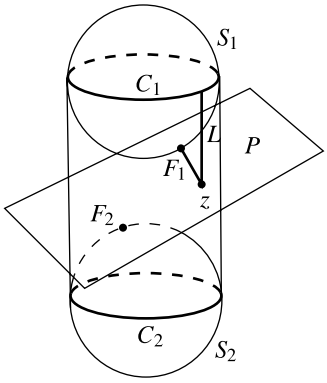


FIGURE 8

that (x, y) lies on a circle:

$$x^2 + y^2 = C^2.$$

Show that the intersection of a plane with this cylinder can be described by an equation of the form

$$(\alpha x + \beta)^2 + y^2 = C^2.$$

What possibilities are there?

2. In Figure 8, the sphere S_1 has the same diameter as the cylinder, so that its equator C_1 lies along the cylinder; it is also tangent to the plane P at F_1. Similarly, the equator C_2 of S_2 lies along the cylinder, and S_2 is tangent to P at F_2.

(a) Let z be any point on the intersection of P and the cylinder. Explain why the length of the line from z to F_1 is equal to the length of the vertical line L from z to C_1.

(b) By proving a similar fact for the length of the line from z to F_2, show that the distance from z to F_1 plus the distance from z to F_2 is a constant, so that the intersection is an ellipse, with foci F_1 and F_2.

3. Similarly, use Figure 9 to prove geometrically that the intersection of a plane and a cone is an ellipse.

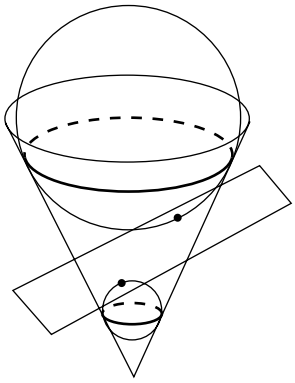


FIGURE 9

APPENDIX 3. POLAR COORDINATES

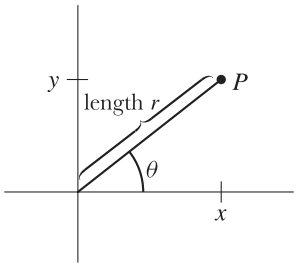
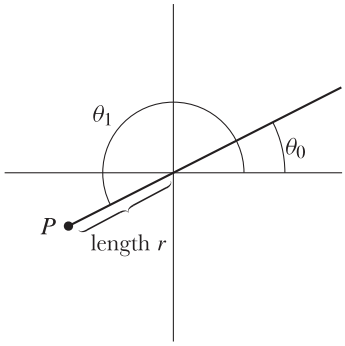


FIGURE 1

In this chapter we've been acting all along as if there's only one way to label points in the plane with pairs of numbers. Actually, there are many different ways, each giving rise to a different "coordinate system." The usual coordinates of a point are called its cartesian coordinates, after the French mathematician and philosopher René Descartes (1596–1650), who first introduced the idea of coordinate systems. In many situations it is more convenient to introduce polar coordinates, which are illustrated in Figure 1. To the point P we assign the polar coordinates (r, θ), where r is the distance from the origin O to P, and θ is the angle between the horizontal axis and the line from O to P. This angle can be measured either in degrees or in radians (Chapter 15), but in either case θ is not determined unambiguously. For example, with degree measurement, points on the right side of the horizontal axis could have either $\theta = 0$ or $\theta = 360$; moreover, θ is completely ambiguous at the origin O. So it is necessary to exclude some ray through the origin if we want to assign a unique pair (r, θ) to each point under consideration.

On the other hand, there is no problem associating a unique point to any pair (r, θ). In fact, we can even associate a point to (r, θ) when $r < 0$, according to the scheme indicated in Figure 2. Thus, it always makes sense to talk about "the point with polar coordinates (r, θ)," even though there is some ambiguity when we talk about "the polar coordinates" of a given point.



P is the point with polar coordinates (r, θ_1) and also the point with polar coordinates $(-r, \theta_2)$.

FIGURE 2

It is clear from Figure 1 (and Figure 2) that the point with polar coordinates (r, θ) has cartesian coordinates (x, y) given by

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

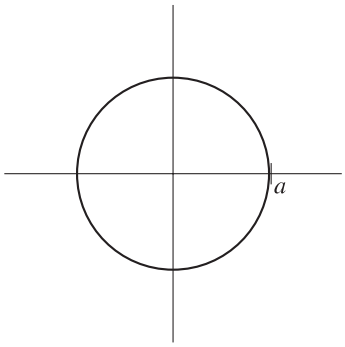


FIGURE 3

Conversely, if a point has cartesian coordinates (x, y), then (any of) its polar coordinates (r, θ) satisfy

$$r = \pm\sqrt{x^2 + y^2}$$
$$\tan\theta = \frac{y}{x} \qquad \text{if } x \neq 0.$$

Now suppose that f is a function. Then by the **graph of f in polar coordinates** we mean the collection of all points P with polar coordinates (r, θ) satisfying $r = f(\theta)$. In other words, the graph of f in polar coordinates is the collection of all points with polar coordinates $(f(\theta), \theta)$. No special significance should be attached to the fact that we are considering pairs $(f(\theta), \theta)$, with $f(\theta)$ first, as opposed to pairs $(x, f(x))$ in the usual graph of f; it is purely a matter of convention that r is considered the first polar coordinate and θ is considered the second.

The graph of f in polar coordinates is often described as "the graph of the equation $r = f(\theta)$." For example, suppose that f is a constant function, $f(\theta) = a$ for all θ. The graph of the equation $r = a$ is simply a circle with center O and radius a (Figure 3). This example illustrates, in a rather blatant way, that polar coordinates are likely to make things simpler in situations that involve symmetry with respect to the origin O.

The graph of the equation $r = \theta$ is shown in Figure 4. The solid line corresponds to all values of $\theta \geq 0$, while the dashed line corresponds to values of $\theta \leq 0$.

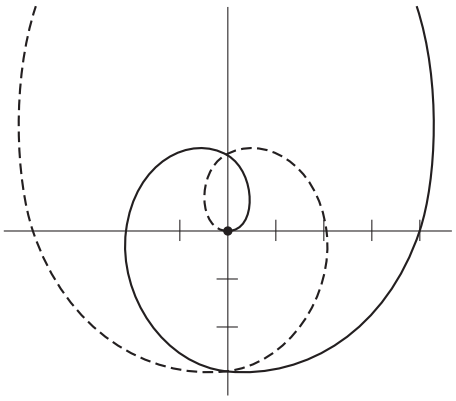


FIGURE 4 Spiral of Archimedes

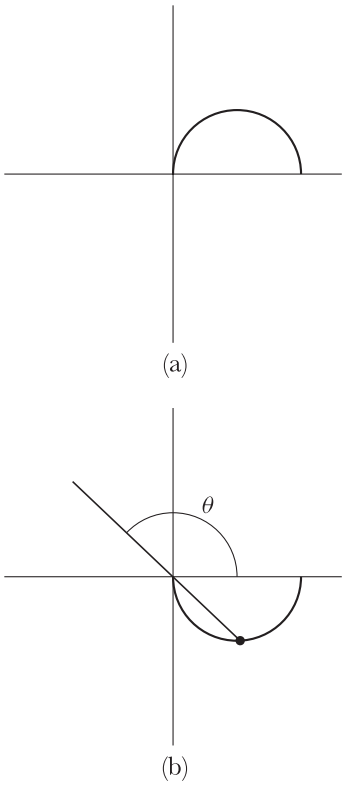


FIGURE 5

As another example involving both positive and negative r, consider the graph of the equation $r = \cos\theta$. Figure 5(a) shows the part that corresponds to $0 \leq \theta \leq 90$ [with θ in degrees]. Figure 5(b) shows the part corresponding to $90 \leq \theta \leq 180$; here $r < 0$. You can check that no new points are added for $\theta > 180$ or $\theta < 0$. It is easy to describe this same graph in terms of the cartesian coordinates of its points. Since the polar coordinates of any point on the graph satisfy

$$r = \cos\theta,$$

and hence

$$r^2 = r\cos\theta,$$

its cartesian coordinates satisfy the equation

$$x^2 + y^2 = x$$

which describes a circle (Problem 3-16). [Conversely, it is clear that if the cartesian coordinates of a point satisfy $x^2 + y^2 = x$, then it lies on the graph of the equation $r = \cos\theta$.]

Although we've now gotten a circle in two different ways, we might well be hesitant about trying to find the equation of an ellipse in polar coordinates. But it turns out that we can get a very nice equation if we choose one of the *foci* as the origin. Figure 6 shows an ellipse with one focus at O, with the sum of the distances of all points from O and the other focus $\mathbf{f}$ being $2a$. We've chosen $\mathbf{f}$ to the left of O, with coordinates written as

$$(-2\varepsilon a, 0).$$

(We have $0 \leq \varepsilon < 1$, since we must have $2a >$ distance from $\mathbf{f}$ to O).

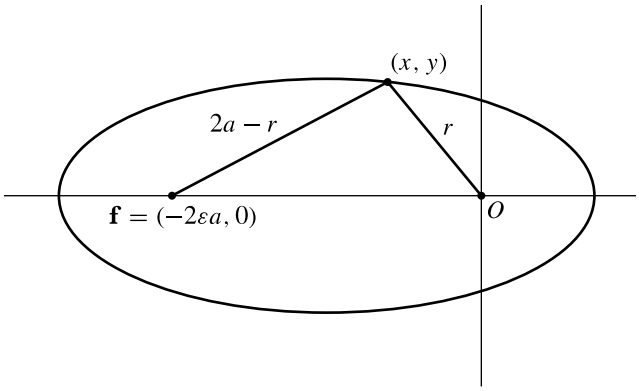


FIGURE 6

The distance r from (x, y) to O is given by

$$r^2 = x^2 + y^2. \tag{1}$$

By assumption, the distance from (x, y) to $\mathbf{f}$ is $2a - r$, hence

$$(2a - r)^2 = (x - [-2\varepsilon a])^2 + y^2,$$

or

$$4a^2 - 4ar + r^2 = x^2 + 4\varepsilon ax + 4\varepsilon^2 a^2 + y^2. \tag{2}$$

Subtracting (1) from (2), and dividing by $4a$, we get

$$a - r = \varepsilon x + \varepsilon^2 a,$$

or

$$\begin{aligned} r &= a - \varepsilon x - \varepsilon^2 a \\ &= (1 - \varepsilon^2)a - \varepsilon x, \end{aligned}$$

which we can write as

$$r = \Lambda - \varepsilon x, \qquad \text{for } \Lambda = (1 - \varepsilon^2)a. \tag{3}$$

Substituting $r \cos\theta$ for x, we have

$$\begin{aligned} r &= \Lambda - \varepsilon r \cos\theta, \\ r(1 + \varepsilon \cos\theta) &= \Lambda, \end{aligned}$$

and thus

$$r = \frac{\Lambda}{1 + \varepsilon \cos\theta}. \tag{4}$$

In Chapter 4 we found that

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{5}$$

is the equation in cartesian coordinates for an ellipse with $2a$ as the sum of the distances to the foci, but with the foci at $(-c, 0)$ and $(c, 0)$, where

$$b = \sqrt{a^2 - c^2}.$$

Since the distance between the foci is $2c$, this corresponds to the ellipse (4) when we take $c = \varepsilon a$ or $\varepsilon = c/a$ (with equation (3) determining Λ). Conversely, given the ellipse described by (4), for the corresponding equation (5) the value of a is determined by (3),

$$a = \frac{\Lambda}{1 - \varepsilon^2},$$

and again using $c = \varepsilon a$, we get

$$b = \sqrt{a^2 - c^2} = \sqrt{a^2 - \varepsilon^2 a^2} = a\sqrt{1 - \varepsilon^2} = \frac{\Lambda}{\sqrt{1 - \varepsilon^2}}.$$

Thus, we can obtain a and b, the lengths of the major and minor axes, immediately from ε and Λ.

The number

$$\varepsilon = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - \left(\frac{b}{a}\right)^2},$$

the *eccentricity* of the ellipse, determines the "shape" of the ellipse (the ratio of the major and minor axes), while the number Λ determines its "size," as shown by (4).

PROBLEMS

1. If two points have polar coordinates (r_1, θ_1) and (r_2, θ_2), show that the distance d between them is given by

$$d^2 = {r_1}^2 + {r_2}^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2).$$

What does this say geometrically?

2. Describe the general features of the graph of f in polar coordinates if

(i) f is even.
(ii) f is odd.
(iii) $f(\theta) = f(\theta + 180)$ [when θ is measured in degrees].

3. Sketch the graphs of the following equations.

(i) $r = a \sin\theta$.
(ii) $r = a \sec\theta$. Hint: It is a very simple graph!
(iii) $r = \cos 2\theta$. Good luck on this one!
(iv) $r = \cos 3\theta$.
(v) $r = |\cos 2\theta|$.
(vi) $r = |\cos 3\theta|$.

4. Find equations for the cartesian coordinates of points on the graphs (i), (ii) and (iii) in Problem 3.

5. Consider a hyperbola, where the difference of the distance between the two foci is the constant $2a$, and choose one focus at O and the other at $(-2\varepsilon a, 0)$. (In this case, we must have $\varepsilon > 1$). Show that we obtain the exact same equation in polar coordinates

$$r = \frac{\Lambda}{1 + \varepsilon \cos\theta}$$

as we obtained for an ellipse.

6. Consider the set of points (x, y) such that the distance (x, y) to O is equal to the distance from (x, y) to the line $y = a$ (Figure 7). Show that the distance to the line is $a - r\cos\theta$, and conclude that the equation can be written

$$a = r(1 + \cos\theta).$$

Notice that this equation for a parabola is again of the same from as (4).

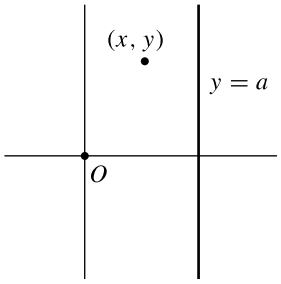


FIGURE 7

7. Now, for any Λ and ε, consider the graph in polar coordinates of the equation (4), which implies (3). Show that the points satisfying this equation satisfy

$$(1 - \varepsilon^2)x^2 + y^2 = \Lambda^2 - 2\Lambda\varepsilon x.$$

Using Problem 4-16, show that this is an ellipse for $\varepsilon < 1$, a parabola for $\varepsilon = 1$, and a hyperbola for $\varepsilon > 1$.

8. (a) Sketch the graph of the *cardioid* $r = 1 - \sin\theta$.
(b) Show that it is also the graph of $r = -1 - \sin\theta$.
(c) Show that it can be described by the equation

$$x^2 + y^2 = \sqrt{x^2 + y^2} - y,$$

and conclude that it can be described by the equation

$$(x^2 + y^2 + y)^2 = x^2 + y^2$$

9. Sketch the graphs of the following equations.

(i) $r = 1 - \frac{1}{2}\sin\theta$.
(ii) $r = 1 - 2\sin\theta$.
(iii) $r = 2 + \cos\theta$.

10. (a) Sketch the graph of the *lemniscate*

$$r^2 = 2a^2 \cos 2\theta.$$

(b) Find an equation for its cartesian coordinates.
(c) Show that it is the collection of all points P in Figure 8 satisfying $d_1 d_2 = a^2$.
(d) Make a guess about the shape of the curves formed by the set of all P satisfying $d_1 d_2 = b$, when $b > a^2$ and when $b < a^2$.

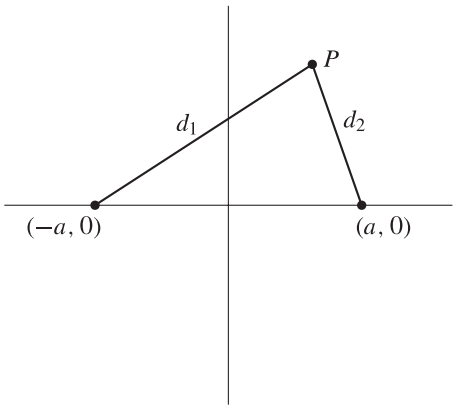


FIGURE 8

CHAPTER 5 LIMITS

The concept of a limit is surely the most important, and probably the most difficult one in all of calculus. The goal of this chapter is the definition of limits, but we are, once more, going to begin with a provisional definition; what we shall define is not the word "limit" but the notion of a function approaching a limit.

PROVISIONAL DEFINITION

The function f approaches the limit l near a, if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a.

Of the six functions graphed in Figure 1, only the first three approach l at a. Notice that although $g(a)$ is not defined, and $h(a)$ is defined "the wrong way," it is still true that g and h approach l near a. This is because we explicitly ruled out, in our definition, the necessity of ever considering the value of the function at a—it is only necessary that $f(x)$ should be close to l for x close to a, but *unequal to* a. We are simply not interested in the value of $f(a)$, or even in the question of whether $f(a)$ is defined.

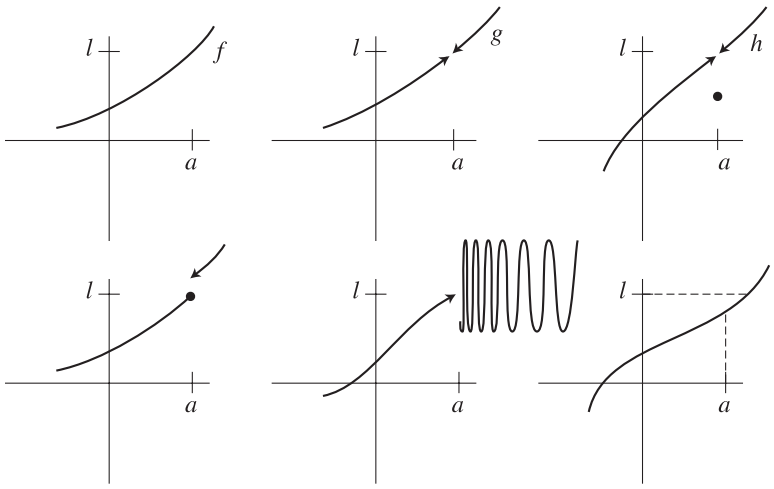
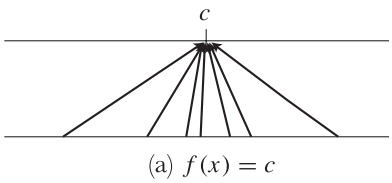
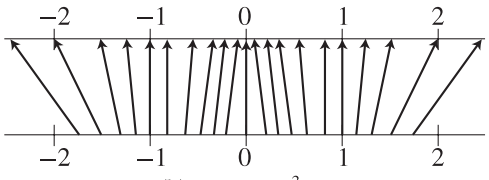


FIGURE 1

One convenient way of picturing the assertion that f approaches l near a is provided by a method of drawing functions that was not mentioned in Chapter 4. In this method, we draw two straight lines, each representing $\mathbf{R}$, and arrows from a point x in one, to $f(x)$ in the other. Figure 2 illustrates such a picture for two different functions.



(a) $f(x) = c$



(b) $f(x) = x^3$

FIGURE 2

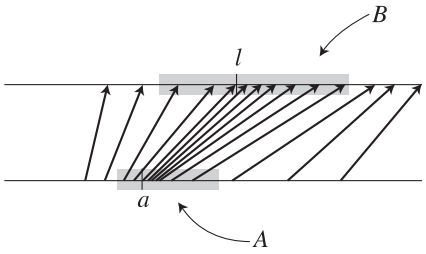


FIGURE 3

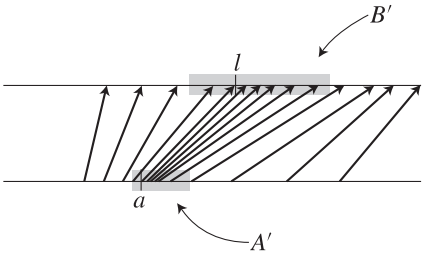


FIGURE 4

Now consider a function f whose drawing looks like Figure 3. Suppose we ask that $f(x)$ be close to l, say within the open interval B which has been drawn in Figure 3. This can be guaranteed if we consider only the numbers x in the interval A of Figure 3. (In this diagram we have chosen the largest interval which will work; any smaller interval containing a could have been chosen instead.) If we choose a smaller interval B' (Figure 4) we will, usually, have to choose a smaller A', but no matter how small we choose the open interval B, there is always supposed to be some open interval A which works.

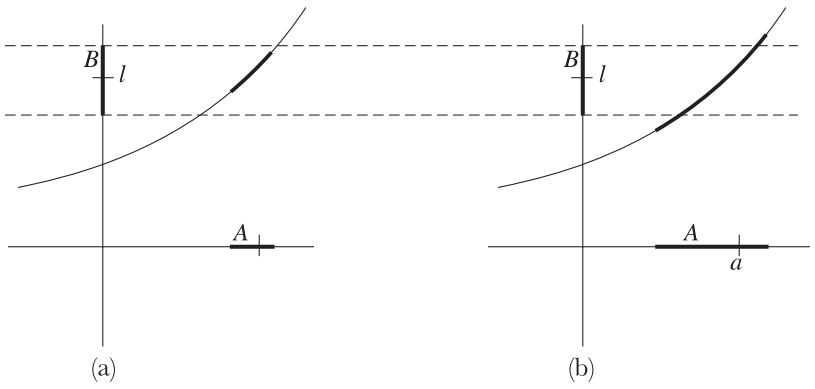
A similar pictorial interpretation is possible in terms of the graph of f, but in this case the interval B must be drawn on the vertical axis, and the set A on the horizontal axis. The fact that $f(x)$ is in B when x is in A means that the part of the graph lying over A is contained in the region which is bounded by the horizontal lines through the end points of B; compare Figure 5(a), where a valid interval A has been chosen, with Figure 5(b), where A is too large.

In order to apply our definition to a particular function, let us consider $f(x) = x \sin 1/x$ (Figure 6). Despite the erratic behavior of this function near 0 it is clear, at least intuitively, that f approaches 0 near 0, and it is certainly to be hoped that our definition will allow us to reach the same conclusion. In the case we are considering, both a and l of the definition are 0, so we must ask if we can get $f(x) = x \sin 1/x$ as close to 0 as desired if we require that x be sufficiently close to 0, but $\neq 0$. To be specific, suppose we wish to get $x \sin 1/x$ within $\frac{1}{10}$ of 0. This means we want

$$-\frac{1}{10} < x \sin \frac{1}{x} < \frac{1}{10},$$

or, more succinctly, $|x \sin 1/x| < \frac{1}{10}$. Now this is easy. Since

$$\left| \sin \frac{1}{x} \right| \leq 1, \qquad \text{for all } x \neq 0,$$



(a) (b)

FIGURE 5

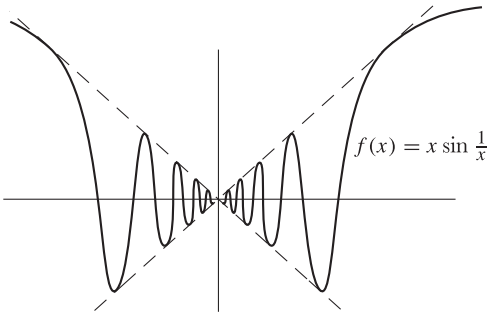


FIGURE 6

we have

$$\left| x \sin \frac{1}{x} \right| \leq |x|, \qquad \text{for all } x \neq 0.$$

This means that if $|x| < \frac{1}{10}$ and $x \neq 0$, then $|x \sin 1/x| < \frac{1}{10}$; in other words, $x \sin 1/x$ is within $\frac{1}{10}$ of 0 provided that x is within $\frac{1}{10}$ of 0, but $\neq 0$. There is nothing special about the number $\frac{1}{10}$; it is just as easy to guarantee that $|f(x)-0| < \frac{1}{100}$ —simply require that $|x| < \frac{1}{100}$, but $x \neq 0$. In fact, if we take any positive number ε we can make $|f(x) - 0| < \varepsilon$ simply by requiring that $|x| < \varepsilon$, and $x \neq 0$.

For the function $f(x) = x^2 \sin 1/x$ (Figure 7) it seems even clearer that f approaches 0 near 0. If, for example, we want

$$\left| x^2 \sin \frac{1}{x} \right| < \frac{1}{10},$$

then we certainly need only require that $|x| < \frac{1}{10}$ and $x \neq 0$, since this implies that $|x^2| < \frac{1}{100}$ and consequently

$$\left| x^2 \sin \frac{1}{x} \right| \leq |x^2| < \frac{1}{100} < \frac{1}{10}.$$

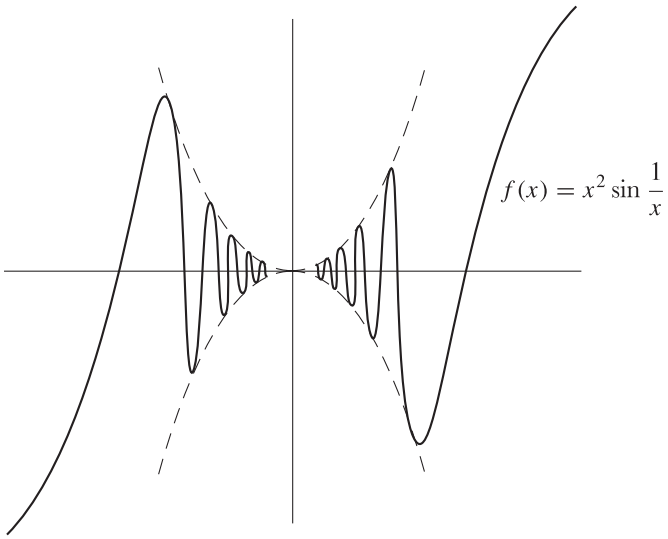


FIGURE 7

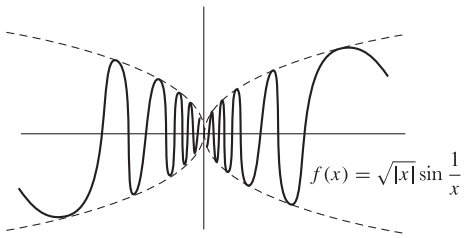


FIGURE 8

(We could do even better, and allow $|x| < 1/\sqrt{10}$ and $x \neq 0$, but there is no particular virtue in being as economical as possible.) In general, if $\varepsilon > 0$, to ensure that

$$\left| x^2 \sin \frac{1}{x} \right| < \varepsilon,$$

we need only require that

$$|x| < \varepsilon \quad \text{and} \quad x \neq 0,$$

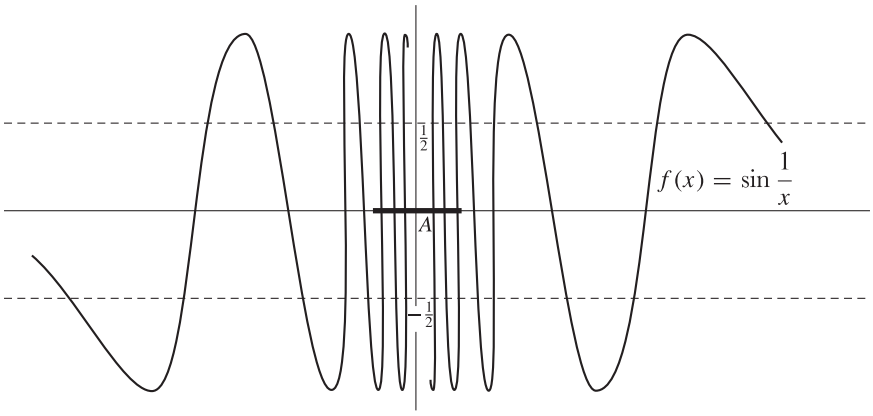


FIGURE 9

provided that $\varepsilon \leq 1$. If we are given an ε which is greater than 1 (it might be, even thought it is "small" ε's which are of interest), then it does not suffice to require that $|x| < \varepsilon$, but it certainly suffices to require that $|x| < 1$ and $x \neq 0$.

As a third example, consider the function $f(x) = \sqrt{|x|} \sin 1/x$ (Figure 8). In order to make $|\sqrt{|x|} \sin 1/x| < \varepsilon$ we can require that

$$|x| < \varepsilon^2 \quad \text{and} \quad x \neq 0$$

(the algebra is left to you).

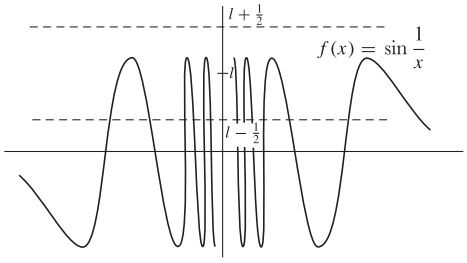


FIGURE 10

Finally, let us consider the function $f(x) = \sin 1/x$ (Figure 9). For this function it is *false* that f approaches 0 near 0. This amounts to saying that it is not true for every number $\varepsilon > 0$ that we can get $|f(x) - 0| < \varepsilon$ by choosing x sufficiently small, and $\neq 0$. To show this we simply have to find *one* $\varepsilon > 0$ for which the condition $|f(x) - 0| < \varepsilon$ cannot be guaranteed, no matter how small we require $|x|$ to be. In fact, $\varepsilon = \frac{1}{2}$ will do: it is impossible to ensure that $|f(x)| < \frac{1}{2}$ no matter how small we require $|x|$ to be; for if A is any interval containing 0, there is some number $x = 1/(90 + 360n)$ which is in this interval, and for this x we have $f(x) = 1$.

This same argument can be used (Figure 10) to show that f does not approach *any* number near 0. To show this we must again find, for any particular number l, some number $\varepsilon > 0$ so that $|f(x) - l| < \varepsilon$ is *not* true, no matter how small x is required to be. The choice $\varepsilon = \frac{1}{2}$ works for any number l; that is, no matter how small we require $|x|$ to be, we cannot ensure that $|f(x) - l| < \frac{1}{2}$. The reason is, that for any interval A containing 0 there is some $x_1 = 1/(90 + 360n)$ in this interval, so that

$$f(x_1) = 1,$$

and also some $x_2 = 1/(270 + 360m)$ in this interval, so that

$$f(x_2) = -1.$$

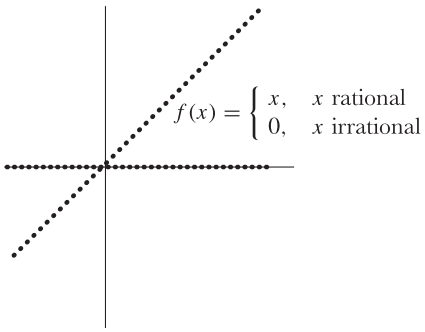


FIGURE 11

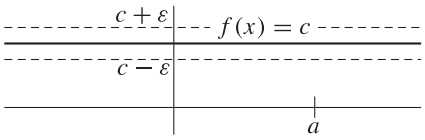


FIGURE 12

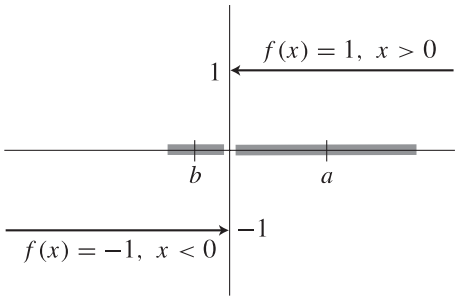


FIGURE 13

But the interval from $l - \frac{1}{2}$ to $l + \frac{1}{2}$ cannot contain both -1 and 1, since its total length is only 1; so we cannot have

$$|1 - l| < \tfrac{1}{2} \quad \text{and also} \quad |-1 - l| < \tfrac{1}{2},$$

no matter what l is.

The phenomenon exhibited by $f(x) = \sin 1/x$ near 0 can occur in many ways. If we consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational,} \end{cases}$$

then, no matter what a is, f does not approach any number l near a. In fact, we cannot make $|f(x) - l| < \frac{1}{4}$ no matter how close we bring x to a, because in any interval around a there are numbers x with $f(x) = 0$, and also numbers x with $f(x) = 1$, so that we would need $|0 - l| < \frac{1}{4}$ and also $|1 - l| < \frac{1}{4}$.

An amusing variation on this behavior is presented by the function shown in Figure 11:

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational.} \end{cases}$$

The behavior of this function is "opposite" to that of $g(x) = \sin 1/x$; it approaches 0 at 0, but does not approach any number at a, if $a \neq 0$. By now you should have no difficulty convincing yourself that this is true.

As a contrast to the functions considered so far, which have been quite pathological, we will now examine some of the simplest functions.

If $f(x) = c$, then f approaches c near a, for every number a. In fact, to ensure that $|f(x) - c| < \varepsilon$ one does not need to restrict x to be near a at all; the condition is automatically satisfied (Figure 12).

As a slight variation, let f be the function shown in Figure 13:

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0. \end{cases}$$

If $a > 0$, then f approaches 1 near a: indeed, to ensure that $|f(x) - 1| < \varepsilon$ it certainly suffices to require that $|x - a| < a$, since this implies

$$\begin{aligned} -a &< x - a \\ \text{or} \quad 0 &< x \end{aligned}$$

so that $f(x) = 1$. Similarly, if $b < 0$, then f approaches -1 near b: to ensure that $|f(x) - (-1)| < \varepsilon$ it suffices to require that $|x - b| < -b$. Finally, as you may easily check, f does not approach any number near 0.

The function $f(x) = x$ is easily dealt with. Clearly f approaches a near a: to ensure that $|f(x) - a| < \varepsilon$ we just have to require that $|x - a| < \varepsilon$.

The function $f(x) = x^2$ requires a little more work. To show that f approaches a^2 near a, we must decide how to ensure that

$$|x^2 - a^2| < \varepsilon.$$

Factoring looks like the most promising procedure: we want

$$|x-a| \cdot |x+a| < \varepsilon.$$

Obviously the factor $|x+a|$ is the one that will cause trouble. On the other hand, there is no need to make $|x+a|$ particularly small; as long as we know *some* bound on the values of $|x+a|$ we will be in good shape. For example, if $|x+a| < 1{,}000{,}000$, then we will just need to require that $|x-a| < \varepsilon/1{,}000{,}000$. Therefore, to begin with, let us require that $|x-a| < 1$ (any positive number other than 1 would do just as well); presumably this will ensure that x is not too large, and consequently that $|x+a|$ is not too large. As a matter of fact, Problem 1-12 shows that

$$|x| - |a| \leq |x-a| < 1,$$

so

$$|x| < 1 + |a|,$$

and consequently

$$|x+a| \leq |x| + |a| < 2|a| + 1.$$

Now we need only the additional requirement that $|x-a| < \varepsilon/(2|a|+1)$. In other words,

$$\text{if } |x-a| < \min\left(1, \frac{\varepsilon}{2|a|+1}\right), \quad \text{then } |x^2 - a^2| < \varepsilon.$$

Naturally, $\min(1, \varepsilon/(2|a|+1))$ will just be $\varepsilon/(2|a|+1)$ for small ε.

Precisely the same sort of trick will show that if $f(x) = x^3$, then f approaches a^3 near a. In fact,

$$\text{if } |x-a| < \min\left(1, \frac{\varepsilon}{(1+|a|)^2 + |a|(1+|a|) + |a|^2}\right), \quad \text{then } |x^3 - a^3| < \varepsilon.$$

The proof of this assertion will show where the weird denominator comes from: If $|x-a| < 1$, then $|x| < |a| + 1$, and consequently

$$\begin{aligned} |x^2 + ax + a^2| &\leq |x|^2 + |a| \cdot |x| + |a|^2 \\ &< (1+|a|)^2 + |a|(1+|a|) + |a|^2. \end{aligned}$$

Therefore

$$\begin{aligned} |x^3 - a^3| &= |x-a| \cdot |x^2 + ax + a^2| \\ &< \frac{\varepsilon}{(1+|a|)^2 + |a|(1+|a|) + |a|^2} \cdot [(1+|a|)^2 + |a|(1+|a|) + |a|^2] \\ &= \varepsilon. \end{aligned}$$

The time has now come to point out that of the many demonstrations about limits which we have given, not one has been a real proof. The fault lies not with our reasoning, but with our definition. If our provisional definition of a function was open to criticism, our provisional definition of approaching a limit is even more vulnerable. This definition is simply not sufficiently precise to be used in proofs. It is hardly clear how one "makes" $f(x)$ close to l (whatever "close" means) by "requiring" x to be sufficiently close to a (however close "suffi-

ciently" close is supposed to be). Despite the criticisms of our definition you may feel (I certainly hope you do) that our arguments were nevertheless quite convincing. In order to present any sort of argument at all, we have been practically forced to invent the real definition. It is possible to arrive at this definition in several steps, each one clarifying some obscure phrase which still remains. Let us begin, once again, with the provisional definition:

> The function f approaches the limit l near a, if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a.

The very first change which we made in this definition was to note that making $f(x)$ close to l meant making $|f(x) - l|$ small, and similarly for x and a:

> The function f approaches the limit l near a, if we can make $|f(x) - l|$ as small as we like by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

The second, more crucial, change was to note that making $|f(x) - l|$ "as small as we like" means making $|f(x) - l| < \varepsilon$ for any $\varepsilon > 0$ that happens to be given us:

> The function f approaches the limit l near a, if for every number $\varepsilon > 0$ we can make $|f(x) - l| < \varepsilon$ by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

There is a common pattern to all the demonstrations about limits which we have given. For each number $\varepsilon > 0$ we found some other positive number, δ say, with the property that if $x \neq a$ and $|x - a| < \delta$, then $|f(x) - l| < \varepsilon$. For the function $f(x) = x \sin 1/x$ (with $a = 0$, $l = 0$), the number δ was just the number ε; for $f(x) = \sqrt{|x|} \sin 1/x$, it was ε^2; for $f(x) = x^2$ it was the minimum of 1 and $\varepsilon/(2|a| + 1)$. In general, it may not be at all clear how to find the number δ, given ε, but it is the condition $|x - a| < \delta$ which expresses how small "sufficiently" small must be:

> The function f approaches the limit l near a, if for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x, if $|x - a| < \delta$ and $x \neq a$, then $|f(x) - l| < \varepsilon$.

This is practically the definition we will adopt. We will make only one trivial change, noting that "$|x - a| < \delta$ and $x \neq a$" can just as well be expressed "$0 < |x - a| < \delta$."

DEFINITION

The function ***f* approaches the limit *l* near *a*** means: for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x, if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$.

This definition is so important (*everything* we do from now on depends on it) that proceeding any further without knowing it is hopeless. If necessary memorize it, like a poem! That, at least, is better than stating it incorrectly; if you do this you are doomed to give incorrect proofs. A good exercise in giving correct proofs is to review every fact already demonstrated about functions approaching limits, giving real proofs of each. This requires writing down the correct definition of what you are proving, but not much more—all the algebraic work has been done already. When proving that f does *not* approach l at a, be sure to negate the definition correctly:

If it is *not* true that

> for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x, if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$,

then

> there is *some* $\varepsilon > 0$ such that for *every* $\delta > 0$ there is *some* x which satisfies $0 < |x - a| < \delta$ but not $|f(x) - l| < \varepsilon$.

Thus, to show that the function $f(x) = \sin 1/x$ does not approach 0 near 0, we consider $\varepsilon = \frac{1}{2}$ and note that for every $\delta > 0$ there is some x with $0 < |x - 0| < \delta$ but not $|\sin 1/x - 0| < \frac{1}{2}$—namely, an x of the form $1/(90 + 360n)$, where n is so large that $1/(90 + 360n) < \delta$.

As an illustration of the use of the definition of a function approaching a limit, we have reserved the function shown in Figure 14, a standard example, but one of the most complicated:

$$f(x) = \begin{cases} 0, & x \text{ irrational, } 0 < x < 1 \\ 1/q, & x = p/q \text{ in lowest terms, } 0 < x < 1. \end{cases}$$

(Recall that p/q is in lowest terms if p and q are integers with no common factor and $q > 0$.)

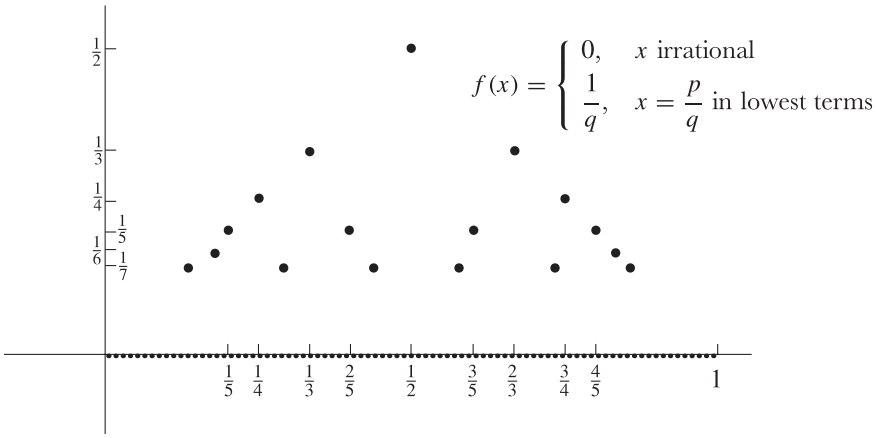


FIGURE 14

For any number a, with $0 < a < 1$, the function f approaches 0 at a. To prove this, consider any number $\varepsilon > 0$. Let n be a natural number so large that $1/n \leq \varepsilon$. Notice that the only numbers x for which $|f(x) - 0| < \varepsilon$ could be false are:

$$\frac{1}{2};\ \frac{1}{3}, \frac{2}{3};\ \frac{1}{4}, \frac{3}{4};\ \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5};\ \ldots;\ \frac{1}{n}, \ldots, \frac{n-1}{n}.$$

(If a is rational, then a might be one of these numbers.) However many of these numbers there may be, there are, at any rate, only finitely many. Therefore, of all these numbers, one is closest to a; that is, $|p/q - a|$ is smallest for one p/q among these numbers. (If a happens to be one of these numbers, then consider only the values $|p/q - a|$ for $p/q \neq a$.) This closest distance may be chosen as the δ. For if $0 < |x - a| < \delta$, then x is *not* one of

$$\frac{1}{2}, \ldots, \frac{n-1}{n}$$

and therefore $|f(x) - 0| < \varepsilon$ *is* true. This completes the proof. Note that our description of the δ which works for a given ε is completely adequate—there is no reason why we must give a formula for δ in terms of ε.

Armed with our definition, we are now prepared to prove our first theorem; you have probably assumed the result all along, which is a very reasonable thing to do. This theorem is really a test case for our definition: if the theorem could not be proved, our definition would be useless.

THEOREM 1 A function cannot approach two different limits near a. In other words, if f approaches l near a, and f approaches m near a, then $l = m$.

PROOF Since this is our first theorem about limits it will certainly be necessary to translate the hypotheses according to the definition.

Since f approaches l near a, we know that for any $\varepsilon > 0$ there is some number $\delta_1 > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \varepsilon.$$

We also know, since f approaches m near a, that there is some $\delta_2 > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta_2, \text{ then } |f(x) - m| < \varepsilon.$$

We have had to use two numbers, δ_1 and δ_2, since there is no guarantee that the δ which works in one definition will work in the other. But, in fact, it is now easy to conclude that for any $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \varepsilon \text{ and } |f(x) - m| < \varepsilon;$$

we simply choose $\delta = \min(\delta_1, \delta_2)$.

To complete the proof we just have to pick a particular $\varepsilon > 0$ for which the two conditions

$$|f(x) - l| < \varepsilon \quad \text{and} \quad |f(x) - m| < \varepsilon$$

cannot both hold, if $l \neq m$. The proper choice is suggested by Figure 15. If $l \neq m$, so that $|l - m| > 0$, we can choose $|l - m|/2$ as our ε. It follows that there is a $\delta > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \frac{|l - m|}{2}$$
$$\text{and } |f(x) - m| < \frac{|l - m|}{2}.$$

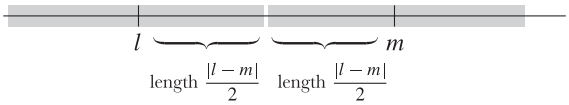


FIGURE 15

This implies that for $0 < |x - a| < \delta$ we have

$$\begin{aligned} |l - m| = |l - f(x) + f(x) - m| &\leq |l - f(x)| + |f(x) - m| \\ &< \frac{|l - m|}{2} + \frac{|l - m|}{2} \\ &= |l - m|, \end{aligned}$$

a contradiction. ∎

The number l which f approaches near a is denoted by $\lim_{x \to a} f(x)$ (read: the limit of $f(x)$ as x approaches a). This definition is possible only because of Theorem 1, which ensures that $\lim_{x \to a} f(x)$ never has to stand for two different numbers. The equation

$$\lim_{x \to a} f(x) = l$$

has exactly the same meaning as the phrase

$$f \text{ approaches } l \text{ near } a.$$

The possibility still remains that f does not approach l near a, for any l, so that $\lim_{x \to a} f(x) = l$ is false for every number l. This is usually expressed by saying that "$\lim_{x \to a} f(x)$ does not exist."

Notice that our new notation introduces an extra, utterly irrelevant letter x, which could be replaced by t, y, or any other letter which does not already appear—the symbols

$$\lim_{x \to a} f(x), \qquad \lim_{t \to a} f(t), \qquad \lim_{y \to a} f(y),$$

all denote precisely the same number, which depends on f and a, and has nothing to do with x, t, or y (these letters, in fact, do not denote anything at all). A more logical symbol would be something like $\lim_{a} f$, but this notation, despite its brevity, is so infuriatingly rigid that almost no one has seriously tried to use it. The notation $\lim_{x \to a} f(x)$ is much more useful because a function f often has no simple name, even

though it might be possible to express $f(x)$ by a simple formula involving x. Thus, the short symbol

$$\lim_{x \to a} (x^2 + \sin x)$$

could be paraphrased only by the awkward expression

$$\lim_{a} f, \text{ where } f(x) = x^2 + \sin x.$$

Another advantage of the standard symbolism is illustrated by the expressions

$$\lim_{x \to a} x + t^3,$$
$$\lim_{t \to a} x + t^3.$$

The first means the number which f approaches near a when

$$f(x) = x + t^3, \quad \text{for all } x;$$

the second means the number which f approaches near a when

$$f(t) = x + t^3, \quad \text{for all } t.$$

You should have little difficulty (especially if you consult Theorem 2) proving that

$$\lim_{x \to a} x + t^3 = a + t^3,$$
$$\lim_{t \to a} x + t^3 = x + a^3.$$

These examples illustrate the main advantage of our notation, which is its flexibility. In fact, the notation $\lim_{x \to a} f(x)$ is so flexible that there is some danger of forgetting what it really means. Here is a simple exercise in the use of this notation, which will be important later: first interpret precisely, and then prove the equality of the expressions

$$\lim_{x \to a} f(x) \quad \text{and} \quad \lim_{h \to 0} f(a + h).$$

An important part of this chapter is the proof of a theorem which will make it easy to find many limits. The proof depends upon certain properties of inequalities and absolute values, hardly surprising when one considers the definition of limit. Although these facts have already been stated in Problems 1-20, 1-21, and 1-22, because of their importance they will be presented once again, in the form of a lemma (a lemma is an auxiliary theorem, a result that justifies its existence only by virtue of its prominent role in the proof of another theorem). The lemma says, roughly, that if x is close to x_0, and y is close to y_0, then $x + y$ will be close to $x_0 + y_0$, and xy will be close to $x_0 y_0$, and $1/y$ will be close to $1/y_0$. This intuitive statement is much easier to remember than the precise estimates of the lemma, and it is not unreasonable to read the proof of Theorem 2 first, in order to see just how these estimates are used.

LEMMA (1) If

$$|x - x_0| < \frac{\varepsilon}{2} \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2},$$

then

$$|(x + y) - (x_0 + y_0)| < \varepsilon.$$

(2) If

$$|x - x_0| < \min\left(1, \frac{\varepsilon}{2(|y_0| + 1)}\right) \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then

$$|xy - x_0y_0| < \varepsilon.$$

(3) If $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon|y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| < \varepsilon.$$

PROOF (1)

$$\begin{aligned} |(x + y) - (x_0 + y_0)| &= |(x - x_0) + (y - y_0)| \\ &\leq |x - x_0| + |y - y_0| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(2) Since $|x - x_0| < 1$ we have

$$|x| - |x_0| \leq |x - x_0| < 1,$$

so that

$$|x| < 1 + |x_0|.$$

Thus

$$\begin{aligned} |xy - x_0y_0| &= |x(y - y_0) + y_0(x - x_0)| \\ &\leq |x| \cdot |y - y_0| + |y_0| \cdot |x - x_0| \\ &< (1 + |x_0|) \cdot \frac{\varepsilon}{2(|x_0| + 1)} + |y_0| \cdot \frac{\varepsilon}{2(|y_0| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(3) We have

$$|y_0| - |y| \leq |y - y_0| < \frac{y_0}{2},$$

so $|y| > |y_0|/2$. In particular, $y \neq 0$, and

$$\frac{1}{|y|} < \frac{2}{|y_0|}.$$

Thus

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| = \frac{|y_0 - y|}{|y| \cdot |y_0|} < \frac{2}{|y_0|} \cdot \frac{1}{|y_0|} \cdot \frac{\varepsilon|y_0|^2}{2} = \varepsilon. \blacksquare$$

THEOREM 2 If $\lim_{x\to a} f(x) = l$ and $\lim_{x\to a} g(x) = m$, then

$$(1) \quad \lim_{x\to a}(f+g)(x) = l + m;$$
$$(2) \quad \lim_{x\to a}(f\cdot g)(x) = l \cdot m.$$

Moreover, if $m \neq 0$, then

$$(3) \quad \lim_{x\to a}\left(\frac{1}{g}\right)(x) = \frac{1}{m}.$$

PROOF The hypothesis means that for every $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$\text{if } 0 < |x-a| < \delta_1, \text{ then } |f(x) - l| < \varepsilon,$$
$$\text{and} \quad \text{if } 0 < |x-a| < \delta_2, \text{ then } |g(x) - m| < \varepsilon.$$

This means (since, after all, $\varepsilon/2$ is also a positive number) that there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$\text{if } 0 < |x-a| < \delta_1, \text{ then } |f(x) - l| < \frac{\varepsilon}{2},$$
$$\text{and} \quad \text{if } 0 < |x-a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2}.$$

Now let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x-a| < \delta$, then $0 < |x-a| < \delta_1$ and $0 < |x-a| < \delta_2$ are both true, so both

$$|f(x) - l| < \frac{\varepsilon}{2} \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2}$$

are true. But by part (1) of the lemma this implies that $|(f+g)(x) - (l+m)| < \varepsilon$. This proves (1).

To prove (2) we proceed similarly, after consulting part (2) of the lemma. If $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$\text{if } 0 < |x-a| < \delta_1, \text{ then } |f(x) - l| < \min\left(1, \frac{\varepsilon}{2(|m|+1)}\right),$$
$$\text{and} \quad \text{if } 0 < |x-a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2(|l|+1)}.$$

Again let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x-a| < \delta$, then

$$|f(x) - l| < \min\left(1, \frac{\varepsilon}{2(|m|+1)}\right) \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2(|l|+1)}.$$

So, by the lemma, $|(f\cdot g)(x) - l\cdot m| < \varepsilon$, and this proves (2).

Finally, if $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x,

$$\text{if } 0 < |x-a| < \delta, \text{ then } |g(x) - m| < \min\left(\frac{|m|}{2}, \frac{\varepsilon|m|^2}{2}\right).$$

But according to part (3) of the lemma this means, first, that $g(x) \neq 0$, so $(1/g)(x)$ makes sense, and second that

$$\left| \left(\frac{1}{g}\right)(x) - \frac{1}{m} \right| < \varepsilon.$$

This proves (3). ∎

Using Theorem 2 we can prove, trivially, such facts as

$$\lim_{x \to a} \frac{x^3 + 7x^5}{x^2 + 1} = \frac{a^3 + 7a^5}{a^2 + 1},$$

without going through the laborious process of finding a δ, given an ε. We must begin with

$$\lim_{x \to a} 7 = 7,$$
$$\lim_{x \to a} 1 = 1,$$
$$\lim_{x \to a} x = a,$$

but these are easy to prove directly. If we *want* to find the δ, however, the proof of Theorem 2 amounts to a prescription for doing this. Suppose, to take a simpler example, that we want to find a δ such that, for all x,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 + x - (a^2 + a)| < \varepsilon.$$

Consulting the proof of Theorem 2(1), we see that we must first find δ_1 and $\delta_2 > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |x^2 - a^2| < \frac{\varepsilon}{2}$$
$$\text{and} \quad \text{if } 0 < |x - a| < \delta_2, \text{ then } |x - a| < \frac{\varepsilon}{2}.$$

Since we have already given proofs that $\lim_{x \to a} x^2 = a^2$ and $\lim_{x \to a} x = a$, we know how to do this:

$$\delta_1 = \min\left(1, \frac{\frac{\varepsilon}{2}}{2|a| + 1}\right),$$
$$\delta_2 = \frac{\varepsilon}{2}.$$

Thus we can take

$$\delta = \min(\delta_1, \delta_2) = \min\left(\min\left(1, \frac{\frac{\varepsilon}{2}}{2|a| + 1}\right), \frac{\varepsilon}{2}\right).$$

If $a \neq 0$, the same method can be used to find a $\delta > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta, \text{ then } \left| \frac{1}{x^2} - \frac{1}{a^2} \right| < \varepsilon.$$

The proof of Theorem 2(3) shows that the second condition will follow if we find a $\delta > 0$ such that, for all x,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 - a^2| < \min\left(\frac{|a|^2}{2}, \frac{\varepsilon |a|^4}{2}\right).$$

Thus we can take

$$\delta = \min\left(1, \frac{\min\left(\frac{|a|^2}{2}, \frac{\varepsilon |a|^4}{2}\right)}{2|a| + 1}\right).$$

Naturally, these complicated expressions for δ can be simplified considerably, after they have been derived.

One technical detail in the proof of Theorem 2 deserves some discussion. In order for $\lim_{x \to a} f(x)$ to be defined it is, as we know, not necessary for f to be defined at a, nor is it necessary for f to be defined at all points $x \neq a$. However, there must be some $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$; otherwise the clause

$$\text{“if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \varepsilon\text{”}$$

would make no sense at all, since the symbol $f(x)$ would make no sense for some x's. If f and g are two functions for which the definition makes sense, it is easy to see that the same is true for $f + g$ and $f \cdot g$. But this is not so clear for $1/g$, since $1/g$ is undefined for x with $g(x) = 0$. However, this fact gets established in the proof of Theorem 2(3).

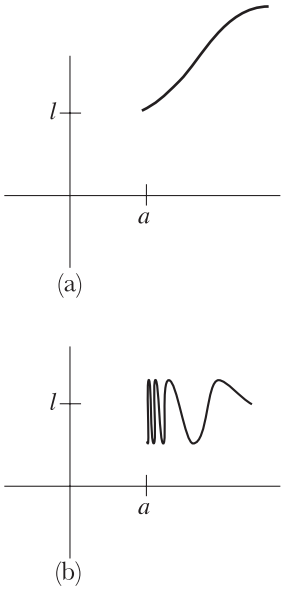


FIGURE 16

There are times when we would like to speak of the limit which f approaches at a, even though there is no $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$. For example, we want to distinguish the behavior of the two functions shown in Figure 16, even though they are not defined for numbers less than a. For the function of Figure 16(a) we write

$$\lim_{x \to a^+} f(x) = l \quad \text{or} \quad \lim_{x \downarrow a} f(x) = l.$$

(The symbols on the left are read: the limit of $f(x)$ as x approaches a from above.) These “limits from above” are obviously closely related to ordinary limits, and the definition is very similar: $\lim_{x \to a^+} f(x) = l$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x,

$$\text{if } 0 < x - a < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

(The condition “$0 < x - a < \delta$” is equivalent to “$0 < |x - a| < \delta$ and $x > a$.”)

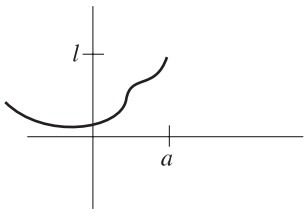


FIGURE 17

“Limits from below” (Figure 17) are defined similarly: $\lim_{x \to a^-} f(x) = l$ (or

$\lim_{x\uparrow a} f(x) = l$) means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x,

$$\text{if } 0 < a - x < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

It is quite possible to consider limits from above and below even if f is defined for numbers both greater and less than a. Thus, for the function f of Figure 13, we have

$$\lim_{x\to 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x\to 0^-} f(x) = -1.$$

It is an easy exercise (Problem 29) to show that $\lim_{x\to a} f(x)$ exists if and only if $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ both exist and are equal.

Like the definitions of limits from above and below, which have been smuggled into the text informally, there are other modifications of the limit concept which will be found useful. In Chapter 4 it was claimed that if x is large, then $\sin 1/x$ is close to 0. This assertion is usually written

$$\lim_{x\to\infty} \sin 1/x = 0.$$

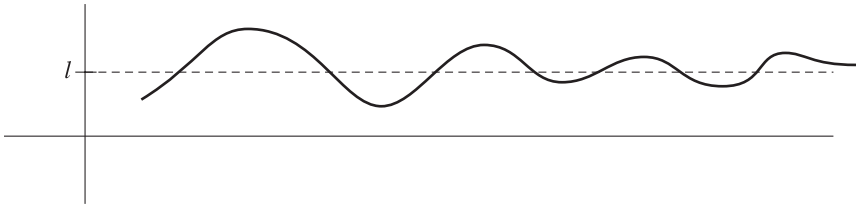


FIGURE 18

The symbol $\lim_{x\to\infty} f(x)$ is read "the limit of $f(x)$ as x approaches ∞," or "as x becomes infinite," and a limit of the form $\lim_{x\to\infty} f(x)$ is often called a limit at infinity. Figure 18 illustrates a general situation where $\lim_{x\to\infty} f(x) = l$. Formally, $\lim_{x\to\infty} f(x) = l$ means that for every $\varepsilon > 0$ there is a number N such that, for all x,

$$\text{if } x > N, \text{ then } |f(x) - l| < \varepsilon.$$

The analogy with the definition of ordinary limits should be clear: whereas the condition "$0 < |x - a| < \delta$" expresses the fact that x is close to a, the condition "$x > N$" expresses the fact that x is large.

We have spent so little time on limits from above and below, and at infinity, because the general philosophy behind the definitions should be clear if you understand the definition of ordinary limits (which are by far the most important). Many exercises on these definitions are provided in the Problems, which also contain several other types of limits which are occasionally useful.

PROBLEMS

1. Find the following limits. (These limits all follow, after some algebraic manipulations, from the various parts of Theorem 2; be sure you know which ones are used in each case, but don't bother listing them.)

(i) $\lim_{x\to 1} \dfrac{x^2-1}{x+1}$.

(ii) $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$.

(iii) $\lim_{x\to 3} \dfrac{x^3-8}{x-2}$.

(iv) $\lim_{x\to y} \dfrac{x^n-y^n}{x-y}$.

(v) $\lim_{y\to x} \dfrac{x^n-y^n}{x-y}$.

(vi) $\lim_{h\to 0} \dfrac{\sqrt{a+h}-\sqrt{a}}{h}$.

2. Find the following limits.

(i) $\lim_{x\to 1} \dfrac{1-\sqrt{x}}{1-x}$.

(ii) $\lim_{x\to 0} \dfrac{1-\sqrt{1-x^2}}{x}$.

(iii) $\lim_{x\to 0} \dfrac{1-\sqrt{1-x^2}}{x^2}$.

3. In each of the following cases, find a δ such that $|f(x)-l| < \varepsilon$ for all x satisfying $0 < |x-a| < \delta$.

(i) $f(x) = x^4$; $l = a^4$.

(ii) $f(x) = \dfrac{1}{x}$; $a = 1$, $l = 1$.

(iii) $f(x) = x^4 + \dfrac{1}{x}$; $a = 1$, $l = 2$.

(iv) $f(x) = \dfrac{x}{1+\sin^2 x}$; $a = 0$, $l = 0$.

(v) $f(x) = \sqrt{|x|}$; $a = 0$, $l = 0$.

(vi) $f(x) = \sqrt{x}$; $a = 1$, $l = 1$.

4. For each of the functions in Problem 4-17, decide for which numbers a the limit $\lim_{x\to a} f(x)$ exists.

***5.** (a) Do the same for each of the functions in Problem 4-19.
(b) Same problem, if we use infinite decimals ending in a string of 0's instead of those ending in a string of 9's.

6. Suppose the functions f and g have the following property: for all $\varepsilon > 0$ and all x,

$$\text{if } 0 < |x-2| < \sin^2\left(\frac{\varepsilon^2}{9}\right) + \varepsilon, \text{ then } |f(x)-2| < \varepsilon,$$

$$\text{if } 0 < |x-2| < \varepsilon^2, \text{ then } |g(x)-4| < \varepsilon.$$

For each $\varepsilon > 0$ find a $\delta > 0$ such that, for all x,

(i) if $0 < |x-2| < \delta$, then $|f(x)+g(x)-6| < \varepsilon$.
(ii) if $0 < |x-2| < \delta$, then $|f(x)g(x)-8| < \varepsilon$.
(iii) if $0 < |x-2| < \delta$, then $\left|\dfrac{1}{g(x)} - \dfrac{1}{4}\right| < \varepsilon$.
(iv) if $0 < |x-2| < \delta$, then $\left|\dfrac{f(x)}{g(x)} - \dfrac{1}{2}\right| < \varepsilon$.

7. Give an example of a function f for which the following assertion is *false*: If $|f(x)-l| < \varepsilon$ when $0 < |x-a| < \delta$, then $|f(x)-l| < \varepsilon/2$ when $0 < |x-a| < \delta/2$.

8. (a) If $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ do not exist, can $\lim_{x\to a}[f(x)+g(x)]$ or $\lim_{x\to a} f(x)g(x)$ exist?
(b) If $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a}[f(x)+g(x)]$ exists, must $\lim_{x\to a} g(x)$ exist?
(c) If $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} g(x)$ does not exist, can $\lim_{x\to a}[f(x)+g(x)]$ exist?
(d) If $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} f(x)g(x)$ exists, does it follow that $\lim_{x\to a} g(x)$ exists?

9. Prove that $\lim_{x\to a} f(x) = \lim_{h\to 0} f(a+h)$. (This is mainly an exercise in understanding what the terms mean.)

10. (a) Prove that $\lim_{x\to a} f(x) = l$ if and only if $\lim_{x\to a}[f(x)-l] = 0$. (First see why the assertion is obvious; then provide a rigorous proof. In this chapter most problems which ask for proofs should be treated in the same way.)
(b) Prove that $\lim_{x\to 0} f(x) = \lim_{x\to a} f(x-a)$.
(c) Prove that $\lim_{x\to 0} f(x) = \lim_{x\to 0} f(x^3)$.
(d) Give an example where $\lim_{x\to 0} f(x^2)$ exists, but $\lim_{x\to 0} f(x)$ does not.

11. Suppose there is a $\delta > 0$ such that $f(x) = g(x)$ when $0 < |x-a| < \delta$. Prove that $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$. In other words, $\lim_{x\to a} f(x)$ depends only on the values of $f(x)$ for x near a—this fact is often expressed by saying that limits are a "local property." (It will clearly help to use δ', or some other letter, instead of δ, in the definition of limits.)

12. (a) Suppose that $f(x) \le g(x)$ for all x. Prove that $\lim_{x\to a} f(x) \le \lim_{x\to a} g(x)$, provided that these limits exist.

(b) How can the hypotheses by weakened?

(c) If $f(x) < g(x)$ for all x, does it necessarily follow that $\lim_{x\to a} f(x) < \lim_{x\to a} g(x)$?

13. Suppose that $f(x) \le g(x) \le h(x)$ and that $\lim_{x\to a} f(x) = \lim_{x\to a} h(x)$. Prove that $\lim_{x\to a} g(x)$ exists, and that $\lim_{x\to a} g(x) = \lim_{x\to a} f(x) = \lim_{x\to a} h(x)$. (Draw a picture!)

***14.** (a) Prove that if $\lim_{x\to 0} f(x)/x = l$ and $b \neq 0$, then $\lim_{x\to 0} f(bx)/x = bl$. Hint: Write $f(bx)/x = b[f(bx)/bx]$.

(b) What happens if $b = 0$?

(c) Part (a) enables us to find $\lim_{x\to 0}(\sin 2x)/x$ in terms of $\lim_{x\to 0}(\sin x)/x$. Find this limit in another way.

15. Evaluate the following limits in terms of the number $\alpha = \lim_{x\to 0}(\sin x)/x$.

(i) $\lim_{x\to 0} \dfrac{\sin 2x}{x}$.

(ii) $\lim_{x\to 0} \dfrac{\sin ax}{\sin bx}$.

(iii) $\lim_{x\to 0} \dfrac{\sin^2 2x}{x}$.

(iv) $\lim_{x\to 0} \dfrac{\sin^2 2x}{x^2}$.

(v) $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$.

(vi) $\lim_{x\to 0} \dfrac{\tan^2 x + 2x}{x + x^2}$.

(vii) $\lim_{x\to 0} \dfrac{x \sin x}{1-\cos x}$.

(viii) $\lim_{h\to 0} \dfrac{\sin(x+h) - \sin x}{h}$.

(ix) $\lim_{x\to 1} \dfrac{\sin(x^2-1)}{x-1}$.

(x) $\lim_{x\to 0} \dfrac{x^2(3+\sin x)}{(x+\sin x)^2}$.

(xi) $\lim_{x\to 1} (x^2-1)^3 \sin\left(\dfrac{1}{x-1}\right)^3$.

16. (a) Prove that if $\lim_{x\to a} f(x) = l$, then $\lim_{x\to a} |f|(x) = |l|$.

(b) Prove that if $\lim_{x\to a} f(x) = l$ and $\lim_{x\to a} g(x) = m$, then $\lim_{x\to a} \max(f, g)(x) = \max(l, m)$ and similarly for min.

17. (a) Prove that $\lim_{x\to 0} 1/x$ does not exist, i.e., show that $\lim_{x\to 0} 1/x = l$ is false for every number l.

(b) Prove that $\lim_{x\to 1} 1/(x-1)$ does not exist.

18. Prove that if $\lim_{x\to a} f(x) = l$, then there is a number $\delta > 0$ and a number M such that $|f(x)| < M$ if $0 < |x-a| < \delta$. (What does this mean pictorially?) Hint: Why does it suffice to prove that $l-1 < f(x) < l+1$ for $0 < |x-a| < \delta$?

19. Prove that if $f(x) = 0$ for irrational x and $f(x) = 1$ for rational x, then $\lim_{x\to a} f(x)$ does not exist for any a.

***20.** Prove that if $f(x) = x$ for rational x, and $f(x) = -x$ for irrational x, then $\lim_{x\to a} f(x)$ does not exist if $a \neq 0$.

21. (a) Prove that if $\lim_{x\to 0} g(x) = 0$, then $\lim_{x\to 0} g(x)\sin 1/x = 0$.

(b) Generalize this fact as follows: If $\lim_{x\to 0} g(x) = 0$ and $|h(x)| \leq M$ for all x, then $\lim_{x\to 0} g(x)h(x) = 0$. (Naturally it is unnecessary to do part (a) if you succeed in doing part (b); actually the statement of part (b) may make it easier than (a)—that's one of the values of generalization.)

22. Consider a function f with the following property: if g is any function for which $\lim_{x\to 0} g(x)$ does not exist, then $\lim_{x\to 0}[f(x) + g(x)]$ also does not exist. Prove that this happens if and only if $\lim_{x\to 0} f(x)$ *does* exist. Hint: This is actually very easy: the assumption that $\lim_{x\to 0} f(x)$ does not exist leads to an immediate contradiction if you consider the right g.

****23.** This problem is the analogue of Problem 22 when $f+g$ is replaced by $f \cdot g$. In this case the situation is considerably more complex, and the analysis requires several steps (those in search of an especially challenging problem can attempt an independent solution).

(a) Suppose that $\lim_{x\to 0} f(x)$ exists and is $\neq 0$. Prove that if $\lim_{x\to 0} g(x)$ does not exist, then $\lim_{x\to 0} f(x)g(x)$ also does not exist.

(b) Prove the same result if $\lim_{x\to 0} |f(x)| = \infty$. (The precise definition of this sort of limit is given in Problem 37.)

(c) Prove that if neither of these two conditions holds, then there is a function g such that $\lim_{x\to 0} g(x)$ does not exist, but $\lim_{x\to 0} f(x)g(x)$ does exist.
Hint: Consider separately the following two cases: (1) for some $\varepsilon > 0$ we have $|f(x)| > \varepsilon$ for all sufficiently small x. (2) For every $\varepsilon > 0$, there are arbitrarily small x with $|f(x)| < \varepsilon$. In the second case, begin by choosing points x_n with $|x_n| < 1/n$ and $|f(x_n)| < 1/n$.

***24.** Suppose that A_n is, for each natural number n, some *finite* set of numbers in $[0, 1]$, and that A_n and A_m have no members in common if $m \neq n$. Define

f as follows:

$$f(x) = \begin{cases} 1/n, & x \text{ in } A_n \\ 0, & x \text{ not in } A_n \text{ for any } n. \end{cases}$$

Prove that $\lim_{x \to a} f(x) = 0$ for all a in $[0, 1]$.

25. Explain why the following definitions of $\lim_{x \to a} f(x) = l$ are all correct:
For every $\delta > 0$ there is an $\varepsilon > 0$ such that, for all x,

(i) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < \delta$.
(ii) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| \leq \delta$.
(iii) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < 5\delta$.
(iv) if $0 < |x - a| < \varepsilon/10$, then $|f(x) - l| < \delta$.

***26.** Give examples to show that the following definitions of $\lim_{x \to a} f(x) = l$ are *not* correct.

(a) For all $\delta > 0$ there is an $\varepsilon > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$.
(b) For all $\varepsilon > 0$ there is a $\delta > 0$ such that if $|f(x) - l| < \varepsilon$, then $0 < |x - a| < \delta$.

27. For each of the functions in Problem 4-17 indicate for which numbers a the one-sided limits $\lim_{x \to a^+} f(x)$ and $\lim_{x \to a^-} f(x)$ exist.

***28.** (a) Do the same for each of the functions in Problem 4-19.
(b) Also consider what happens if decimals ending in 0's are used instead of decimals ending in 9's.

29. Prove that $\lim_{x \to a} f(x)$ exists if $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$.

30. Prove that

(i) $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(-x)$.
(ii) $\lim_{x \to 0} f(|x|) = \lim_{x \to 0^+} f(x)$.
(iii) $\lim_{x \to 0} f(x^2) = \lim_{x \to 0^+} f(x)$.

(These equations, and others like them, are open to several interpretations. They might mean only that the two limits are equal if they both exist; or that if a certain one of the limits exists, the other also exists and is equal to it; or that if either limit exists, then the other exists and is equal to it. Decide for yourself which interpretations are suitable.)

***31.** Suppose that $\lim_{x \to a^-} f(x) < \lim_{x \to a^+} f(x)$. (Draw a picture to illustrate this assertion.) Prove that there is some $\delta > 0$ such that $f(x) < f(y)$ whenever $x < a < y$ and $|x - a| < \delta$ and $|y - a| < \delta$. Is the converse true?

***32.** Prove that $\lim_{x\to\infty}(a_nx^n+\cdots+a_0)/(b_mx^m+\cdots+b_0)$ exists if and only if $m \geq n$. What is the limit when $m = n$? When $m > n$? Hint: the one easy limit is $\lim_{x\to\infty} 1/x^k = 0$; do some algebra so that this is the only information you need.

33. Find the following limits.

(i) $\lim_{x\to\infty} \dfrac{x+\sin^3 x}{5x+6}$.

(ii) $\lim_{x\to\infty} \dfrac{x\sin x}{x^2+5}$.

(iii) $\lim_{x\to\infty} \sqrt{x^2+x}-x$.

(iv) $\lim_{x\to\infty} \dfrac{x^2(1+\sin^2 x)}{(x+\sin x)^2}$.

34. Prove that $\lim_{x\to 0^+} f(1/x) = \lim_{x\to\infty} f(x)$.

35. Find the following limits in terms of the number $\alpha = \lim_{x\to 0}(\sin x)/x$.

(i) $\lim_{x\to\infty} \dfrac{\sin x}{x}$.

(ii) $\lim_{x\to\infty} x\sin\dfrac{1}{x}$.

36. Define "$\lim_{x\to-\infty} f(x) = l$."

(a) Find $\lim_{x\to-\infty}(a_nx^n+\cdots+a^0)/(b_mx^m+\cdots+b_0)$.

(b) Prove that $\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(-x)$.

(c) Prove that $\lim_{x\to 0^-} f(1/x) = \lim_{x\to-\infty} f(x)$.

37. We define $\lim_{x\to a} f(x) = \infty$ to mean that for all N there is a $\delta > 0$ such that, for all x, if $0 < |x-a| < \delta$, then $f(x) > N$. (Draw an appropriate picture!)

(a) Show that $\lim_{x\to 3} 1/(x-3)^2 = \infty$.

(b) Prove that if $f(x) > \varepsilon > 0$ for all x, and $\lim_{x\to a} g(x) = 0$, then

$$\lim_{x\to a} f(x)/|g(x)| = \infty.$$

38. (a) Define $\lim_{x\to a^+} f(x) = \infty$, $\lim_{x\to a^-} f(x) = \infty$, and $\lim_{x\to a} f(x) = \infty$. (Or at least convince yourself that you could write down the definitions if you had the energy. How many other such symbols can you define?)

(b) Prove that $\lim_{x\to 0^+} 1/x = \infty$.

(c) Prove that $\lim_{x\to 0^+} f(x) = \infty$ if and only if $\lim_{x\to\infty} f(1/x) = \infty$.

39. Find the following limits, when they exist.

(i) $\lim_{x\to\infty} \dfrac{x^3+4x-7}{7x^2-x+1}$

(ii) $\lim_{x\to\infty} x(1+\sin^2 x)$.

(iii) $\lim_{x\to\infty} x\sin^2 x$.

(iv) $\lim_{x\to\infty} x^2 \sin\dfrac{1}{x}$.

(v) $\lim_{x\to\infty} \sqrt{x^2+2x} - x$.

(vi) $\lim_{x\to\infty} x(\sqrt{x+2} - \sqrt{x}\,)$.

(vii) $\lim_{x\to\infty} \dfrac{\sqrt{|x|}}{x}$.

40. (a) Find the perimeter of a regular n-gon inscribed in a circle of radius r; use radian measure for any trigonometric functions involved. [Answer: $2rn\sin(\pi/n)$.]

(b) What value does this perimeter approach as n becomes very large?

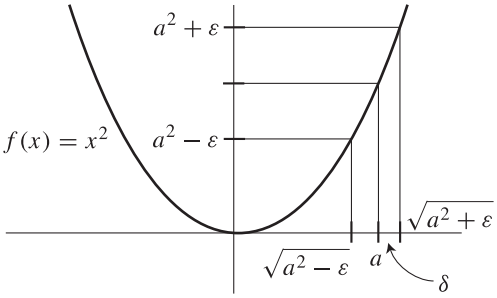


FIGURE 19

41. After sending the manuscript for the first edition of this book off to the printer, I thought of a much simpler way to prove that $\lim_{x\to a} x^2 = a^2$ and $\lim_{x\to a} x^3 = a^3$, without going through all the factoring tricks on page 95. Suppose, for example, that we want to prove that $\lim_{x\to a} x^2 = a^2$, where $a > 0$. Given $\varepsilon > 0$, we simply let δ be the minimum of $\sqrt{a^2+\varepsilon} - a$ and $a - \sqrt{a^2-\varepsilon}$ (see Figure 19); then $|x-a| < \delta$ implies that $\sqrt{a^2-\varepsilon} < x < \sqrt{a^2+\varepsilon}$, so $a^2-\varepsilon < x^2 < a^2+\varepsilon$, or $|x^2-a^2| < \varepsilon$. It is fortunate that these pages had already been set, so that I couldn't make these changes, because this "proof" is completely fallacious. Wherein lies the fallacy?

CHAPTER 6 CONTINUOUS FUNCTIONS

If f is an arbitrary function, it is not necessarily true that

$$\lim_{x \to a} f(x) = f(a).$$

In fact, there are many ways this can fail to be true. For example, f might not even be defined at a, in which case the equation makes no sense (Figure 1).

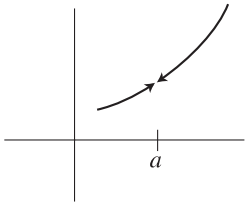


FIGURE 1

Again, $\lim_{x \to a} f(x)$ might not exist (Figure 2). Finally, as illustrated in Figure 3, even if f is defined at a and $\lim_{x \to a} f(x)$ exists, the limit might not equal $f(a)$.

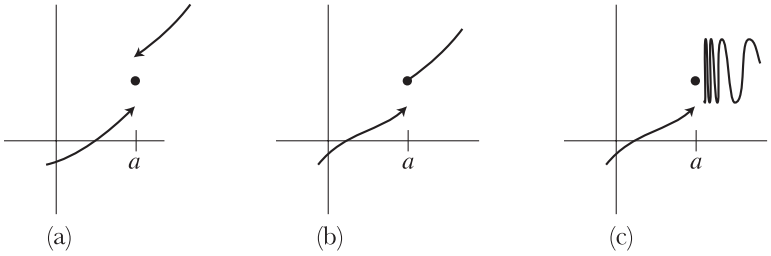


FIGURE 2

We would like to regard all behavior of this type as abnormal and honor, with some complimentary designation, functions which do not exhibit such peculiarities. The term which has been adopted is "continuous." Intuitively, a function f is continuous if the graph contains no breaks, jumps, or wild oscillations. Although this description will usually enable you to decide whether a function is continuous simply by looking at its graph (a skill well worth cultivating) it is easy to be fooled, and the precise definition is *very* important.

DEFINITION

The function f is **continuous at *a*** if

$$\lim_{x \to a} f(x) = f(a).$$

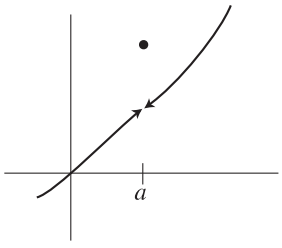


FIGURE 3

We will have no difficulty finding many examples of functions which are, or are not, continuous at some number a—every example involving limits provides an example about continuity, and Chapter 5 certainly provides enough of these.

The function $f(x) = \sin 1/x$ is not continuous at 0, because it is not even defined at 0, and the same is true of the function $g(x) = x \sin 1/x$. On the other hand, if we are willing to extend the second of these functions, that is, if we wish to define

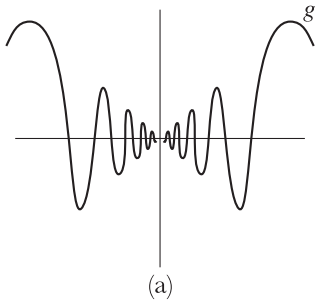
a new function G by

$$G(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then the choice of $a = G(0)$ can be made in such a way that G will be continuous at 0—to do this we can (if fact, we must) define $G(0) = 0$ (Figure 4). This sort of extension is not possible for f; if we define

$$F(x) = \begin{cases} \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then F will not be continuous at 0, no matter what a is, because $\lim_{x \to 0} f(x)$ does not exist.



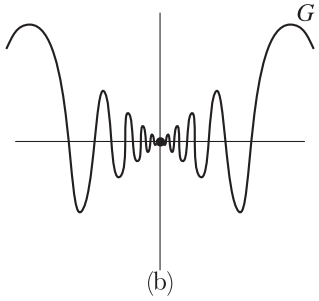


FIGURE 4

The function

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

is not continuous at a, if $a \neq 0$, since $\lim_{x \to a} f(x)$ does not exist. However, $\lim_{x \to 0} f(x) = 0 = f(0)$, so f is continuous at precisely one point, 0.

The functions $f(x) = c$, $g(x) = x$, and $h(x) = x^2$ are continuous at all numbers a, since

$$\lim_{x \to a} f(x) = \lim_{x \to a} c = c = f(a),$$
$$\lim_{x \to a} g(x) = \lim_{x \to a} x = a = g(a),$$
$$\lim_{x \to a} h(x) = \lim_{x \to a} x^2 = a^2 = h(a).$$

Finally, consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms.} \end{cases}$$

In Chapter 5 we showed that $\lim_{x \to a} f(x) = 0$ for all a. Since $0 = f(a)$ only when a is irrational, this function is continuous at a if a is irrational, but not if a is rational.

It is even easier to give examples of continuity if we prove two simple theorems.

THEOREM 1 If f and g are continuous at a, then

(1) $f + g$ is continuous at a,
(2) $f \cdot g$ is continuous at a.

Moreover, if $g(a) \neq 0$, then

(3) $1/g$ is continuous at a.

PROOF Since f and g are continuous at a,

$$\lim_{x\to a} f(x) = f(a) \quad \text{and} \quad \lim_{x\to a} g(x) = g(a).$$

By Theorem 2(1) of Chapter 5 this implies that

$$\lim_{x\to a}(f+g)(x) = f(a) + g(a) = (f+g)(a),$$

which is just the assertion that $f + g$ is continuous at a. The proofs of parts (2) and (3) are left to you. ∎

Starting with the functions $f(x) = c$ and $f(x) = x$, which are continuous at a, for every a, we can use Theorem 1 to conclude that a function

$$f(x) = \frac{b_n x^n + b_{n-1}x^{n-1} + \cdots + b_0}{c_m x^m + c_{m-1}x^{m-1} + \cdots + c_0}$$

is continuous at every point in its domain. But it is harder to get much further than that. When we discuss the sine function in detail it will be easy to prove that sin is continuous at a for all a; let us assume this fact meanwhile. A function like

$$f(x) = \frac{\sin^2 x + x^2 + x^4 \sin x}{\sin^{27} x + 4x^2 \sin^2 x}$$

can now be proved continuous at every point in its domain. But we are still unable to prove the continuity of a function like $f(x) = \sin(x^2)$; we obviously need a theorem about the composition of continuous functions. Before stating this theorem, the following point about the definition of continuity is worth noting. If we translate the equation $\lim_{x\to a} f(x) = f(a)$ according to the definition of limits, we obtain

$$\begin{aligned}&\text{for every } \varepsilon > 0 \text{ there is } \delta > 0 \text{ such that, for all } x,\\ &\quad \text{if } 0 < |x-a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.\end{aligned}$$

But in this case, where the limit is $f(a)$, the phrase

$$0 < |x - a| < \delta$$

may be changed to the simpler condition

$$|x - a| < \delta,$$

since if $x = a$ it is certainly true that $|f(x) - f(a)| < \varepsilon$.

THEOREM 2 If g is continuous at a, and f is continuous at $g(a)$, then $f \circ g$ is continuous at a. (Notice that f is required to be continuous at $g(a)$, not at a.)

PROOF Let $\varepsilon > 0$. We wish to find a $\delta > 0$ such that for all x,

$$\begin{aligned}&\text{if } |x-a| < \delta, \text{ then } |(f\circ g)(x) - (f\circ g)(a)| < \varepsilon,\\ &\qquad \text{i.e., } \quad |f(g(x)) - f(g(a))| < \varepsilon.\end{aligned}$$

We first use continuity of f to estimate how close $g(x)$ must be to $g(a)$ in order for this inequality to hold. Since f is continuous at $g(a)$, there is a $\delta' > 0$ such that for all y,

$$(1) \qquad \text{if } |y - g(a)| < \delta', \text{ then } |f(y) - f(g(a))| < \varepsilon.$$

In particular, this means that

$$(2) \qquad \text{if } |g(x) - g(a)| < \delta', \text{ then } |f(g(x)) - f(g(a))| < \varepsilon.$$

We now use continuity of g to estimate how close x must be to a in order for the inequality $|g(x) - g(a)| < \delta'$ to hold. The number δ' is a positive number just like any other positive number; we can therefore take δ' as the ε (!) in the definition of continuity of g at a. We conclude that there is a $\delta > 0$ such that, for all x,

$$(3) \qquad \text{if } |x - a| < \delta, \text{ then } |g(x) - g(a)| < \delta'.$$

Combining (2) and (3) we see that for all x,

$$\text{if } |x - a| < \delta, \text{ then } |f(g(x)) - f(g(a))| < \varepsilon. \blacksquare$$

We can now reconsider the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

We have already noted that f is continuous at 0. A few applications of Theorems 1 and 2, together with the continuity of sin, show that f is also continuous at a, for $a \neq 0$. Functions like $f(x) = \sin(x^2 + \sin(x + \sin^2(x^3)))$ should be equally easy for you to analyze.

The few theorems of this chapter have all been related to continuity of functions at a single point, but the concept of continuity doesn't begin to be really interesting until we focus our attention on functions which are continuous at all points of some interval. If f is continuous at x for all x in (a, b), then f is called **continuous on** (a, b). Continuity on a closed interval must be defined a little differently; a function f is called **continuous on** $[a, b]$ if

(1) f is continuous at x for all x in (a, b),
(2) $\lim_{x \to a^+} f(x) = f(a)$ and $\lim_{x \to b^-} f(x) = f(b)$.

Functions which are continuous on an interval are usually regarded as especially well behaved; indeed continuity might be specified as the first condition which a "reasonable" function ought to satisfy. A continuous function is sometimes described, intuitively, as one whose graph can be drawn without lifting your pencil from the paper. Consideration of the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

shows that this description is a little too optimistic, but it is nevertheless true that there are many important results involving functions which are continuous on an interval. There theorems are generally much harder than the ones in this chapter,

but there is a simple theorem which forms a bridge between the two kinds of results. The hypothesis of this theorem requires continuity at only a single point, but the conclusion describes the behavior of the function on some interval containing the point. Although this theorem is really a lemma for later arguments, it is included here as a preview of things to come.

THEOREM 3 Suppose f is continuous at a, and $f(a) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $|x - a| < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $|x - a| < \delta$.

PROOF Consider the case $f(a) > 0$. Since f is continuous at a, if $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.$$

Since $f(a) > 0$ we can take $f(a)$ as the ε. Thus there is $\delta > 0$ so that for all x,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < f(a),$$

and this last inequality implies $f(x) > 0$.

A similar proof can be given in the case $f(a) < 0$; take $\varepsilon = -f(a)$. Or one can apply the first case to the function $-f$. ∎

PROBLEMS

1. For which of the following functions f is there a continuous function F with domain $\mathbf{R}$ such that $F(x) = f(x)$ for all x in the domain of f?

(i) $f(x) = \dfrac{x^2 - 4}{x - 2}$.

(ii) $f(x) = \dfrac{|x|}{x}$.

(iii) $f(x) = 0, x$ irrational.

(iv) $f(x) = 1/q, x = p/q$ rational in lowest terms.

2. At which points are the functions of Problems 4-17 and 4-19 continuous?

3. (a) Suppose that f is a function satisfying $|f(x)| \leq |x|$ for all x. Show that f is continuous at 0. (Notice that $f(0)$ must equal 0.)

(b) Give an example of such a function f which is not continuous at any $a \neq 0$.

(c) Suppose that g is continuous at 0 and $g(0) = 0$, and $|f(x)| \leq |g(x)|$. Prove that f is continuous at 0.

4. Give an example of a function f such that f is continuous nowhere, but $|f|$ is continuous everywhere.

5. For each number a, find a function which is continuous at a, but not at any other points.

6. (a) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ but continuous at all other points.
(b) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$, and at 0, but continuous at all other points.

7. Suppose that f satisfies $f(x+y) = f(x) + f(y)$, and that f is continuous at 0. Prove that f is continuous at a for all a.

8. Suppose that f is continuous at a and $f(a) = 0$. Prove that if $\alpha \neq 0$, then $f + \alpha$ is nonzero in some open interval containing a.

9. (a) Suppose f is *not* continuous at a. Prove that for some number $\varepsilon > 0$ there are numbers x arbitrarily close to a with $|f(x) - f(a)| > \varepsilon$. Illustrate graphically.
(b) Conclude that for some number $\varepsilon > 0$ *either* there are numbers x arbitrarily close to a with $f(x) < f(a) - \varepsilon$ *or* there are numbers x arbitrarily close to a with $f(x) > f(a) + \varepsilon$.

10. (a) Prove that if f is continuous at a, then so is $|f|$.
(b) Prove that every continuous f can be written $f = E + O$, where E is even and continuous and O is odd and continuous.
(c) Prove that if f and g are continuous, then so are $\max(f, g)$ and $\min(f, g)$.
(d) Prove that every continuous f can be written $f = g - h$, where g and h are nonnegative and continuous.

11. Prove Theorem 1(3) by using Theorem 2 and continuity of the function $f(x) = 1/x$.

***12.** (a) Prove that if f is continuous at l and $\lim_{x \to a} g(x) = l$, then $\lim_{x \to a} f(g(x)) = f(l)$. (You can go right back to the definitions, but it is easier to consider the function G with $G(x) = g(x)$ for $x \neq a$, and $G(a) = l$.)
(b) Show that if continuity of f at l is not assumed, then it is not generally true that $\lim_{x \to a} f(g(x)) = f(\lim_{x \to a} g(x))$. Hint: Try $f(x) = 0$ for $x \neq l$, and $f(l) = 1$.

13. (a) Prove that if f is continuous on $[a, b]$, then there is a function g which is continuous on $\mathbf{R}$, and which satisfies $g(x) = f(x)$ for all x in $[a, b]$. Hint: Since you obviously have a great deal of choice, try making g constant on $(-\infty, a]$ and $[b, \infty)$.
(b) Give an example to show that this assertion is false if $[a, b]$ is replaced by (a, b).

14. (a) Suppose that g and h are continuous at a, and that $g(a) = h(a)$. Define $f(x)$ to be $g(x)$ if $x \geq a$ and $h(x)$ if $x \leq a$. Prove that f is continuous at a.
(b) Suppose g is continuous on $[a, b]$ and h is continuous on $[b, c]$ and $g(b) = h(b)$. Let $f(x)$ be $g(x)$ for x in $[a, b]$ and $h(x)$ for x in $[b, c]$.

Show that f is continuous on $[a, c]$. (Thus, continuous functions can be "pasted together".)

15. (a) Prove the following version of Theorem 3 for "right-hand continuity": Suppose that $\lim_{x\to a^+} f(x) = f(a)$, and $f(a) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $0 \leq x - a < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $0 \leq x - a < \delta$.

(b) Prove a version of Theorem 3 when $\lim_{x\to b^-} f(x) = f(b)$.

16. If $\lim_{x\to a} f(x)$ exists, but is $\neq f(a)$, then f is said to have a **removable discontinuity** at a.

(a) If $f(x) = \sin 1/x$ for $x \neq 0$ and $f(0) = 1$, does f have a removable discontinuity at 0? What if $f(x) = x \sin 1/x$ for $x \neq 0$, and $f(0) = 1$?

(b) Suppose f has a removable discontinuity at a. Let $g(x) = f(x)$ for $x \neq a$, and let $g(a) = \lim_{x\to a} f(x)$. Prove that g is continuous at a. (Don't work very hard; this is quite easy.)

(c) Let $f(x) = 0$ if x is irrational, and let $f(p/q) = 1/q$ if p/q is in lowest terms. What is the function g defined by $g(x) = \lim_{y\to x} f(y)$?

*(d) Let f be a function with the property that every point of discontinuity is a removable discontinuity. This means that $\lim_{y\to x} f(y)$ exists for all x, but f may be discontinuous at some (even infinitely many) numbers x. Define $g(x) = \lim_{y\to x} f(y)$. Prove that g is continuous. (This is not quite so easy as part (b).)

**(e) Is there a function f which is discontinuous at every point, and which has only removable discontinuities? (It is worth thinking about this problem now, but mainly as a test of intuition; even if you suspect the correct answer, you will almost certainly be unable to prove it at the present time. See Problem 22-33.)

CHAPTER 7 THREE HARD THEOREMS

This chapter is devoted to three theorems about continuous functions, and some of their consequences. The proofs of the three theorems themselves will not be given until the next chapter, for reasons which are explained at the end of this chapter.

THEOREM 1 If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some x in $[a, b]$ such that $f(x) = 0$.

(Geometrically, this means that the graph of a continuous function which starts below the horizontal axis and ends above it must cross this axis at some point, as in Figure 1.)

THEOREM 2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$, that is, there is some number N such that $f(x) \leq N$ for all x in $[a, b]$.

(Geometrically, this theorem means that the graph of f lies below some line parallel to the horizontal axis, as in Figure 2.)

THEOREM 3 If f is continuous on $[a, b]$, then there is some number y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$ (Figure 3).

These three theorems differ markedly from the theorems of Chapter 6. The hypotheses of those theorems always involved continuity at a single point, while

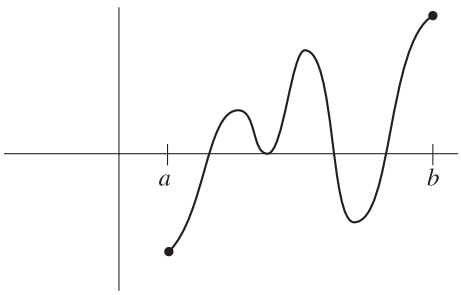


FIGURE 1

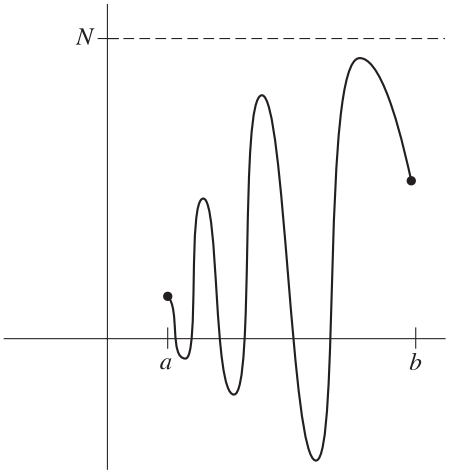


FIGURE 2

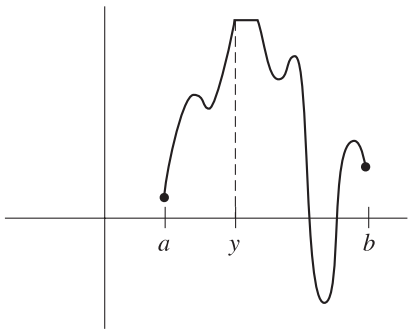


FIGURE 3

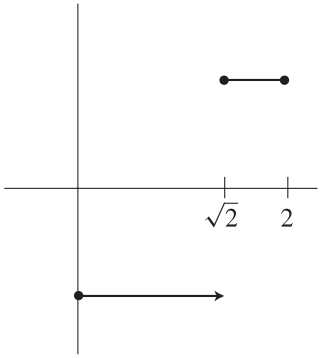


FIGURE 4

the hypotheses of the present theorems require continuity on a whole interval $[a, b]$—if continuity fails to hold at a single point, the conclusions may fail. For example, let f be the function shown in Figure 4,

$$f(x) = \begin{cases} -1, & 0 \le x < \sqrt{2} \\ 1, & \sqrt{2} \le x \le 2. \end{cases}$$

Then f is continuous at every point of $[0, 2]$ except $\sqrt{2}$, and $f(0) < 0 < f(2)$, but there is no point x in $[0, 2]$ such that $f(x) = 0$; the discontinuity at the single point $\sqrt{2}$ is sufficient to destroy the conclusion of Theorem 1.

Similarly, suppose that f is the function shown in Figure 5,

$$f(x) = \begin{cases} 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

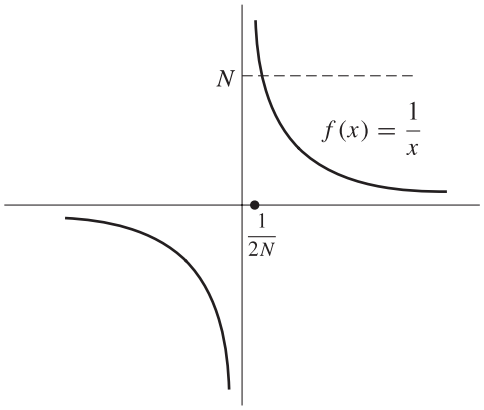


FIGURE 5

Then f is continuous at every point of $[0, 1]$ except 0, but f is not bounded above on $[0, 1]$. In fact, for any number $N > 0$ we have $f(1/2N) = 2N > N$.

This example also shows that the closed interval $[a, b]$ in Theorem 2 cannot be replaced by the open interval (a, b), for the function f is continuous on $(0, 1)$, but is not bounded there.

Finally, consider the function shown in Figure 6,

$$f(x) = \begin{cases} x^2, & x < 1 \\ 0, & x \ge 1. \end{cases}$$

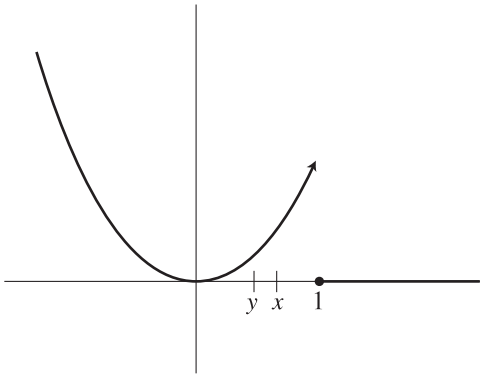


FIGURE 6

On the interval $[0, 1]$ the function f is bounded above, so f does satisfy the conclusion of Theorem 2, even though f is not continuous on $[0, 1]$. But f does not satisfy the conclusion of Theorem 3—there is no y in $[0, 1]$ such that $f(y) \ge f(x)$ for all x in $[0, 1]$; in fact, it is certainly not true that $f(1) \ge f(x)$ for all x in $[0, 1]$ so we cannot choose $y = 1$, nor can we choose $0 \le y < 1$ because $f(y) < f(x)$ if x is any number with $y < x < 1$.

This example shows that Theorem 3 is considerably stronger than Theorem 2. Theorem 3 is often paraphrased by saying that a continuous function on a closed interval "takes on its maximum value" on that interval.

As a compensation for the stringency of the hypotheses of our three theorems, the conclusions are of a totally different order than those of previous theorems. They describe the behavior of a function, not just near a point, but on a whole interval; such "global" properties of a function are always significantly more difficult to prove than "local" properties, and are correspondingly of much greater power. To illustrate the usefulness of Theorems 1, 2, and 3, we will soon deduce some important consequences, but it will help to first mention some simple generalizations of these theorems.

THEOREM 4 If f is continuous on $[a, b]$ and $f(a) < c < f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

PROOF Let $g = f - c$. Then g is continuous, and $g(a) < 0 < g(b)$. By Theorem 1, there is some x in $[a, b]$ such that $g(x) = 0$. But this means that $f(x) = c$. ∎

THEOREM 5 If f is continuous on $[a, b]$ and $f(a) > c > f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

PROOF The function $-f$ is continuous on $[a, b]$ and $-f(a) < -c < -f(b)$. By Theorem 4 there is some x in $[a, b]$ such that $-f(x) = -c$, which means that $f(x) = c$. ∎

Theorems 4 and 5 together show that f takes on any value between $f(a)$ and $f(b)$. We can do even better than this: if c and d are in $[a, b]$, then f takes on any value between $f(c)$ and $f(d)$. The proof is simple: if, for example, $c < d$, then just apply Theorems 4 and 5 to the interval $[c, d]$. Summarizing, if a continuous function on an interval takes on two values, it takes on every value in between; this slight generalization of Theorem 1 is often called the Intermediate Value Theorem.

THEOREM 6 If f is continuous on $[a, b]$, then f is bounded below on $[a, b]$, that is, there is some number N such that $f(x) \geq N$ for all x in $[a, b]$.

PROOF The function $-f$ is continuous on $[a, b]$, so by Theorem 2 there is a number M such that $-f(x) \leq M$ for all x in $[a, b]$. But this means that $f(x) \geq -M$ for all x in $[a, b]$, so we can let $N = -M$. ∎

Theorems 2 and 6 together show that a continuous function f on $[a, b]$ is bounded on $[a, b]$, that is, there is a number N such that $|f(x)| \leq N$ for all x in $[a, b]$. In fact, since Theorem 2 ensures the existence of a number N_1 such that $f(x) \leq N_1$ for all x in $[a, b]$, and Theorem 6 ensures the existence of a number N_2 such that $f(x) \geq N_2$ for all x in $[a, b]$, we can take $N = \max(|N_1|, |N_2|)$.

THEOREM 7 If f is continuous on $[a, b]$, then there is some y in $[a, b]$ such that $f(y) \leq f(x)$ for all x in $[a, b]$.
(A continuous function on a closed interval takes on its minimum value on that interval.)

PROOF The function $-f$ is continuous on $[a, b]$; by Theorem 3 there is some y in $[a, b]$ such that $-f(y) \geq -f(x)$ for all x in $[a, b]$, which means that $f(y) \leq f(x)$ for all x in $[a, b]$. ∎

Now that we have derived the trivial consequences of Theorems 1, 2, and 3, we can begin proving a few interesting things.

THEOREM 8 Every positive number has a square root. In other words, if $\alpha > 0$, then there is some number x such that $x^2 = \alpha$.

PROOF Consider the function $f(x) = x^2$, which is certainly continuous. Notice that the statement of the theorem can be expressed in terms of f: "the number α has a square root" means that f takes on the value α. The proof of this fact about f will be an easy consequence of Theorem 4.

There is obviously a number $b > 0$ such that $f(b) > \alpha$ (as illustrated in Figure 7); in fact, if $\alpha > 1$ we can take $b = \alpha$, while if $\alpha < 1$ we can take $b = 1$. Since $f(0) < \alpha < f(b)$, Theorem 4 applied to $[0, b]$ implies that for some x (in $[0, b]$), we have $f(x) = \alpha$, i.e., $x^2 = \alpha$. ∎

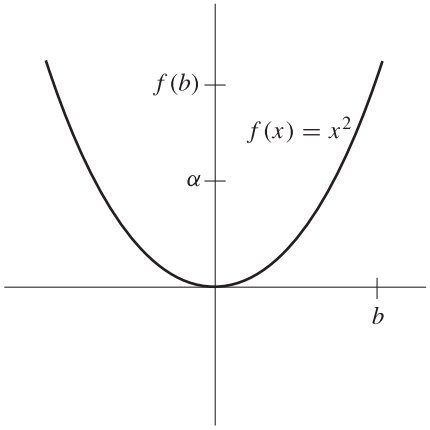


FIGURE 7

Precisely the same argument can be used to prove that a positive number has an nth root, for any natural number n. If n happens to be odd, one can do better: *every* number has an nth root. To prove this we just note that if the positive number α has the nth root x, i.e., if $x^n = \alpha$, then $(-x)^n = -\alpha$ (since n is odd), so $-\alpha$ has the nth root $-x$. The assertion, that for odd n any number α has an nth root, is equivalent to the statement that the equation

$$x^n - \alpha = 0$$

has a root if n is odd. Expressed in this way the result is susceptible of great generalization.

THEOREM 9 If n is odd, then any equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

has a root.

PROOF We obviously want to consider the function

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0;$$

we would like to prove that f is sometimes positive and sometimes negative. The intuitive idea is that for large $|x|$, the function is very much like $g(x) = x^n$ and, since n is odd, this function is positive for large positive x and negative for large negative x. A little algebra is all we need to make this intuitive idea work.

The proper analysis of the function f depends on writing

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 = x^n\left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}\right).$$

Note that

$$\left|\frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n}\right| \leq \frac{|a_{n-1}|}{|x|} + \cdots + \frac{|a_0|}{|x^n|}.$$

Consequently, if we choose x satisfying

$$(*) \qquad |x| > 1, 2n|a_{n-1}|, \ldots, 2n|a_0|,$$

then $|x^k| > |x|$ and

$$\frac{|a_{n-k}|}{|x^k|} < \frac{|a_{n-k}|}{|x|} < \frac{|a_{n-k}|}{2n|a_{n-k}|} = \frac{1}{2n},$$

so

$$\left|\frac{a_{n-1}}{x}+\frac{a_{n-2}}{x^2}+\cdots+\frac{a_0}{x^n}\right| \leq \underbrace{\frac{1}{2n}+\cdots+\frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}.$$

In other words,

$$-\frac{1}{2} \leq \frac{a_{n-1}}{x}+\cdots+\frac{a_0}{x^n} \leq \frac{1}{2},$$

which implies that

$$\frac{1}{2} \leq 1+\frac{a_{n-1}}{x}+\cdots+\frac{a_0}{x^n}.$$

Therefore, if we choose an $x_1 > 0$ which satisfies $(*)$, then

$$\frac{(x_1)^n}{2} \leq (x_1)^n\left(1+\frac{a_{n-1}}{x_1}+\cdots+\frac{a_0}{(x_1)^n}\right) = f(x_1),$$

so that $f(x_1) > 0$. On the other hand, if $x_2 < 0$ satisfies $(*)$, then $(x_2)^n < 0$ and

$$\frac{(x_2)^n}{2} \geq (x_2)^n\left(1+\frac{a_{n-1}}{x_2}+\cdots+\frac{a_0}{(x_2)^n}\right) = f(x_2),$$

so that $f(x_2) < 0$.

Now applying Theorem 1 to the interval $[x_2, x_1]$ we conclude that there is an x in $[x_2, x_1]$ such that $f(x) = 0$. ∎

Theorem 9 disposes of the problem of odd degree equations so happily that it would be frustrating to leave the problem of even degree equations completely undiscussed. At first sight, however, the problem seems insuperable. Some equations, like $x^2 - 1 = 0$, have a solution, and some, like $x^2 + 1 = 0$, do not—what more is there to say? If we are willing to consider a more general question, however, something interesting *can* be said. Instead of trying to solve the equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

let us ask about the possibility of solving the equations

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c$$

for all possible numbers c. This amount to allowing the constant term a_0 to vary. The information which can be given concerning the solution of these equations depends on a fact which is illustrated in Figure 8.

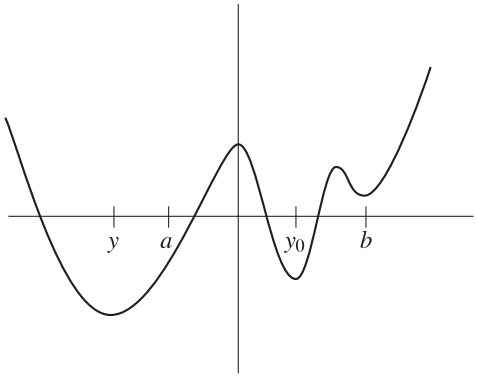


FIGURE 8

The graph of the function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, with n even, contains, at least the way we have drawn it, a lowest point. In other words, there is a number y such that $f(y) \leq f(x)$ for all numbers x—the function f takes on a minimum value, not just on each closed interval, but on the whole line. (Notice that this is false if n is odd.) The proof depends on Theorem 7, but a tricky application will be required. We can apply Theorem 7 to any interval $[a, b]$, and obtain a point y_0 such that $f(y_0)$ is the minimum value of f on $[a, b]$; but if $[a, b]$ happens to be the interval shown in Figure 8, for example, then the point y_0 will not be the place where f has its minimum value for the whole line. In the next

theorem the entire point of the proof is to choose an interval $[a, b]$ in such a way that this cannot happen.

THEOREM 10 If n is even and $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then there is a number y such that $f(y) \leq f(x)$ for all x.

PROOF As in the proof of Theorem 9, if

$$M = \max(1, 2n|a_{n-1}|, \ldots, 2n|a_0|),$$

then for all x with $|x| \geq M$, we have

$$\frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}.$$

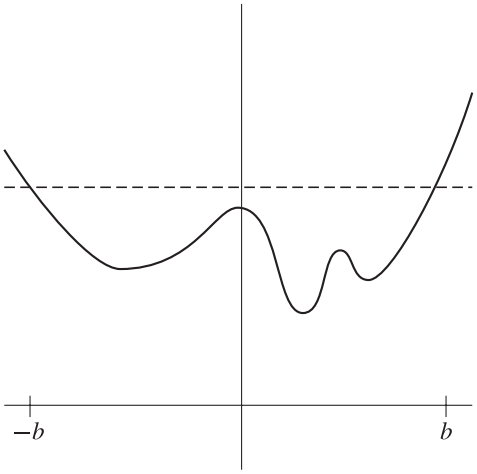


FIGURE 9

Since n is even, $x^n \geq 0$ for all x, so

$$\frac{x^n}{2} \leq x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}\right) = f(x),$$

provided that $|x| \geq M$. Now consider the number $f(0)$. Let $b > 0$ be a number such that $b^n \geq 2f(0)$ and also $b > M$. Then, if $x \geq b$, we have (Figure 9)

$$f(x) \geq \frac{x^n}{2} \geq \frac{b^n}{2} \geq f(0).$$

Similarly, if $x \leq -b$, then

$$f(x) \geq \frac{x^n}{2} \geq \frac{(-b)^n}{2} = \frac{b^n}{2} \geq f(0).$$

Summarizing:

$$\text{if } x \geq b \text{ or } x \leq -b, \text{ then } f(x) \geq f(0).$$

Now apply Theorem 7 to the function f on the interval $[-b, b]$. We conclude that there is a number y such that

$$(1) \qquad \text{if } -b \leq x \leq b, \text{ then } f(y) \leq f(x).$$

In particular, $f(y) \leq f(0)$. Thus

$$(2) \qquad \text{if } x \leq -b \text{ or } x \geq b, \text{ then } f(x) \geq f(0) \geq f(y).$$

Combining (1) and (2) we see that $f(y) \leq f(x)$ for all x. ∎

Theorem 10 now allows us to prove the following result.

THEOREM 11 Consider the equation

$$(*) \qquad x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c,$$

and suppose n is even. Then there is a number m such that $(*)$ has a solution for $c \geq m$ and has no solution for $c < m$.

PROOF Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ (Figure 10).

According to Theorem 10 there is a number y such that $f(y) \leq f(x)$ for all x. Let $m = f(y)$. If $c < m$, then the equation $(*)$ obviously has no solution, since the left side always has a value $\geq m$. If $c = m$, then $(*)$ has y as a solution. Finally, suppose $c > m$. Let b be a number such that $b > y$ and $f(b) > c$. Then $f(y) = m < c < f(b)$. Consequently, by Theorem 4, there is some number x in $[y, b]$ such that $f(x) = c$, so x is a solution of $(*)$. ∎

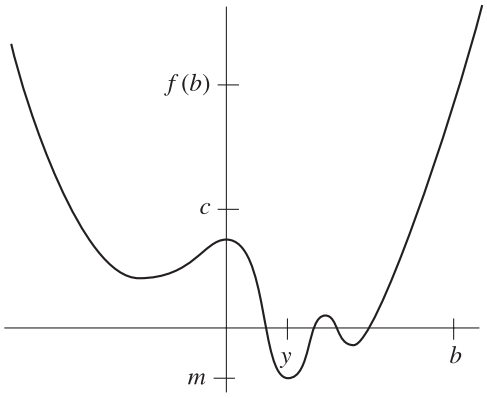


FIGURE 10

These consequences of Theorems 1, 2, and 3 are the only ones we will derive now (these theorems will play a fundamental role in everything we do later, however). Only one task remains—to prove Theorems 1, 2, and 3. Unfortunately, we cannot hope to do this—on the basis of our present knowledge about the real numbers (namely, P1–P12) a proof is *impossible*. There are several ways of convincing ourselves that this gloomy conclusion is actually the case. For example, the proof of Theorem 8 relies only on the proof of Theorem 1; if we could prove Theorem 1, then the proof of Theorem 8 would be complete, and we would have a proof that every positive number has a square root. As pointed out in Part I, it is impossible to prove this on the basis of P1–P12. Again, suppose we consider the function

$$f(x) = \frac{1}{x^2 - 2}$$

If there were no number x with $x^2 = 2$, then f would be continuous, since the denominator would never $= 0$. But f is not bounded on $[0, 2]$. So Theorem 2 depends essentially on the existence of numbers other than rational numbers, and therefore on some property of the real numbers other than P1–P12.

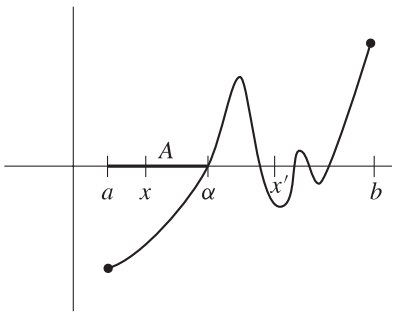


FIGURE 11

Despite our inability to prove Theorems 1, 2, and 3, they are certainly results which we want to be true. If the pictures we have been drawing have any connection with the mathematics we are doing, if our notion of continuous function corresponds to any degree with our intuitive notion, Theorems 1, 2, and 3 have got to be true. Since a proof of any of these theorems must require some new property of **R** which has so far been overlooked, our present difficulties suggest a way to discover that property: let us try to construct a proof of Theorem 1, for example, and see what goes wrong.

One idea which seems promising is to locate the first point where $f(x) = 0$, that is, the smallest x in $[a, b]$ such that $f(x) = 0$. To find this point, first consider the set A which contains all numbers x in $[a, b]$ such that f is negative on $[a, x]$. In Figure 11, x is such a point, while x' is not. The set A itself is indicated by a heavy line. Since f is negative at a, and positive at b, the set A contains some points greater than a, while all points sufficiently close to b are not in A. (We are here using the continuity of f on $[a, b]$, as well as Problem 6-15.)

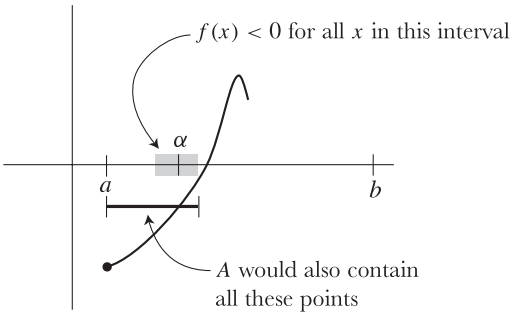


FIGURE 12

Now suppose α is the smallest number which is greater than all members of A; clearly $a < \alpha < b$. We claim that $f(\alpha) = 0$, and to prove this we only have to eliminate the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

Suppose first that $f(\alpha) < 0$. Then, by Theorem 6-3, $f(x)$ would be less than 0 for all x in a small interval containing α, in particular for some numbers bigger than α (Figure 12); but this contradicts the fact that α is bigger than every member of A, since the larger numbers would also be in A. Consequently, $f(\alpha) < 0$ is false.

On the other hand, suppose $f(\alpha) > 0$. Again applying Theorem 6-3, we see that $f(x)$ would be positive for all x in a small interval containing α, in particular for some numbers smaller than α (Figure 13). This means that these smaller numbers are all *not* in A. Consequently, one could have chosen an even smaller α which would be greater than all members of A. Once again we have a contradiction; $f(\alpha) > 0$ is also false. Hence $f(\alpha) = 0$ and, we are tempted to say, Q.E.D.

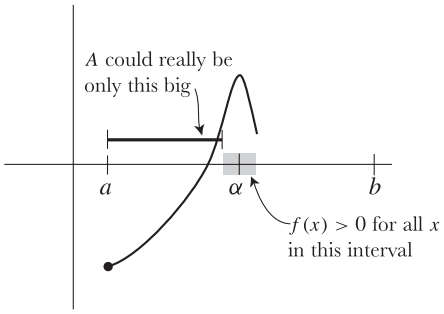


FIGURE 13

We know, however, that something must be wrong, since no new properties of **R** were ever used, and it does not require much scrutiny to find the dubious point. It is clear that we can choose a number α which is greater than all members of A (for example, we can choose $\alpha = b$), but it is not so clear that we can choose a *smallest* one. In fact, suppose A consists of all numbers $x \geq 0$ such that $x^2 < 2$. If the number $\sqrt{2}$ did not exist, there would not be a least number greater than all the members of A; for any $y > \sqrt{2}$ we chose, we could always choose a still smaller one.

Now that we have discovered the fallacy, it is almost obvious what additional property of the real numbers we need. All we must do is say it properly and use it. That is the business of the next chapter.

PROBLEMS

1. For each of the following functions, decide which are bounded above or below on the indicated interval, and which take on their maximum or minimum value. (Notice that f *might* have these properties even if f is not continuous, and even if the interval is not a closed interval.)

(i) $f(x) = x^2$ on $(-1, 1)$.
(ii) $f(x) = x^3$ on $(-1, 1)$.
(iii) $f(x) = x^2$ on **R**.
(iv) $f(x) = x^2$ on $[0, \infty)$.
(v) $f(x) = \begin{cases} x^2, & x \leq a \\ a + 2, & x > a \end{cases}$ on $(-a - 1, a + 1)$. (It will be necessary to consider several possibilities for a.)
(vi) $f(x) = \begin{cases} x^2, & x < a \\ a + 2, & x \geq a \end{cases}$ on $[-a - 1, a + 1]$.
(vii) $f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.

(viii) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ 1/q & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.

(ix) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ -1/q & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.

(x) $f(x) = \begin{cases} x, & x \text{ rational} \\ 0 & x \text{ irrational} \end{cases}$ on $[0, a]$.

(xi) $f(x) = \sin^2(\cos x + \sqrt{a + a^2})$ on $[0, a^3]$.

(xii) $f(x) = [x]$ on $[0, a]$.

2. For each of the following polynomial functions f, find an integer n such that $f(x) = 0$ for some x between n and $n + 1$.

(i) $f(x) = x^3 - x + 3$.

(ii) $f(x) = x^5 + 5x^4 + 2x + 1$.

(iii) $f(x) = x^5 + x + 1$.

(iv) $f(x) = 4x^2 - 4x + 1$.

3. Prove that there is some number x such that

(i) $x^{179} + \dfrac{163}{1 + x^2 + \sin^2 x} = 119$.

(ii) $\sin x = x - 1$.

4. This problem is a continuation of Problem 3-7.

(a) If $n - k$ is even, and ≥ 0, find a polynomial function of degree n with exactly k roots.

(b) A root a of the polynomial function f is said to have **multiplicity** m if $f(x) = (x - a)^m g(x)$, where g is a polynomial function that does *not* have a as a root. Let f be a polynomial function of degree n. Suppose that f has k roots, counting multiplicities, i.e., suppose that k is the sum of the multiplicities of all the roots. Show that $n - k$ is even.

5. Suppose that f is continuous on $[a, b]$ and that $f(x)$ is always rational. What can be said about f?

6. Suppose that f is a *continuous* function on $[-1, 1]$ such that $x^2 + (f(x))^2 = 1$ for all x. (This means that $(x, f(x))$ always lies on the unit circle.) Show that either $f(x) = \sqrt{1 - x^2}$ for all x, or else $f(x) = -\sqrt{1 - x^2}$ for all x.

7. How many continuous functions f are there which satisfy $(f(x))^2 = x^2$ for all x?

8. Suppose that f and g are continuous, that $f^2 = g^2$, and that $f(x) \neq 0$ for all x. Prove that either $f(x) = g(x)$ for all x, or else $f(x) = -g(x)$ for all x.

9. (a) Suppose that f is continuous, that $f(x) = 0$ only for $x = a$, and that $f(x) > 0$ for some $x > a$ as well as for some $x < a$. What can be said about $f(x)$ for all $x \neq a$?

(b) Again assume that f is continuous and that $f(x) = 0$ only for $x = a$, but suppose, instead, that $f(x) > 0$ for some $x > a$ and $f(x) < 0$ for some $x < a$. Now what can be said about $f(x)$ for $x \neq a$?

*(c) Discuss the sign of $x^3 + x^2y + xy^2 + y^3$ when x and y are not both 0.

10. Suppose f and g are continuous on $[a, b]$ and that $f(a) < g(a)$, but $f(b) > g(b)$. Prove that $f(x) = g(x)$ for some x in $[a, b]$. (If your proof isn't very short, it's not the right one.)

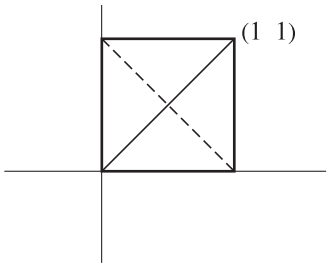


FIGURE 14

11. Suppose that f is a continuous function on $[0, 1]$ and that $f(x)$ is in $[0, 1]$ for each x (draw a picture). Prove that $f(x) = x$ for some number x.

12. (a) Problem 11 shows that f intersects the diagonal of the square in Figure 14 (solid line). Show that f must also intersect the other (dashed) diagonal.

(b) Prove the following more general fact: If g is continuous on $[0, 1]$ and $g(0) = 0$, $g(1) = 1$ or $g(0) = 1$, $g(1) = 0$, then $f(x) = g(x)$ for some x.

13. (a) Let $f(x) = \sin 1/x$ for $x \neq 0$ and let $f(0) = 0$. Is f continuous on $[-1, 1]$? Show that f satisfies the conclusion of the Intermediate Value Theorem on $[-1, 1]$; in other words, if f takes on two values somewhere on $[-1, 1]$, it also takes on every value in between.

*(b) Suppose that f satisfies the conclusion of the Intermediate Value Theorem, and that f takes on each value *only once*. Prove that f is continuous.

*(c) Generalize to the case where f takes on each value only finitely many times.

14. If f is a continuous function on $[0, 1]$, let $\|f\|$ be the maximum value of $|f|$ on $[0, 1]$.

(a) Prove that for any number c we have $\|cf\| = |c| \cdot \|f\|$.

*(b) Prove that $\|f + g\| \leq \|f\| + \|g\|$. Give an example where $\|f + g\| \neq \|f\| + \|g\|$.

(c) Prove that $\|h - f\| \leq \|h - g\| + \|g - f\|$.

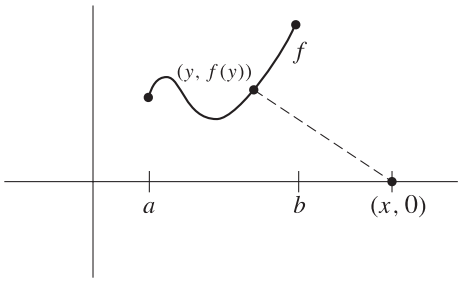


FIGURE 15

***15.** Suppose that ϕ is continuous and $\lim_{x\to\infty} \phi(x)/x^n = 0 = \lim_{x\to-\infty} \phi(x)/x^n$.

(a) Prove that if n is odd, then there is a number x such that $x^n + \phi(x) = 0$.

(b) Prove that if n is even, then there is a number y such that $y^n + \phi(y) \leq x^n + \phi(x)$ for all x.

Hint: Of which proofs does this problem test your understanding?

***16.** Let f be any polynomial function. Prove that there is some number y such that $|f(y)| \leq |f(x)|$ for all x.

***17.** Suppose that f is a continuous function with $f(x) > 0$ for all x, and $\lim_{x\to\infty} f(x) = 0 = \lim_{x\to-\infty} f(x)$. (Draw a picture.) Prove that there is some number y such that $f(y) \geq f(x)$ for all x.

***18.** (a) Suppose that f is continuous on $[a, b]$, and let x by any number. Prove that there is a point on the graph of f which is closest to $(x, 0)$; in other words there is some y in $[a, b]$ such that the distance from $(x, 0)$ to $(y, f(y))$ is $\leq$ distance from $(x, 0)$ to $(z, f(z))$ for all z in $[a, b]$. (See Figure 15.)

(b) Show that this same assertion is not necessarily true if $[a, b]$ is replaced by (a, b) throughout.

(c) Show that the assertion *is* true if $[a, b]$ is replaced by $\mathbf{R}$ throughout.

(d) In cases (a) and (c), let $g(x)$ be the minimum distance from $(x, 0)$ to a point on the graph of f. Prove that $g(y) \leq g(x) + |x - y|$, and conclude that g is continuous.

(e) Prove that there are numbers x_0 and x_1 in $[a, b]$ such that the distance from $(x_0, 0)$ to $(x_1, f(x_1))$ is $\leq$ the distance from $(x_0', 0)$ to $(x_1', f(x_1'))$ for any x_0', x_1' in $[a, b]$.

****19.** (a) Suppose that f is continuous on $[0, 1]$ and $f(0) = f(1)$. Let n be any natural number. Prove that there is some number x such that $f(x) = f(x+1/n)$, as shown in Figure 16 for $n = 4$. Hint: Consider the function $g(x) = f(x) - f(x + 1/n)$; what would be true if $g(x) \neq 0$ for all x?

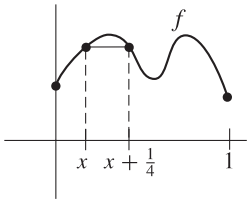


FIGURE 16

(b) Suppose $0 < a < 1$, but that a is not equal to $1/n$ for any natural number n. Find a function f which is continuous on $[0, 1]$ and which satisfies $f(0) = f(1)$, but which does not satisfy $f(x) = f(x + a)$ for any x.

****20.** (a) Prove that there does not exist a continuous function f defined on $\mathbf{R}$ which takes on every value exactly twice. Hint: If $f(a) = f(b)$ for $a < b$, then either $f(x) > f(a)$ for all x in (a, b) or $f(x) < f(a)$ for all x in (a, b). Why? In the first case all values close to $f(a)$, but slightly larger than $f(a)$, are taken on somewhere in (a, b); this implies that $f(x) < f(a)$ for $x < a$ and $x > b$.

(b) Refine part (a) by proving that there is no continuous function f which takes on each value either 0 times or 2 times, i.e., which takes on exactly twice each value that it does take on. Hint: The previous hint implies that f has either a maximum or a minimum value (which must be taken on twice). What can be said about values close to the maximum value?

(c) Find a continuous function f which takes on every value exactly 3 times. More generally, find one which takes on every value exactly n times, if n is odd.

(d) Prove that if n is even, then there is no continuous f which takes on every value exactly n times. Hint: To treat the case $n = 4$, for example, let $f(x_1) = f(x_2) = f(x_3) = f(x_4)$. Then either $f(x) > 0$ for all x in two of the three intervals (x_1, x_2), (x_2, x_3), (x_3, x_4), or else $f(x) < 0$ for all x in two of these three intervals.

CHAPTER 8 LEAST UPPER BOUNDS

This chapter reveals the most important property of the real numbers. Nevertheless, it is merely a sequel to Chapter 7; the path which must be followed has already been indicated, and further discussion would be useless delay.

DEFINITION

A set A of real numbers is **bounded above** if there is a number x such that

$$x \geq a \quad \text{for every } a \text{ in } A.$$

Such a number x is called an **upper bound** for A.

Obviously A is bounded above if and only if there is a number x which is an upper bound for A (and in this case there will be lots of upper bounds for A); we often say, as a concession to idiomatic English, that "A has an upper bound" when we mean that there is a number which is an upper bound for A.

Notice that the term "bounded above" has now been used in two ways—first, in Chapter 7, in reference to functions, and now in reference to sets. This dual usage should cause no confusion, since it will always be clear whether we are talking about a set of numbers or a function. Moreover, the two definitions are closely connected: if A is the set $\{f(x) : a \leq x \leq b\}$, then the function f is bounded above on $[a, b]$ if and only if the set A is bounded above.

The entire collection $\mathbf{R}$ of real numbers, and the natural numbers $\mathbf{N}$, are both examples of sets which are *not* bounded above. An example of a set which *is* bounded above is

$$A = \{x : 0 \leq x < 1\}.$$

To show that A is bounded above we need only name some upper bound for A, which is easy enough; for example, 138 is an upper bound for A, and so are 2, $1\frac{1}{2}$, $1\frac{1}{4}$, and 1. Clearly, 1 is the least upper bound of A; although the phrase just introduced is self-explanatory, in order to avoid any possible confusion (in particular, to ensure that we all know what the superlative of "less" means), we define this explicitly.

DEFINITION

A number x is a **least upper bound** of A if

(1) x is an upper bound of A,
and (2) if y is an upper bound of A, then $x \leq y$.

The use of the indefinite article "a" in this definition was merely a concession to temporary ignorance. Now that we have made a precise definition, it is easily seen that if x and y are both least upper bounds of A, then $x = y$. Indeed, in this case

$$\begin{aligned} x \leq y, &\quad \text{since } y \text{ is an upper bound, and } x \text{ is a least upper bound,} \\ \text{and } y \leq x, &\quad \text{since } x \text{ is an upper bound, and } y \text{ is a least upper bound;} \end{aligned}$$

it follows that $x = y$. For this reason we speak of *the* least upper bound of A. The term **supremum** of A is synonymous and has one advantage. It abbreviates quite nicely to

$$\sup A \qquad \text{(pronounced "soup } A\text{")}$$

and saves us from the abbreviation

$$\operatorname{lub} A$$

(which is nevertheless used by some authors).

There is a series of important definitions, analogous to those just given, which can now be treated more briefly. A set A of real numbers is **bounded below** if there is a number x such that

$$x \leq a \qquad \text{for every } a \text{ in } A.$$

Such a number x is called a **lower bound** for A. A number x is the **greatest lower bound** of A if

$$\begin{aligned} &(1) \quad x \text{ is a lower bound of } A, \\ \text{and} \quad &(2) \quad \text{if } y \text{ is a lower bound of } A, \text{ then } x \geq y. \end{aligned}$$

The greatest lower bound of A is also called the **infimum** of A, abbreviated

$$\inf A;$$

some authors use the abbreviation

$$\operatorname{glb} A.$$

One detail has been omitted from our discussion so far—the question of which sets have at least one, and hence exactly one, least upper bound or greatest lower bound. We will consider only least upper bounds, since the question for greatest lower bounds can then be answered easily (Problem 2).

If A is not bounded above, then A has no upper bound at all, so A certainly cannot be expected to have a least upper bound. It is tempting to say that A does have a least upper bound if it has *some* upper bound, but, like the principle of mathematical induction, this assertion can fail to be true in a rather special way. If $A = \emptyset$, then A is bounded above. Indeed, any number x is an upper bound for $\emptyset$:

$$x \geq y \quad \text{for every } y \text{ in } \emptyset$$

simply because there is no y in $\emptyset$. Since *every* number is an upper bound for $\emptyset$, there is surely no least upper bound for $\emptyset$. With this trivial exception however,

our assertion is true—and very important, definitely important enough to warrant consideration of details. We are finally ready to state the last property of the real numbers which we need.

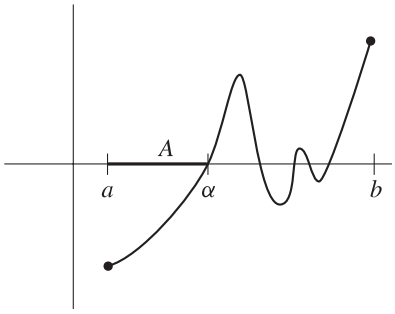


FIGURE 1

(P13) (The least upper bound property) If A is a set of real numbers, $A \neq \emptyset$, and A is bounded above, then A has a least upper bound.

Property P13 may strike you as anticlimactic, but that is actually one of its virtues. To complete our list of basic properties for the real numbers we require no particularly abstruse proposition, but only a property so simple that we might feel foolish for having overlooked it. Of course, the least upper bound property is not really so innocent as all that; after all, it does *not* hold for the rational numbers $\mathbf{Q}$. For example, if A is the set of all rational numbers x satisfying $x^2 < 2$, then there is no *rational* number y which is an upper bound for A and which is less than or equal to every other *rational* number which is an upper bond for A. It will become clear only gradually how significant P13 is, but we are already in a position to demonstrate its power, by supplying the proofs which were omitted in Chapter 7.

THEOREM 7-1 If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some number x in $[a, b]$ such that $f(x) = 0$.

PROOF Our proof is merely a rigorous version of the outline developed at the end of Chapter 7—we will locate the smallest number x in $[a, b]$ with $f(x) = 0$.

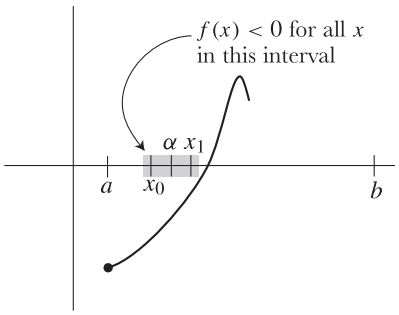


FIGURE 2

Define the set A, shown in Figure 1, as follows:

$$A = \{x : a \leq x \leq b, \text{ and } f \text{ is negative on the interval } [a, x]\}.$$

Clearly $A \neq \emptyset$, since a is in A; in fact, there is some $\delta > 0$ such that A contains all points x satisfying $a \leq x < a + \delta$; this follows from Problem 6-15, since f is continuous on $[a, b]$ and $f(a) < 0$. Similarly, b is an upper bound for A and, in fact, there is a $\delta > 0$ such that all points x satisfying $b - \delta < x \leq b$ are upper bounds for A; this also follows from Problem 6-15, since $f(b) > 0$.

From these remarks it follows that A has a least upper bound α and that $a < \alpha < b$. We now wish to show that $f(\alpha) = 0$, by eliminating the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

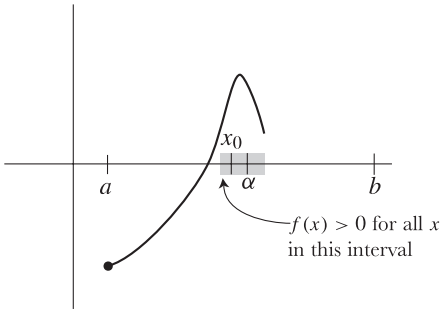


FIGURE 3

Suppose first that $f(\alpha) < 0$. By Theorem 6-3, there is a $\delta > 0$ such that $f(x) < 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 2). Now there is some number x_0 in A which satisfies $\alpha - \delta < x_0 < \alpha$ (because otherwise α would not be the *least* upper bound of A). This means that f is negative on the whole interval $[a, x_0]$. But if x_1 is a number between α and $\alpha + \delta$, then f is also negative on the whole interval $[x_0, x_1]$. Therefore f is negative on the interval $[a, x_1]$, so x_1 is in A. But this contradicts the fact that α is an upper bound for A; our original assumption that $f(\alpha) < 0$ must be false.

Suppose, on the other hand, that $f(\alpha) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 3). Once again we know that there is an x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$; but this means that f is negative on $[a, x_0]$,

which is impossible, since $f(x_0) > 0$. Thus the assumption $f(\alpha) > 0$ also leads to a contradiction, leaving $f(\alpha) = 0$ as the only possible alternative. ∎

The proofs of Theorems 2 and 3 of Chapter 7 require a simple preliminary result, which will play much the same role as Theorem 6-3 played in the previous proof.

THEOREM 1 If f is continuous at a, then there is a number $\delta > 0$ such that f is bounded above on the interval $(a - \delta, a + \delta)$ (see Figure 4).

PROOF Since $\lim_{x \to a} f(x) = f(a)$, there is, for every $\varepsilon > 0$, a $\delta > 0$ such that, for all x,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.$$

It is only necessary to apply this statement to some particular ε (any one will do), for example, $\varepsilon = 1$. We conclude that there is a $\delta > 0$ such that, for all x,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < 1.$$

It follows, in particular, that if $|x - a| < \delta$, then $f(x) - f(a) < 1$. This completes the proof: on the interval $(a - \delta, a + \delta)$ the function f is bounded above by $f(a) + 1$. ∎

It should hardly be necessary to add that we can now also prove that f is bounded below on some interval $(a - \delta, a + \delta)$, and, finally, that f is bounded on some open interval containing a.

A more significant point is the observation that if $\lim_{x \to a^+} f(x) = f(a)$, then there

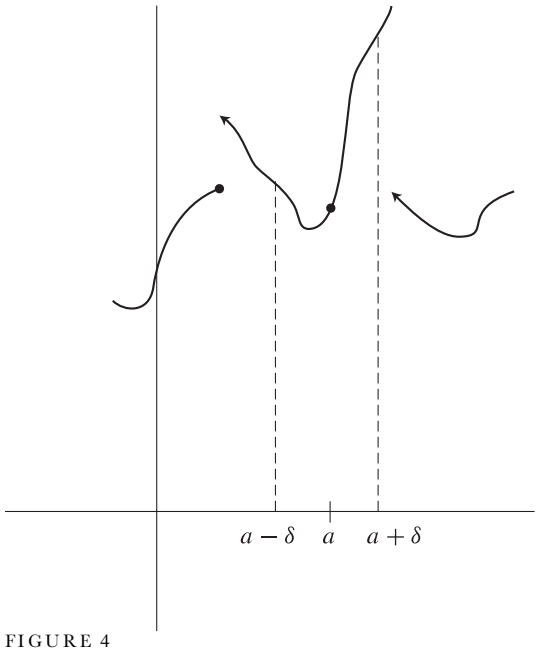


FIGURE 4

is a $\delta > 0$ such that f is bounded on the set $\{x : a \le x < a + \delta\}$, and a similar observation holds if $\lim_{x \to b^-} f(x) = f(b)$. Having made these observations (and assuming that you will supply the proofs), we tackle our second major theorem.

THEOREM 7-2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$.

PROOF Let

$$A = \{x : a \le x \le b \text{ and } f \text{ is bounded above on } [a, x]\}.$$

Clearly $A \neq \emptyset$ (since a is in A), and A is bounded above (by b), so A has a least upper bound α. Notice that we are here applying the term "bounded above" both to the set A, which can be visualized as lying on the horizontal axis, and to f, i.e., to the sets $\{f(y) : a \le y \le x\}$, which can be visualized as lying on the vertical axis (Figure 5).

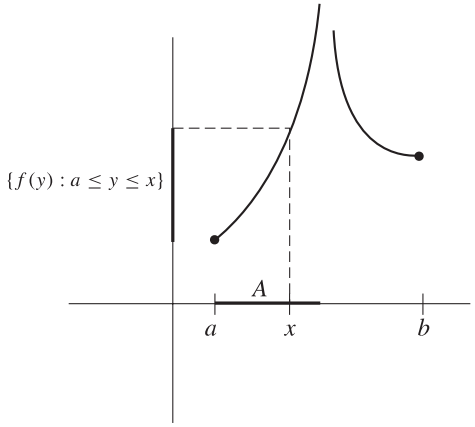


FIGURE 5

Our first step is to prove that we actually have $\alpha = b$. Suppose, instead, that $\alpha < b$. By Theorem 1 there is $\delta > 0$ such that f is bounded on $(\alpha - \delta, \alpha + \delta)$. Since α is the least upper bound of A there is some x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$. This means that f is bounded on $[a, x_0]$. But if x_1 is any number with $\alpha < x_1 < \alpha + \delta$, then f is also bounded on $[x_0, x_1]$. Therefore f is bounded on $[a, x_1]$, so x_1 is in A, contradicting the fact that α is an upper bound for A. This contradiction shows that $\alpha = b$. One detail should be mentioned: this demonstration implicitly assumed that $a < \alpha$ [so that f would be defined on some interval $(\alpha - \delta, \alpha + \delta)$]; the possibility $a = \alpha$ can be ruled out similarly, using the existence of a $\delta > 0$ such that f is bounded on $\{x : a \le x < a + \delta\}$.

The proof is not quite complete—we only know that f is bounded on $[a, x]$ for every $x < b$, not necessarily that f is bounded on $[a, b]$. However, only one small argument needs to be added.

There is a $\delta > 0$ such that f is bounded on $\{x : b - \delta < x \le b\}$. There is x_0 in A such that $b - \delta < x_0 < b$. Thus f is bounded on $[a, x_0]$ and also on $[x_0, b]$, so f is bounded on $[a, b]$. ∎

To prove the third important theorem we resort to a trick.

THEOREM 7-3 If f is continuous on $[a, b]$, then there is a number y in $[a, b]$ such that $f(y) \ge f(x)$ for all x in $[a, b]$.

PROOF We already know that f is bounded on $[a, b]$, which means that the set

$$\{f(x) : x \text{ in } [a, b]\}$$

is bounded. This set is obviously not $\emptyset$, so it has a least upper bound α. Since $\alpha \ge f(x)$ for x in $[a, b]$ it suffices to show that $\alpha = f(y)$ for some y in $[a, b]$.

Suppose instead that $\alpha \neq f(y)$ for all y in $[a, b]$. Then the function g defined by

$$g(x) = \frac{1}{\alpha - f(x)}, \quad x \text{ in } [a, b]$$

is continuous on $[a, b]$, since the denominator of the right side is never 0. On the other hand, α is the least upper bound of $\{f(x) : x \text{ in } [a, b]\}$; this means that

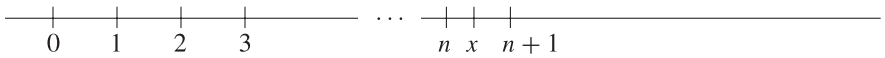
$$\text{for every } \varepsilon > 0 \text{ there is } x \text{ in } [a, b] \text{ with } \alpha - f(x) < \varepsilon.$$

This, in turn, means that

$$\text{for every } \varepsilon > 0 \text{ there is } x \text{ in } [a, b] \text{ with } g(x) > 1/\varepsilon.$$

But *this* means that g is not bounded on $[a, b]$, contradicting the previous theorem. ∎

At the beginning of this chapter the set of natural numbers **N** was given as an example of an unbounded set. We are now going to *prove* that **N** is unbounded. After the difficult theorems proved in this chapter you may be startled to find such an "obvious" theorem winding up our proceedings. If so, you are, perhaps, allowing the geometrical picture of **R** to influence you too strongly. "Look," you may say, "the real numbers look like



so every number x is between two integers n, $n + 1$ (unless x is itself an integer)." Basing the argument on a geometric picture is not a proof, however, and even the geometric picture contains an assumption: that if you place unit segments end-to-end you will eventually get a segment larger than any given segment. This axiom, often omitted from a first introduction to geometry, is usually attributed (not quite justly) to Archimedes, and the corresponding property for numbers, that **N** is not bounded, is called the *Archimedian property* of the real numbers. This property is *not* a consequence of P1–P12 (see reference [17] of the Suggested Reading), although it does hold for **Q**, of course. Once we have P13 however, there are no longer any problems.

THEOREM 2 **N** is not bounded above.

PROOF Suppose **N** were bounded above. Since $\mathbf{N} \neq \emptyset$, there would be a least upper bound α for **N**. Then

$$\alpha \geq n \qquad \text{for all } n \text{ in } \mathbf{N}.$$

Consequently,

$$\alpha \geq n + 1 \quad \text{for all } n \text{ in } \mathbf{N},$$

since $n + 1$ is in **N** if n is in **N**. But this means that

$$\alpha - 1 \geq n \quad \text{for all } n \text{ in } \mathbf{N},$$

and *this* means that $\alpha - 1$ is also an upper bound for **N**, contradicting the fact that α is the least upper bound. ∎

There is a consequence of Theorem 2 (actually an equivalent formulation) which we have very often assumed implicitly.

THEOREM 3 For any $\varepsilon > 0$ there is a natural number n with $1/n < \varepsilon$.

PROOF Suppose not; then $1/n \geq \varepsilon$ for all n in $\mathbf{N}$. Thus $n \leq 1/\varepsilon$ for all n in $\mathbf{N}$. But this means that $1/\varepsilon$ is an upper bound for $\mathbf{N}$, contradicting Theorem 2. ∎

A brief glance through Chapter 6 will show you that the result of Theorem 3 was used in the discussion of many examples. Of course, Theorem 3 was not available at the time, but the examples were so important that in order to give them some cheating was tolerated. As partial justification for this dishonesty we can claim that this result was never used in the proof of a *theorem*, but if your faith has been shaken, a review of all the proofs given so far is in order. Fortunately, such deception will not be necessary again. We have now stated every property of the real numbers that we will ever need. Henceforth, no more lies.

PROBLEMS

1. Find the least upper bound and the greatest lower bound (if they exist) of the following sets. Also decide which sets have greatest and least elements (i.e., decide when the least upper bound and greatest lower bound happens to belong to the set).

(i) $\left\{\dfrac{1}{n} : n \text{ in } \mathbf{N}\right\}$.

(ii) $\left\{\dfrac{1}{n} : n \text{ in } \mathbf{Z} \text{ and } n \neq 0.\right\}$.

(iii) $\{x : x = 0 \text{ or } x = 1/n \text{ for some } n \text{ in } \mathbf{N}\}$.

(iv) $\{x : 0 \leq x \leq \sqrt{2} \text{ and } x \text{ is rational}\}$.

(v) $\{x : x^2 + x + 1 \geq 0\}$.

(vi) $\{x : x^2 + x - 1 < 0\}$.

(vii) $\{x : x < 0 \text{ and } x^2 + x - 1 < 0\}$.

(viii) $\left\{\dfrac{1}{n} + (-1)^n : n \text{ in } \mathbf{N}\right\}$.

2. (a) Suppose $A \neq \emptyset$ is bounded below. Let $-A$ denote the set of all $-x$ for x in A. Prove that $-A \neq \emptyset$, that $-A$ is bounded above, and that $-\sup(-A)$ is the greatest lower bound of A.

(b) If $A \neq \emptyset$ is bounded below, let B be the set of all lower bounds of A. Show that $B \neq \emptyset$, that B is bounded above, and that $\sup B$ is the greatest lower bound of A.

3. Let f be a continuous function on $[a, b]$ with $f(a) < 0 < f(b)$.

(a) The proof of Theorem 1 showed that there is a smallest x in $[a, b]$ with $f(x) = 0$. Is there necessarily a second smallest x in $[a, b]$ with

$f(x) = 0$? Show that there is a largest x in $[a, b]$ with $f(x) = 0$. (Try to give an easy proof by considering a new function closely related to f.)

(b) The proof of Theorem 1 depended upon consideration of $A = \{x : a \leq x \leq b \text{ and } f \text{ is negative on } [a, x]\}$. Give another proof of Theorem 1, which depends upon consideration of $B = \{x : a \leq x \leq b \text{ and } f(x) < 0\}$. Which point x in $[a, b]$ with $f(x) = 0$ will this proof locate? Give an example where the sets A and B are not the same.

***4.** (a) Suppose that f is continuous on $[a, b]$ and that $f(a) = f(b) = 0$. Suppose also that $f(x_0) > 0$ for some x_0 in $[a, b]$. Prove that there are numbers c and d with $a \leq c < x_0 < d \leq b$ such that $f(c) = f(d) = 0$, but $f(x) > 0$ for all x in (c, d). Hint: The previous problem can be used to good advantage.

(b) Suppose that f is continuous on $[a, b]$ and that $f(a) < f(b)$. Prove that there are numbers c and d with $a \leq c < d \leq b$ such that $f(c) = f(a)$ and $f(d) = f(b)$ and $f(a) < f(x) < f(d)$ for all x in (c, d).

5. (a) Suppose that $y - x > 1$. Prove that there is an integer k such that $x < k < y$. Hint: Let l by the largest integer satisfying $l \leq x$, and consider $l + 1$.

(b) Suppose $x < y$. Prove that there is a rational number r such that $x < r < y$. Hint: If $1/n < y - x$, then $ny - nx > 1$. (Query: Why have parts (a) and (b) been postponed until this problem set?)

(c) Suppose that $r < s$ are rational numbers. Prove that there is an irrational number between r and s. Hint: As a start, you know that there is an irrational number between 0 and 1.

(d) Suppose that $x < y$. Prove that there is an irrational number between x and y. Hint: It is unnecessary to do any more work; this follows from (b) and (c).

***6.** A set A of real numbers is said to be **dense** if every open interval contains a point of A. For example, Problem 5 shows that the set of rational numbers and the set of irrational numbers are each dense.

(a) Prove that if f is continuous and $f(x) = 0$ for all numbers x in a dense set A, then $f(x) = 0$ for all x.

(b) Prove that if f and g are continuous and $f(x) = g(x)$ for all x in a dense set A, then $f(x) = g(x)$ for all x.

(c) If we assume instead that $f(x) \geq g(x)$ for all x in A, show that $f(x) \geq g(x)$ for all x. Can $\geq$ be replaced by $>$ throughout?

7. Prove that if f is continuous and $f(x + y) = f(x) + f(y)$ for all x and y, then there is a number c such that $f(x) = cx$ for all x. (This conclusion can be demonstrated simply by combining the results of two previous problems.) Point of information: There *do* exist *noncontinuous* functions f satisfying $f(x + y) = f(x) + f(y)$ for all x and y, but we cannot prove this now; in fact, this simple question involves ideas that are usually never mentioned in any undergraduate course. The Suggested Reading contains references.

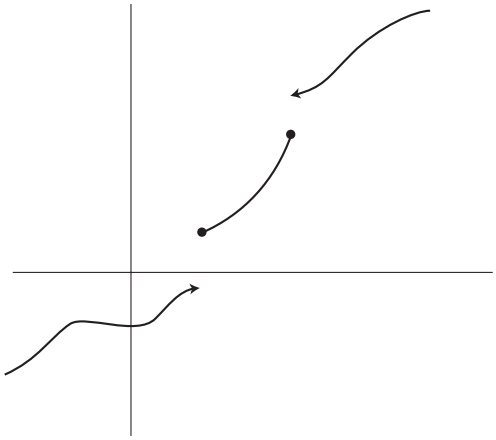


FIGURE 6

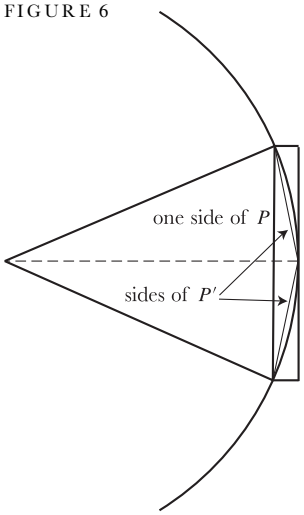


FIGURE 7

***8.** Suppose that f is a function such that $f(a) \leq f(b)$ whenever $a < b$ (Figure 6).

(a) Prove that $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist. Hint: Why is this problem in this chapter?
(b) Prove that f never has a removable discontinuity (this terminology comes from Problem 6-16).
(c) Prove that if f satisfies the conclusions of the Intermediate Value Theorem, then f is continuous.

***9.** If f is a bounded function on $[0, 1]$, let $|||f||| = \sup\{|f(x)| : x \text{ in } [0, 1]\}$. Prove analogues of the properties of $\| \ \|$ in Problem 7-14.

10. Suppose $\alpha > 0$. Prove that every number x can be written uniquely in the form $x = k\alpha + x'$, where k is an integer, and $0 \leq x' < \alpha$.

11. (a) Suppose that $a_1, a_2, a_3, \ldots$ is a sequence of positive numbers with $a_{n+1} \leq a_n/2$. Prove that for any $\varepsilon > 0$ there is some n with $a_n < \varepsilon$.
(b) Suppose P is a regular polygon inscribed inside a circle. If P' is the inscribed regular polygon with twice as many sides, show that the difference between the area of the circle and the area of P' is less than half the difference between the area of the circle and the area of P (use Figure 7).
(c) Prove that there is a regular polygon P inscribed in a circle with area as close as desired to the area of the circle. In order to do part (c) you will need part (a). This was clear to the Greeks, who used part (a) as the basis for their entire treatment of proportion and area. By calculating the areas of polygons, this method ("the method of exhaustion") allows computations of π to any desired accuracy; Archimedes used it to show that $\frac{223}{71} < \pi < \frac{22}{7}$. But it has far greater theoretical importance:
*(d) Using the fact that the areas of two regular polygons with the same number of sides have the same ratio as the square of their sides, prove that the areas of two circles have the same ratios as the square of their radii. Hint: Deduce a contradiction from the assumption that the ratio of the areas is greater, or less, than the ratio of the square of the radii by inscribing appropriate polygons.

12. Suppose that A and B are two nonempty sets of numbers such that $x \leq y$ for all x in A and all y in B.

(a) Prove that $\sup A \leq y$ for all y in B.
(b) Prove that $\sup A \leq \inf B$.

13. Let A and B be two nonempty sets of numbers which are bounded above, and let $A+B$ denote the set of all numbers $x+y$ with x in A and y in B. Prove that $\sup(A+B) = \sup A + \sup B$. Hint: The inequality $\sup(A+B) \leq \sup A + \sup B$ is easy. Why? To prove that $\sup A + \sup B \leq \sup(A + B)$ it suffices to prove that $\sup A + \sup B \leq \sup(A + B) + \varepsilon$ for all $\varepsilon > 0$; begin by choosing x in A

and y in B with $\sup A - x < \varepsilon/2$ and $\sup B - y < \varepsilon/2$.

FIGURE 8

14. (a) Consider a sequence of closed intervals $I_1 = [a_1, b_1], I_2 = [a_2, b_2], \ldots$. Suppose that $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all n (Figure 8). Prove that there is a point x which is in every I_n.

(b) Show that this conclusion is false if we consider open intervals instead of closed intervals.

The simple result of Problem 14(a) is called the "Nested Interval Theorem." It may be used to give alternative proofs of Theorems 1 and 2. The appropriate reasoning, outlined in the next two problems, illustrates a general method, called a "bisection argument."

***15.** Suppose f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$. Then either $f((a+b)/2) = 0$, or f has different signs at the end points of the interval $[a, (a+b)/2]$, or f has different signs at the end points of $[(a+b)/2, b]$. Why? If $f((a+b)/2) \neq 0$, let I_1 be one of the two intervals on which f changes sign. Now bisect I_1. Either f is 0 at the midpoint, or f changes sign on one of the two intervals. Let I_2 be such an interval. Continue in this way, to define I_n for each n (unless f is 0 at some midpoint). Use the Nested Interval Theorem to find a point x where $f(x) = 0$.

***16.** Suppose f were continuous on $[a, b]$, but not bounded on $[a, b]$. Then f would be unbounded on either $[a, (a+b)/2]$ or $[(a+b)/2, b]$. Why? Let I_1 be one of these intervals on which f is unbounded. Proceed as in Problem 15 to obtain a contradiction.

17. (a) Let $A = \{x : x < \alpha\}$. Prove the following (they are all easy):

(i) If x is in A and $y < x$, then y is in A.
(ii) $A \neq \emptyset$.
(iii) $A \neq \mathbf{R}$.
(iv) If x is in A, then there is some number x' in A such that $x < x'$.

(b) Suppose, conversely, that A satisfies (i)–(iv). Prove that $A = \{x : x < \sup A\}$.

***18.** A number x is called an **almost upper bound** for A if there are only finitely many numbers y in A with $y \geq x$. An **almost lower bound** is defined similarly.

(a) Find all almost upper bounds and almost lower bounds of the sets in Problem 1.

(b) Suppose that A is a bounded infinite set. Prove that the set B of all almost upper bounds of A is nonempty, and bounded below.

(c) It follows from part (b) that inf B exists; this number is called the **limit superior** of A, and denoted by $\overline{\lim}\, A$ or $\limsup A$. Find $\overline{\lim}\, A$ for each set A in Problem 1.

(d) Define $\underline{\lim}\, A$, and find it for all A in Problem 1.

***19.** If A is a bounded infinite set prove

(a) $\underline{\lim}\, A \leq \overline{\lim}\, A$.

(b) $\overline{\lim}\, A \leq \sup A$.

(c) If $\overline{\lim}\, A < \sup A$, then A contains a largest element.

(d) The analogues of parts (b) and (c) for $\underline{\lim}$.

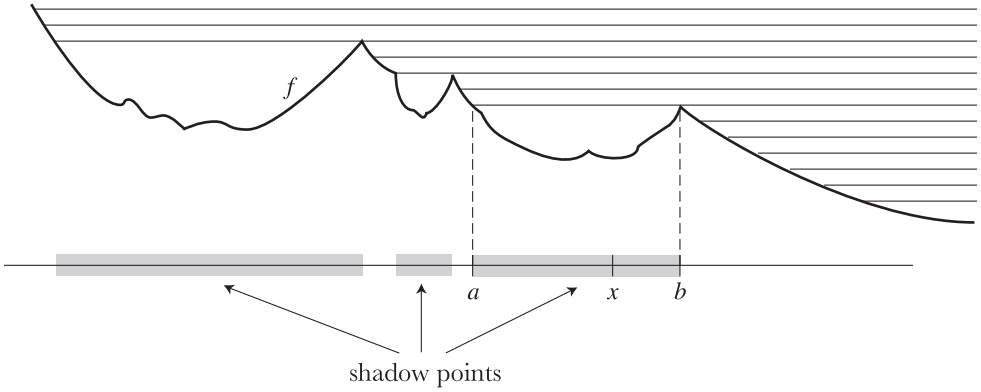


FIGURE 9

***20.** Let f be a continuous function on $\mathbf{R}$. A point x is called a **shadow point** of f if there is a number $y > x$ with $f(y) > f(x)$. The rationale for this terminology is indicated in Figure 9; the parallel lines are the rays of the sun rising in the east (you are facing north). Suppose that all points of (a, b) are shadow points, but that a and b are not shadow points.

(a) For x in (a, b), prove that $f(x) \leq f(b)$. Hint: Let $A = \{y : x \leq y \leq b$ and $f(x) \leq f(y)\}$. If $\sup A$ were less than b, then $\sup A$ would be a shadow point. Use this fact to obtain a contradiction to the fact that b is not a shadow point.

(b) Now prove that $f(a) \leq f(b)$. (This is a simple consequence of continuity.)

(c) Finally, using the fact that a is not a shadow point, prove that $f(a) = f(b)$.

This result is known as the Rising Sun Lemma. Aside from serving as a good illustration of the use of least upper bounds, it is instrumental in proving several beautiful theorems that do not appear in this book; see page 443.

APPENDIX. UNIFORM CONTINUITY

Now that we've come to the end of the "foundations," it might be appropriate to slip in one further fundamental concept. This notion is not used crucially in the rest of the book, but it can help clarify many points later on.

We know that the function $f(x) = x^2$ is continuous at a for all a. In other words,

> if a is any number, then for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x, if $|x - a| < \delta$, then $|x^2 - a^2| < \varepsilon$.

Of course, δ depends on ε. But δ *also depends* on a—the δ that works at a might not work at b (Figure 1). Indeed, it's clear that given $\varepsilon > 0$ there is no one $\delta > 0$ that works for all a, or even for all positive a. In fact, the number $a + \delta/2$ will certainly satisfy $|x - a| < \delta$, but if $a > 0$, then

$$\left|\left(a + \frac{\delta}{2}\right)^2 - a^2\right| = \left|a\delta + \frac{\delta^2}{4}\right| \geq a\delta,$$

and this won't be $< \varepsilon$ once $a > \varepsilon/\delta$. (This is just an admittedly confusing computational way of saying that f is growing faster and faster!)

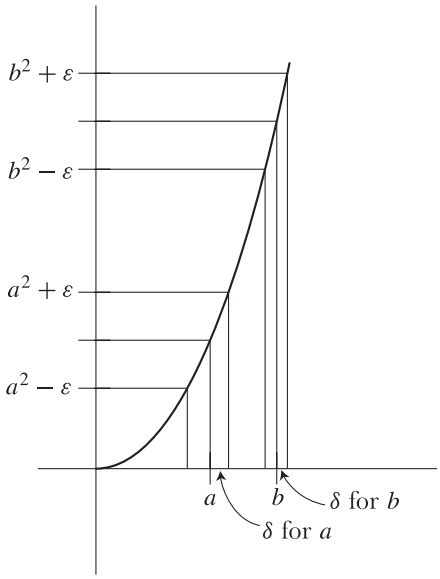


FIGURE 1

On the other hand, for any $\varepsilon > 0$ there *will* be one $\delta > 0$ that works for all a in any interval $[-N, N]$. In fact, the δ which works at N or $-N$ will also work everywhere else in the interval.

As a final example, consider the function $f(x) = \sin 1/x$, or the function whose graph appears in Figure 18 on page 62. It is easy to see that, so long as $\varepsilon < 1$, there will not be one $\delta > 0$ that works for these functions at all points a in the open interval $(0, 1)$.

These examples illustrate important distinctions between the behavior of various continuous functions on certain intervals, and there is a special term to signal this distinction.

DEFINITION

The function f is **uniformly continuous on an interval** A if for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x and y in A,

$$\text{if } |x - y| < \delta, \text{ then } |f(x) - f(y)| < \varepsilon.$$

We've seen that a function can be continuous on the whole line, or on an open interval, without being uniformly continuous there. On the other hand, the function $f(x) = x^2$ did turn out to be uniformly continuous on any closed interval. This shouldn't be too surprising—it's the same sort of thing that occurs when we ask whether a function is bounded on an interval—and we would be led to suspect that any continuous function on a closed interval is also uniformly continuous on that interval. In order to prove this, we'll need to deal first with one subtle point.

Suppose that we have two intervals $[a, b]$ and $[b, c]$ with the common endpoint b, and a function f that is continuous on $[a, c]$. Let $\varepsilon > 0$ and suppose that

the following two statements hold:

(i) if x and y are in $[a, b]$ and $|x - y| < \delta_1$, then $|f(x) - f(y)| < \varepsilon$,
(ii) if x and y are in $[b, c]$ and $|x - y| < \delta_2$, then $|f(x) - f(y)| < \varepsilon$.

We'd like to know if there is some $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever x and y are points in $[a, c]$ with $|x - y| < \delta$. Our first inclination might be to choose δ as the minimum of δ_1 and δ_2. But it is easy to see what goes wrong (Figure 2): we might have x in $[a, b]$ and y in $[b, c]$, and then neither (i) nor (ii) tells us anything about $|f(x) - f(y)|$. So we have to be a little more cagey, and also use continuity of f at b.

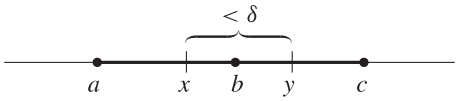


FIGURE 2

LEMMA Let $a < b < c$ and let f be continuous on the interval $[a, c]$. Let $\varepsilon > 0$, and suppose that statements (i) and (ii) hold. Then there is a $\delta > 0$ such that,

$$\text{if } x \text{ and } y \text{ are in } [a, c] \text{ and } |x - y| < \delta, \text{ then } |f(x) - f(y)| < \varepsilon.$$

PROOF Since f is continuous at b, there is a $\delta_3 > 0$ such that,

$$\text{if } |x - b| < \delta_3, \text{ then } |f(x) - f(b)| < \frac{\varepsilon}{2}.$$

It follows that

(iii) if $|x - b| < \delta_3$ and $|y - b| < \delta_3$, then $|f(x) - f(y)| < \varepsilon$.

Choose δ to be the minimum of δ_1, δ_2, and δ_3. We claim that this δ works. In fact, suppose that x and y are any two points in $[a, c]$ with $|x - y| < \delta$. If x and y are both in $[a, b]$, then $|f(x) - f(y)| < \varepsilon$ by (i); and if x and y are both in $[b, c]$, then $|f(x) - f(y)| < \varepsilon$ by (ii). The only other possibility is that

$$x < b < y \qquad \text{or} \qquad y < b < x.$$

In either case, since $|x - y| < \delta$, we also have $|x - b| < \delta$ and $|y - b| < \delta$. So $|f(x) - f(y)| < \varepsilon$ by (iii). ∎

THEOREM 1 If f is continuous on $[a, b]$, then f is uniformly continuous on $[a, b]$.

PROOF It's the usual trick, but we've got to be a little bit careful about the mechanism of the proof. For $\varepsilon > 0$ let's say that f is *ε-good* on $[a, b]$ if there is some $\delta > 0$ such that, for all y and z in $[a, b]$,

$$\text{if } |y - z| < \delta, \text{ then } |f(y) - f(z)| < \varepsilon.$$

Then we're trying to prove that f is ε-good on $[a, b]$ for all $\varepsilon > 0$.

Consider any particular $\varepsilon > 0$. Let

$$A = \{x : a \le x \le b \text{ and } f \text{ is } \varepsilon\text{-good on } [a, x]\}.$$

Then $A \neq \emptyset$ (since a is in A), and A is bounded above (by b), so A has a least upper bound α. We really should write α_ε, since A and α might depend on ε. But we won't since we intend to prove that $\alpha = b$, no matter what ε is.

Suppose that we had $\alpha < b$. Since f is continuous at α, there is some $\delta_0 > 0$ such that, if $|y - \alpha| < \delta_0$, then $|f(y) - f(\alpha)| < \varepsilon/2$. Consequently, if $|y - \alpha| < \delta_0$ and $|z - \alpha| < \delta_0$, then $|f(y) - f(z)| < \varepsilon$. So f is surely ε-good on the interval $[\alpha - \delta_0, \alpha + \delta_0]$. On the other hand, since α is the least upper bound of A, it is also clear that f is ε-good on $[a, \alpha - \delta_0]$. Then the Lemma implies that f is ε-good on $[a, a + \delta_0]$, so $a + \delta_0$ is in A, contradicting the fact that α is an upper bound.

To complete the proof we just have to show that $\alpha = b$ is actually in A. The argument for this is practically the same: Since f is continuous at b, there is some $\delta_0 > 0$ such that, if $|b - y| < \delta_0$, then $|f(y) - f(b)| < \varepsilon/2$. So f is ε-good on $[b - \delta_0, b]$. But f is also ε-good on $[a, b - \delta_0]$, so the Lemma implies that f is ε-good on $[a, b]$. ∎

PROBLEMS

1. (a) For which of the following values of α is the function $f(x) = x^\alpha$ uniformly continuous on $[0, \infty)$: $\alpha = 1/3, 1/2, 2, 3$?
(b) Find a function f that is continuous and bounded on $(0, 1]$, but not uniformly continuous on $(0, 1]$.
(c) Find a function f that is continuous and bounded on $[0, \infty)$ but which is not uniformly continuous on $[0, \infty)$.

2. (a) Prove that if f and g are uniformly continuous on A, then so is $f + g$.
(b) Prove that if f and g are uniformly continuous and bounded on A, then fg is uniformly continuous on A.
(c) Show that this conclusion does not hold if one of them isn't bounded.
(d) Suppose that f is uniformly continuous on A, that g is uniformly continuous on B, and that $f(x)$ is in B for all x in A. Prove that $g \circ f$ is uniformly continuous on A.

3. Use a "bisection argument" (page 140) to give another proof of Theorem 1.

4. Derive Theorem 7-2 as a consequence of Theorem 1.

PART 3
DERIVATIVES AND INTEGRALS

In 1604, at the height of
his scientific career, Galileo argued
that for a rectilinear motion
in which speed increases proportionally
to distance covered,
the law of motion should be
just that ($x = ct^2$)
which he had discovered
in the investigation of falling bodies.
Between 1695 and 1700
not a single one of the monthly issues
of Leipzig's Acta Eruditorum was published
without articles of Leibniz,
the Bernoulli brothers
or the Marquis de l'Hôpital treating,
with notation only slightly different from
that which we use today,
the most varied problems of
differential calculus, integral calculus
and the calculus of variations.
Thus in the space of almost precisely
one century
infinitesimal calculus or,
as we now call it in English,
The Calculus,
the calculating tool par excellence,
had been forged;
and nearly three centuries of
constant use have not completely dulled
this incomparable instrument.

NICHOLAS BOURBAKI

CHAPTER 9 DERIVATIVES

The derivative of a function is the first of the two major concepts of this section. Together with the integral, it constitutes the source from which calculus derives its particular flavor. While it is true that the concept of a function is fundamental, that you cannot do anything without limits or continuity, and that least upper bounds are essential, everything we have done until now has been preparation—if adequate, this section will be easier than the preceding ones—for the really exciting ideas to come, the powerful concepts that are truly characteristic of calculus.

Perhaps (some would say "certainly") the interest of the ideas to be introduced in this section stems from the intimate connection between the mathematical concepts and certain physical ideas. Many definitions, and even some theorems, may be described in terms of physical problems, often in a revealing way. In fact, the demands of physics were the original inspiration for these fundamental ideas of calculus, and we shall frequently mention the physical interpretations. But we shall always first define the ideas in precise mathematical form, and discuss their significance in terms of mathematical problems.

The collection of all functions exhibits such diversity that there is almost no hope of discovering any interesting general properties pertaining to all. Because continuous functions form such a restricted class, we might expect to find some nontrivial theorems pertaining to them, and the sudden abundance of theorems after Chapter 6 shows that this expectation is justified. But the most interesting and most powerful results about functions will be obtained only when we restrict our attention even further, to functions which have even greater claim to be called "reasonable," which are even better behaved than most continuous functions.

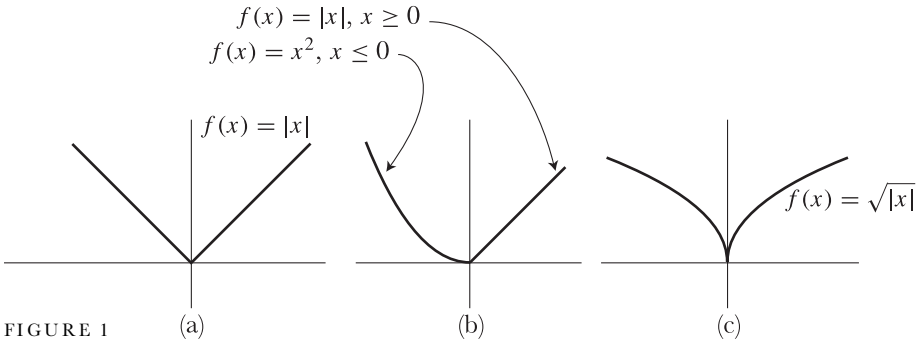


FIGURE 1

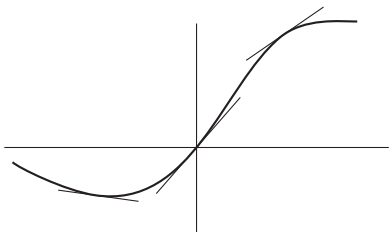


FIGURE 2

Figure 1 illustrates certain types of misbehavior which continuous functions can display. The graphs of these functions are "bent" at $(0, 0)$, unlike the graph of Figure 2, where it is possible to draw a "tangent line" at each point. The quotation marks have been used to avoid the suggestion that we have defined "bent" or

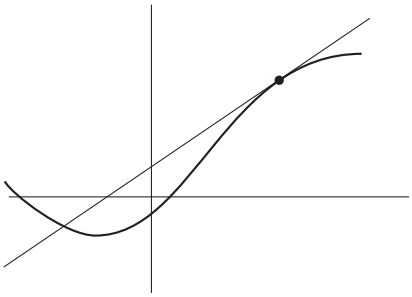


FIGURE 3

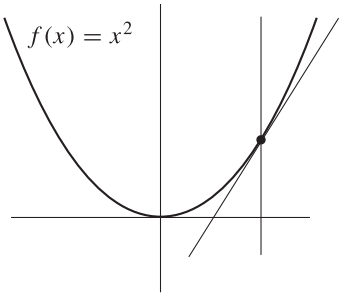


FIGURE 4

"tangent line," although we are suggesting that the graph might be "bent" at a point where a "tangent line" cannot be drawn. You have probably already noticed that a tangent line cannot be defined as a line which intersects the graph only once—such a definition would be both too restrictive and too permissive. With such a definition, the straight line shown in Figure 3 would not be a tangent line to the graph in that picture, while the parabola would have two tangent lines at each point (Figure 4), and the three functions in Figure 5 would have more than one tangent line at the points where they are "bent."

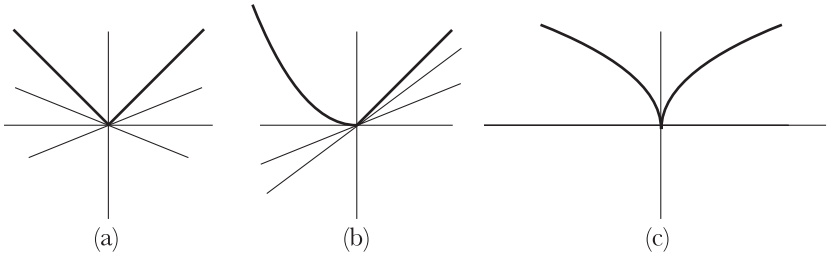


FIGURE 5

A more promising approach to the definition of a tangent line might start with "secant lines," and use the notion of limits. If $h \neq 0$, then the two distinct points $(a, f(a))$ and $(a + h, f(a + h))$ determine, as in Figure 6, a straight line whose slope is

$$\frac{f(a+h) - f(a)}{h}.$$

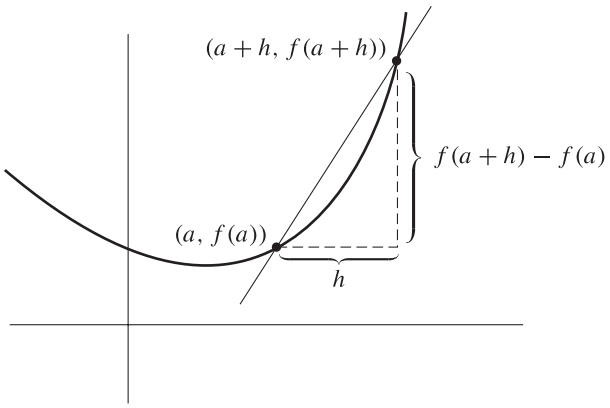


FIGURE 6

As Figure 7 illustrates, the "tangent line" at $(a, f(a))$ seems to be the limit, in some sense, of these "secant lines," as h approaches 0. We have never before talked about a "limit" of lines, but we *can* talk about the limit of their slopes: the

slope of the tangent line through $(a, f(a))$ should be

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}.$$

We are ready for a definition, and some comments.

DEFINITION

The function f is **differentiable at a** if

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \text{ exists.}$$

In this case the limit is denoted by $f'(a)$ and is called the **derivative of f at a.** (We also say that f is **differentiable** if f is differentiable at a for every a in the domain of f.)

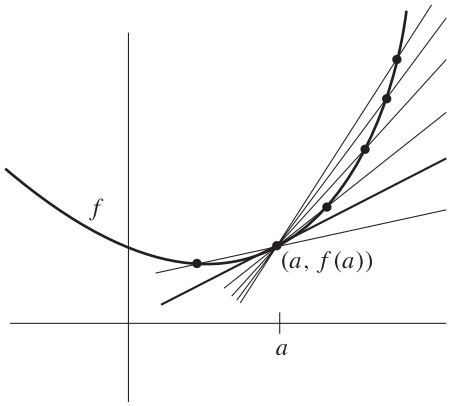


FIGURE 7

The first comment on our definition is really an addendum; we define the **tangent line** to the graph of f at $(a, f(a))$ to be the line through $(a, f(a))$ with slope $f'(a)$. This means that the tangent line at $(a, f(a))$ is defined only if f is differentiable at a.

The second comment refers to notation. The symbol $f'(a)$ is certainly reminiscent of functional notation. In fact, for any function f, we denote by f' the function whose domain is the set of all numbers a such that f is differentiable at a, and whose value at such a number a is

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}.$$

(To be very precise: f' is the collection of all pairs

$$\left(a, \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}\right)$$

for which $\lim_{h\to 0}[f(a+h)-f(a)]/h$ exists.) The function f' is called the **derivative** of f.

Our third comment, somewhat longer than the previous two, refers to the physical interpretation of the derivative. Consider a particle which is moving along a straight line (Figure 8(a)) on which we have chosen an "origin" point O, and a direction in which distances from O shall be written as positive numbers, the distance from O of points in the other direction being written as negative numbers. Let $s(t)$ denote the distance of the particle from O, at time t. The suggestive notation $s(t)$ has been chosen purposely; since a distance $s(t)$ is determined for each

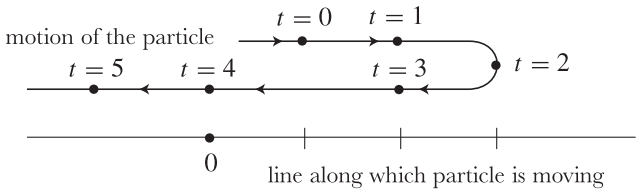


FIGURE 8(a)

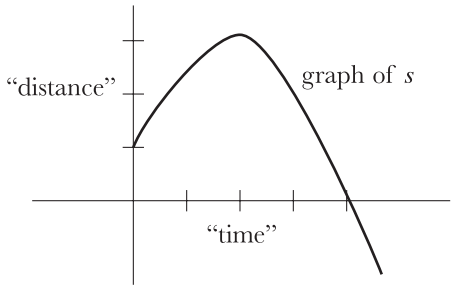


FIGURE 8(b)

number t, the physical situation automatically supplies us with a certain function s. The graph of s indicates the distance of the particle from O, on the vertical axis, in terms of the time, indicated on the horizontal axis (Figure 8(b)).

The quotient

$$\frac{s(a+h)-s(a)}{h}$$

has a natural physical interpretation. It is the "average velocity" of the particle during the time interval from a to $a+h$. For any particular a, this average speed depends on h, of course. On the other hand, the limit

$$\lim_{h\to 0}\frac{s(a+h)-s(a)}{h}$$

depends only on a (as well as the particular function s) and there are important physical reasons for considering this limit. We would like to speak of the "velocity of the particle at time a," but the usual definition of velocity is really a definition of average velocity; the only reasonable definition of "velocity at time a" (so-called "instantaneous velocity") is the limit

$$\lim_{h\to 0}\frac{s(a+h)-s(a)}{h}$$

Thus we *define* the (**instantaneous**) **velocity** of the particle at a to be $s'(a)$. Notice that $s'(a)$ could easily be negative; the absolute value $|s'(a)|$ is sometimes called the (**instantaneous**) **speed**.

It is important to realize that instantaneous velocity is a theoretical concept, an abstraction which does not correspond precisely to any observable quantity. While it would not be fair to say that instantaneous velocity has nothing to do with average velocity, remember that $s'(t)$ is not

$$\frac{s(t+h)-s(t)}{h}$$

for any particular h, but merely the limit of these average velocities as h approaches 0. Thus, when velocities are measured in physics, what a physicist really measures is an average velocity over some (very small) time interval; such a procedure cannot be expected to give an exact answer, but this is really no defect, because physical measurements can never be exact anyway.

The velocity of a particle is often called the "rate of change of its position." This notion of the derivative, as a rate of change, applies to any other physical situation in which some quantity varies with time. For example, the "rate of change of mass" of a growing object means the derivative of the function m, where $m(t)$ is the mass at time t.

In order to become familiar with the basic definitions of this chapter, we will spend quite some time examining the derivatives of particular functions. Before proving the important theoretical results of Chapter 11, we want to have a good idea of what the derivative of a function looks like. The next chapter is devoted exclusively to one aspect of this problem—calculating the derivative of complicated functions. In this chapter we will emphasize the concepts, rather than the

calculations, by considering a few simple examples. Simplest of all is a constant function, $f(x) = c$. In this case

$$\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h\to 0} \frac{c-c}{h} = 0.$$

Thus f is differentiable at a for every number a, and $f'(a) = 0$. This means that the tangent line to the graph of f always has slope 0, so the tangent line always coincides with the graph.

Constant functions are not the only ones whose graphs coincide with their tangent lines—this happens for any linear function $f(x) = cx + d$. Indeed

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h\to 0} \frac{c(a+h) + d - [ca+d]}{h} \\ &= \lim_{h\to 0} \frac{ch}{h} = c; \end{aligned}$$

the slope of the tangent line is c, the same as the slope of the graph of f.

A refreshing difference occurs for $f(x) = x^2$. Here

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h\to 0} \frac{(a+h)^2 - a^2}{h} \\ &= \lim_{h\to 0} \frac{a^2 + 2ah + h^2 - a^2}{h} \\ &= \lim_{h\to 0} 2a + h \\ &= 2a. \end{aligned}$$

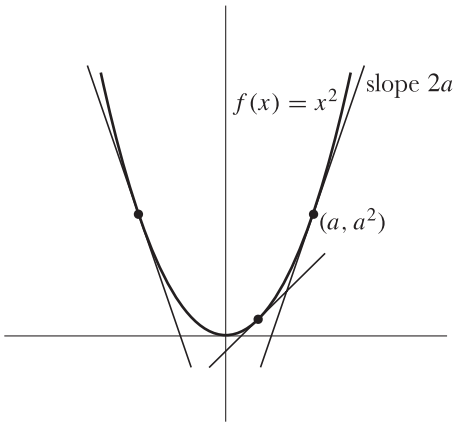


FIGURE 9

Some of the tangent lines to the graph of f are shown in Figure 9. In this picture each tangent line appears to intersect the graph only once, and this fact can be checked fairly easily: Since the tangent line through (a, a^2) has slope $2a$, it is the graph of the function

$$\begin{aligned} g(x) &= 2a(x-a) + a^2 \\ &= 2ax - a^2. \end{aligned}$$

Now, if the graphs of f and g intersect at a point $(x, f(x)) = (x, g(x))$, then

$$\begin{aligned} & x^2 = 2ax - a^2 \\ \text{or} \quad & x^2 - 2ax + a^2 = 0; \\ \text{so} \quad & (x-a)^2 = 0 \\ \text{or} \quad & x = a. \end{aligned}$$

In other words, (a, a^2) is the only point of intersection.

The function $f(x) = x^2$ happens to be quite special in this regard; usually a tangent line will intersect the graph more than once. Consider, for example, the function $f(x) = x^3$. In this case

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h\to 0} \frac{(a+h)^3 - a^3}{h} \\ &= \lim_{h\to 0} \frac{a^3 + 3a^2h + 3ah^2 + h^3 - a^3}{h} \\ &= \lim_{h\to 0} \frac{3a^2h + 3ah^2 + h^3}{h} \\ &= \lim_{h\to 0} 3a^2 + 3ah + h^2 \\ &= 3a^2. \end{aligned}$$

Thus the tangent line to the graph of f at (a, a^3) has slope $3a^2$. This means that the tangent line is the graph of

$$\begin{aligned} g(x) &= 3a^2(x-a) + a^3 \\ &= 3a^2x - 2a^3. \end{aligned}$$

The graphs of f and g intersect at the point $(x, f(x)) = (x, g(x))$ when

$$\begin{aligned} x^3 &= 3a^2x - 2a^3 \\ \text{or} \quad x^3 - 3a^2x + 2a^3 &= 0. \end{aligned}$$

This equation is easily solved if we remember that one solution of the equation has got to be $x = a$, so that $(x - a)$ is a factor of the left side; the other factor can then be found by dividing. We obtain

$$(x-a)(x^2 + ax - 2a^2) = 0.$$

It so happens that $x^2 + ax - 2a^2$ also has $x - a$ as a factor; we obtain finally

$$(x-a)(x-a)(x+2a) = 0.$$

Thus, as illustrated in Figure 10, the tangent line through (a, a^3) also intersects the graph at the point $(-2a, -8a^3)$. These two points are always distinct, except when $a = 0$.

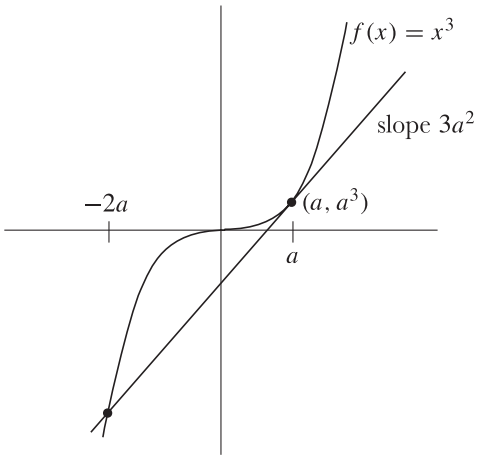


FIGURE 10

We have already found the derivative of sufficiently many functions to illustrate the classical, and still very popular, notation for derivatives. For a given function f, the derivative f' is often denoted by

$$\frac{df(x)}{dx}.$$

For example, the symbol

$$\frac{dx^2}{dx}$$

denotes the derivative of the function $f(x) = x^2$. Needless to say, the separate parts of the expression

$$\frac{df(x)}{dx}$$

are not supposed to have any sort of independent existence—the d's are *not* numbers, they *cannot* be canceled, and the entire expression is *not* the quotient of two other numbers "$df(x)$" and "dx." This notation is due to Leibniz (generally considered an independent co-discoverer of calculus, along with Newton), and is affectionately referred to as Leibnizian notation.* Although the notation $df(x)/dx$ seems very complicated, in concrete cases it may be shorter; after all, the symbol dx^2/dx is actually more concise than the phrase "the derivative of the function $f(x) = x^2$."

The following formulas state in standard Leibnizian notation all the information that we have found so far:

$$\begin{aligned} \frac{dc}{dx} &= 0, \\ \frac{d(ax+b)}{dx} &= a, \\ \frac{dx^2}{dx} &= 2x, \\ \frac{dx^3}{dx} &= 3x^2. \end{aligned}$$

Although the meaning of these formulas is clear enough, attempts at literal interpretation are hindered by the reasonable stricture that an equation should not contain a function on one side and a number on the other. For example, if the third equation is to be true, then either $df(x)/dx$ must denote $f'(x)$, rather than f', or else $2x$ must denote, not a number, but the function whose value at x is $2x$. It is really impossible to assert that one or the other of these alternatives is intended; in practice $df(x)/dx$ sometimes means f' and sometimes means $f'(x)$, while $2x$ may denote either a number or a function. Because of this ambiguity, most authors are reluctant to denote $f'(a)$ by

$$\frac{df(x)}{dx}(a);$$

instead $f'(a)$ is usually denoted by the barbaric, but unambiguous, symbol

$$\left.\frac{df(x)}{dx}\right|_{x=a}.$$

*Leibniz was led to this symbol by his intuitive notion of the derivative, which he considered to be, not the limit of quotients $[f(x+h)-f(x)]/h$, but the "value" of this quotient when h is an "infinitely small" number. This "infinitely small" quantity was denoted by dx and the corresponding "infinitely small" difference $f(x+dx)-f(x)$ by $df(x)$. Although this point of view is impossible to reconcile with properties (P1)–(P13) of the real numbers, some people find this notion of the derivative congenial.

In addition to these difficulties, Leibnizian notation is associated with one more ambiguity. Although the notation dx^2/dx is absolutely standard, the notation $df(x)/dx$ is often replaced by df/dx. This, of course, is in conformity with the practice of confusing a function with its value at x. So strong is this tendency that functions are often indicated by a phrase like the following: "consider the function $y = x^2$." We will sometimes follow classical practice to the extent of using y as the name of a function, but we will nevertheless carefully distinguish between the function and its values—thus we will always say something like "consider the function (defined by) $y(x) = x^2$."

Despite the many ambiguities of Leibnizian notation, it is used almost exclusively in older mathematical writing, and is still used very frequently today. The staunchest opponents of Leibnizian notation admit that it will be around for quite some time, while its most ardent admirers would say that it will be around forever, and a good thing too! In any case, Leibnizian notation cannot be ignored completely.

The policy adopted in this book is to disallow Leibnizian notation within the text, but to include it in the Problems; several chapters contain a few (immediately recognizable) problems which are expressly designed to illustrate the vagaries of Leibnizian notation. Trusting that these problems will provide ample practice in this notation, we return to our basic task of examining some simple examples of derivatives.

The few functions examined so far have all been differentiable. To fully appreciate the significance of the derivative it is equally important to know some examples of functions which are *not* differentiable. The obvious candidates are the three functions first discussed in this chapter, and illustrated in Figure 1; if they turn out to be differentiable at 0 something has clearly gone wrong.

Consider first $f(x) = |x|$. In this case

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

Now $|h|/h = 1$ for $h > 0$, and $|h|/h = -1$ for $h < 0$. This shows that

$$\lim_{h \to 0} \frac{f(h) - f(0)}{h} \text{ does not exist.}$$

In fact,

$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = 1$$

$$\text{and} \quad \lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = -1.$$

(These two limits are sometimes called the **right-hand derivative** and the **left-hand derivative**, respectively, of f at 0.)

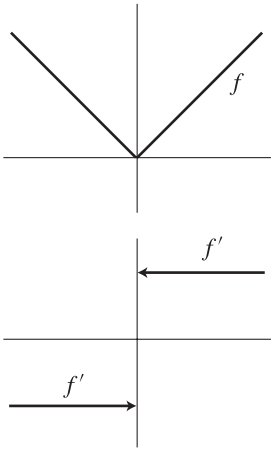


FIGURE 11

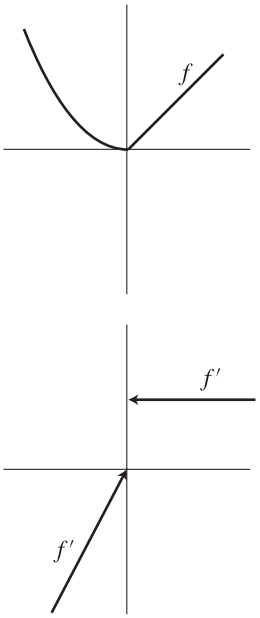


FIGURE 12

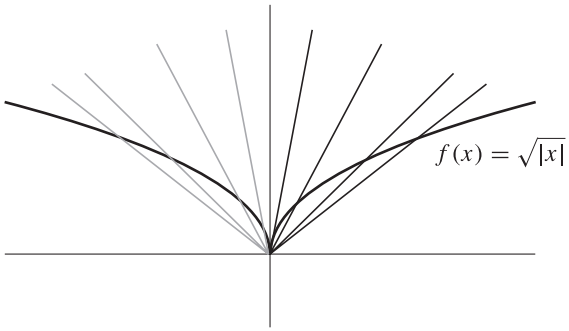


FIGURE 13

If $a \neq 0$, then $f'(a)$ does exist. In fact,

$$\begin{aligned} f'(x) &= 1 \qquad \text{if } x > 0, \\ f'(x) &= -1 \qquad \text{if } x < 0. \end{aligned}$$

The proof of this fact is left to you (it is easy if you remember the derivative of a linear function). The graphs of f and of f' are shown in Figure 11.

For the function

$$f(x) = \begin{cases} x^2, & x \leq 0 \\ x, & x \geq 0, \end{cases}$$

a similar difficulty arises in connection with $f'(0)$. We have

$$\frac{f(h) - f(0)}{h} = \begin{cases} \dfrac{h^2}{h} = h, & h < 0 \\ \dfrac{h}{h} = 1, & h > 0. \end{cases}$$

Therefore,

$$\begin{aligned} \lim_{h \to 0^-} \frac{f(h) - f(0)}{h} &= 0, \\ \text{but} \quad \lim_{h \to 0^+} \frac{f(h) - f(0)}{h} &= 1. \end{aligned}$$

Thus $f'(0)$ does not exist; f is not differentiable at 0. Once again, however, $f'(x)$ exists for $x \neq 0$—it is easy to see that

$$f'(x) = \begin{cases} 2x, & x < 0 \\ 1, & x > 0. \end{cases}$$

The graphs of f and f' are shown in Figure 12.

Even worse things happen for $f(x) = \sqrt{|x|}$. For this function

$$\frac{f(h) - f(0)}{h} = \begin{cases} \dfrac{\sqrt{h}}{h} = \dfrac{1}{\sqrt{h}}, & h > 0 \\ \dfrac{\sqrt{-h}}{h} = -\dfrac{1}{\sqrt{-h}}, & h < 0. \end{cases}$$

In this case the right-hand limit

$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^+} \frac{1}{\sqrt{h}}$$

does not exist; instead $1/\sqrt{h}$ becomes arbitrarily large as h approaches 0. And, what's more, $-1/\sqrt{-h}$ becomes arbitrarily large in absolute value, but *negative* (Figure 13).

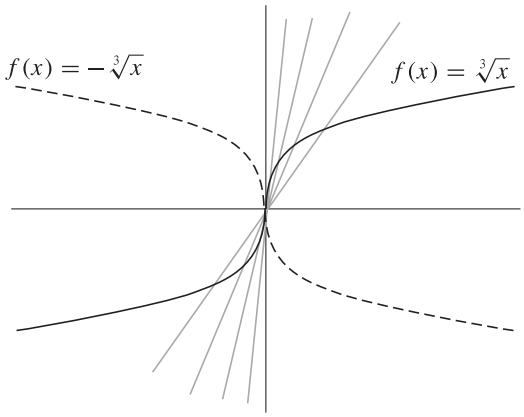


FIGURE 14

The function $f(x) = \sqrt[3]{x}$, although not differentiable at 0, is at least a little better behaved than this. The quotient

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{h^{1/3}}{h} = \frac{1}{h^{2/3}} = \frac{1}{\left(\sqrt[3]{h}\right)^2}$$

simply becomes arbitrarily large as h goes to 0. Sometimes one says that f has an "infinite" derivative at 0. Geometrically this means that the graph of f has a "tangent line" which is parallel to the vertical axis (Figure 14). Of course, $f(x) = -\sqrt[3]{x}$ has the same geometric property, but one would say that f has a derivative of "negative infinity" at 0.

Remember that differentiability is supposed to be an improvement over mere continuity. This idea is supported by the many examples of functions which are continuous, but not differentiable; however, one important point remains to be noted:

THEOREM 1 If f is differentiable at a, then f is continuous at a.

PROOF

$$\begin{aligned}\lim_{h\to 0} f(a+h) - f(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \cdot h \\ &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h\to 0} h \\ &= f'(a) \cdot 0 \\ &= 0.\end{aligned}$$

As we pointed out in Chapter 5, the equation $\lim_{h\to 0} f(a+h) - f(a) = 0$ is equivalent to $\lim_{x\to a} f(x) = f(a)$; thus f is continuous at a. ∎

It is very important to remember Theorem 1, and just as important to remember that the converse is not true. A differentiable function is continuous, but a continuous function need not be differentiable (keep in mind the function $f(x) = |x|$, and you will never forget which statement is true and which false).

The continuous functions examined so far have been differentiable at all points with at most one exception, but it is easy to give examples of continuous functions which are not differentiable at several points, even an infinite number (Figure 15). Actually, one can do much worse than this. There is a function which is *continuous*

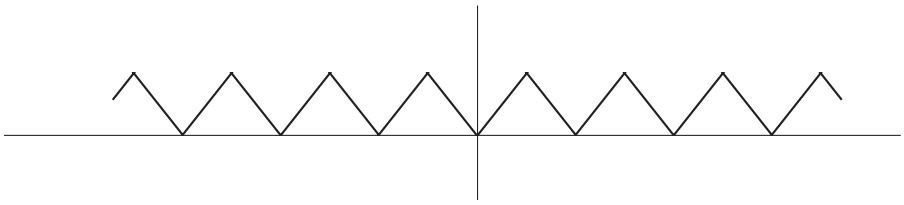


FIGURE 15

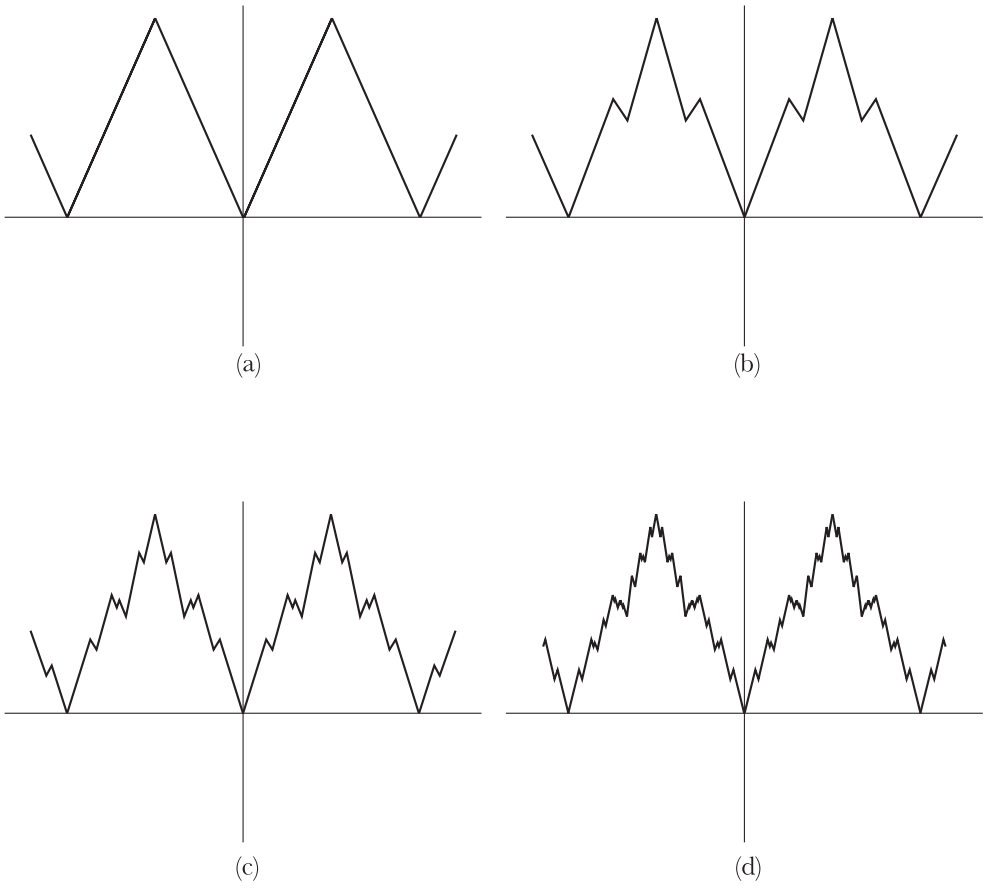


FIGURE 16

everywhere and *differentiable nowhere!* Unfortunately, the definition of this function will be inaccessible to us until Chapter 24, and I have been unable to persuade the artist to draw it (consider carefully what the graph should look like and you will sympathize with her point of view). It is possible to draw some rough approximations to the graph, however; several successively better approximations are shown in Figure 16.

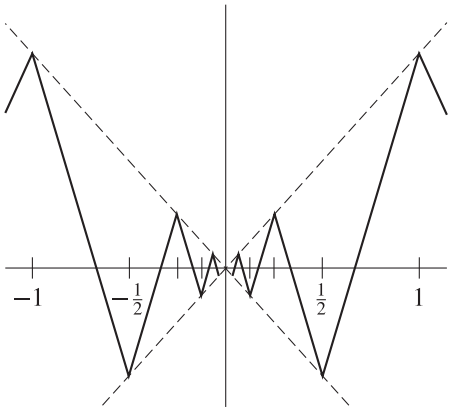


FIGURE 17

Although such spectacular examples of nondifferentiability must be postponed, we can, with a little ingenuity, find a continuous function which is not differentiable at infinitely many points, *all of which are in* [0, 1]. One such function is illustrated in Figure 17. The reader is given the problem of defining it precisely; it is a straight line version of the function

$$f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

This particular function f is itself quite sensitive to the question of differentiability. Indeed, for $h \neq 0$ we have

$$\frac{f(h) - f(0)}{h} = \frac{h \sin \frac{1}{h} - 0}{h} = \sin \frac{1}{h}.$$

Long ago we proved that $\lim_{h \to 0} \sin 1/h$ does not exist, so f is not differentiable at 0. Geometrically, one can see that a tangent line cannot exist, by noting that the secant line through $(0, 0)$ and $(h, f(h))$ in Figure 18 can have any slope between -1 and 1, no matter how small we require h to be.

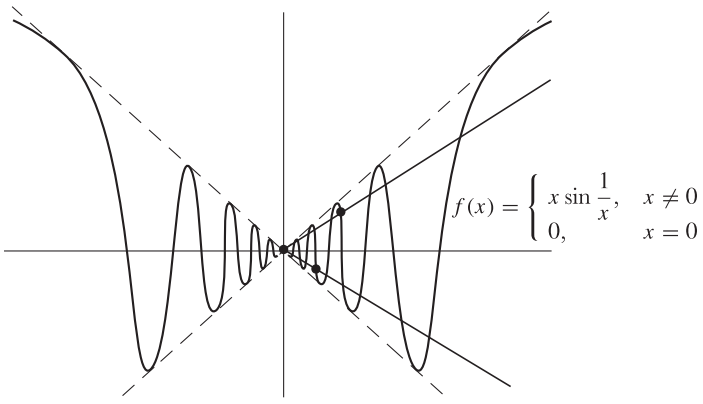


FIGURE 18

This finding represents something of a triumph; although continuous, the function f seems somehow quite unreasonable, and we can now enunciate one mathematically undesirable feature of this function—it is not differentiable at 0. Nevertheless, one should not become too enthusiastic about the criterion of differentiability. For example, the function

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is differentiable at 0; in fact $g'(0) = 0$:

$$\begin{aligned} \lim_{h \to 0} \frac{g(h) - g(0)}{h} &= \lim_{h \to 0} \frac{h^2 \sin \frac{1}{h}}{h} \\ &= \lim_{h \to 0} h \sin \frac{1}{h} \\ &= 0. \end{aligned}$$

The tangent line to the graph of g at $(0, 0)$ is therefore the horizontal axis (Figure 19).

This example suggests that we should seek even more restrictive conditions on a function than mere differentiability. We can actually use the derivative to formulate such conditions if we introduce another set of definitions, the last of this chapter.

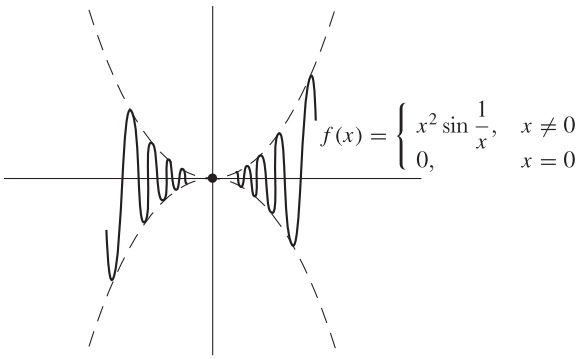


FIGURE 19

For any function f, we obtain, by taking the derivative, a new function f' (whose domain may be considerably smaller than that of f). The notion of differentiability can be applied to the function f', of course, yielding another function $(f')'$, whose domain consists of all points a such that f' is differentiable at a. The function $(f')'$ is usually written simply f'' and is called the **second derivative** of f. If $f''(a)$ exists, then f is said to be 2-times differentiable at a, and the number $f''(a)$ is called the **second derivative of f at a.**

In physics the second derivative is particularly important. If $s(t)$ is the position at time t of a particle moving along a straight line, then $s''(t)$ is called the **acceleration** at time t. Acceleration plays a special role in physics, because, as stated in Newton's laws of motion, the force on a particle is the product of its mass and its acceleration. Consequently you can feel the second derivative when you sit in an accelerating car.

There is no reason to stop at the second derivative—we can define $f''' = (f'')'$, $f'''' = (f''')'$, etc. This notation rapidly becomes unwieldy, so the following abbreviation is usually adopted (it is really a recursive definition):

$$\begin{aligned} f^{(1)} &= f', \\ f^{(k+1)} &= (f^{(k)})'. \end{aligned}$$

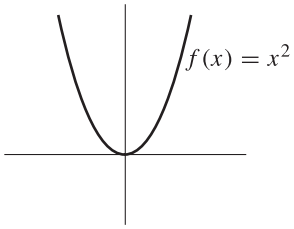
Thus

$$\begin{aligned} f^{(1)} &= f' \\ f^{(2)} &= f'' = (f')', \\ f^{(3)} &= f''' = (f'')', \\ f^{(4)} &= f'''' = (f''')', \\ &\text{etc.} \end{aligned}$$

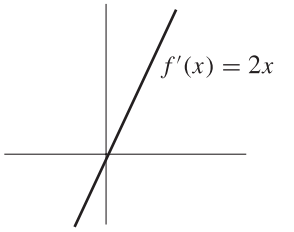
The various functions $f^{(k)}$, for $k \geq 2$, are sometimes called **higher-order derivatives** of f.

Usually, we resort to the notation $f^{(k)}$ only for $k \geq 4$, but it is convenient to have $f^{(k)}$ defined for smaller k also. In fact, a reasonable definition can be made for $f^{(0)}$, namely,

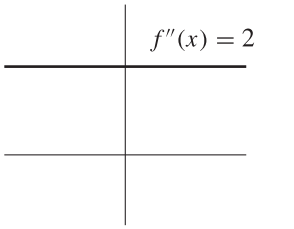
$$f^{(0)} = f.$$



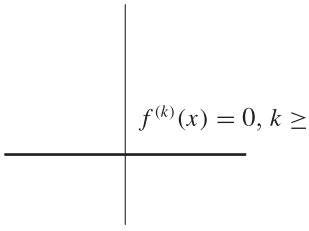
(a)



(b)



(c)



(d)

FIGURE 20

Leibnizian notation for higher-order derivatives should also be mentioned. The natural Leibnizian symbol for $f''(x)$, namely,

$$\frac{d\left(\dfrac{df(x)}{dx}\right)}{dx},$$

is abbreviated to

$$\frac{d^2 f(x)}{(dx)^2}, \qquad \text{or more frequently to} \qquad \frac{d^2 f(x)}{dx^2}$$

Similar notation is used for $f^{(k)}(x)$.

The following example illustrates the notation $f^{(k)}$, and also shows, in one very simple case, how various higher-order derivatives are related to the original function. Let $f(x) = x^2$. Then, as we have already checked,

$$\begin{aligned} f'(x) &= 2x, \\ f''(x) &= 2, \\ f'''(x) &= 0, \\ f^{(k)}(x) &= 0, \qquad \text{if } k \geq 3. \end{aligned}$$

Figure 20 shows the function f, together with its various derivatives.

A rather more illuminating example is presented by the following function, whose graph is shown in Figure 21(a):

$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x \leq 0. \end{cases}$$

It is easy to see that

$$\begin{aligned} f'(a) &= 2a \quad \text{if } a > 0, \\ f'(a) &= -2a \quad \text{if } a < 0. \end{aligned}$$

Moreover,

$$\begin{aligned} f'(0) &= \lim_{h \to 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \to 0} \frac{f(h)}{h}. \end{aligned}$$

Now

$$\lim_{h \to 0^+} \frac{f(h)}{h} = \lim_{h \to 0^+} \frac{h^2}{h} = 0$$

$$\text{and} \quad \lim_{h \to 0^-} \frac{f(h)}{h} = \lim_{h \to 0^-} \frac{-h^2}{h} = 0,$$

so

$$f'(0) = \lim_{h \to 0} \frac{f(h)}{h} = 0.$$

This information can all be summarized as follows:

$$f'(x) = 2|x|.$$

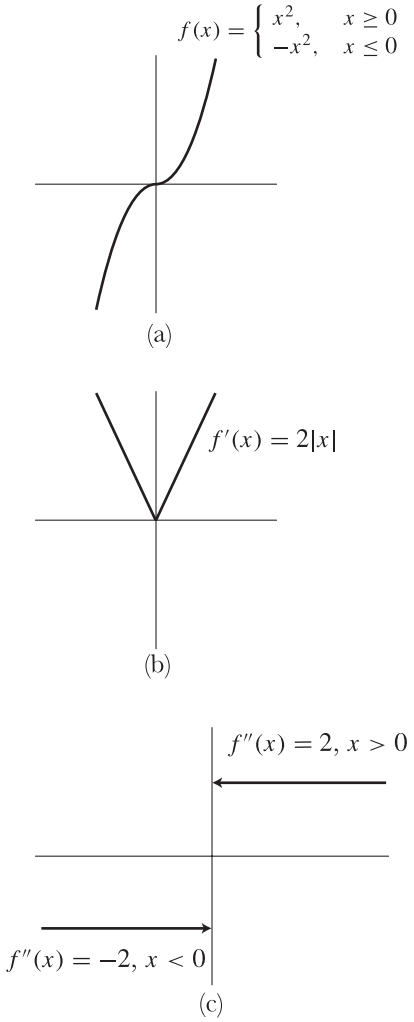


FIGURE 21

It follows that $f''(0)$ does not exist! Existence of the second derivative is thus a rather strong criterion for a function to satisfy. Even a "smooth looking" function like f reveals some irregularity when examined with the second derivative. This suggests that the irregular behavior of the function

$$g(x) = \begin{cases} x^2 \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

might also be revealed by the second derivative. At the moment we know that $g'(0) = 0$, but we do not know $g'(a)$ for any $a \neq 0$, so it is hopeless to begin computing $g''(0)$. We will return to this question at the end of the next chapter, after we have perfected the technique of finding derivatives.

PROBLEMS

1. (a) Prove, working directly from the definition, that if $f(x) = 1/x$, then $f'(a) = -1/a^2$, for $a \neq 0$.
(b) Prove that the tangent line to the graph of f at $(a, 1/a)$ does not intersect the graph of f, except at $(a, 1/a)$.

2. (a) Prove that if $f(x) = 1/x^2$, then $f'(a) = -2/a^3$ for $a \neq 0$.
(b) Prove that the tangent line to f at $(a, 1/a^2)$ intersects f at one other point, which lies on the opposite side of the vertical axis.

3. Prove that if $f(x) = \sqrt{x}$, then $f'(a) = 1/(2\sqrt{a})$, for $a > 0$. (The expression you obtain for $[f(a + h) - f(a)]/h$ will require some algebraic face lifting, but the answer should suggest the right trick.)

4. For each natural number n, let $S_n(x) = x^n$. Remembering that $S_1'(x) = 1$, $S_2'(x) = 2x$, and $S_3'(x) = 3x^2$, conjecture a formula for $S_n'(x)$. Prove your conjecture. (The expression $(x + h)^n$ may be expanded by the binomial theorem.)

5. Find f' if $f(x) = [x]$.

6. Prove, starting from the definition (and drawing a picture to illustrate):

(a) if $g(x) = f(x) + c$, then $g'(x) = f'(x)$;
(b) if $g(x) = cf(x)$, then $g'(x) = cf'(x)$.

7. Suppose that $f(x) = x^3$.

(a) What is $f'(9)$, $f'(25)$, $f'(36)$?
(b) What is $f'(3^2)$, $f'(5^2)$, $f'(6^2)$?
(c) What is $f'(a^2)$, $f'(x^2)$?

If you do not find this problem silly, you are missing a very important point: $f'(x^2)$ means the derivative of f at the number which we happen to be calling x^2; it is *not* the derivative at x of the function $g(x) = f(x^2)$. Just to drive the point home:

(d) For $f(x) = x^3$, compare $f'(x^2)$ and $g'(x)$ where $g(x) = f(x^2)$.

8. (a) Suppose $g(x) = f(x+c)$. Prove (starting from the definition) that $g'(x) = f'(x+c)$. Draw a picture to illustrate this. To do this problem you must write out the definitions of $g'(x)$ and $f'(x+c)$ correctly. The purpose of Problem 7 was to convince you that although this problem is easy, it is not an utter triviality, and there is something to prove: you cannot simply put prime marks into the equation $g(x) = f(x+c)$. To emphasize this point:

(b) Prove that if $g(x) = f(cx)$, then $g'(x) = c \cdot f'(cx)$. Try to see pictorially why this should be true, also.

(c) Suppose that f is differentiable and periodic, with period a (i.e., $f(x+a) = f(x)$ for all x). Prove that f' is also periodic.

9. Find $f'(x)$ and also $f'(x+3)$ in the following cases. Be very methodical, or you will surely slip up somewhere. Consult the answers (after you do the problem, naturally).

(i) $f(x) = (x+3)^5$.
(ii) $f(x+3) = x^5$.
(iii) $f(x+3) = (x+5)^7$.

10. Find $f'(x)$ if $f(x) = g(t+x)$, and if $f(t) = g(t+x)$. The answers will *not* be the same.

11. (a) Prove that Galileo was wrong: if a body falls a distance $s(t)$ in t seconds, and s' is proportional to s, then s cannot be a function of the form $s(t) = ct^2$.

(b) Prove that the following facts are true about s if $s(t) = (a/2)t^2$ (the first fact will show why we switched from c to $a/2$):

(i) $s''(t) = a$ (the acceleration is constant).
(ii) $[s'(t)]^2 = 2as(t)$.

(c) If s is measured in feet, the value of a is 32. How many seconds do you have to get out of the way of a chandelier which falls from a 400-foot ceiling? If you don't make it, how fast will the chandelier be going when it hits you? Where was the chandelier when it was moving with half that speed?

12. Imagine a road on which the speed limit is specified at every single point. In other words, there is a certain function L such that the speed limit x miles from the beginning of the road is $L(x)$. Two cars, A and B, are driving along this road; car A's position at time t is $a(t)$, and car B's is $b(t)$.

(a) What equation expresses the fact that car A always travels at the speed limit? (The answer is *not* $a'(t) = L(t)$.)

(b) Suppose that A always goes at the speed limit, and that B's position at time t is A's position at time $t-1$. Show that B is also going at the speed limit at all times.

(c) Suppose B always stays a constant distance behind A. Under what conditions will B still always travel at the speed limit?

13. Suppose that $f(a) = g(a)$ and that the left-hand derivative of f at a equals the right-hand derivative of g at a. Define $h(x) = f(x)$ for $x \leq a$, and $h(x) = g(x)$ for $x \geq a$. Prove that h is differentiable at a.

14. Let $f(x) = x^2$ if x is rational, and $f(x) = 0$ if x is irrational. Prove that f is differentiable at 0. (Don't be scared by this function. Just write out the definition of $f'(0)$.)

***15.** (a) Let f be a function such at $|f(x)| \leq x^2$ for all x. Prove that f is differentiable at 0. (If you have done Problem 14 you should be able to do this.)

(b) This result can be generalized if x^2 is replaced by $|g(x)|$, where g has what property?

16. Let $\alpha > 1$. If f satisfies $|f(x)| \leq |x|^\alpha$, prove that f is differentiable at 0.

17. Let $0 < \beta < 1$. Prove that if f satisfies $|f(x)| \geq |x|^\beta$ and $f(0) = 0$, then f is not differentiable at 0.

***18.** Let $f(x) = 0$ for irrational x, and $1/q$ for $x = p/q$ in lowest terms. Prove that f is not differentiable at a for any a. Hint: It obviously suffices to prove this for irrational a. Why? If $a = m.a_1a_2a_3\ldots$ is the decimal expansion of a, consider $[f(a+h) - f(a)]/h$ for h rational, and also for

$$h = -0.00\ldots0a_{n+1}a_{n+2}\ldots.$$

19. (a) Suppose that $f(a) = g(a) = h(a)$, that $f(x) \leq g(x) \leq h(x)$ for all x, and that $f'(a) = h'(a)$. Prove that g is differentiable at a, and that $f'(a) = g'(a) = h'(a)$. (Begin with the definition of $g'(a)$.)

(b) Show that the conclusion does not follow if we omit the hypothesis $f(a) = g(a) = h(a)$.

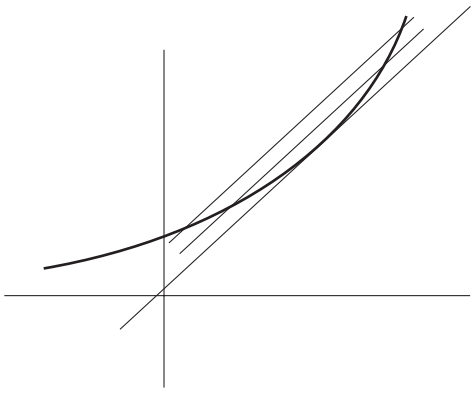


FIGURE 22

20. Let f be any polynomial function; we will see in the next chapter that f is differentiable. The tangent line to f at $(a, f(a))$ is the graph of $g(x) = f'(a)(x-a) + f(a)$. Thus $f(x) - g(x)$ is the polynomial function $d(x) = f(x) - f'(a)(x-a) - f(a)$. We have already seen that if $f(x) = x^2$, then $d(x) = (x-a)^2$, and if $f(x) = x^3$, then $d(x) = (x-a)^2(x+2a)$.

(a) Find $d(x)$ when $f(x) = x^4$, and show that it is divisible by $(x-a)^2$.

(b) There certainly seems to be some evidence that $d(x)$ is always divisible by $(x-a)^2$. Figure 22 provides an intuitive argument: usually, lines parallel to the tangent line will intersect the graph at two points; the tangent line intersects the graph only once near the point, so the intersection should be a "double intersection." To give a rigorous proof, first note that

$$\frac{d(x)}{x-a} = \frac{f(x) - f(a)}{x-a} - f'(a).$$

Now answer the following questions. Why is $f(x) - f(a)$ divisible by $(x-a)$? Why is there a polynomial function h such that $h(x) =$

$d(x)/(x-a)$ for $x \neq a$? Why is $\lim_{x\to a} h(x) = 0$? Why is $h(a) = 0$? Why does this solve the problem?

21. (a) Show that $f'(a) = \lim_{x\to a}[f(x) - f(a)]/(x-a)$. (Nothing deep here.)

(b) Show that derivatives are a "local property": if $f(x) = g(x)$ for all x in some open interval containing a, then $f'(a) = g'(a)$. (This means that in computing $f'(a)$, you can ignore $f(x)$ for any particular $x \neq a$. Of course you can't ignore $f(x)$ for all such x at once!)

***22.** (a) Suppose that f is differentiable at x. Prove that

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x-h)}{2h}.$$

Hint: Remember an old algebraic trick—a number is not changed if the same quantity is added to and then subtracted from it.

****(b)** Prove, more generally, that

$$f'(x) = \lim_{h,k\to 0^+} \frac{f(x+h) - f(x-k)}{h+k}.$$

***23.** Prove that if f is even, then $f'(x) = -f'(-x)$. (In order to minimize confusion, let $g(x) = f(-x)$; find $g'(x)$ and *then* remember what other thing g is.) Draw a picture!

***24.** Prove that if f is odd, then $f'(x) = f'(-x)$. Once again, draw a picture.

25. Problems 23 and 24 say that f' is even if f is odd, and odd if f is even. What can therefore be said about $f^{(k)}$?

26. Find $f''(x)$ if

(i) $f(x) = x^3$.
(ii) $f(x) = x^5$.
(iii) $f'(x) = x^4$.
(iv) $f(x+3) = x^5$.

27. If $S_n(x) = x^n$, and $0 \leq k \leq n$, prove that

$$S_n{}^{(k)}(x) = \frac{n!}{(n-k)!}x^{n-k}$$
$$= k!\binom{n}{k}x^{n-k}.$$

***28.** (a) Find $f'(x)$ if $f(x) = |x|^3$. Find $f''(x)$. Does $f'''(x)$ exist for all x?

(b) Analyze f similarly if $f(x) = x^4$ for $x \geq 0$ and $f(x) = -x^4$ for $x \leq 0$.

***29.** Let $f(x) = x^n$ for $x \geq 0$ and let $f(x) = 0$ for $x \leq 0$. Prove that $f^{(n-1)}$ exists (and find a formula for it), but that $f^{(n)}(0)$ does not exist.

30. Interpret the following specimens of Leibnizian notation; each is a restatement of some fact occurring in a previous problem.

(i) $\dfrac{dx^n}{dx} = nx^{n-1}$

(ii) $\dfrac{dz}{dy} = -\dfrac{1}{y^2}$ if $z = \dfrac{1}{y}$.

(iii) $\dfrac{d[f(x)+c]}{dx} = \dfrac{df(x)}{dx}$.

(iv) $\dfrac{d[cf(x)]}{dx} = c\dfrac{df(x)}{dx}$.

(v) $\dfrac{dz}{dx} = \dfrac{dy}{dx}$ if $z = y + c$.

(vi) $\left.\dfrac{dx^3}{dx}\right|_{x=a^2} = 3a^4$.

(vii) $\left.\dfrac{df(x+a)}{dx}\right|_{x=b} = \left.\dfrac{df(x)}{dx}\right|_{x=b+a}$.

(viii) $\left.\dfrac{df(cx)}{dx}\right|_{x=b} = c \cdot \left.\dfrac{df(x)}{dx}\right|_{x=cb}$.

(ix) $\dfrac{df(cx)}{dx} = c \cdot \left.\dfrac{df(y)}{dy}\right|_{y=cx}$.

(x) $\dfrac{d^k x^n}{dx^k} = k!\binom{n}{k}x^{n-k}$.

CHAPTER 10 DIFFERENTIATION

The process of finding the derivative of a function is called *differentiation*. From the previous chapter you may have the impression that this process is usually laborious, requires recourse to the definition of the derivative, and depends upon successfully recognizing some limit. It is true that such a procedure is often the only possible approach—if you forget the definition of the derivative you are likely to be lost. Nevertheless, in this chapter we will learn to differentiate a large number of functions, without the necessity of even recalling the definition. A few theorems will provide a mechanical process for differentiating a large class of functions, which are formed from a few simple functions by the process of addition, multiplication, division, and composition. This description should suggest what theorems will be proved. We will first find the derivative of a few simple functions, and then prove theorems about the sum, products, quotients, and compositions of differentiable functions. The first theorem is merely a formal recognition of a computation carried out in the previous chapter.

THEOREM 1 If f is a constant function, $f(x) = c$, then

$$f'(a) = 0 \quad \text{for all numbers } a.$$

PROOF

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{c - c}{h} = 0. \blacksquare$$

The second theorem is also a special case of a computation in the last chapter.

THEOREM 2 If f is the identity function, $f(x) = x$, then

$$f'(a) = 1 \quad \text{for all numbers } a.$$

PROOF

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \to 0} \frac{a + h - a}{h} \\ &= \lim_{h \to 0} \frac{h}{h} = 1. \blacksquare \end{aligned}$$

The derivative of the sum of two functions is just what one would hope—the sum of the derivatives.

THEOREM 3 If f and g are differentiable at a, then $f+g$ is also differentiable at a, and

$$(f+g)'(a) = f'(a) + g'(a).$$

PROOF

$$\begin{aligned}(f+g)'(a) &= \lim_{h\to 0} \frac{(f+g)(a+h) - (f+g)(a)}{h} \\ &= \lim_{h\to 0} \frac{f(a+h) + g(a+h) - [f(a) + g(a)]}{h} \\ &= \lim_{h\to 0} \left[\frac{f(a+h) - f(a)}{h} + \frac{g(a+h) - g(a)}{h} \right] \\ &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} + \lim_{h\to 0} \frac{g(a+h) - g(a)}{h} \\ &= f'(a) + g'(a). \blacksquare\end{aligned}$$

The formula for the derivative of a product is not as simple as one might wish, but it is nevertheless pleasantly symmetric, and the proof requires only a simple algebraic trick, which we have found useful before—a number is not changed if the same quantity is added to and subtracted from it.

THEOREM 4 If f and g are differentiable at a, then

$$(f \cdot g)'(a) = f'(a) \cdot g(a) + f(a) \cdot g'(a).$$

PROOF

$$\begin{aligned}(f \cdot g)'(a) &= \lim_{h\to 0} \frac{(f \cdot g)(a+h) - (f \cdot g)(a)}{h} \\ &= \lim_{h\to 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\ &= \lim_{h\to 0} \left[\frac{f(a+h)[g(a+h) - g(a)]}{h} + \frac{[f(a+h) - f(a)]g(a)}{h} \right] \\ &= \lim_{h\to 0} f(a+h) \cdot \lim_{h\to 0} \frac{g(a+h) - g(a)}{h} + \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h\to 0} g(a) \\ &= f(a) \cdot g'(a) + f'(a) \cdot g(a).\end{aligned}$$

(Notice that we have used Theorem 9-1 to conclude that $\lim_{h\to 0} f(a+h) = f(a)$.) $\blacksquare$

In one special case Theorem 4 simplifies considerably:

THEOREM 5 If $g(x) = cf(x)$ and f is differentiable at a, then g is differentiable at a, and

$$g'(a) = c \cdot f'(a).$$

PROOF If $h(x) = c$, so that $g = h \cdot f$, then by Theorem 4,

$$\begin{aligned}g'(a) &= (h \cdot f)'(a) \\ &= h(a) \cdot f'(a) + h'(a) \cdot f(a) \\ &= c \cdot f'(a) + 0 \cdot f(a) \\ &= c \cdot f'(a). \blacksquare\end{aligned}$$

Notice, in particular, that $(-f)'(a) = -f'(a)$, and consequently $(f - g)'(a) = (f + [-g])'(a) = f'(a) - g'(a)$.

To demonstrate what we have already achieved, we will compute the derivative of some more special functions.

THEOREM 6 If $f(x) = x^n$ for some natural number n, then

$$f'(a) = na^{n-1} \qquad \text{for all } a.$$

PROOF The proof will be by induction on n. For $n = 1$ this is simply Theorem 2. Now assume that the theorem is true for n, so that if $f(x) = x^n$, then

$$f'(a) = na^{n-1} \qquad \text{for all } a.$$

Let $g(x) = x^{n+1}$. If $I(x) = x$, the equation $x^{n+1} = x^n \cdot x$ can be written

$$g(x) = f(x) \cdot I(x) \qquad \text{for all } x;$$

thus $g = f \cdot I$. It follows from Theorem 4 that

$$\begin{aligned} g'(a) = (f \cdot I)'(a) &= f'(a) \cdot I(a) + f(a) \cdot I'(a) \\ &= na^{n-1} \cdot a + a^n \cdot 1 \\ &= na^n + a^n \\ &= (n+1)a^n, \qquad \text{for all } a. \end{aligned}$$

This is precisely the case $n + 1$ which we wished to prove. ∎

Putting together the theorems proved so far we can now find f' for f of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0.$$

We obtain

$$f'(x) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + 2a_2 x + a_1.$$

We can also find f'':

$$f''(x) = n(n-1)a_n x^{n-2} + (n-1)(n-2)a_{n-1}x^{n-3} + \cdots + 2a_2.$$

This process can be continued easily. Each differentiation reduces the highest power of x by 1, and eliminates one more a_i. It is a good idea to work out the derivatives f''', $f^{(4)}$, and perhaps $f^{(5)}$, until the pattern becomes quite clear. The last interesting derivative is

$$f^{(n)}(x) = n!a_n;$$

for $k > n$ we have

$$f^{(k)}(x) = 0.$$

Clearly, the next step in our program is to find the derivative of a quotient f/g. It is quite a bit simpler, and, because of Theorem 4, obviously sufficient to find the derivative of $1/g$.

THEOREM 7 If g is differentiable at a, and $g(a) \neq 0$, then $1/g$ is differentiable at a, and

$$\left(\frac{1}{g}\right)'(a) = \frac{-g'(a)}{[g(a)]^2}.$$

PROOF Before we even write

$$\frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h}$$

we must be sure that this expression makes sense—it is necessary to check that $(1/g)(a+h)$ is defined for sufficiently small h. This requires only two observations. Since g is, by hypothesis, differentiable at a, it follows from Theorem 9-1 that g is continuous at a. Since $g(a) \neq 0$, it follows from Theorem 6-3 that there is some $\delta > 0$ such that $g(a+h) \neq 0$ for $|h| < \delta$. Therefore $(1/g)(a+h)$ *does* make sense for small enough h, and we can write

$$\begin{aligned}
\lim_{h\to 0} \frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h} &= \lim_{h\to 0} \frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{h} \\
&= \lim_{h\to 0} \frac{g(a) - g(a+h)}{h[g(a)\cdot g(a+h)]} \\
&= \lim_{h\to 0} \frac{-[g(a+h) - g(a)]}{h} \cdot \frac{1}{g(a)g(a+h)} \\
&= \lim_{h\to 0} \frac{-[g(a+h) - g(a)]}{h} \cdot \lim_{h\to 0} \frac{1}{g(a)\cdot g(a+h)} \\
&= -g'(a) \cdot \frac{1}{[g(a)]^2}.
\end{aligned}$$

(Notice that we have used continuity of g at a once again.) ∎

The general formula for the derivative of a quotient is now easy to derive. Though not particularly appealing, it is important, and must simply be memorized (I always use the incantation: "bottom times derivative of top, minus top times derivative of bottom, over bottom squared.")

THEOREM 8 If f and g are differentiable at a and $g(a) \neq 0$, then f/g is differentiable at a, and

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a)\cdot f'(a) - f(a)\cdot g'(a)}{[g(a)]^2}.$$

PROOF Since $f/g = f \cdot (1/g)$ we have

$$\begin{aligned}\left(\frac{f}{g}\right)'(a) &= \left(f \cdot \frac{1}{g}\right)'(a) \\ &= f'(a) \cdot \left(\frac{1}{g}\right)(a) + f(a) \cdot \left(\frac{1}{g}\right)'(a) \\ &= \frac{f'(a)}{g(a)} + \frac{f(a)(-g'(a))}{[g(a)]^2} \\ &= \frac{f'(a) \cdot g(a) - f(a) \cdot g'(a)}{[g(a)]^2}. \blacksquare\end{aligned}$$

We can now differentiate a few more functions. For example,

$$\text{if } f(x) = \frac{x^2-1}{x^2+1}, \text{ then } f'(x) = \frac{(x^2+1)(2x) - (x^2-1)(2x)}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2};$$

$$\text{if } f(x) = \frac{x}{x^2+1}, \text{ then } f'(x) = \frac{(x^2+1) - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2};$$

$$\text{if } f(x) = \frac{1}{x}, \qquad \text{then } f'(x) = -\frac{1}{x^2} = (-1)x^{-2}.$$

Notice that the last example can be generalized: if

$$f(x) = x^{-n} = \frac{1}{x^n}, \qquad \text{for some natural number } n,$$

then

$$f'(x) = \frac{-nx^{n-1}}{x^{2n}} = (-n)x^{-n-1};$$

thus Theorem 6 actually holds both for positive and negative integers. If we interpret $f(x) = x^0$ to mean $f(x) = 1$, and $f'(x) = 0 \cdot x^{-1}$ to mean $f'(x) = 0$, then Theorem 6 is true for $n = 0$ also. (The word "interpret" is necessary because it is not clear how 0^0 should be defined and, in any case, $0 \cdot 0^{-1}$ is meaningless.)

Further progress in differentiation requires the knowledge of the derivatives of certain special functions to be studied later. One of these is the sine function. For the moment we shall divulge, and use, the following information, without proof:

$$\begin{aligned}\sin'(a) &= \cos a \qquad \text{for all } a, \\ \cos'(a) &= -\sin a \qquad \text{for all } a,\end{aligned}$$

This information allows us to differentiate many other functions. For example, if

$$f(x) = x \sin x,$$

then

$$\begin{aligned}f'(x) &= x \cos x + \sin x, \\ f''(x) &= -x \sin x + \cos x + \cos x \\ &= -x \sin x + 2 \cos x;\end{aligned}$$

if

$$g(x) = \sin^2 x = \sin x \cdot \sin x,$$

then

$$\begin{aligned} g'(x) &= \sin x \cos x + \cos x \sin x \\ &= 2 \sin x \cos x, \\ g''(x) &= 2[(\sin x)(-\sin x) + \cos x \cos x] \\ &= 2[\cos^2 x - \sin^2 x]; \end{aligned}$$

if

$$h(x) = \cos^2 x = \cos x \cdot x,$$

then

$$\begin{aligned} h'(x) &= (\cos x)(-\sin x) + (-\sin x)\cos x \\ &= -2 \sin x \cos x, \\ h''(x) &= -2[\cos^2 x - \sin^2 x]. \end{aligned}$$

Notice that

$$g'(x) + h'(x) = 0,$$

hardly surprising, since $(g + h)(x) = \sin^2 x + \cos^2 x = 1$. As we would expect, we also have $g''(x) + h''(x) = 0$.

The examples above involved only products of two functions. A function involving triple products can be handled by Theorem 4 also; in fact it can be handled in two ways. Remember that $f \cdot g \cdot h$ is an abbreviation for

$$(f \cdot g) \cdot h \quad \text{or} \quad f \cdot (g \cdot h).$$

Choosing the first of these, for example, we have

$$\begin{aligned} (f \cdot g \cdot h)'(x) &= (f \cdot g)'(x) \cdot h(x) + (f \cdot g)(x)h'(x) \\ &= [f'(x)g(x) + f(x)g'(x)]h(x) + f(x)g(x)h'(x) \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x). \end{aligned}$$

The choice of $f \cdot (g \cdot h)$ would, of course, have given the same result, with a different intermediate step. The final answer is completely symmetric and easily remembered:

> $(f \cdot g \cdot h)'$ is the sum of the three terms obtained by differentiating each of f, g, and h and multiplying by the other two.

For example, if

$$f(x) = x^3 \sin x \cos x,$$

then

$$f'(x) = 3x^2 \sin x \cos x + x^3 \cos x \cos x + x^3(\sin x)(-\sin x).$$

Products of more than 3 functions can be handled similarly. For example, you should have little difficulty deriving the formula

$$\begin{aligned} (f \cdot g \cdot h \cdot k)'(x) = {} & f'(x)g(x)h(x)k(x) + f(x)g'(x)h(x)k(x) \\ & + f(x)g(x)h'(x)k(x) + f(x)g(x)h(x)k'(x). \end{aligned}$$

You might even try to prove (by induction) the general formula:

$$(f_1 \cdot \cdots \cdot f_n)'(x) = \sum_{i=1}^{n} f_1(x) \cdot \cdots \cdot f_{i-1}(x) f_i'(x) f_{i+1}(x) \cdot \cdots \cdot f_n(x).$$

Differentiating the most interesting functions obviously requires a formula for $(f \circ g)'(x)$ in terms of f' and g'. To ensure that $f \circ g$ be differentiable at a, one reasonable hypothesis would seem to be that g be differentiable at a. Since the behavior of $f \circ g$ near a depends on the behavior of f near $g(a)$ (not near a), it also seems reasonable to assume that f is differentiable at $g(a)$. Indeed we shall prove that if g is differentiable at a and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a, and

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

This extremely important formula is called the *Chain Rule*, presumable because a composition of functions might be called a "chain" of functions. Notice that $(f \circ g)'$ is practically the product of f' and g', but not quite: f' must be evaluated at $g(a)$ and g' at a. Before attempting to prove this theorem we will try a few applications. Suppose

$$f(x) = \sin x^2.$$

Let us, temporarily, use S to denote the ("squaring") function $S(x) = x^2$. Then

$$f = \sin \circ S.$$

Therefore we have

$$\begin{aligned} f'(x) &= \sin'(S(x)) \cdot S'(x) \\ &= \cos x^2 \cdot 2x. \end{aligned}$$

Quite a different result is obtained if

$$f(x) = \sin^2 x.$$

In this case

$$f = S \circ \sin,$$

so

$$\begin{aligned} f'(x) &= S'(\sin x) \cdot \sin'(x) \\ &= 2 \sin x \cdot \cos x. \end{aligned}$$

Notice that this agrees (as it should) with the result obtained by writing $f = \sin \cdot \sin$ and using the product formula.

Although we have invented a special symbol, S, to name the "squaring" function, it does not take much practice to do problems like this without bothering to write down special symbols for functions, and without even bothering to write down the particular composition which f is—one soon becomes accustomed to taking f apart in one's head. The following differentiations may be used as practice for such mental gymnastics—if you find it necessary to work a few out on paper, by all means do so, but try to develop the knack of writing f' immediately after seeing

the definition of f; problems of this sort are so simple that, if you just remember the Chain Rule, there is no thought necessary.

if $f(x) = \sin x^3$	then $f'(x) = \cos x^3 \cdot 3x^2$
$f(x) = \sin^3 x$	$f'(x) = 3\sin^2 x \cdot \cos x$
$f(x) = \sin \dfrac{1}{x}$	$f'(x) = \cos \dfrac{1}{x} \cdot \left(\dfrac{-1}{x^2}\right)$
$f(x) = \sin(\sin x)$	$f'(x) = \cos(\sin x) \cdot \cos x$
$f(x) = \sin(x^3 + 3x^2)$	$f'(x) = \cos(x^3 + 3x^2) \cdot (3x^2 + 6x)$
$f(x) = (x^3 + 3x^2)^{53}$	$f'(x) = 53(x^3 + 3x^2)^{52} \cdot (3x^2 + 6x)$.

A function like

$$f(x) = \sin^2 x^2 = [\sin x^2]^2,$$

which is the composition of three functions,

$$f = S \circ \sin \circ S,$$

can also be differentiated by the Chain Rule. It is only necessary to remember that a triple composition $f \circ g \circ h$ means $(f \circ g) \circ h$ or $f \circ (g \circ h)$. Thus if

$$f(x) = \sin^2 x^2$$

we can write

$$f = (S \circ \sin) \circ S,$$
$$f = S \circ (\sin \circ S).$$

The derivative of either expression can be found by applying the Chain Rule twice; the only doubtful point is whether the two expressions lead to equally simple calculations. As a matter of fact, as any experienced differentiator knows, it is much better to use the second:

$$\boxed{f = S \circ (\sin \circ S).}$$

We can now write down $f'(x)$ in one fell swoop. To begin with, note that the first function to be differentiated is S, so the formula for $f'(x)$ begins

$$f'(x) = 2(\quad) \cdot \square.$$

Inside the parentheses we must put $\sin x^2$, the value at x of the second function, $\sin \circ S$. Thus we begin by writing

$$f'(x) = 2\sin x^2 \cdot \square$$

(the parentheses weren't really necessary, after all). We must now multiply this much of the answer by the derivative of $\sin \circ S$ at x; this part is easy—it involves a composition of two functions, which we already know how to handle. We obtain, for the final answer,

$$f'(x) = 2\sin x^2 \cdot \cos x^2 \cdot 2x.$$

The following example is handled similarly. Suppose

$$\boxed{f(x) = \sin(\sin x^2).}$$

Without even bothering to write down f as a composition $g \circ h \circ k$ of three functions, we can see that the left-most one will be sin, so our expression for $f'(x)$ begins

$$f'(x) = \cos(\quad) \cdot \square.$$

Inside the parentheses we must put the value of $h \circ k(x)$; this is simply $\sin x^2$ (what you get from $\sin(\sin x^2)$ by deleting the first sin). So our expression for $f'(x)$ begins

$$f'(x) = \cos(\sin x^2) \cdot \square.$$

We can now forget about the first sin in $\sin(\sin x^2)$; we have to multiply what we have so far by the derivative of the function whose value at x is $\sin x^2$—which is again a problem we already know how to solve:

$$f'(x) = \cos(\sin x^2) \cdot \cos x^2 \cdot 2x.$$

Finally, here are the derivatives of some other functions which are the composition of sin and S, as well as some other triple compositions. You can probably just "see" that the answers are correct—if not, try writing out f as a composition:

if $f(x) = \sin((\sin x)^2)$	then $f'(x) = \cos((\sin x)^2) \cdot 2 \sin x \cdot \cos x$
$f(x) = [\sin(\sin x)]^2$	$f'(x) = 2 \sin(\sin x) \cdot \cos(\sin x) \cdot \cos x$
$f(x) = \sin(\sin(\sin x))$	$f'(x) = \cos(\sin(\sin x)) \cdot \cos(\sin x) \cdot \cos x$
$f(x) = \sin^2(x \sin x)$	$f'(x) = 2 \sin(x \sin x) \cdot \cos(x \sin x) \cdot [\sin x + x \cos x]$
$f(x) = \sin(\sin(x^2 \sin x))$	$f'(x) = \cos(\sin(x^2 \sin x)) \cdot \cos(x^2 \sin x) \cdot [2x \sin x + x^2 \cos x].$

The rule for treating compositions of four (or even more) functions is easy—always (mentally) put in parentheses starting from the right,

$$f \circ (g \circ (h \circ k)),$$

and start reducing the calculation to the derivative of a composition of a smaller number of functions:

$$f'(g(h(k(x)))) \cdot \square.$$

For example, if

$$f(x) = \sin^2(\sin^2(x)) \qquad [f = S \circ \sin \circ S \circ \sin = S \circ (\sin \circ (S \circ \sin))]$$

then

$$f'(x) = 2 \sin(\sin^2 x) \cdot \cos(\sin^2 x) \cdot 2 \sin x \cdot \cos x;$$

THEOREM If F is a complete ordered field, then F is isomorphic to R.

PROOF Since two fields are defined to be isomorphic if there is an isomorphism between them, we must actually construct a function f from R to F which is an isomorphism. We begin by defining f on the integers as follows:

$$f(0) = 0,$$
$$f(n) = \underbrace{1 + \cdots + 1}_{n \text{ times}} \quad \text{for } n > 0,$$
$$f(n) = -\underbrace{(1 + \cdots + 1)}_{|n| \text{ times}} \quad \text{for } n < 0.$$

It is easy to check that

$$f(m + n) = f(m) + f(n),$$
$$f(m \cdot n) = f(m)\, f(n),$$

for all integers m and n, and it is convenient to denote $f(n)$ by n. We then define f on the rational numbers by

$$f(m/n) = m \div n = m\, n^{-1}$$

(notice that the n-fold sum $1 + \cdots + 1 = 0$ if $n > 0$, since F is an ordered field). This definition makes sense because if $m/n = k/l$, then $ml = nk$, so $m\, n^{-1} = k\, l^{-1}$. It is easy to check that

$$f(r_1 + r_2) = f(r_1) + f(r_2),$$
$$f(r_1 \cdot r_2) = f(r_1)\, f(r_2),$$

for all rational numbers r_1 and r_2, and that $f(r_1) \le f(r_2)$ if $r_1 < r_2$.

The definition of $f(x)$ for arbitrary x is based on the now familiar idea that any real number is determined by the rational numbers less than it. For x any in R, let A_x be the subset of F consisting of all $f(r)$, for all rational numbers $r < x$. The set A_x is certainly not empty, and it is also bounded above, for if r_0 is a rational number with $r_0 > x$, then $f(r_0) \ge f(r)$ for all $f(r)$ in A_x. Since F is a complete ordered field, the set A_x has a least upper bound; we define $f(x)$ as $\sup A_x$.

We now have $f(x)$ defined in two different ways, first for rational x, and then for any x. Before proceeding further, it is necessary to show that these two definitions agree for rational x. In other words, if x is a rational number, we want to show that

$$\sup A_x = f(x),$$

where $f(x)$ here denotes $m \div n$, for $x = m/n$. This is not automatic, but depends on the completeness of F; a slight digression is thus required.

Since F is complete, the elements

$$\underbrace{1 + \cdots + 1}_{n \text{ times}} \quad \text{for natural numbers } n$$

form a set which is not bounded above; the proof is exactly the same as the proof for R (Theorem 8-2). The consequences of this fact for F have exact analogues in F: in particular, if a and b are elements of F with $a \leq b$, then there is a rational number r such that

$$a \leq f(r) \leq b.$$

Having made this observation, we return to the proof that the two definitions of $f(x)$ agree for rational x. If y is a rational number with $y < x$, then we have already seen that $f(y) \leq f(x)$. Thus every element of A_x is $\leq f(x)$. Consequently,

$$\sup A_x \leq f(x).$$

On the other hand, suppose that we had

$$\sup A_x \leq f(x).$$

Then there would be a rational number r such that

$$\sup A_x \leq f(r) \leq f(x).$$

But the condition $f(r) \leq f(x)$ means that $r < x$, which means that $f(r)$ is in the set A_x; this clearly contradicts the condition $\sup A_x \leq f(r)$. This shows that the original assumption is false, so

$$\sup A_x = f(x).$$

We thus have a certain well-defined function f from R to F. In order to show that f is an isomorphism we must verify conditions (1)–(4) of the definition. We will begin with (4).

If x and y are real numbers with $x < y$, then clearly A_x is contained in A_y. Thus

$$f(x) = \sup A_x \leq \sup A_y = f(y).$$

To rule out the possibility of equality, notice that there are rational numbers r and s with

$$x < r < s < y.$$

We know that $f(r) \leq f(s)$. It follows that

$$f(x) < f(r) \leq f(s) < f(y).$$

This proves (4).

Condition (1) follows immediately from (4): if $x = y$, then either $x < y$ or $y < x$; in the first case $f(x) \leq f(y)$, and in the second case $f(y) \leq f(x)$; in either case $f(x) = f(y)$.

To prove (2), let a be an element of F, and let B be the set of all rational numbers r with $f(r) \leq a$. The set B is not empty, and it is also bounded above, because there is a rational number s with $f(s) > a$, so that $f(s) > f(r)$ for r in B, which implies that $s > r$. Let x be the least upper bound of B; we claim that $f(x) = a$. In order to prove this it suffices to eliminate the alternatives

$$f(x) \leq a,$$
$$a \leq f(x).$$

In the first case there would be a rational number r with

$$f(x) \le f(r) \le a.$$

But this means that $x < r$ and that r is in B, which contradicts the fact that $x = \sup B$. In the second case there would be a rational number r with

$$a \le f(r) \le f(x).$$

This implies that $r < x$. Since $x = \sup B$, this means that $r < s$ for some s in B. Hence

$$f(r) \le f(s) \le a,$$

again a contradiction. Thus $f(x) = a$, proving (2).

To check (3), let x and y be real numbers and suppose that $f(x + y) \neq f(x) + f(y)$. Then either

$$f(x + y) < f(x) + f(y) \qquad \text{or} \qquad f(x) + f(y) < f(x + y).$$

In the first case there would be a rational number r such that

$$f(x + y) < f(r) < f(x) + f(y).$$

But this would mean that

$$x + y < r.$$

Therefore r could be written as the sum of two rational numbers

$$r = r_1 + r_2, \qquad \text{where } x < r_1 \text{ and } y < r_2.$$

Then, using the facts checked above for rational numbers, it would follow that

$$f(r) = f(r_1 + r_2) = f(r_1) + f(r_2) < f(x) + f(y),$$

a contradiction. The other case is handled similarly.

Finally, if x and y are positive real numbers, the same sort of reasoning shows that

$$f(x \cdot y) = f(x) \cdot f(y);$$

the general case is then a simple consequence.

This theorem brings to an end our investigation of the real numbers, and resolves any doubts about them: There is a complete ordered field and, up to isomorphism, only one complete ordered field. It is an important part of a mathematical education to follow a construction of the real numbers in detail, but it is not necessary to refer ever again to this particular construction. It is utterly irrelevant that a real number happens to be a collection of rational numbers, and such a fact should never enter the proof of any important theorem about the real numbers. Reasonable proofs should use only the fact that the real numbers are a complete ordered field, because this property of the real numbers characterizes them up to isomorphism, and any significant mathematical property of the real numbers will be true for all isomorphic fields. To be candid I should admit that this last assertion is just a prejudice of the author, but it is one shared by almost all other mathematicians.

PROBLEMS

1. Let f be an isomorphism from F_1 to F_2.
   (a) Show that $f(0) = 0$ and $f(1) = 1$. (Here 0 and 1 on the left denote elements in F_1, while 0 and 1 on the right denote elements of F_2.)
   (b) Show that $f(-a) = -f(a)$ and $f(a^{-1}) = f(a)^{-1}$, for $a \neq 0$.

2. Here is an opportunity to convince yourself that any significant property of a field is shared by any field isomorphic to it. The point of this problem is to write out very formal proofs until you are certain that all statements of this sort are obvious. F_1 and F_2 will be two fields which are isomorphic; for simplicity we will denote the operations in both by $\cdot$ and $+$. Show that:
   (a) If the equation $x^2 + 1 = 0$ has a solution in F_1, then it has a solution in F_2.
   (b) If every polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ with $a_0, \ldots, a_{n-1}$ in F_1, has a root in F_1, then every polynomial equation $x^n + b_{n-1}x^{n-1} + \cdots + b_0 = 0$ with $b_0, \ldots, b_{n-1}$ in F_2 has a root in F_2.
   (c) If $1 + \cdots + 1$ (summed n times) $= 0$ in F_1, then the same is true in F_2.
   (d) If F_1 and F_2 are ordered fields (and the isomorphism f satisfies $f(x) \geq f(y)$ for $x \geq y$) and F_1 is complete, then F_2 is complete.

3. Let f be an isomorphism from F_1 to F_2 and g an isomorphism from F_2 to F_3. Define the function $g \circ f$ from F_1 to F_3 by $(g \circ f)(x) = g(f(x))$. Show that $g \circ f$ is an isomorphism.

4. Suppose that F is a complete ordered field, so that there is an isomorphism from R to F. Show that there is actually only one isomorphism from R to F. Hint: In case $F = R$, this is Problem 3-17. Now if f and g are two isomorphisms from R to F consider $g^{-1} \circ f$.

5. Find an isomorphism from C to C other than the identity function.

SUGGESTED READING

A man ought to read
just as inclination leads him;
for what he reads as a task
will do him little good.

SAMUEL JOHNSON

One purpose of this bibliography is to guide the reader to other sources, but the most important function it can serve is to indicate the variety of mathematical reading available. Consequently, there is an attempt to achieve diversity, but no pretense of being complete. The present plethora of mathematics books would make such an undertaking almost hopeless in any case, and since I have tried to encourage independent reading, the more standard a text, the less likely it is to appear here. In some cases, this philosophy may seem to have been carried to extremes, as some entries in the list cannot be read by a student just nishing a rst course of calculus until several years have elapsed. Nevertheless, there are many selections which can be read now, and I can't believe that it hurts to have some idea of what lies ahead.

Many of these books have gone through numerous editions and printings, which will be re ected in more recent publication dates. Many of the books with older publications dates are out of print, though that generally doesn't apply to books from the redoubtable Dover Publications, or from the Mathematical Association of America. Those that are no longer in print can still often be found in well-stocked academic libraries.

One of the most elementary unproved theorems mentioned in this book is the fact that every natural number can be written as a product of primes in only one way. A proof of this basic theorem will be found near the beginning of almost any book on elementary number theory. Few books have won so enthusiastic an audience as

[1] An Introduction to the Theory of Numbers (fth edition), by G. H. Hardy and E. M. Wright; Oxford University Press, 1980.

The Pergamon Press published a series, Popular Lectures in Mathematics, with several titles worth investigating, among them

[2] A Selection of Problems in the Theory of Numbers, by W. Sierpinski; Macmillan (Pergamon), 1964.

Finally, I will mention an intriguing little book, now out of print I fear,

[3] Three Pearls of Number Theory, by A. Khinchin; Graylock Press, 1952.

The subject of irrational numbers straddles the elds of number theory and analysis. An excellent introduction will be found in

[4] Irrational Numbers, by I. M. Niven; Mathematical Association of America, 1956.

Together with many historical notes, there are references to some fairly elementary articles in journals. There is also a proof that e is transcendental (see also [51]) and, nally, a proof of the "Gelfond-Schneider theorem": if a and b are algebraic, with $a = 0$ or 1, and b is irrational, then a^b is transcendental.

All the books listed so far begin with natural numbers, but whenever necessary take for granted the irrational numbers, not to mention the integers and rational

numbers. Several books present a construction of the rational numbers from the natural numbers, but one of the most lucid treatments is still to be found in

[5] Foundations of Analysis (second edition), by E. Landau; Chelsea, 1960.

Incidentally, the original German edition,

[6] Grundlagen der Analysis (fourth edition), by E. Landau; Chelsea, 1965.

has been printed in paper back, together with a complete German-English dictionary (of about 300 words) for the whole book—an excellent way to begin reading mathematical German. The basic idea for constructing the real numbers is derived from Dedekind, whose contributions can be found in

[7] Essays on the Theory of Numbers, by R. Dedekind; Dover, 1963.

While many mathematicians are content to accept the natural numbers as a natural starting point, numbers can be de ned in terms of sets, the most basic starting point of all. A charming exposition of set theory can be found in a sophisticated little book called

[8] Naive Set Theory, by P. R. Halmos; Springer-Verlag, 1991.

Another very good introduction is

[9] Theory of Sets, by E. Kamke; Dover, 1950.

Perhaps it is necessary to assure some victims of the \new math" that set theory does have some mathematical content (in fact, some very deep theorems). Using these deep results, Kamke proves that there is a discontinuous function f such that $f(x + y) = f(x) + f(y)$ for all x and y. For those who enjoy reading the classics, the most important notions of set theory were rst introduced by Cantor, whose work is reproduced in

[10] Contributions to the Founding of the Theory of Transfinite Numbers, by G. Cantor; Dover, 1952.

Inequalities, which were treated as an elementary topic in Chapters 1 and 2, actually form a specialized eld. A good elementary introduction is provided by

[11] Analytic Inequalities, by N. Kazarinoff; Mathematical Association of America, 1961.

Twelve different proofs that the geometric mean is less than or equal to the arithmetic mean, each based on a different principle, can be found in the beginning of the more advanced book

[12] An Introduction to Inequalities, by E. Beckenbach and R. Bellman; Mathematical Association of America, 1961.

The classic work on inequalities is

[13] Inequalities (second edition), by G. H. Hardy, J. E. Littlewood, and G. Polya; Cambridge University Press, 1988.

Each of the authors of this triple collaboration has provided his own contribution to the sparse literature about the nature of mathematical thinking, written from a mathematician's point of view. My favorite is

[14] A Mathematician's Apology, by G. H. Hardy; Cambridge University Press, 1992.

Littlewood's anecdotal selections are entitled

[15] A Mathematician's Miscellany, by J. E. Littlewood; Methuen, 1953.

Polya's contribution is pedagogy at the highest level:

[16] Mathematics and Plausible Reasoning (Vol. I: Induction and Analogy in Mathematics; Vol. II: Patterns of Plausible Inference), by G. Polya; Princeton University Press, 1990.

Geometry is the other main field which can be considered as background for calculus. Euclid's *Elements* is still a masterful mathematical work, but should perhaps be postponed until some preparation has been made, with a modern work on "classical geometry," like

[17] Elementary Geometry from an Advanced Standpoint (second edition), by E. Moise; Addison-Wesley, 1974.

This beautiful book provides excellent historical perspectives and contains a thorough discussion of the role of the "Archimedean axiom" in geometry; in addition, Chapter 28 describes an ordered field in which the Archimedean axiom does not hold. Speaking of beautiful geometry books, all sorts of fascinating things can be found in

[18] Introduction to Geometry (second edition), by H. S. Coxeter; Wiley, 1989.

Almost all treatments of geometry at least mention convexity, which forms another specialized topic. I cannot imagine a better introduction to convexity, or a better mathematical experience in general, than reading and working through

[19] Convex Figures, by I. M. Yaglom and W. G. Boltyanskii; Holt, Rinehart and Winston, 1961.

This book contains a carefully arranged sequence of definitions, statements of theorems, whose proofs are to be supplied by the reader (worked-out proofs are supplied in the back of the book). Another geometry book has been modeled on the same principle:

[20] Combinatorial Geometry in the Plane, by H. Hadwiger and H. Debrunner; Holt, Rinehart and Winston, 1964.

Along with these two out-of-the-ordinary books, I might mention an extremely valuable little book, also of a specialized sort,

[21] Counterexamples in Analysis, by B. Gelbaum and J. Olmsted; Holden-Day, 1964.

Many of the example in this book come from more advanced topics in analysis, but quite a few can be appreciated by someone who knows calculus.

Of calculus books I will mention only two, each something of a classic:

[22] A Course of Pure Mathematics (tenth edition), by G. H. Hardy; Cambridge University Press, 1952.

[23] Differential and Integral Calculus (two volumes), by R. Courant; Wiley (Interscience), 1988.

Courant is especially strong on applications to physics. Speaking of such applications, an elegant exposition of the material in Chapter 17, together with much further discussion, can be found in the article

[24] On the Geometry of the Kepler Problem, by John Milnor; in The American Mathematical Monthly, Volume 90 (1983), pp. 353–365.

(In this paper the curve c of Chapter 17 is denoted by v, and the derivative of the important composition $v \circ s^{-1}$ (page 331) is introduced quite off-handedly as $dv/d\theta$.) A "straight-forward" derivation of Kepler's laws, together with numerous references, can be found in another article in this same journal,

[25] The Mathematical Relationship Between Kepler's Laws and Newton's, by Andrew T. Hyman; in The American Mathematical Monthly, Volume 100 (1993), pp. 932–936.

The latter parts of Volume I of Courant contain material usually found in advanced calculus, including differential equations and Fourier series. An introduction to Fourier series (requiring a little advanced calculus) will also be found in

[26] An Introduction to Fourier Series and Integrals, by R. E. Seeley; W. A. Benjamin, 1966.

The second volume of Courant (advanced calculus in earnest) contains additional material on differential equations, as well as an introduction to the calculus of variations. A widely admired, though somewhat more advanced, book on differential equations is

[27] Lectures on Ordinary Differential Equations, by W. Hurewicz; Dover, 1990.

I will bypass the more or less standard advanced calculus books (which can easily be found by the reader) since nowadays there is a movement to revise the whole presentation of advanced calculus, basing it upon linear algebra. One of the first, and still one of the nicest, treatments of advanced calculus using linear algebra is

[28] Calculus of Vector Functions, by R. H. Crowell and R. E. Williamson; Prentice-Hall, 1962.

Several recent books on advanced calculus attempt to acquaint undergraduates with very large areas of modern mathematics. My favorite, of course, is

[29] Calculus on Manifolds, by M. Spivak; W. A. Benjamin, 1965.

There are three other topics which are somewhat out of place in this bibliography because they are rapidly becoming established as part of a standard undergraduate curriculum. The purposeful study of fields and related systems is the domain of "algebra." One of the favorite texts is

[30] Topics in Algebra (second edition), by I. N. Herstein; Wiley, 1975.

A more advanced book is the great classic:

[31] Algebra, by B. L. van der Waerden; Springer-Verlag, 1990.

By the way, this book contains a proof of the partial fraction decomposition of a rational function.

There are now several introductions to complex analysis, as well as many elementary books on topology. Although the latter subject has not been mentioned before, it has really been in the background of many discussions, since it is the natural generalization of the ideas about limits and continuity which play such a prominent role in Part II of this book.

The next few topics, ranging from elementary to very difficult, are included in this bibliography because they have been alluded to in the text. The proof that a nondecreasing function is differentiable at almost all points (and an explanation of just what this means) receives a beautiful exposition in

[32] Functional Analysis, by F. Riesz and B. Sz.-Nagy; Ungar, 1955.

(After this elementary beginning, the book moves on to quite advanced material.) The gamma function has an elegant little book devoted entirely to its properties, most of them proved by using the theorem of Bohr and Mollerup which was mentioned in Problem 19-39:

[33] The Gamma Function, by E. Artin; Holt, Rinehart and Winston, 1964.

The gamma function is only one of several important improper integrals in mathematics. In particular, the calculation of $\int_0^\infty e^{-x^2}\,dx$ (see Problem 19-41) is important in probability theory, where the "normal distribution function"

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}y^2}\,dy$$

plays a fundamental role. A classic book on probability theory is

[34] An Introduction to Probability Theory and Its Applications (third edition), by W. Feller; Wiley, 1968.

The impossibility of integrating certain functions in elementary terms (among them $f(x) = e^{-x^2}$) is one of the most esoteric subjects in mathematics. An interesting discussion of the possibilities of integrating in elementary terms, with an

outline of the impossibility proofs, and references to the original papers of Liouville, will be found in

[35] The Integration of Functions of a Single Variable (second edition), by G. H. Hardy; Cambridge University Press, 1958.

A complete presentation of the impossibility proofs will be found in

[36] Integration in Finite Terms, by J. Ritt; Columbia University Press, 1948.

Oddly enough, a related but seemingly more difficult problem has a much neater solution. There are simple differential equations ($y' + xy = 0$ is a specific example) whose solutions cannot be expressed even in terms of indefinite integrals of elementary functions. This fact is proved on page 43 of the (60-page) book:

[37] An Introduction to Differential Algebra, by I. Kaplansky; Hermann, 1957.

To read this book you will need to know quite a bit of algebra, however.

A few words should also be said in defense of the process of integrating in elementary terms, which many mathematicians look upon as an art (unlike differentiation, which is merely a skill). You are probably already aware that the process of integration can be expedited by tables of indefinite integrals. For those who enjoy pursuing tables there is a really beautiful collection, that includes indefinite integrals, definite improper integrals, and a great deal more besides (if you should ever happen to need the value of the thirty-fourth Bernoulli number, this is the place to look):

[38] Tables of Integrals, Series, and Products, by I. S. Gradschteyn et al.; Academic Press, 1980.

For the thrifty, there is a paperback table of integrals:

[39] Tables of Indefinite Integrals, by G. Petit Bois; Dover, 1961.

The remaining references are of a somewhat different sort. They fall into three categories, of which the first is historical. The letter of H. A. Schwarz referred to in Problem 11-65 will be found in

[40] Ways of Thought of Great Mathematicians, by H. Meschkowski; Holden-Day, 1964.

Some historical remarks, and an attempt to incorporate them into the teaching of calculus, will be found in

[41] The Calculus: A Genetic Approach, by O. Toeplitz; University of Chicago Press, 1981.

An admirable textbook on the history of mathematics is

[42] An Introduction to the History of Mathematics (sixth edition), by H. Eves; Saunders College Publishing, 1990.

Three good scholarly works are

[43] History of Analytic Geometry, by C. Boyer; Scholar's Bookshelf, 1988.
[44] A History of the Calculus, and Its Conceptual Development, by C. Boyer; Dover, 1959.
[45] The Mathematics of Great Amateurs, by J. L. Coolidge; Oxford University Press, 1990

and extracts from original sources will be found in

[46] A Source Book in Mathematics (2 vols.), by D. Smith; Dover, 1959.

Despite the impression that might be given by the large number of books listed here, it is often hard to find specific concrete information about the origins of calculus. For example, it is almost impossible to find out who first proved the Mean Value Theorem (according to the Encyklopädie der Mathematischen Wissenschaften, Volume II, it was O. Bonnet, whose name is familiar to students of differential geometry from the "Gauss-Bonnet Theorem"). Similarly, though many history books tell us that Wallis proved Wallis' formula by a "complicated method of interpolation," most never bother to mention what it was, even though it inspired Euler's investigations of the gamma function (a description is given in the answer book, along with the solution to Problem 19-40).

The second category in this final group of books might be described as "popularizations." There are a surprisingly large number of first-rate ones by real mathematicians:

[47] What is Mathematics? (fourth edition), by R. Courant and H. Robbins; Oxford University Press, 1979.
[48] Geometry and the Imagination, by D. Hilbert and S. Cohn-Vossen; Chelsea, 1952.
[49] The Enjoyment of Mathematics, by H. Rademacher and O. Toeplitz; Dover, 1990.
[50] Famous Problems of Mathematics (second edition), by H. Tietze; Graylock Press, 1965.

One of the most renowned "popularizations" is especially concerned with the teaching of mathematics:

[51] Elementary Mathematics from an Advanced Standpoint (vol. 1: Arithmetic, Algebra, Analysis; vol. 2: Geometry), by F. Klein; Dover, 1948.

Volume 1 contains a proof of the transcendence of e, which, although not so elementary as the one in [4], is a direct analogue of the proof that π is transcendental, replacing integrals with complex line integrals. It can be read as soon as the basic facts about complex analysis are known.

The third category is the very opposite extreme—original papers. The difficulties encountered here are formidable, and I have only had the courage to list one such paper, the source of the quotation for Part IV. It is not even in English,

although you do have a choice of foreign languages. The article in the original French is in

[52] Oeuvres Completes dÕAbel, Christiania. Johnson Reprint Corporation, New York, 1965.

It rst appeared in a German translation in the Journal fr¬ die reine und angewandte Mathematik, Volume 1, 1826. To compound the dif culties, these references will usually be available only in university libraries. Yet the study of this paper will probably be as valuable as any other reading mentioned here. The reason is suggested by a remark of Abel himself, who attributed his profound knowledge of mathematics to the fact that he read the masters, rather than the pupils.

ANSWERS TO SELECTED PROBLEMS

CHAPTER 1

1. (i) $1 = a^{-1}a = a^{-1}(ax) = (a^{-1}a)x = 1 \cdot x = x$.
   (iii) If $x^2 = y^2$, then $0 = x^2 - y^2 = (x - y)(x + y)$, so either $x - y = 0$ or $x + y = 0$, that is, either $x = y$ or $x = -y$.
   (vi) Replace y by $-y$ in (iv).
2. One step requires dividing by $x - y = 0$.
3. (i) $a/b = ab^{-1} = (ac)(b^{-1}c^{-1}) = (ac)(bc)^{-1}$ (by (iii)) $= ac/bc$.
   (ii) $(ad + bc)/(bd) = (ad + bc)(bd)^{-1} = (ad + bc)(b^{-1}d^{-1})$ (by (iii)) $= ab^{-1} + cd^{-1} = a/b + c/d$.
   (iii) $ab(a^{-1}b^{-1}) = (a \cdot a^{-1})(b \cdot b^{-1}) = 1$, so $a^{-1} \cdot b^{-1} = (ab)^{-1}$.
   (v) $(a/b)/(c/d) = (a/b)(c/d)^{-1} = (a \cdot b^{-1})(c \cdot d^{-1})^{-1} = (a \cdot b^{-1})(c^{-1} \cdot d) = ad(b^{-1} \cdot c^{-1}) = ad(bc)^{-1} = (ad)/(bc)$.
4. (i) $x < -1$.
   (iii) $x > 7$ or $x < -7$.
   (v) All x, since $x^2 - 2x + 2 = (x - 1)^2 + 1$.
   (vii) $x > 3$ or $x < -2$, since 3 and -2 are the roots of $x^2 - x - 6 = 0$.
   (ix) $x > 0$ or $-5 < x < 3$.
   (xi) $x < 3$.
   (xiii) $x > 1$ or $0 < x < 1$.
5. (i) $b - a$ and $d - c$ are in P, so $(b - a) + (d - c) = (b + d) - (a + c)$ is in P. Thus, $b + d > a + c$.
   (iii) Using (ii), $-c < -d$; then (i) implies that $a + (-c) < b + (-d)$.
   (v) $(b - a)$ and $-c$ are in P, so $-c(b - a) = ac - bc$ is in P, that is, $ac > bc$.
   (vii) Using (iv), $a > 0$ and $a < 1$, so $a^2 < a$.
   (ix) Substitute a for c and b for d in (viii).
9. (i) $|2| + |-3| \geq |-5| + |7|$.
   (iii) $|a + b| + |c| \geq |a + b + c|$.
   (v) $|2| + |3| + |5| \geq |-7|$.
10. (i) a if $a \geq b$ and $b \geq 0$;
    $-a$ if $a \leq b$ and $b \geq 0$;
    $a + 2b$ if $a \geq -b$ and $b \leq 0$;
    $-a - 2b$ if $a \leq -b$ and $b \leq 0$.
    (iii) $x - x^2$ if $x \geq 0$;
    $-x - x^2$ if $x \leq 0$.
11. (i) $x = 11, -5$.
    (iii) $-6 < x < -2$.
    (v) No x (the distance from x to 1 plus the distance from x to -1 is at least 2).
    (vii) $x = 1, -1$.
12. (i) $(|xy|)^2 = (xy)^2 = x^2y^2 = |x|^2|y|^2 = (|x| \cdot |y|)^2$; since $|xy|$ and $|x| \cdot |y|$ are both ≥ 0, this proves that $|xy| = |x| \cdot |y|$.
    (iii) $|x|/|y| = |x| \cdot |y|^{-1} = |x| \cdot |y^{-1}|$ by (ii)) $= |xy^{-1}|$ (by (i)) $= |x/y|$.
    (v) It follows from (iv) that $|x| = |y - (y - x)| \leq |y| + |y - x|$, so $|x| - |y| \leq |x - y|$.
    (vii) $|x + y + z| \leq |x + y| + |z| \leq |x| + |y| + |z|$. If equality holds, then $|x + y| = |x| + |y|$, so x and y have the same sign. Moreover, z must

have the same sign as $x + y$, so x, y, and z must all have the same sign (unless one is 0).

CHAPTER 2

1. (i) Since $1^2 = 1 \cdot (2) \cdot (2 \cdot 1 + 1)/6$, the formula is true for $n = 1$. Suppose that the formula is true for k. Then

$$\begin{aligned} 1^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)}{6}[k(2k+1) + 6(k+1)] \\ &= \frac{(k+1)}{6}[(k+2)(2k+3)] \\ &= \frac{(k+1)(k+2)(2[k+1]+1)}{6}, \end{aligned}$$

so the formula is true for $k + 1$.

2. (i)

$$\begin{aligned} \sum_{i=1}^{n} (2i - 1) &= 1 + 3 + 5 + \cdots + (2n - 1) \\ &= 1 + 2 + 3 + \cdots + 2n - 2(1 + \cdots + n) \\ &= \frac{(2n)(2n+1)}{2} - n(n+1) \\ &= n^2. \end{aligned}$$

5. (a) Since

$$1 + r = \frac{1 - r^2}{1 - r},$$

the formula is true for $n = 1$. Suppose that

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Then

$$\begin{aligned} 1 + r + \cdots + r^n + r^{n+1} &= \frac{1 - r^{n+1}}{1 - r} + r^{n+1} \\ &= \frac{1 - r^{n+1} + r^{n+1}(1 - r)}{1 - r} \\ &= \frac{1 - r^{n+2}}{1 - r}. \end{aligned}$$

(b)

$$\begin{aligned} S &= 1 + r + \cdots + r^n \\ rS &= \quad r + \cdots + r^n + r^{n+1}. \end{aligned}$$

Thus

$$S(1 - r) = S - rS = 1 - r^{n+1},$$

so

$$S = \frac{1 - r^{n+1}}{1 - r}.$$

6. (i) From

$$(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1, \qquad k = 1, \ldots, n$$

we obtain

$$(n+1)^4 - 1 = 4\sum_{k=1}^{n} k^3 + 6\sum_{k=1}^{n} k^2 + 4\sum_{k=1}^{n} k + n,$$

so

$$\sum_{k=1}^{n} k^3 = \frac{(n+1)^4 - 1 - 6\dfrac{n(n+1)(2n+1)}{6} - 4\dfrac{n(n+1)}{2} - n}{4}$$

$$= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}.$$

(iii) From

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}, \qquad k = 1, \ldots, n$$

we obtain

$$1 - \frac{1}{n+1} = \sum_{k=1}^{n} \frac{1}{k(k+1)}.$$

x

8. 1 is either even or odd, in fact it is odd. Suppose n is either even or odd; then n can be written either as $2k$ or $2k+1$. In the first case $n+1 = 2k+1$ is odd; in the second case $n+1 = 2k+1+1 = 2(k+1)$ is even. In either case, $n+1$ is either even or odd. (Admittedly, this looks shy, but it is really correct.)

9. Let B be the set of all natural numbers l such that $n_0 - 1 + l$ is in A. Then 1 is in B, and $l+1$ is in B if l is in B, so B contains all natural numbers, which means that A contains all natural numbers $\geq n_0$.

12. (a) Yes, for if $a+b$ were rational, then $b = (a+b) - a$ would be rational. If a and b are irrational, then $a+b$ could be rational, for b could be $r - a$ for some rational number r.
    (b) If $a = 0$, then ab is rational. But if $a = 0$, then ab could not be rational, for then $b = (ab) \cdot a^{-1}$ would be rational.
    (c) Yes; for example, $4, 2$.
    (d) Yes; for example, 2 and -2.

13. (a) Since

$$(3n+1)^2 = 9n^2 + 6n + 1 = 3(3n^2 + 2n) + 1,$$
$$(3n+2)^2 = 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1,$$

it follows that if k^2 is divisible by 3, then k must also be divisible by 3. Now suppose that 3 were rational, and let $3 = p/q$ where p and

q have no common factor. Then $p^2 = 3q^2$, so p^2 is divisible by 3, so p must be. Thus, $p = 3p'$ for some natural number p', and consequently $(3p')^2 = 3q^2$, or $3(p')^2 = q^2$. Thus, q is also divisible by 3, a contradiction.

The same proofs work for 5 and 6, because the equations

$$(5n+1)^2 = 25n^2 + 10n + 1 = 5(5n^2 + 2n) + 1,$$
$$(5n+2)^2 = 25n^2 + 20n + 4 = 5(5n^2 + 4n) + 4,$$
$$(5n+3)^2 = 25n^2 + 30n + 9 = 5(5n^2 + 6n + 1) + 4,$$
$$(5n+4)^2 = 25n^2 + 40n + 16 = 5(5n^2 + 8n + 3) + 1,$$

and the corresponding equations for numbers of the form $6n$, show that if k^2 is divisible by 5 or 6, then k must be. The proof fails for 4, because $(4n+2)^2$ is divisible by 4. (For precisely this reason this proof cannot be used to show that in general $\sqrt{a}$ is irrational if a is not a perfect square—we have no guarantee that $(an+m)^2$ might not be a multiple of a for some $m < a$. Actually, this assertion is true, but the proof requires the information in Problem 17.)

(b) Since

$$(2n+1)^3 = 8n^3 + 12n^2 + 6n + 1 = 2(4n^3 + 6n^2 + 3n) + 1,$$

it follows that if k^3 is even, then k is even. If $\sqrt[3]{2} = p/q$ where p and q have no common factors, then $p^3 = 2q^3$, so p^3 is divisible by 2, so p must be. Thus $p = 2p'$ for some natural number p', and consequently $(2p')^3 = 2q^3$, or $4(p')^3 = q^3$. Thus, q is also even, a contradiction.

The proof for $\sqrt[3]{3}$ is similar, using the equations

$$(3n+1)^3 = 27n^3 + 27n^2 + 9n + 1 = 3(9n^3 + 9n^2 + 3n) + 1,$$
$$(3n+2)^3 = 27n^3 + 54n^2 + 36n + 8 = 3(9n^3 + 18n^2 + 12n + 2) + 2.$$

19. If $n = 1$, then $(1+h)^n = 1 + nh$. Suppose that $(1+h)^n \geq 1 + nh$. Then

$$\begin{aligned}(1+h)^{n+1} &= (1+h)(1+h)^n \geq (1+h)(1+nh), \quad \text{since } 1+h > 0\\ &= 1 + (n+1)h + nh^2 \geq 1 + (n+1)h.\end{aligned}$$

For $h > 0$, the inequality follows directly from the binomial theorem, since all the other terms appearing in the expansion of $(1+h)^n$ are positive.

CHAPTER 3

1. (i) $(x+1)/(x+2)$; the expression $f(f(x))$ makes sense only when $x \neq -1$ and $x \neq -2$.
   (iii) $1/(1+cx)$ (for $x \neq -1/c$ if $c \neq 0$).
   (v) $(x+y+2)/(x+1)(y+1)$ (for $x, y \neq -1$).
   (vii) Only $c = 1$, since $f(x) = f(cx)$ implies that $x = cx$, and this must be true for at least one $x \neq 0$.
2. (i) $y \geq 0$ and rational, or $y \geq 1$.
   (iii) 0.
   (v) $-1, 0, 1$.

3. (i) $\{x : -1 \le x \le 1\}$.
   (iii) $\{x : x = 1 \text{ and } x = 2\}$.
   (v) .
4. (i) 2^{2y}.
   (iii) $2^{2\sin t} + \sin(2^t)$.
5. (i) P s.
   (iii) s S.
   (v) P P.
   (vii) s s s P P P s.
11. (a) y.
   (b) H(y).
   (c) H(y).
12. (a)

	even	odd
even	even	neither
odd	neither	odd

   (b)

	even	odd
even	even	odd
odd	odd	even

   (c)

	f even	f odd
g even	even	even
g odd	even	odd

   (d) Let $g(x) = f(x)$ for $x \ge 0$ and de neg arbitrarily for $x < 0$.
21. (i) Let $g(x) = h(x) = 1$ and let f be a function for which $f(2) = f(1) + f(1)$. Then $f \circ (g + h) = f \circ g + f \circ h$.
   (ii) $[(g + h) \circ f](x) = (g + h)(f(x)) = g(f(x)) + h(f(x)) = (g \circ f)(x) + (h \circ f)(x) = [(g \circ f) + (h \circ f)](x)$.
   (iii) $\frac{1}{f \circ g}(x) = \frac{1}{f(g(x))} = \frac{1}{f}(g(x)) = \left(\frac{1}{f} \circ g\right)(x)$.
   (iv) Let $g(x) = 2$ and let f be a function for which $f(\frac{1}{2}) = 1/f(2)$. Then $1/(f \circ g) = f \circ (1/g)$.

CHAPTER 4

1. (i) $(2, 4)$.
   (iii) $[2, 4]$.
   (v) $(-2, 2)$.
   (vii) $(-\infty, 1] \cup [1, \infty)$.
3. (i) All points below the graph of $f(x) = x$.
   (iii) All points below the graph of $f(x) = x^2$.
   (v) All points between the graphs of $f(x) = x + 1$ and $f(x) = x - 1$.
   (vii) A collection of straight lines parallel to the graph of $f(x) = -x$, intersecting the horizontal axis at the points $(n, 0)$ for integers n.
   (ix) All points inside the circle of radius 1 and around $(1, 2)$.
4. (i) A square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.
   (iii) The union of the graph of $f(x) = x$ and of $f(x) = 2 - x$.
   (v) The point $(0, 0)$.
   (vii) The circle of radius 5 around $(1, 0)$, since $x^2 - 2x + y^2 = (x - 1)^2 + y^2 - 1$.
6. (a) Simply observe that the graph of $f(x) = m(x - a) + b = mx + (b - ma)$ is a straight line with slope m, which goes through the point (a, b). (The important point about this exercise is simply to remember the point slope form.)
   (b) The straight line through (a, b) and (c, d) has slope $(d - b)/(c - a)$, so the equation follows from part (a).
   (c) When $m = m$ and $b = b$. In that case, there is clearly no number x with $f(x) = g(x)$, while such a number x always exists if $m = m$, namely, $x = (b - b)/(m - m)$.
7. (a) If $B = 0$ and $A = 0$, then the set is the vertical straight line formed by all points (x, y) with $x = -C/A$. If $B = 0$, the set is the graph of $f(x) = (-A/B)x + (-C/A)$.
   (b) The points (x, y) on the vertical line with $x = a$ are precisely the ones which satisfy $1 \cdot x + 0 \cdot y + (-a) = 0$. The points (x, y) on the graph of $f(x) = mx + b$ are precisely the ones which satisfy $(-m)x + 1 \cdot y + (-b) = 0$.
11. (i) The graph of f is symmetric with respect to the vertical axis.
    (ii) The graph of f is symmetric with respect to the origin. Equivalently, the part of the graph to the left of the vertical axis is obtained by reflecting first through the vertical axis, and then through the horizontal axis.
    (iii) The graph of f lies above or on the horizontal axis.
    (iv) The graph of f repeats the part between 0 and 1 over and over.
21. (a) The square of the distance from (x, x^2) to $(0, \frac{1}{4})$ is

$$\begin{aligned} x^2 + \left(x^2 - \frac{1}{4}\right)^2 &= x^2 + x^4 - \frac{x^2}{2} + \frac{1}{16} \\ &= x^4 + \frac{x^2}{2} + \frac{1}{16} \\ &= (x^2 + \tfrac{1}{4})^2, \end{aligned}$$

which is the square of the distance from (x, x^2) to the graph of g.

(b) The point (x, y) satisfies this condition if and only if

$$(x - a)^2 + (y - b)^2 = (y - c)^2,$$

or

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = y^2 - 2cy + c^2,$$

or

$$y = \frac{1}{2(b - c)}x^2 + \frac{-a}{b - c}x + \frac{a^2 + b^2 - c^2}{2(b - c)}.$$

(This solution works only for $b \neq c$, which is just the condition that P is not on L. If P is on L, then the solution is the vertical line through P.)

CHAPTER 5

1. (ii)

$$\lim_{x \to 2} \frac{x^3 - 8}{x - 2} = \lim_{x \to 2}(x^2 + 2x + 4) = 12.$$

(iv)

$$\lim_{x \to y} \frac{x^n - y^n}{x - y} = \lim_{x \to y} x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1}$$
$$= y^{n-1} + y^{n-1} + \cdots + y^{n-1} = ny^{n-1}.$$

(vi)

$$\lim_{h \to 0} \frac{\sqrt{a + h} - \sqrt{a}}{h} = \lim_{h \to 0} \frac{(\sqrt{a + h} - \sqrt{a})(\sqrt{a + h} + \sqrt{a})}{h(\sqrt{a + h} + \sqrt{a})}$$
$$= \lim_{h \to 0} \frac{1}{\sqrt{a + h} + \sqrt{a}}$$
$$= \frac{1}{2\sqrt{a}}.$$

3. (i) It is possible to find δ by beginning with the equation

$$x^4 - a^4 = (x - a)(x^3 + ax^2 + a^2x + a^3).$$

If $|x - a| < 1$, then $|x| < 1 + |a|$, so

$$|x^3 + ax^2 + a^2x + a^3| \leq |x|^3 + |a| \cdot |x|^2 + |a|^2 \cdot |x| + |a|^3$$
$$< (1 + |a|)^3 + |a|(1 + |a|)^2 + |a|^2(1 + |a|) + |a|^3;$$

therefore we can choose

$$\delta = \min\left(1, \frac{\varepsilon}{(1 + |a|)^3 + |a|(1 + |a|)^2 + |a|^2(1 + |a|) + |a|^3}\right).$$

It is instructive, and probably easier, to use part (2) of the lemma. This shows that $|x^4 - a^4| < \varepsilon$ when

$$|x^2 - a^2| < \min\left(1, \frac{\varepsilon}{2(|a|^2 + 1)}\right),$$

which is true when

$$|x - a| < \min\left(1, \frac{\min\left(1, \frac{\varepsilon}{2(|a|^2+1)}\right)}{2(|a|+1)}\right)$$

$$= \min\left(1, \frac{\varepsilon}{4(|a|^2+1)(|a|+1)}\right) = \delta.$$

(ii) By part (3) of the lemma, $|1/x - 1| < \varepsilon$ when

$$|x - 1| < \min\left(\frac{1}{2}, \varepsilon\right) = \delta.$$

(iii) By part (1) of the lemma, $|(x^4 + 1/x) - 2| < \varepsilon$ when $|1/x - 1| < \varepsilon/2$ and $|x^4 - 1| < \varepsilon/2$. According to parts (i) and (ii) of this problem, this happens when

$$|x - 1| < \min\left(\frac{1}{2}, \frac{\varepsilon}{4}, \frac{\varepsilon}{8 \cdot 2 \cdot 2}\right) = \min\left(\frac{1}{2}, \frac{\varepsilon}{32}\right) = \delta.$$

(v) Let $\delta = \varepsilon^2$, since $0 < |x| < \varepsilon^2$ implies that $|x| < \varepsilon$.

6. (i) We need $|f(x) - 2| < \varepsilon/2$ and $|g(x) - 4| < \varepsilon/2$, so we need

$$0 < |x - 2| < \min\left(\sin^2\frac{\varepsilon^2}{36} + 2, \frac{\varepsilon}{4}\right) = \delta.$$

(iii) We need

$$|g(x) - 4| < \min\left(\frac{|4|}{2}, \frac{|4|^2}{2}\right),$$

so we need

$$0 < |x - 2| < [\min(2, 8\varepsilon)]^2 = \delta.$$

9. Let $l = \lim_{x \to a} f(x)$ and define $g(h) = f(a + h)$. Then for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x, if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. Now, if $0 < |h| < \delta$, then $0 < |(h + a) - a| < \delta$, so $|f(a + h) - l| < \varepsilon$. This inequality can be written $|g(h) - l| < \varepsilon$. Thus, $\lim_{h \to 0} g(h) = l$, which can also be written $\lim_{h \to 0} f(a + h) = l$. The same sort of argument shows that if $\lim_{h \to 0} f(a + h) = m$, then $\lim_{x \to a} f(x) = m$. So either limit exists if the other does, and in this case they are equal.

10. (a) Intuitively, we can get $f(x)$ as close to 0 as we like if and only if we can get $f(x) - l$ as close to 0 as we like. The formal proof is so trivial that it takes a bit of work to make it look like a proof at all. To be very precise, suppose $\lim_{x \to a} f(x) = l$ and let $g(x) = f(x) - l$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x, if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. This last inequality can be written $|g(x) - 0| < \varepsilon$, so $\lim_{x \to a} g(x) = 0$. The argument in the other direction is similarly uninteresting.

(b) Intuitively, making x close to a is the same as making $x - a$ close to 0. Formally: Suppose that $\lim_{x \to a} f(x) = l$, and let $g(x) = f(x - a)$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x, if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. Now, if $0 < |y| < \delta$, then $0 < |(y + a) - a| < \delta$, so $|f(y + a) - l| < \varepsilon$. But this last inequality can be written $|g(y) - l| < \varepsilon$. So $\lim_{y \to 0} g(y) = l$. The argument in the reverse direction is similar.

(c) Intuitively, x is close to 0 if and only if x^3 is. Formally: Let $\lim_{x \to 0} f(x) = l$. For every $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x| < \delta$, then $|f(x) - l| < \varepsilon$. Then if $0 < |x| < \min(1, \delta)$, we have $0 < |x^3| < \delta$, so $|f(x^3) - l| < \varepsilon$. Thus, $\lim_{x \to 0} f(x) = l$. On the other hand, if we assume that $\lim_{x \to 0} f(x^3)$ exists, say $\lim_{x \to 0} f(x^3) = m$, then for all $\varepsilon > 0$ there is a such that if $0 < |x| < \delta$, then $|f(x^3) - m| < \varepsilon$. Then if $0 < |x| < \delta^3$, we have $0 < |\sqrt[3]{x}| < \delta$, so $|f([\sqrt[3]{x}]^3) - m| < \varepsilon$, or $|f(x) - m| < \varepsilon$. Thus $\lim_{x \to 0} f(x) = m$.

(d) Let $f(x) = 1$ for $x \geq 0$, and $f(x) = -1$ for $x < 0$. Then $\lim_{x \to 0} f(x^2) = 1$, but $\lim_{x \to 0} f(x)$ does not exist.

17. (a) The function $f(x) = 1/x$ cannot approach a limit at 0, since it becomes arbitrarily large near 0. In fact, no matter what $\varepsilon > 0$ may be, there is some x satisfying $0 < |x| < \delta$, but $1/x > |l| + \varepsilon$, namely, $x = \min(\delta, 1/(|l| + \varepsilon))$. This x does not satisfy $|1/(x - l)| < \varepsilon$.

(b) No matter what $\delta > 0$ may be, there is some x satisfying $0 < |x - 1| < \delta$, but $1/(x - 1) > |l| + \varepsilon$, namely, $x = \min(1 + \delta, 1 + 1/(|l| + \varepsilon))$. This x does not satisfy $|1/(x - 1) - l| < \varepsilon$. (It is also possible to apply Problem 10(b): $\lim_{x \to 0} 1/x = \lim_{x \to 1} 1/(x - 1)$ if the latter exists, so this limit does not exist, because of part (a).)

25. (i) This is the usual definition, simply calling the numbers a and ε, instead of a and ε.

(ii) This is a minor modification of (i): if the condition is true for $\varepsilon > 0$, then it applies to $\varepsilon/2$, so there is an $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon/2 < \varepsilon$.

(iii) This is a similar modification: apply it to $\varepsilon/5$ to obtain (i).

(iv) This is also a modification: it says the same thing as (i), since $10\varepsilon > 0$, and it is only the existence of some $\delta > 0$ that is in question.

29. If $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = l$, then for every $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x,

$$\text{if } a < x < a + \delta_1, \text{ then } |f(x) - l| < \varepsilon,$$
$$\text{if } a - \delta_2 < x < a, \text{ then } |f(x) - l| < \varepsilon.$$

Let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then either $a - \delta_2 < a - \delta < x < a$ or else $a < x < a + \delta < a + \delta_1$, so $|f(x) - l| < \varepsilon$.

30. (i) If $l = \lim_{x \to 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. If $-\delta < x < 0$, then $0 < -x < \delta$, so $|f(-x) - l| < \varepsilon$. Thus $\lim_{x \to 0^-} f(-x) = l$. Similarly, if $\lim_{x \to 0^-} f(x)$ exists, then $\lim_{x \to 0^+} f(x)$ exists and has the same value. (Intuitively, $-x$ is close to 0 and positive if and only if x is close to 0 and negative.)

(ii) If $l = \lim_{x \to 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. So if $0 < |x| < \delta$, then $|f(|x|) - l| < \varepsilon$. Thus $\lim_{x \to 0} f(|x|) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then $|x|$ is close to 0 and positive.)

(iii) If $l = \lim_{x \to 0^+} f(x)$, then for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \varepsilon$ for $0 < x < \delta$. If $0 < |x| < \delta$, then $0 < x^2 < \delta$, so $|f(x^2) - l| < \varepsilon$. Thus $\lim_{x \to 0} f(x^2) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then x^2 is close to 0 and positive.)

34. If $l = \lim_{x} f(x)$, then for every $\varepsilon > 0$ there is some N such that $|f(x) - l| < \varepsilon$ for $x > N$, and we can clearly assume that $N > 0$. Now, if $0 < x < 1/N$, then $1/x > N$, so $|f(1/x) - l| < \varepsilon$. Thus $\lim_{x \to 0^+} f(1/x) = l$. The reverse direction is similar.

CHAPTER 6

1. (i) $F(x) = x + 2$ for all x.
   (iii) $F(x) = 0$ for all x.

CHAPTER 7

1. (i) Bounded above and below; minimum value 0; no maximum value.
   (iii) Bounded below but not above; minimum value 0.
   (v) Bounded above and below. It is understood that $a > -1$ (so that $-a - 1 < a + 1$). If $-1 < a \leq -\frac{1}{2}$, then $a \leq -a - 1$, so $f(x) = a + 2$ for all x in $(-a - 1, a + 1)$, so $a + 2$ is the maximum and minimum value. If $-\frac{1}{2} < a \leq 0$, then f has the minimum value a^2, and if $a \geq 0$, then f has the minimum value 0. Since $a + 2 > (a + 1)^2$ only for $[-1 - \sqrt{5}]/2 < a < [1 + \sqrt{5}]/2$, when $a \leq -\frac{1}{2}$ the function f has a maximum value only for $a \leq [1 + \sqrt{5}]/2$ (the maximum value being $a + 2$).
   (vii) Bounded above and below; maximum value 1; minimum value 0.
   (ix) Bounded above and below; maximum value 1; minimum value -1.
   (xi) f has a maximum and minimum value, since f is continuous.
2. (i) $n = -2$, since $f(-2) < 0 < f(-1)$.
   (iii) $n = -1$, since $f(-1) = -1 < 0 < f(0)$.
3. (i) If $f(x) = x^{179} + 163/(1 + x^2 + \sin^2 x)$, then f is continuous on $\mathbb{R}$ and $f(-1) > 0$, while $f(-2) < 0$, so $f(x) = 0$ for some x in $(-2, 1)$.
5. f is constant, for if f took on two different values, then f would take on all values in between, which would include irrational values.
7. (1) $f(x) = x$;
   (2) $f(x) = -x$;

(3) $f(x) = |x|$;
(4) $f(x) = -|x|$.

10. Apply Theorem 1 to $f - g$.
11. If $f(0) = 0$ or $f(1) = 1$, choose $x = 0$ or 1. If $f(0) > 0 = I(0)$ and $f(1) < 1 = I(1)$, then Problem 10 applied to f and I implies that $f(x) = x$ for some x.

CHAPTER 8

1. (i) 1 is the greatest element, and the greatest lower bound is 0, which is not in the set.
(iii) 1 is the greatest element, and 0 is the least element.
(v) Since $\{x : x^2 + x + 1 \quad 0\} = R$, there is no least upper bound or greatest lower bound.
(vii) Since $\{x : x < 0 \text{ and } x^2 + x - 1 < 0\} = ([-1 - \quad 5]/2, 0)$, the greatest lower bound is $[-1 - \quad 5]/2$, and the least upper bound is 0; neither belongs to the set.

2. (a) Since $A = \quad$, there is some x in A. Then $-x$ is in $-A$, so $-A = \quad$. Since A is bounded above, there is some y such that $y \quad x$ for all x in A. Then $-y \quad -x$ for all x in A, so $-y \quad z$ for all z in $-A$, so $-A$ is bounded below. Let $= \sup(-A)$. Then $\quad$ is an upper bound for $-A$, so, reversing the argument just given, $-$ is a lower bound for A. Moreover, if $\quad$ is any lower bound for A, then $-\quad$ is an upper bound for $-A$, so $-\quad$, so $\quad -$. Thus $-$ is the greatest lower bound for A.

5. (a) If l is the largest integer with $l \quad x$, then $l + 1 > x$, but $l + 1 \quad x + 1 < y$. So we can let $k = l + 1$. (Proof that a largest such integer exists: Since N is not bounded above, there is some natural number n with $-n < x < n$. There are consequently only a finite number of integers with $-n \quad l \quad x$. Pick the largest.)
(b) Since $y - x > 0$, there is some natural number n with $1/n < y - x$. Since $ny - nx > 1$, there is, by part (a), an integer k with $nx < k < ny$, which means that $x < k/n < y$.
(c) Choose $r = \quad 2(s - r)/2$.
(d) By part (b), there is a rational number r with $x < r < y$, and therefore a rational number s with $x < r < s < y$. Apply part (c) to $r < s$.

10. Let k be the largest integer $x/$ (the solution to Problem 5 shows that such a k exists), and let $x = x - k \quad 0$. If $x - k = x \quad$, then $x \quad (k + 1)$, so $k + 1 \quad x/$, contradicting the choice of k. So $0 \quad x < \quad$.

12. (a) Since any y in B satisfies $y \quad x$ for all x in A, any y in B is an upper bound for A, so $y \quad \sup A$.
(b) Part (a) shows that $\sup A$ is a lower bound for B, so $\sup A \quad \inf B$.

13. Since $x \quad \sup A$ and $y \quad \sup B$ for every x in A, and y in B, it follows that $x + y \quad \sup A + \sup B$. Thus, $\sup A + \sup B$ is an upper bound for $A + B$, so $\sup(A + B) \quad \sup A + \sup B$. If x and y are chosen in A and B, respectively, so that $\sup A - x < \quad / 2$ and $\sup B - y < \quad / 2$, then $\sup A + \sup B - (x + y) < \quad$. Hence,

$$\sup(A + B) \quad x + y > \sup A + \sup B - \quad .$$

CHAPTER 9

1. (a)

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h}$$
$$= \lim_{h \to 0} \frac{-1}{a(a+h)} = -\frac{1}{a^2}.$$

(b) The tangent line through $(a, 1/a)$ is the graph of

$$g(x) = -\frac{1}{a^2}(x - a) + \frac{1}{a}$$
$$= -\frac{x}{a^2} + \frac{2}{a}.$$

If $f(x) = g(x)$, then

$$\frac{1}{x} = -\frac{x}{a^2} + \frac{2}{a}$$

or

$$x^2 - 2ax + a^2 = 0,$$

so $x = a$.

2. (a)

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{\frac{1}{(a+h)^2} - \frac{1}{a^2}}{h}$$
$$= \lim_{h \to 0} \frac{(-2ah - h^2)}{ha^2(a+h)^2} = -\frac{2}{a^3}.$$

(b) The tangent line through $(a, 1/a^2)$ is the graph of

$$g(x) = -\frac{2}{a^3}(x - a) + \frac{1}{a^2}$$
$$= -\frac{2x}{a^3} + \frac{3}{a^2}.$$

If $f(x) = g(x)$, then

$$\frac{1}{x^2} = -\frac{2x}{a^3} + \frac{3}{a^2},$$

or

$$2x^3 - 3ax^2 + a^3 = 0,$$

or

$$0 = (x - a)(2x^2 - ax - a^2) = (x - a)(2x + a)(x - a).$$

So $x = a$ or $x = -a/2$; the point $(-a/2, 4/a^2)$ lies on the opposite side of the vertical axis from $(a, 1/a^2)$.

3.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}$$

$$= \lim_{h \to 0} \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})} = \lim_{h \to 0} \frac{h}{h(\sqrt{a+h} + \sqrt{a})}$$

$$= \frac{1}{2\sqrt{a}}.$$

4. Conjecture: $S_n'(x) = nx^{n-1}$. Proof:

$$S_n'(x) = \lim_{h \to 0} \frac{S_n(x+h) - S_n(x)}{h} = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h \to 0} \frac{\sum_{j=0}^{n} \binom{n}{j} x^{n-j} h^j - x^n}{h}$$

$$= \lim_{h \to 0} \sum_{j=1}^{n} \binom{n}{j} x^{n-j} h^{j-1}$$

$$= \binom{n}{1} x^{n-1} = nx^{n-1}, \quad \text{since } \lim_{h \to 0} h^{j-1} = 0 \text{ for } j > 1.$$

5. $f'(x) = 0$ for x not an integer, and $f'(x)$ is not defined if x is an integer.

6. (a)

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \to 0} \frac{[f(x+h) + c] - [f(x) + c]}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = f'(x).$$

(b)

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \to 0} \frac{cf(x+h) - cf(x)}{h}$$

$$= c \cdot \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = cf'(x).$$

7. (a) $f'(9) = 3 \cdot 9^2$; $f'(25) = 3 \cdot (25)^2$; $f'(36) = 3 \cdot (36)^2$.
(b) $f'(3^2) = f'(9) = 3 \cdot 9^2$; $f'(5^2) = f'(25) = 3 \cdot (25)^2$; $f'(6^2) = f'(36) = 3 \cdot (36)^2$.
(c) $f'(a^2) = 3(a^2)^2 = 3a^4$; $f'(x^2) = 3(x^2)^2 = 3x^4$.
(d) $f'(x^2) = 3x^4$; but $g(x) = x^6$, so $g'(x) = 6x^5$.

8. (a)

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \to 0} \frac{f(x+h+c) - f(x+c)}{h}$$

$$= \lim_{h \to 0} \frac{f([x+c]+h) - f(x+c)}{h} = f'(x+c).$$

(b)

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \to 0} \frac{f(cx+ch) - f(cx)}{h}$$
$$= \lim_{h \to 0} \frac{c[f(cx+ch) - f(cx)]}{ch} = \lim_{k \to 0} \frac{c[f(cx+k) - f(cx)]}{k}$$
$$= c \cdot \lim_{k \to 0} \frac{f(cx+k) - f(cx)}{k} = c \cdot f'(cx).$$

(Compare the manipulations in this calculation with Problem 5-14.)

(c) If $g(x) = f(x+a)$, then $g'(x) = f'(x+a)$, by part (a). But $g = f$, so $f'(x) = g'(x) = f'(x+a)$ for all x, which means that f' is periodic, with period a.

9. (i) If $g(x) = x^5$, then $g'(x) = 5x^4$. Now $f(x) = g(x+3)$, so by Problem 8(a), $f'(x) = g'(x+3) = 5(x+3)^4$. And $f'(x+3) = 5(x+6)^4$.

(ii) $f(x) = (x-3)^5$, so $f'(x) = 5(x-3)^4$, as in part (i). And $f'(x+3) = 5 \cdot 0^4 = 0$.

(iii) $f(x) = (x+2)^7$, so $f'(x) = 7(x+2)^6$, as in part (i). And $f'(x+3) = 7(x+5)^6$.

10. If $f(x) = g(t+x)$, then $f'(x) = g'(t+x)$, by Problem 8(a). If $h(t) = g(t+x)$, then $h'(t) = g'(t+x)$, by Problem 8(a), so $h'(x) = g'(2x)$.

11. (a) If $s(t) = ct^2$, then $s'(t) = 2ct$, and there is no number k such that $s'(t) = ks(t)$ [that is, $2ct = kct^2$] for all t.
(By the way, at this point we do not know any nonzero function f for which f' is proportional to f. After Chapter 18 it might be amusing to determine what the world would be like if Galileo were correct.)

(b) (i) If $s(t) = (a/2)t^2$, then $s'(t) = at$, so $s''(t) = a$.

(ii) $[s'(t)]^2 = (at)^2 = 2a \cdot (a/2)t^2 = 2as(t)$.

(c) The chandelier falls $s(t) = 16t^2$ feet in t seconds, so it falls 400 feet in t seconds, if $400 = 16t^2$, or $t = 5$. After 5 seconds the velocity will be $s'(5) = 5a = 5 \cdot 32 = 160$ feet per second. The speed was half this amount when $80 = s'(t) = 32t$, or $t = \frac{5}{2}$.

21. (a) This is another way of writing the definition (see Problem 5-9).

(b) This follows from Problem 5-11, applied to the functions $r(h) = [f(a+h) - f(a)]/h$ and $s(h) = [g(a+h) - g(a)]/h$.

26. (i) $f'(x) = 6x$.

(iii) $f'(x) = 4x^3$.

30. (i) means that $f'(a) = na^{n-1}$ if $f(x) = x^n$.

(iii) means that $g'(a) = f'(a)$ if $g(x) = f(x) + c$.

(v) means the same as (iii).

(vii) means that $g'(b) = f'(b+a)$ if $g(x) = f(x+a)$.

(ix) means that $g'(b) = cf'(cb)$ if $g(x) = f(cx)$.

CHAPTER 10

1. (i) $(1+2x) \cdot \cos(x+x^2)$.

(iii) $(-\sin x) \cdot \cos'(\cos x)$.

(v) $\cos\left(\frac{\cos x}{x}\right) \cdot \frac{-x\sin x - \cos x}{x^2}$.

(vii) $(\cos(x + \sin x)) \cdot (1 + \cos x)$.

2. (i) $(\cos((x + 1)^2(x + 2))) \cdot [2(x + 1)(x + 2) + (x + 1)^2]$.

(iii) $[2\sin((x + \sin x)^2)\cos((x + \sin x)^2)] \cdot 2(x + \sin x)(1 + \cos x)$.

(v) $(\cos(x \sin x)) \cdot (\sin x + x\cos x) + (\cos(\sin x^2)(\cos x^2)) \cdot 2x$.

(vii) $(2\sin x \cos x \sin x^2 \sin^2 x^2) + (2x\cos x^2 \sin^2 x \sin^2 x^2)$
$+ (4x \sin x^2 \cos x^2 \sin^2 x \sin x^2)$.

(ix) $6(x + \sin^5 x)^5(1 + 5\sin^4 x \cos x)$.

(xi) $\cos(\sin^7 x^7 + 1)^7 \cdot 7(\sin^7 x^7 + 1)^6 \cdot (7\sin^6 x^7 \cdot \cos x^7 \cdot 7x^6)$.

(xiii) $\cos(x^2 + \sin(x^2 + \sin x^2)) \cdot [(2x + \cos(x^2 + \sin x^2) \cdot (2x + 2x\cos x^2))]$.

(xv) $\dfrac{(1 + \sin x)(2x\cos x^2 \cdot \sin^2 x + \sin x^2 \cdot 2\sin x \cos x) - \cos x \sin x^2 \sin^2 x}{(1 + \sin x)^2}$.

(xvii) $\cos\left(\sin\left(\frac{x^3}{\sin x^3}\right)\right)$.

$$3x^2 \sin\left(\frac{x^3}{\sin x}\right) - x^3 \cos\left(\frac{x^3}{\sin x}\right) \cdot \frac{3x^2 \sin x - x^3 \cos x}{\sin^2 x}$$

$$\sin^2\left(\frac{x^3}{\sin x}\right)$$

4. (i) $\left(\frac{(x + 1)^2}{(x + 2)^2}\right)$.

(iii) $2x^2$.

5. (i) $-x^2$.

(iii) 17.

6. (i) $f'(x) = g'(x + g(a))$.

(iii) $f'(x) = g'(x + g(x)) \cdot (1 + g'(x))$.

(v) $f'(x) = g(a)$.

7. (a) $A(t) = 2\pi r(t)r'(t)$. Since $r'(t) = 4$ for that t with $r(t) = 6$, it follows that $A'(t) = 2\pi \cdot 6 \cdot 4 = 48\pi$ when $r(t) = 6$.

(b) If $V(t)$ is the volume at time t, then $V(t) = \frac{4}{3}\pi r(t)^3$, so $V'(t) = 4\pi r(t)^2 r'(t) = 4\pi \cdot 6^2 \cdot 4 = 576\pi$ when $r(t) = 6$.

(c) First method: Since $A'(t) = 2\pi r(t)r'(t)$, and $A'(t) = 5$ for $r(t) = 3$, it follows that

$$r'(t) = \frac{A'(t)}{2\pi r(t)} = \frac{5}{6\pi} \quad \text{when } r(t) = 3.$$

Thus

$$V'(t) = 4\pi r(t)^2 r'(t)$$
$$= 4\pi \cdot 9 \cdot \frac{5}{6}$$
$$= 30\pi \quad \text{when } r(t) = 3.$$

To apply the second method, we first note that if

$$f(t) = A(t)^{3/2} = \sqrt{A(t)^3},$$

then, using Problem 9-3 and the Chain Rule,

$$f'(t) = \frac{1}{2\sqrt{A(t)^3}} \cdot 3A(t)^2 A'(t)$$
$$= \frac{1}{2A(t)^{3/2}} \cdot 3A(t)^2 A'(t)$$
$$= \frac{3}{2} A(t)^{1/2} A'(t) \quad \text{(just as we might have guessed).}$$

Now

$$V(t) = \frac{4\pi r(t)^3}{3} = \frac{4\pi [r(t)^2]^{3/2}}{3}$$
$$= \frac{4\pi [\pi r(t)^2]^{3/2}}{\pi^{1/2} 3}$$
$$= \frac{4\pi A(t)^{3/2}}{\pi^{1/2} 3}.$$

So

$$V'(t) = \frac{4}{3\pi^{1/2}} \cdot \frac{3}{2} \pi A(t) A'(t)$$
$$= \frac{2}{1/2} \cdot \pi^{1/2} r(t) A'(t)$$
$$= 2 \cdot 3 \cdot 5 = 30.$$

10. (i) $(f \circ h)'(0) = f'(h(0)) \cdot h'(0) = f'(3) \cdot \sin^2(\sin 1) =$
$[6 \sin \frac{1}{3} \cdot \cos \frac{1}{3}] \sin^2(\sin 1)$.

(iii) $(x^2)' = h'(x^4) \cdot 2x^2 = \sin^2(\sin(x^4 + 1)) \cdot 2x^2$.

12. The Chain Rule implies that

$$\left(\frac{1}{g}\right)'(x) = (f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$
$$= -\frac{1}{g(x)^2} \cdot g'(x).$$

33. (i) $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} = (\cos y) \cdot (1 + 2x) = (\cos(x + x^2)) \cdot (1 + 2x)$.

(iii) $\dfrac{dz}{dx} = \dfrac{dz}{du}\cdot\dfrac{du}{dx} = (-\sin u)\cdot(\cos x) = (-\cos(\sin x))\cdot(\cos x)$.

CHAPTER 11

1. (i) $0 = f'(x) = 3x^2 - 2x - 8$ for $x = 2$ and $x = -\frac{4}{3}$, both of which are in $[-2, 2]$;
$f(-2) = 5, f(2) = -11, f(-\frac{4}{3}) = \frac{203}{27}$;
maximum $= \frac{203}{27}$, minimum $= -11$.

(iii) $0 = f'(x) = 12x^3 - 24x^2 + 12x = 12x(x^2 - 2x + 1)$ for $x = 0$ and $x = 1$, of which only 0 is in $[-\frac{1}{2}, \frac{1}{2}]$;
$f(-\frac{1}{2}) = \frac{43}{16}, f(\frac{1}{2}) = \frac{11}{16}, f(0) = 0$;
maximum $= \frac{43}{16}$, minimum $= 0$.

(v) $0 = f'(x) =$

$$\frac{x^2+1-(x+1)2x}{(x^2+1)^2} = \frac{1-2x-x^2}{(x^2+1)^2}$$

for $x = -1+\sqrt{2}$ and $x = -1-\sqrt{2}$, of which only $-1+\sqrt{2}$ is in $[-1, \frac{1}{2}]$;
$f(-1) = 0, f(\frac{1}{2}) = \frac{6}{5}, f(-1+\sqrt{2}) = (1+\sqrt{2})/2$;
maximum $= (1+\sqrt{2})/2$, minimum $= 0$.

2. (i) $-\frac{4}{3}$ is a local maximum point, and 2 is a local minimum point.

(iii) 0 is a local minimum point, and there are no local maximum points.

(v) $-1+\sqrt{2}$ is a local maximum point, and $-1-\sqrt{2}$ is a local minimum point.

4. (a) Notice that f actually has a minimum value, since f is a polynomial function of even degree. The minimum occurs at a point x with

$$0 = f'(x) = 2\sum_{i=1}^{n}(x - a_i),$$

so $x = (a_1 + \cdots + a_n)/n$.

5. (i) 3 and 7 are local maximum points, and 5 and 9 are local minimum points.

(iii) All irrational $x > 0$ are local minimum points, and all irrational $x < 0$ are local maximum points.

(v) x is a local maximum (minimum) point if the decimal expansion contains (does not contain) a 5.

8. If $f(x)$ is the total length of the path, then

$$f(x) = \sqrt{x^2 + a^2} + \sqrt{(1-x)^2 + b^2}.$$

The positive function f clearly has a minimum, since $\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = \infty$, and f is differentiable everywhere, so the minimum occurs at a point x with $f'(x) = 0$. Now, $f'(x) = 0$ when

$$\frac{x}{\sqrt{x^2+a^2}} - \frac{(1-x)}{\sqrt{(1-x)^2+b^2}} = 0.$$

This equation says that $\cos = \cos$.

It is also possible to notice that $f(x)$ is equal to the sum of the lengths of the dashed line segment and the line segment from $(x, 0)$ to $(1, b)$. This is shortest when the two line segments lie along a line (because of Problem 4-9(b), if a rigorous reason is required); a little plane geometry shows that this happens when $=$.

9. If x is the length of one side of a rectangle of perimeter P, then the length of the other side is $(P - 2x)/2$, so the area is

$$A(x) = \frac{x(P - 2x)}{2}.$$

So the rectangle with greatest area occurs when x is the maximum point for A on $(0, P/2)$. Since A is continuous on $[0, P/2]$, and $A(0) = A(P/2) = 0$, and $A(x) > 0$ for x in $(0, P/2)$, the maximum exists. Since A is differentiable on $(0, P/2)$, the minimum point x satisfies

$$0 = A'(x) = \frac{P - 2x}{2} - x$$
$$= \frac{P - 4x}{2},$$

so $x = P/4$.

10. Let $S(r)$ be the surface area of the right circular cylinder of volume V with radius r. Since

$$V = \pi r^2 h \quad \text{where } h \text{ is the height,}$$

we have $h = V/\pi r^2$, so

$$S(r) = 2\pi r^2 + 2\pi rh$$
$$= 2\pi r^2 + \frac{2V}{r}.$$

We want the minimum point of S on $(0, \infty)$; this exists, since $\lim_{r \to 0} S(r) = \infty$ $\lim_{r \to \infty} S(r) = \infty$. Since S is differentiable on $(0, \infty)$, the minimum point r satisfies

$$0 = S'(r) = 4\pi r - \frac{2V}{r^2}$$
$$= \frac{4\pi r^3 - 2V}{r^2},$$

or

$$r = \sqrt[3]{\frac{V}{2\pi}}.$$

19. 1 is a local maximum point, and 3 is a local minimum point.

25. (a) We have

$$\frac{f(b) - f(a)}{b - a} = f'(x) \quad \text{for some } x \text{ in } (a, b)$$
$$M,$$

so $f(b) - f(a) \le M(b - a)$.

(b) We have

$$\frac{f(b) - f(a)}{b - a} = f'(x) \quad \text{for some } x \text{ in } (a, b)$$

$$\ge m,$$

so $f(b) - f(a) \ge m(b - a)$.

(c) If $|f'(x)| \le M$ for all x in $[a, b]$, then $-M \le f'(x) \le M$, so

$$f(a) - M(b - a) \le f(b) - f(a) + M(b - a),$$

or

$$|f(b) - f(a)| \le M(b - a).$$

28. (a) $f(x) = -\cos x + a$ for some number a (because $f(x) = -\cos x$ is one such function, and any two such functions differ by a constant function).

(b) $f'(x) = x^4/4 + a$ for some number a, so $f(x) = x^5/20 + ax + b$ for some numbers a and b.

(c) $f'(x) = x^2 + x^3/3 + a$ for some a, so $f'(x) = x^3/6 + x^4/12 + ax + b$ for some a and b, so $f(x) = x^4/24 + x^5/60 + ax^2/2 + bx + c$ for some numbers a, b, and c. Equivalently, and more simply, $f(x) = x^4/24 + x^5/60 + ax^2 + bx + c$ for some numbers a, b, and c.

29. (a) Since $s''(t) = -32$, we have $s'(t) = -32t + \quad$ for some , so $s(t) = -16t^2 + t + \quad$ for some and .

(b) Clearly, $s(0) = 0 + 0 + \quad$ and $s'(0) = 0 + \quad$. Thus, $= v_0$ and $= s_0$.

(c) In this case $s_0 = 0$ and $v_0 = v$, so $s(t) = -16t^2 + vt$. The maximum value of s occurs when $0 = s'(t) = -32t + v$, or $t = v/32$, so the maximum value is

$$s\left(\frac{v}{32}\right) = -16\left(\frac{v}{32}\right)^2 + v \cdot \frac{v}{32}$$

$$= -\frac{v^2}{64} + \frac{v^2}{32}$$

$$= \frac{v^2}{64}.$$

At that moment the velocity is clearly 0, but the acceleration is -32 (as at any time). The weight hits the ground at time $t > 0$ when

$$0 = s(t) = -16t^2 + vt,$$

or $t = v/16$ (it takes as long to fall back down as it took to reach the top). The velocity is then

$$s'(v/16) = -32\left(\frac{v}{16}\right) + v$$

$$= -v$$

(the same velocity with which it was initially moving upward).

44. Apply the Mean Value Theorem to $f(x) = \sqrt{x}$ on $[64, 66]$:

$$\frac{66 - 64}{66 - 64} = f'(x) = \frac{1}{2\sqrt{x}} \quad \text{for some } x \text{ in } [64, 66].$$

Since $64 < x < 81$, we have $8 < \sqrt{x} < 9$, so

$$\frac{1}{2 \cdot 9} < \frac{66 - 8}{2} < \frac{1}{2 \cdot 8}.$$

48. l'Hôpital's Rule does not lead to the equation

$$\lim_{x \to 1} \frac{3x^2 + 1}{2x - 3} = \lim_{x \to 1} \frac{6x}{2}$$

because $\lim_{x \to 1} 3x^2 + 1 = 0$.

49. (i)

$$\lim_{x \to 0} \frac{x}{\tan x} = \lim_{x \to 0} \frac{1}{\sec^2 x} = \lim_{x \to 0} \cos^2 x = 1.$$

(ii)

$$\lim_{x \to 0} \frac{\cos^2 x - 1}{x^2} = \lim_{x \to 0} \frac{-2 \sin x \cos x}{2x} = -1.$$

CHAPTER 12

1. (i) $f^{-1}(x) = (x - 1)^{1/3}$. (If $y = f^{-1}(x)$, then $x = f(y) = y^3 + 1$, so $y = (x - 1)^{1/3}$.)

(iii) $f^{-1} = f$. (If $y = f^{-1}(x)$, then

$$x = f(y) = \begin{cases} y, & y \text{ rational} \\ -y, & y \text{ irrational;} \end{cases}$$

since $\pm y$ is rational or irrational if and only if y is, we have $y = x$ if x is rational and $y = -x$ if x is irrational, so $y = f(x)$.)

(v)

$$f^{-1}(x) = \begin{cases} x, & x = a_1, \ldots, a_n \\ a_{i-1}, & x = a_i, \quad i = 2, \ldots, n \\ a_n, & x = a_1. \end{cases}$$

(vii) $f^{-1} = f$.

2. (i) f^{-1} is increasing and $f^{-1}(x)$ is not defined for $x < 0$.

(iii) f^{-1} is decreasing and $f^{-1}(x)$ is not defined for $x \geq 0$.

3. Suppose f is increasing. Let $a < b$. Then $f^{-1}(a) \neq f^{-1}(b)$, since f^{-1} is one-one. So either $f^{-1}(a) < f^{-1}(b)$ or $f^{-1}(a) > f^{-1}(b)$. But if $f^{-1}(a) > f^{-1}(b)$, then

$$b = f(f^{-1}(b)) < f(f^{-1}(a)) = a,$$

a contradiction. The proof is similar for decreasing f, or one can consider $-f$ instead.

4. Clearly, $f + g$ is increasing, for if $f(a) < f(b)$ and $g(a) < g(b)$, then $(f + g)(a) = f(a) + g(a) < f(b) + g(b) = (f + g)(b)$.
$f \cdot g$ is not necessarily increasing; for example, if $f(x) = g(x) = x$. (But $f \cdot g$

is increasing if $f'(x) > 0$ for all x.)
$f \circ g$ is increasing, for if $a < b$, then $g(a) < g(b)$, so $f(g(a)) < f(g(b))$.

5. (a) If $(f \circ g)(x) = (f \circ g)(y)$, so that $f(g(x)) = f(g(y))$, then $g(x) = g(y)$, since f is one-one, so $x = y$, since g is one-one.
$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$: for if $y = (f \circ g)^{-1}(x)$, then $x = (f \circ g)(y) = f(g(y))$, so $g(y) = f^{-1}(x)$, so $y = g^{-1}(f^{-1}(x))$.

6. If $f(x) = f(y)$, then

$$\frac{ax + b}{cx + d} = \frac{ay + b}{cy + d},$$

so

$$acxy + bcy + adx + bd = acxy + ady + bcx + bd,$$

or

$$ad(x - y) = bc(x - y).$$

If $ad = bc$, this implies that $x - y = 0$. (But if $ad = bc$, then $f(x) = f(y)$ for all x and y in the domain of f.)
If $y = f^{-1}(x)$, then $x = f(y)$, so

$$x = \frac{ay + b}{cy + d}$$

so

$$f^{-1}(x) = y = \frac{-dx + b}{cx - a} \qquad \text{for } x = a/c.$$

7. (i) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, 2]$ or $[2, \infty)$, since f is increasing on $(-\infty, 0]$ and $[2, \infty)$, and decreasing on $[0, 2]$.
(iii) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, \infty)$, since f is increasing on $(-\infty, 0]$ and decreasing on $[0, \infty)$.

17. The formula for the derivative reads:

$$\frac{dx}{dy} = \frac{1}{\dfrac{dy}{dx}}.$$

(In this formula, it is understood that dx/dy means $(f^{-1})'(y)$, while dy/dx is an "expression involving x," and in the final answer x must be replaced by y, by means of the equation $y = f(x)$.)
The computation in Problem 17, when completed, shows that

$$\frac{dx^{1/n}}{dx} = \frac{1}{n(x^{1/n})^{n-1}} = \frac{1}{nx^{1-(1/n)}}$$
$$= \frac{1}{n}x^{(1/n)-1}.$$

18.
$$\begin{aligned} G'(x) &= x(f^{-1})'(x) + f^{-1}(x) - F'(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - f(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - x(f^{-1})'(x) \\ &= f^{-1}(x). \end{aligned}$$

19. (i)
$$(h^{-1})'(3) = \frac{1}{h'(h^{-1}(3))} = \frac{1}{h'(0)} = \frac{1}{\sin^2(\sin 1)}.$$

20. Since
$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))},$$
we have
$$\begin{aligned} (f^{-1})'(x) &= \frac{-f'(f^{-1}(x)) \cdot (f^{-1})'(x)}{[f'(f^{-1}(x))]^2} \\ &= \frac{-f'(f^{-1}(x))}{[f'(f^{-1}(x))]^3}. \end{aligned}$$

CHAPTER 13

1. If $P_n = \{t_0, \ldots, t_n\}$ is the partition with $t_i = ib/n$, then
$$\begin{aligned} L(f, P_n) &= \sum_{i=1}^{n} (t_{i-1})^3 \cdot (t_i - t_{i-1}) \\ &= \sum_{i=1}^{n} (i-1)^3 \cdot \frac{b^3}{n^3} \cdot \frac{b}{n} \\ &= \frac{b^4}{n^4} \sum_{j=0}^{n-1} j^3 \\ &= \frac{b^4}{n^4} \left[\frac{(n-1)^4}{4} + \frac{(n-1)^3}{2} + \frac{(n-1)^2}{4} \right], \end{aligned}$$
and similarly
$$\begin{aligned} U(f, P_n) &= \frac{b^4}{n^4} \sum_{j=1}^{n} j^3 \\ &= \frac{b^4}{n^4} \left[\frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} \right]. \end{aligned}$$
Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large, so $U(f, P_n) - L(f, P_n)$ can be made as small as desired, by choosing n large enough. This shows that f is integrable. Moreover, there is only one number a with $L(f, P_n) \le a \le U(f, P_n)$ for all n; since $\int_0^b x^3\,dx$ has this property, the proof that $\int_0^b x^3\,dx = b^4/4$ will be complete once we show that $L(f, P_n) \le b^4/4 \le U(f, P_n)$ for all n. This

can be done by a straightforward computation, but it actually follows from the fact that $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large. In fact, if it were true that $b^4/4 < \int_0^b x^3\,dx$, then it would not be possible to make $L(f, P_n)$ as close as desired to $b^4/4$ by choosing n large enough, since each $L(f, P_n) \le \int_0^b x^3\,dx$, and similarly we cannot have $b^4/4 > \int_0^b x^3\,dx$.

2. We have

$$L(f, P_n) = \frac{b^5}{n^5}\left[\frac{(n-1)^5}{5} + \frac{(n-1)^4}{2} + \frac{(n-1)^3}{3} - \frac{(n-1)}{30}\right],$$

$$U(f, P_n) = \frac{b^5}{n^5}\left[\frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30}\right].$$

Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^5/5$ as desired by choosing n large enough. As in Problem 1, this implies that $\int_0^b x^4\,dx = b^5/5$.

7. (i) $\int_0^2 f = 0$.
(iii) $\int_0^2 f = 3$.
(v) f is not integrable.
(vii) $\int_0^2 f = 1$.
(For a rigorous proof that the functions in (i), (iii), and (vii) are integrable, see Problem 20. The values of the integrals, which are clear from the geometric picture, can also be deduced rigorously by using the ideas in the proof of Problem 20, together with known integrals.)

8. (i)
$$\int_{-2}^{2}\left[\frac{x^2}{2} + 2 - x^2\right]dx = \frac{16}{3}.$$

(iii)
$$\int_{-2/2}^{2/2}[(1-x^2) - x^2]\,dx = \frac{2}{3}\sqrt{2}.$$

(v)
$$\int_0^2[(x^2 - 2x + 4) - x^2]\,dx = 4.$$

9.
$$\int_a^b\left(\int_c^d f(x)g(y)\,dy\right)dx = \int_a^b f(x)\left(\int_c^d g(y)\,dy\right)dx \quad \text{(here } f(x) \text{ is the constant)}$$
$$= \int_c^d g(y)\,dy \cdot \int_a^b f(x)\,dx$$
(here $\int_c^d g(y)\,dy$ is the constant).

13. (a) Clearly $L(f, P) \le 0$ for every partition P.

(b) Apply part (a) to $f - g$, and use the fact that

$$\int_a^b (f - g) = \int_a^b f - \int_a^b g.$$

23. (a) Clearly

$$m(b - a) \le L(f, P) \le U(f, P) \le M(b - a)$$

for all partitions P of $[a, b]$. Consequently,

$$m(b - a) \le \int_a^b f(x)\,dx \le M(b - a).$$

Thus

$$\mu = \frac{\int_a^b f(x)\,dx}{b - a}$$

satisfies $m \le \mu \le M$.

(b) Let m and M be the minimum and maximum values of f on $[a, b]$. Since f is continuous, it takes on the values m and M, and consequently the number μ of part (a).

33. (a) 0.
(b) $\frac{1}{2}$.

37. Since

$$-|f| \le f \le |f|,$$

we have

$$-\int_a^b |f| \le \int_a^b f \le \int_a^b |f|,$$

so

$$\left|\int_a^b f\right| \le \int_a^b |f|.$$

(Problem 36 implies that $\int_a^b |f|$ makes sense.)

CHAPTER 14

1. (i) $(\sin^3 x^3) \cdot 3x^2$.
(iii) $\int_8^x \frac{1}{1 + t^2 + \sin^2 t}\,dt$.
(v) $\int_a^b \frac{1}{1 + t^2 + \sin^2 t}\,dt$.
(vii) $(F^{-1})'(x) = \frac{1}{F'(F^{-1}(x))} = F^{-1}(x)$.

2. (i) All $x = 1$.
(iii) All $x = 1$.
(v) All x.
(vii) All $x = 0$. (F is not differentiable at 0 because $F(x) = 0$ for $x \le 0$, but there are $x > 0$ arbitrarily close to 0 with $\frac{F(x)}{x} = \frac{1}{2}$.)

5. (i)

$$(f^{-1})'(0) = \frac{1}{f'(f^{-1}(0))} = \frac{1}{1+\sin(\sin(f^{-1}(0)))}$$
$$= \frac{1}{1+\sin(\sin 0)} = 1.$$

11. $F(x) = x\int_0^x f(t)\,dt$, so

$$F'(x) = xf(x) + \int_0^x f(t)\,dt.$$

14.

$$f(x) = \int_0^x \int_0^y \int_0^x \frac{1}{1+\sin^2 t}\,dt\,dz\,dy.$$

16. We can choose

$$f(x) = \frac{x^{(1/n)+1}}{\frac{1}{n}+1}.$$

Then

$$\int_0^b x^n\,dx = f(b) - f(0) = \frac{b^{(1/n)+1}}{\frac{1}{n}+1}.$$

CHAPTER 15

1. (i)

$$\frac{1}{1+\arctan^2(\arctan x)} \cdot \frac{1}{1+\arctan^2 x} \cdot \frac{1}{1+x^2}.$$

(iii)

$$\frac{1}{1+(\tan x \arctan x)^2} \cdot \sec^2 x \arctan x + \frac{\tan x}{1+x^2}.$$

2. (i) 0.
(iii) 0.
(v) 0.

7. (a)

$$\sin 2x = \sin(x+x) = \sin x\cos x + \cos x \sin x = 2\sin x\cos x.$$
$$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x.$$
$$\sin 3x = \sin(2x+x) = \sin 2x\cos x + \cos 2x \sin x$$
$$= 2\sin x\cos^2 x + (\cos^2 x - \sin^2 x)\sin x$$
$$= 3\sin x\cos^2 x - \sin^3 x.$$
$$\cos 3x = \cos(2x+x) = \cos 2x\cos x - \sin 2x \sin x$$
$$= (\cos^2 x - \sin^2 x)\cos x - 2\sin^2 x\cos x$$
$$= \cos^3 x - \sin^2 x\cos x - 2\sin^2 x\cos x$$
$$= \cos^3 x - 3\sin^2 x\cos x$$
$$= 4\cos^3 x - 3\cos x.$$

(b) Since $\cos \pi/4 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 2 \cdot \frac{\pi}{4} = 2\cos^2 \frac{\pi}{4} - 1,$$

we have $\cos \pi/4 = \sqrt{2}/2$. It follows, since $\sin \pi/4 > 0$ and $\sin^2 + \cos^2 = 1$, that $\sin \pi/4 = \sqrt{2}/2$, and consequently $\tan \pi/4 = 1$. Similarly, since $\cos \pi/6 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 3 \cdot \frac{\pi}{6} = 4\cos^3 \frac{\pi}{6} - 3\cos \frac{\pi}{6},$$

we have $\cos \pi/6 = \sqrt{3}/2$. It follows, since $\sin \pi/6 > 0$, that $\sin \pi/6 = \sqrt{1 - (\sqrt{3}/2)^2} = \frac{1}{2}$.

9. (a)

$$\begin{aligned}
\tan(x + y) &= \frac{\sin(x + y)}{\cos(x + y)} \\
&= \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y} \\
&= \frac{\dfrac{\sin x \cos y}{\cos x \cos y} + \dfrac{\cos x \sin y}{\cos x \cos y}}{\dfrac{\cos x \cos y}{\cos x \cos y} - \dfrac{\sin x \sin y}{\cos x \cos y}} \\
&= \frac{\tan x + \tan y}{1 - \tan x \tan y}.
\end{aligned}$$

(b) From part (a) we have

$$\begin{aligned}
\tan(\arctan x + \arctan y) &= \frac{\tan(\arctan x) + \tan(\arctan y)}{1 - \tan(\arctan x)\tan(\arctan y)} \\
&= \frac{x + y}{1 - xy},
\end{aligned}$$

provided that $\arctan x$, $\arctan y$, and $\arctan x + \arctan y = k\pi + \pi/2$. Since $-\pi/2 < \arctan x, \arctan y < \pi/2$, this is always the case except when $\arctan x + \arctan y = \pm\pi/2$, which is equivalent to $xy = 1$. From this equation we can conclude that

$$\arctan x + \arctan y = \arctan \frac{x + y}{1 - xy}$$

provided that $\arctan x + \arctan y$ lies in $(-\pi/2, \pi/2)$, which is true whenever $xy < 1$. (If $x, y > 0$ and $xy > 1$, so that $\arctan x + \arctan y > \pi/2$, then we must add π to the right side, and if $x, y < 0$ and $xy > 1$, so that $\arctan x + \arctan y < -\pi/2$, then we must subtract.)

11. The first formula is derived by subtracting the second of the following two equations from the first:

$$\begin{aligned}\cos(m-n)x &= \cos(mx-nx) = \cos mx\cos(-nx) - \sin mx\sin(-nx)\\ &= \cos mx\cos nx + \sin mx\sin nx,\\ \cos(m+n)x &= \cos mx\cos nx - \sin mx\sin nx.\end{aligned}$$

The other formulas are derived similarly.

12. It follows from Problem 11 that if $m = n$, then

$$\begin{aligned}\int_{-\pi}^{\pi}\sin mx\sin nx\,dx &= \frac{1}{2}\int_{-\pi}^{\pi}[\cos(m-n)x - \cos(m+n)x]\,dx\\ &= \frac{1}{2}\left[\frac{\sin(m-n)}{m-n} - \frac{\sin(m+n)}{m+n}\right]_{-\pi}^{\pi}\\ &= 0.\end{aligned}$$

But if $m = n$, then

$$\begin{aligned}\int_{-\pi}^{\pi}\sin mx\sin nx\,dx &= \frac{1}{2}\int_{-\pi}^{\pi}1 - \cos(m+n)x\,dx\\ &= \frac{1}{2}\left[\pi - \cos(m+n)\pi\right] - \left[-\pi - \cos(m+n)\pi\right]\\ &= \pi.\end{aligned}$$

The other formulas are proved similarly.

15. (a) We have

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x\\ &= 1 - 2\sin^2 x\\ &= 2\cos^2 x - 1.\end{aligned}$$

So

$$\sin^2 x = \frac{1-\cos 2x}{2},$$
$$\cos^2 x = \frac{1+\cos 2x}{2}.$$

(b) These formulas follow from part (a), because $\cos 2x \geq 0$ and $\sin x \geq 0$ (since $0 \leq x \leq \pi/2$).

(c)

$$\int_a^b \sin^2 x\,dx = \int_a^b \frac{1-\cos 2x}{2}\,dx = \frac{1}{2}(b-a) - \frac{1}{4}(\sin 2b - \sin 2a).$$
$$\int_a^b \cos^2 x\,dx = \int_a^b \frac{1+\cos 2x}{2}\,dx = \frac{1}{2}(b-a) + \frac{1}{4}(\sin 2b - \sin 2a).$$

19. (a) $\arctan 1 - \arctan 0 = \pi/4$.

(b) $\lim_x \arctan x - \arctan 0 = \pi/2$.

20. $\lim_x x \sin\frac{1}{x} = \lim_{x \to 0^+} \frac{1}{x}\sin x = 1$.

21. (a)
$$(\sin^\circ)(x) = \frac{\pi}{180}\cos\frac{\pi x}{180} = \frac{\pi}{180}\cos^\circ(x).$$
$$(\cos^\circ)(x) = \frac{\pi}{180}\cdot -\sin\frac{\pi x}{180} = -\frac{\pi}{180}\sin^\circ(x).$$

(b) $\lim_{x \to 0} \frac{\sin^\circ x}{x} = \lim_{x \to 0} \frac{\sin(\pi x/180)}{x} = \lim_{x \to 0} \frac{\pi}{180}\cdot\frac{\sin(\pi x/180)}{\pi x/180} = \frac{\pi}{180}$.
$\lim_x x \sin\frac{1}{x} = \lim_{x \to 0^+} \frac{\sin^\circ x}{x} = \frac{\pi}{180}$.

CHAPTER 18

1. (i) $e^{e^{e^x}} \cdot e^{e^{e^x}} \cdot e^{e^x} \cdot e^x$.

(iii) $(\sin x)^{\sin(\sin x)}[(\log(\sin x)) \cdot \cos(\sin x) \cdot \cos x + (\cos x/\sin x) \cdot \sin(\sin x)]$.

(v) $\sin x^{\sin x^{\sin x}}[(\log(\sin x)) \cdot \sin x^{\sin x} \cdot \{(\log(\sin x)) \cdot \cos x + (\cos x/\sin x) \cdot \sin x\} + (\cos x/\sin x) \cdot \sin x^{\sin x}]$.

(vii) $\left(\arcsin\frac{x}{\sin x}\right)^{\log(\sin e^x)} \cdot \left[\log\left(\arcsin\frac{x}{\sin x}\right) \cdot (\cos e^x)e^x \cdot \frac{\sin x - x\cos x}{\sin^2 x}\cdot\frac{1}{\sqrt{1-\left(\frac{x}{\sin x}\right)^2}} + \log(\sin e^x) \cdot \frac{\sin x - x\cos x}{\left(\arcsin\frac{x}{\sin x}\right)\sqrt{1-\left(\frac{x}{\sin x}\right)^2}\cdot\sin^2 x}\right] < 0$.

(ix) $(\log x)^{\log x} \cdot \left[\log(\log x) \cdot \frac{1}{x} + \log x \cdot \frac{1}{\log x}\cdot\frac{1}{x}\right]$.

5. (i) 0.

(iii) $\frac{1}{6}$.

(v) $\frac{1}{3}$.

7. (a)
$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$$
$$= \left(\frac{e^{2x}}{4} + \frac{1}{2} + \frac{e^{-2x}}{4}\right) - \left(\frac{e^{2x}}{4} - \frac{1}{2} + \frac{e^{-2x}}{4}\right)$$
$$= 1.$$

(c)

$$\sinh x\cosh y + \cosh x\sinh y = \frac{e^x - e^{-x}}{2}\cdot\frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2}\cdot\frac{e^y - e^{-y}}{2}$$

$$= \frac{e^{x+y}}{4} + \frac{e^{-x-y}}{4} - \frac{e^{-x+y}}{4} + \frac{e^{x-y}}{4} + \frac{e^{x+y}}{4} + \frac{e^{-x-y}}{4} + \frac{e^{-x+y}}{4} - \frac{e^{x-y}}{4}$$

$$= \frac{e^{x+y} + e^{-(x+y)}}{2} = \sinh(x + y).$$

(e) Since

$$\sinh x = \frac{e^x + e^{-x}}{2},$$

we have

$$\sinh(-x) = \frac{e^{-x} - e^{-(-x)}}{2} = \cosh x.$$

(g) Since

$$\tanh x = \frac{\sinh x}{\cosh x},$$

we have

$$\tanh(-x) = \frac{(\cosh x)^2 - (\sinh x)^2}{\cosh^2 x}$$

$$= \frac{1}{\cosh^2 x} \quad \text{by part (a).}$$

8. (a) If $y = \arg\cosh x$, then $x \geq 0$ and

$$x = \cosh y = \sqrt{1 + \sinh^2 y} \quad \text{by part (a).}$$

So

$$\sinh(\arg\cosh x) = \sinh y = \sqrt{x^2 - 1} \quad \text{since } \sinh y \geq 0 \text{ for } y \geq 0.$$

(c)

$$(\arg\sinh)'(x) = \frac{1}{\sinh'(\arg\sinh(x))}$$

$$= \frac{1}{\cosh(\arg\sinh(x))}$$

$$= \frac{1}{\sqrt{1 + x^2}} \quad \text{by part (b).}$$

(e)

$$(\arg\tanh)'(x) = \frac{1}{\tanh'(\arg\tanh(x))},$$

$$= \cosh^2(\arg\tanh(x)).$$

Now,

$$\tanh^2 y + \frac{1}{\cosh^2 y} = 1 \quad \text{by Problem 7(b)}$$

so

$$\tanh^2(\arg\tanh(x)) + \frac{1}{\cosh^2(\arg\tanh(x))} = 1,$$

or

$$\cosh^2(\arg\tanh(x)) = \frac{1}{1 - x^2}.$$

9. (a) If $y = \arg\sinh x$, then

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

so

$$e^y - e^{-y} = 2x,$$
$$e^{2y} - 2xe^y - 1 = 0,$$
$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

so

$$e^y = x + \sqrt{1 + x^2} \quad \text{since} \quad e^y > 0$$

or

$$y = \arg\sinh x = \log(x + \sqrt{1 + x^2}).$$

Similarly,

$$\arg\cosh x = \log(x + \sqrt{x^2 - 1}),$$
$$\arg\tanh x = \tfrac{1}{2}\log(1 + x) - \tfrac{1}{2}\log(1 - x).$$

(b)

$$\int_a^b \frac{1}{\sqrt{1 + x^2}}\,dx = \arg\sinh b - \arg\sinh a \quad \text{by Problem 8(c)}$$
$$= \log(b + \sqrt{1 + b^2}) - \log(a + \sqrt{1 + a^2}).$$

$$\int_a^b \frac{1}{\sqrt{x^2 - 1}}\,dx = \log(b + \sqrt{b^2 - 1}) - \log(a + \sqrt{a^2 - 1}).$$

$$\int_a^b \frac{1}{1 - x^2}\,dx = \frac{1}{2}[\log(1 + b) - \log(1 - b) - \log(1 + a) + \log(1 - a)].$$

12. (a) $\lim_x a^x = \lim_x e^{x \log a}$. Since $\log a < 0$, we have $\lim_x x \log a = -\infty$, so $\lim_x e^{x \log a} = 0$.

(c) $\lim_x \frac{(\log x)^n}{x} = \lim_y \frac{y^n}{e^y} = 0.$

(e) $\lim_{x \to 0^+} x^x = \lim_{x \to 0^+} e^{x \log x}$. Now, $\lim_{x \to 0^+} x \log x = 0$ by part (d), so $\lim_{x \to 0^+} x^x = 1$.

16. (a) $\lim_{y \to 0} \log(1+y)/y = \log(1) = 1.$

(b) $\lim_{x} x \log(1+1/x) = \lim_{y \to 0^+} \log(1+y)/y = 1.$

(c)

$$\begin{aligned} e = \exp(1) &= \exp(\lim_x x \log(1+1/x)) \\ &= \lim_x \exp(x \log(1+1/x)) \\ &= \lim_x (1+1/x)^x. \end{aligned}$$

(The starred equality depends on the continuity of exp at 1, and can be justified as follows. For every $\varepsilon > 0$ there is some $\delta > 0$ such that $|e - \exp y| < \varepsilon$ for $|y - 1| < \delta$. Moreover, there is some N such that $|x \log(1+1/x) - 1| < \delta$ for $x > N$. So $|e - \exp(x \log(1+1/x))| < \varepsilon$ for $x > N$.

(d)

$$\begin{aligned} e^a = [\lim_x (1+1/x)^x]^a &= \lim_x (1+1/x)^{ax} \\ &= \lim_{ax} (1+1/x)^{ax} \\ &= \lim_y (1+a/y)^y. \end{aligned}$$

18. After one year the number of dollars yielded by an initial investment of one dollar will be

$$\lim_x (1 + a/100x)^x = e^{a/100}.$$

19. (a) Clearly $f'(x) = 1/x$ for $x > 0$. If $x < 0$, then $f(x) = \log(-x)$, so $f'(x) = (-1) \cdot 1/(-x) = 1/x$.

(b) We can write $\log|f|$ as $g \circ f$ where $g(x) = \log|x|$ is the function of part (a). So $(\log|f|)' = (g' \circ f) \cdot f' = 1/f \cdot f'$.

20. (c) Let $g(x) = f(x)/e^{cx}$. Then

$$g'(x) = \frac{e^{cx} f'(x) - f(x) c e^{cx}}{e^{2cx}} = 0,$$

so there is some number k such that $g(x) = k$ for all x.

21. (a) According to Problem 20, there is some k such that $A(t) = ke^{ct}$. Then $k = ke^{0 \cdot t} = A_0$. So $A(t) = A_0 e^{ct}$.

(b) If $A(t + \alpha) = A(t)/2$, then

$$A_0 e^{ct + c\alpha} = \frac{A_0 e^{ct}}{2},$$

so $e^{ct} e^{c\alpha} = e^{ct}/2$ or $e^{c\alpha} = \frac{1}{2}$, so $\alpha = -(\log 2)/c$. It is easy to check that this α does work.

22. Newton's law states that, for a certain (positive) number c,

$$T'(t) = c(T - M),$$

which can be written

$$(T - M)' = c(T - M).$$

So by Problem 20 there is some number k such that

$$T(t) - M = ke^{ct},$$

and $k = ke^{0 \cdot t} = T(0) = T_0$. So $T(t) = M + T_0 e^{ct}$.

CHAPTER 19

1. (i) $\sqrt[5]{x^3} + \sqrt[6]{x^4} = x^{1/10} + x^{-1/3}$.

(ii) $$\frac{1}{x - 1 + \sqrt{x+1}} = \frac{x - 1 - \sqrt{x+1}}{-2}.$$

(iii) $(e^x + e^{2x} + e^{3x})/e^{4x} = e^{-3x} + e^{-2x} + e^{-x}$.

(iv) $a^x/b^x = (a/b)^x = e^{x \log(a/b)}$.

(v) $\tan^2 x = \sec^2 x - 1$.

(vi) $$\frac{1}{a^2 + x^2} = \frac{1/a^2}{\left(\frac{x}{a}\right)^2 + 1}.$$

(vii) $$\frac{1}{a^2 - x^2} = \frac{1/a}{1 - (x/a)^2}.$$

(viii) $$\frac{1}{1 + \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x.$$

(ix) $$\frac{8x^2 + 6x + 4}{x + 1} = 8x - 2 + \frac{6}{x+1}.$$

(x) $$\frac{1}{2x - x^2} = \frac{1}{1 - (x-1)^2}.$$

2. (i) $-\cos e^x$. (Let $u = e^x$.)

(iii) $(\log x)^2/2$. (Let $u = \log x$.)

(v) e^{e^x}. (Let $u = e^{e^x}$.)

(vii) $2e^{\sqrt{x}}$. (Let $u = \sqrt{x}$.)

(ix) $-(\log(\cos x))^2/2$. (Let $u = \log(\cos x)$.)

3. (i) $$\int x^2 e^x\,dx = x^2 e^x - \int 2x e^x\,dx = x^2 e^x - 2x e^x - \int e^x\,dx$$
$$= x^2 e^x - 2x e^x + 2e^x.$$

(iii) We have

$$\int e^{ax} \sin bx\,dx = \frac{e^{ax}\sin bx}{a} - \frac{b}{a}\int e^{ax}\cos bx\,dx$$
$$= \frac{e^{ax}\sin bx}{a} - \frac{b}{a}\left(\frac{e^{ax}\cos bx}{a} - \frac{b}{a}\int e^{ax}(-\sin bx)\,dx\right),$$

so

$$\int e^{ax}\sin bx\,dx = \frac{a}{a^2+b^2}e^{ax}\sin bx - \frac{b}{a^2+b^2}e^{ax}\cos bx.$$

(v) Using the result $\int (\log x)^2\,dx = x(\log x)^2 - 2x(\log x) + 2x$ from the text, we have

$$\begin{aligned}\int (\log x)^3\,dx &= [x(\log x)^2 - 2x(\log x) + 2x]\log x \\ &\quad - \int \frac{1}{x}[x(\log x)^2 - 2x(\log x) + 2x]\,dx \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x\log x \\ &\quad - \int (\log x)^2\,dx + 2[x\log x - x] - 2x \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x\log x \\ &\quad - [x(\log x)^2 - 2x(\log x) + 2x] + 2[x\log x - x] - 2x \\ &= x(\log x)^3 - 3x(\log x)^2 + 6x\log x - 6x.\end{aligned}$$

(vii)

$$\begin{aligned}\int \sec^3 x\,dx &= \int (\sec^2 x)(\sec x)\,dx = \tan x\sec x - \int (\tan x)(\sec x\tan x)\,dx \\ &= \tan x\sec x - \int \sec x(\sec^2 x - 1)\,dx \\ &= \tan x\sec x - \int \sec^3 x\,dx + \int \sec x\,dx,\end{aligned}$$

so

$$\int \sec^3 x\,dx = \tfrac{1}{2}[\tan x\sec x + \log(\sec x + \tan x)].$$

(ix)

$$\begin{aligned}\int x\log x\,dx &= \frac{2x^{3/2}}{3}\log x - \frac{2}{3}\int x^{3/2}\cdot\frac{1}{x}\,dx \\ &= \frac{2x^{3/2}}{3}\log x - \frac{2}{3}\int x^{1/2}\,dx \\ &= \frac{2x^{3/2}}{3}\log x - \frac{4}{9}x^{3/2}.\end{aligned}$$

4. (i) Let $x = \sin u$, $dx = \cos u\,du$. The integral becomes

$$\int \frac{\cos u\,du}{1 - \sin^2 u} = \int 1\,du = u = \arcsin x.$$

(iii) Let $x = \sec u$, $dx = \sec u\tan u\,du$. The integral becomes

$$\int \frac{\sec u\tan u\,du}{\sec^2 u - 1} = \int \sec u\,du = \log(\sec u + \tan u)$$
$$= \log(x + \sqrt{x^2 - 1}).$$

(v) Let $x = \sin u$, $dx = \cos u\,du$. The integral becomes

$$\int \frac{\cos u\,du}{\sin u\sqrt{1-\sin^2 u}} = \int \csc u\,du = -\log(\csc u + \cot u)$$
$$= -\log\left(\frac{1}{x} + \frac{\sqrt{1-x^2}}{x}\right).$$

(vii) Let $x = \sin u$, $dx = \cos u\,du$. The integral becomes

$$\int (\sin^3 u\cos u)\cos u\,du = \int \sin^3 u\cos^2 u\,du = \int (\sin u)(1-\cos^2 u)\cos^2 u\,du$$
$$= \int (\sin u)(\cos^2 u - \cos^4 u)\,du = -\frac{\cos^3 u}{3} + \frac{\cos^5 u}{5}$$
$$= -\frac{(1-x^2)^{3/2}}{3} + \frac{(1-x^2)^{5/2}}{5}.$$

(ix) Let $x = \tan u$, $dx = \sec^2 u\,du$. The integral becomes

$$\int \sec u\sec^2 u\,du = \int \sec^3 u\,du$$
$$= \tfrac{1}{2}[\tan u\sec u + \log(\sec u + \tan u)] \quad \text{by Problem 3(vii)}$$
$$= \tfrac{1}{2}[x\sqrt{1+x^2} + \log(x + \sqrt{1+x^2})].$$

5. (i) Let $u = \sqrt{x+1}$, $x = u^2 - 1$, $dx = 2u\,du$. The integral becomes

$$\int \frac{2u\,du}{1+u} = \int \left(2 + \frac{-2}{1+u}\right) du$$
$$= 2u - 2\log(1+u) = 2\sqrt{x+1} - 2\log(1+\sqrt{x+1}).$$

(iii) Let $u = x^{1/6}$, $x = u^6$, $dx = 6u^5\,du$. The integral becomes

$$\int \frac{6u^5\,du}{u^3+u^2} = 6\int \left(u^2 - u + 1 - \frac{1}{u+1}\right) du = 2u^3 - 3u^2 + 6u - 6\log(u+1)$$
$$= 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\log\sqrt[6]{x}+1.$$

(v) Let $u = \tan x$, $x = \arctan u$, $dx = du/(1+u^2)$. The integral becomes

$$\int \frac{du}{(1+u^2)(2+u)} = \frac{1}{5}\int \left(\frac{1}{2+u} - \frac{u-2}{1+u^2}\right) du$$
$$= \frac{1}{5}\int \frac{du}{2+u} - \frac{1}{10}\int \frac{2u}{1+u^2}\,du + \frac{2}{5}\int \frac{du}{1+u^2}$$
$$= \frac{1}{5}\log(2+u) - \frac{1}{10}\log(1+u^2) + \frac{2}{5}\arctan u$$
$$= \frac{1}{5}\log(2+\tan x) - \frac{1}{10}\log(1+\tan^2 x) + \frac{2}{5}x.$$

(vii) Let $u = 2^x$, $x = (\log u)/(\log 2)$, $dx = du/(u \log 2)$. The integral becomes

$$\int \frac{u^2 + 1}{(u+1)u}\,du = \frac{1}{\log 2}\int \left(1 + \frac{1 - u}{u(u+1)}\right) du$$

$$= \frac{1}{\log 2}\int \left(1 + \frac{1}{u} - \frac{2}{u+1}\right) du$$

$$= \frac{1}{\log 2}[u + \log u - 2\log(u+1)]$$

$$= \frac{1}{\log 2}[2^x + x\log 2 - 2\log(2^x + 1)].$$

(ix) Let $u = \sqrt{x}$, $x = u^2$, $dx = 2u$. The integral becomes

$$\int \frac{1 - u^2}{1 - u}\,2u\,du.$$

Now let $u = \sin y$, $du = \cos y\,dy$. The integral becomes

$$\int \frac{2\cos y \sin y \cos y}{1 - \sin y}\,dy = 2\int \frac{(1 - \sin^2 y)\sin y}{1 - \sin y}\,dy$$

$$= 2\int (1 + \sin y)\sin y\,dy$$

$$= 2\int \sin y\,dy + \int 1 - \cos 2y\,dy$$

$$= -2\cos y + y - \frac{\sin 2y}{2} = -2\cos y + y - \sin y \cos y$$

$$= -2\sqrt{1 - u^2} + \arcsin u - u\sqrt{1 - u^2}$$

$$= -2\sqrt{1 - x} + \arcsin\sqrt{x} - \sqrt{x}\sqrt{1 - x}.$$

The substitution $u = \sqrt{1 - x}$, $x = 1 - u^2$, $dx = -2u\,du$ leads to

$$\int \frac{-2u^2\,du}{1 - \sqrt{1 - u^2}}$$

and the substitution $u = \sin y$ then leads to

$$\int \frac{-2\sin^2 y \cos y\,dy}{1 - \cos y} = -2\sin y - y - \sin y \cos y$$

$$= -2u - \arcsin u - u\sqrt{1 - u^2}$$

$$= -2\sqrt{1 - x} - \arcsin\sqrt{1 - x} - \sqrt{1 - x}\sqrt{x}.$$

These answers agree, since

$$\arcsin\sqrt{x} = \frac{\pi}{2} - \arcsin\sqrt{1 - x}$$

(check this by comparing their derivatives and their values for $x = 0$).

6. In these problems s will denote the original integral.

(i)
$$I = \int \frac{2}{x-1}\,dx + \int \frac{3}{(x+1)^2}\,dx$$
$$= 2\log(x-1) - \frac{3}{x+1}.$$

(iii)
$$I = \int \frac{1}{(x-1)^2}\,dx + \int \frac{4}{(x+1)^3}\,dx$$
$$= -\frac{1}{(x-1)} - \frac{2}{(x+1)^2}.$$

(v)
$$I = \frac{1}{2}\int \frac{2x}{x^2+1}\,dx + \int \frac{4}{x^2+1}\,dx$$
$$= \tfrac{1}{2}\log(x^2+1) + 4\arctan x.$$

(vii)
$$I = \int \frac{1}{(x+1)}\,dx + \int \frac{2x}{(x^2+x+1)}\,dx$$
$$= \int \frac{1}{x+1}\,dx + \int \frac{2x+1}{x^2+x+1}\,dx - \int \frac{1}{x^2+x+1}\,dx.$$

Now
$$\int \frac{1}{x^2+x+1}\,dx = \int \frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}\,dx$$
$$= \frac{4}{3}\int \frac{1}{\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right)^2+1}\,dx$$
$$= \frac{4}{3}\cdot\frac{\sqrt{3}}{2}\arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right)$$
$$= \frac{2\sqrt{3}}{3}\arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right),$$

so
$$I = \log(x+1) + \log(x^2+x+1) - \frac{2\sqrt{3}}{3}\arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right).$$

(ix)
$$I = \int \frac{2x+1}{(x^2+x+1)^2}\,dx - \int \frac{1}{(x^2+x+1)^2}\,dx$$
$$= \int \frac{2x+1}{(x^2+x+1)^2}\,dx - \frac{16}{9}\int \frac{1}{\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right)^2+1}^2\,dx.$$

Now the substitution
$$u = \frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right), \quad dx = \frac{\sqrt{3}}{2}\,du$$

changes the second integral to

$$\int_{16/9}^{3} \frac{du}{(u^2+1)^2}.$$

Using the reduction formula, this can be written

$$\int_{8/9}^{3} \frac{u}{2(u^2+1)} + \frac{1}{2}\int_{8/9}^{3} \frac{du}{u^2+1} = \int_{8/9}^{3} \frac{\log(u^2+1)}{4} + \frac{1}{2}\arctan u,$$

so

$$I = \int \frac{1}{x^2+x+1} = \int_{2/9}^{3} \log\frac{4}{3}(x^2+x+1) - \int_{4/9}^{3} \arctan\left(\frac{2}{3}\left(x+\frac{1}{2}\right)\right).$$

13. The equation $\int e^x \sin x\,dx = e^x \sin x - \int e^x \cos x - \int e^x \sin x\,dx$ means that any function F with $F'(x) = e^x \sin x$ can be written $F(x) = e^x \sin x - e^x \cos x - G(x)$ where G is another function with $G'(x) = e^x \sin x$. Of course, $G = F + c$ for some number c, but it is not necessarily true that $F = G$.

15. (a)
$$\int \arcsin x\,dx = \int 1 \cdot \arcsin x\,dx = x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\,dx$$
$$= x\arcsin x + \sqrt{1-x^2}.$$

16. (a)
$$\int \sin^4 x\,dx = -\frac{\sin^3 x\cos x}{4} + \frac{3}{4}\int \sin^2 x\,dx$$
$$= -\frac{\sin^3 x\cos x}{4} + \frac{3}{4}\int \frac{\sin x\cos x}{2} + \frac{1}{2}\int 1\,dx$$
$$= -\frac{\sin^3 x\cos x}{4} - \frac{3\sin x\cos x}{8} + \frac{3}{8}x.$$
$$\int \sin^4 x\,dx = \int \left(\frac{1-\cos 2x}{2}\right)^2 dx = \int \frac{1}{4}\left(\cos^2 2x + \cos^2 2x\right)\,dx$$
$$= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4}\int \frac{1+\cos 4x}{2}\,dx$$
$$= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4}\left(\frac{x}{2} + \frac{\sin 4x}{8}\right)$$
$$= \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}.$$

(b) It follows that these two answers are the same, since they have the same value for $x = 0$.

20. (a)

$$\int \sin^n x\,dx = \int (\sin x)(\sin^{n-1} x)\,dx$$
$$= -\cos x \sin^{n-1} x + (n-1)\int \cos x(\sin^{n-2} x)\cos x\,dx$$
$$= -\cos x \sin^{n-1} x + (n-1)\int (\sin^{n-2} x - \sin^n x)\,dx,$$

so

$$\int \sin^n x\,dx = -\frac{1}{n}\cos x \sin^{n-1} x + \frac{n-1}{n}\int \sin^{n-2} x\,dx.$$

(b)

$$\int \cos^n x\,dx = \int (\cos x)(\cos^{n-1} x)\,dx$$
$$= \sin x \cos^{n-1} x + (n-1)\int \sin x(\cos^{n-2} x)\sin x\,dx$$
$$= \sin x \cos^{n-1} x + (n-1)\int (\cos^{n-2} x - \cos^n x)\,dx,$$

so

$$\int \cos^n x\,dx = \frac{1}{n}\sin x \cos^{n-1} x + \frac{n-1}{n}\int \cos^{n-2} x\,dx.$$

(c)

$$\int \frac{dx}{(1+x^2)^n} = \int \frac{dx}{(1+x^2)^{n-1}} - \int \frac{x^2\,dx}{(1+x^2)^n}$$
$$= \int \frac{dx}{(1+x^2)^{n-1}} - \int x\cdot\frac{x}{(1+x^2)^n}\,dx$$
$$= \int \frac{dx}{(1+x^2)^{n-1}} - \frac{x}{2(1-n)(1+x^2)^{n-1}} - \int \frac{dx}{2(1-n)(1+x^2)^{n-1}}$$

so

$$\int \frac{dx}{(1+x^2)^n} = \frac{1}{2(n-1)}\frac{x}{(x^2+1)^{n-1}} - \frac{(2n-3)}{2(n-1)}\int \frac{1}{(x^2+1)^{n-1}}\,dx.$$

We can also use the substitution $x = \tan u$, $dx = \sec^2 u\,du$, which changes the integral to

$$\int \frac{\sec^2 u\,du}{\sec^{2n} u} = \int \cos^{2n-2} u\,du$$
$$= \frac{1}{2n-2}\cos^{2n-3} u \sin u + \frac{2n-3}{2n-2}\int \cos^{2n-4} u\,du$$
$$= \frac{1}{2n-2}\cdot\frac{1}{(1+x^2)^{2n-3}}\cdot\frac{x}{1+x^2} + \frac{2n-3}{2n-2}\int \frac{dx}{(1+x^2)^{n-1}}$$
$$= \frac{1}{2(n-1)}\frac{x}{(1+x^2)^{n-1}} + \frac{2n-3}{2n-2}\int \frac{dx}{(1+x^2)^{n-1}}.$$

CHAPTER 20

1. (i) $P_{3,0}(x) = e + ex + ex^2 + (5e/3!)x^3$.

   (iii) $P_{2n,\pi/2}(x) = 1 - \dfrac{(x-\pi/2)^2}{2!} + \dfrac{(x-\pi/2)^4}{4!} - \cdots + \dfrac{(-1)^n(x-\pi/2)^{2n}}{(2n)!}$.

   (v) $P_{n,1}(x) = e + e(x-1) + \dfrac{e(x-1)^2}{2!} + \cdots + \dfrac{e(x-1)^n}{n!}$.

   (vii) $P_{4,0}(x) = x + x^3$.

   (ix) $P_{2n+1,0}(x) = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$.

2. If f is a polynomial function of degree n, then $f^{(n+1)} = 0$. It follows from Taylor's Theorem that $R_{n,a}(x) = 0$, so $f(x) = P_{n,a}(x)$.

   (i) $-12 + 2(x-3) + (x-3)^2$.

   (iii) $243 + 405(x-3) + 270(x-3)^2 + 90(x-3)^3 + 15(x-3)^4 + (x-3)^5$.

3. (i) $\displaystyle\sum_{i=0}^{9} \frac{(-1)^i}{(2i+1)!}$ since $\dfrac{1}{(2n+2)!} < 10^{-17}$ for $2n+2 \geq 19$, or $n \geq 9$.

   (iii) $\displaystyle\sum_{i=0}^{8} \frac{(-1)^i}{2^i(2i+1)!}$ since $\dfrac{1}{2^{2n+2}(2n+2)!} < 10^{-20}$ for $2n+2 \geq 18$, or $n \geq 8$.

   (v) $\displaystyle\sum_{i=0}^{11} \frac{2^i}{i!}$ since $\dfrac{3^2 \cdot 2^{n+1}}{(n+1)!} < 10^{-5}$ for $n+1 \geq 12$, or $n \geq 11$.

8. (i) $c_i = a_i + b_i$.

   (iii) $c_i = (i+1)a_i$.

   (v) $c_0 = \displaystyle\int_0^a f(t)\,dt$; $c_i = a_{i-1}/i$ for $i > 0$.

CHAPTER 22

1. (i) $1 - n/(n+1) = 1/(n+1) < \varepsilon$ for $n+1 > 1/\varepsilon$.

   (iii) $\lim_n \sqrt{n^2+1} - \sqrt{n} + 1 = \lim_n \sqrt{n^2+1} - \sqrt{n^2} + \lim_n \sqrt{n} - \sqrt{n+1}$
   $= 0 + 0 = 0$. (Each of these two limits can be proved in the same way that $\lim_n \sqrt{n+1} - \sqrt{n} = 0$ was proved in the text.)

(v) Clearly $\lim_n (\log a)/n = 0$. So $\lim_n \sqrt[n]{a} = \lim_n e^{(\log a)/n} = e^0$ (by Theorem 1) $= 1$.

(vii) $\sqrt[n]{n^2} \le \sqrt[n]{n^2 + n} \le \sqrt[n]{2n^2}$, so $(\sqrt[n]{n})^2 \le \sqrt[n]{n^2 + n} \le \sqrt[n]{2}(\sqrt[n]{n})^2$, and $\lim_n (\sqrt[n]{n})^2 = \lim_n \sqrt[n]{2}(\sqrt[n]{n})^2 = 1$ by parts (v) and (vi).

(ix) Clearly $(n) \le \log_2 n$, and $\lim_n (\log_2 n)/n = 0$.

5. (a) If $0 < a < 2$, then $a^2 < 2a < 4$, so $a < \sqrt{2a} < 2$.

(b) Part (a) shows that

$$2 < \sqrt{2} \cdot 2 < \sqrt{2} \cdot \sqrt{2} \cdot 2 < \cdots < 2,$$

so the sequence converges by Theorem 2.

(c) If this sequence is denoted by $\{a_n\}$, then the sequence $\{\sqrt{2a_n}\}$ is the same as $\{a_{n+1}\}$. So the hint shows that $l = \sqrt{2}l$, or $l = 2$.

8. If x is rational, then $n!\,x$ is a multiple of $2k$ for sufficiently large n, so $(\cos n!\,x)^{2k} = 1$ for all such n, so $\lim_n (\lim_k (\cos n!\,x)^{2k}) = 1$. If x is irrational, then $n!\,x$ is not a multiple of $2k$ for any n, so $|\cos n!\,x| < 1$, so $\lim_k (\cos n!\,x)^{2k} = 0$, so $f(x) = 0$.

9. (i) $\int_0^1 e^x\,dx = e - 1$. (Use partitions of $[0, 1]$ into n equal parts.)

(iii) $\int_0^1 \frac{1}{1+x}\,dx = \log 2$.

(v) $\int_0^1 \frac{1}{(1+x)^2}\,dx = \frac{1}{2}$.

CHAPTER 23

1. (i) (Absolutely) convergent, since $|(\sin n)/n^2| \le 1/n^2$.

(iii) Divergent, since the first n terms have sum $\frac{1}{2} + \cdots + 1/n$. (Leibniz's Theorem does not apply since the terms are not decreasing in absolute value.)

(v) Divergent, since

$$\frac{1}{\sqrt[3]{n^2 - 1}} \ge \frac{1}{2n^{2/3}}$$

for sufficiently large n.

(vii) Convergent, since

$$\lim_n \frac{(n+1)^2/(n+1)!}{n^2/n!} = \lim_n \left(\frac{n+1}{n}\right)^2 \cdot \frac{1}{n+1} = 0.$$

(ix) Divergent, since $1/(1 + \log n) > 1/n$.

(xi) Convergent, since $1/(1 + \log n)^n < \frac{1}{2^n}$ for $n > 9$.

(xiii) Divergent, since

$$\frac{n^2}{n^3 + 1} > \frac{1}{2n}$$

for large enough n.

(xv) Divergent, since

$$\int_2^N \frac{1}{x \log x}\,dx = \log(\log N) - \log(\log 2) \qquad \text{as } N \to \infty.$$

(Notice that $f(x) = 1/(x \log x)$ is decreasing on $[2, \infty)$, since

$$f'(x) = \frac{-[1 + \log x]}{(x \log x)^2} < 0 \qquad \text{for } x > 1.$$

(xvii) Convergent, since $e^n/n^2(\log n) < 1/n^2$ for $n > 2$.

(xix) Convergent, since

$$\lim_n \frac{2^{n+1}(n+1)!/(n+1)^{n+1}}{2^n n!/n^n} = \lim_n \frac{2(n+1)n^n}{(n+1)^{n+1}}$$
$$= \lim_n \frac{2}{\left(1 + \frac{1}{n}\right)^n} = \frac{2}{e},$$

by Problem 18-16.

5. (a) For each N we clearly have

$$0 \le \sum_{n=1}^{N} a_n 10^{-n} < 9 \sum_{n=1}^{N} 10^{-n} = 1,$$

so $\sum_{n=1}^{\infty} a_n 10^{-n}$ converges by the boundedness criterion, and lies between 0 and 1. (Actually, this number is denoted by $.a_1a_2a_3a_4\ldots$ only when the sequence $\{a_n\}$ is not eventually 0.)

17. The area of the shaded region is $\frac{1}{2}$. The integral is

$$\tfrac{1}{2}([1 - \tfrac{1}{2}] + [\tfrac{1}{4} - \tfrac{1}{8}] + [\tfrac{1}{16} - \tfrac{1}{32}] + \cdots) - \tfrac{1}{2}([\tfrac{1}{2} - \tfrac{1}{4}] + [\tfrac{1}{8} - \tfrac{1}{16}] + \cdots)$$
$$= \tfrac{1}{2}(\tfrac{1}{2} + \tfrac{1}{8} + \tfrac{1}{32} + \tfrac{1}{128} + \cdots) - \tfrac{1}{2}(\tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{64} + \tfrac{1}{256} + \cdots)$$
$$= \tfrac{1}{4}(1 + \tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{64} + \cdots) - \tfrac{1}{8}(1 + \tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{64} + \cdots)$$
$$= \frac{1}{8}\left(1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \cdots\right)$$
$$= \frac{1}{8} \cdot \frac{1}{1 - \frac{1}{4}}$$
$$= \frac{1}{6}.$$

CHAPTER 24

1. (i)

$$f(x) = \lim_n f_n(x) = \begin{cases} 0, & x = 0 \\ 1, & 0 < x \le 1. \end{cases}$$

$\{f_n\}$ does not converge uniformly to f.

(iii) $f(x) = \lim_n f_n(x) = 0$ (since $\lim_n x^n =$ for $x > 1$). The sequence $\{f_n\}$ does not converge uniformly to; in fact, for any n we have $f_n(x)$ large for suf ciently large x.

(v) $f(x) = \lim_n f_n(x) = 0$, and $\{f_n\}$ converges uniformly to 0, since $|f_n(x)| \quad 1/n$ for all x.

3. (i) $\check{S}\frac{1}{a}\check{S}\frac{x}{a^2}\check{S}\frac{x^2}{a^3}\check{S}\cdots$.

(iii) $\sum_{k=0}^{\infty}(\check{S}1)^k\binom{\check{S}\frac{1}{2}}{k}x^k$.

(v) $\sum_{k=0}^{\infty}(\check{S}1)^k\frac{\binom{\check{S}\frac{1}{2}}{k}}{2k+1}x^{2k+1}$.

4. (i) $e^{\check{S}x}$.

(iii) If

$$f(x) = \frac{x^2}{2}\check{S}\frac{x^3}{3\cdot 2}+\frac{x^4}{4\cdot 3}\check{S}\cdots, \qquad |x| \quad 1$$

then

$$f\ (x) = x\check{S}\frac{x^2}{2}+\frac{x^3}{3}\check{S}\cdots$$
$$= \log(1+x) \qquad |x| < 1,$$

so for $|x| < 1$ we have $f(x) = (1+x)\log(1+x)\check{S}(1+x)+c$ for some number c. Since $f(\ 0) = 0$, we have $c = 1$, so $f(x) = (1+x)\cdot\log(1+x)\check{S}x$ for $|x| < 1$.

6. Since

$$\sin x = \sum_{n=0}^{\infty}\frac{(\check{S}1)^n x^{2n+1}}{(2n+1)!}$$

we have

$$f(x) = \sum_{n=0}^{\infty}\frac{(\check{S}1)^n x^{2n}}{(2n+1)!}$$

(notice that the right side is 1 for $x = 0$). So

$$f^{(k)}(0) = \begin{cases}\dfrac{(\check{S}1)^n}{(2n+1)!}, & k = 2n\\ 0, & k \text{ odd.}\end{cases}$$

CHAPTER 25

1. (i) $|3+4i| = 5; \quad = \arctan\frac{4}{3}$.

(iii) $|(1+i)^5| = (|1+i|)^5 = (\ 2)^5$; since $/4 = \arctan 1$, 1 is an argument for $1+i$, an argument for $(1+i)^5$ is $5/4$.

(v) $|(|3+4i|)| = |5| = 5; \quad = 0$.

2. (i)

$$x = \frac{-i \pm \sqrt{-1-4}}{2} = \frac{-i \pm \sqrt{-5}\,i}{2} = \frac{(-1+\sqrt{5})i}{2} \text{ or } \frac{(-1-\sqrt{5})i}{2}.$$

(iii) $x^2 + 2ix - 1 = (x+i)^2$, so the only solution is $x = -i$.

(v) $x^3 - x^2 - x - 2 = (x-2)(x^2+x+1)$. The solutions are

$$2, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

3. (i) All $z = iy$ with y real.

(iii) All z on the perpendicular bisector of the line segment between a and b.

(v) For $z = x + iy$, we clearly need $1 - x = 1 -$ real part of $z > 0$, or $x < 1$. For such x, our condition $x^2 + y^2 < 1 - x$ is equivalent to $x^2 + y^2 < (1-x)^2$, or $x < (1 - y^2)/2$. The set of points with $x = (1 - y^2)/2$ is the parabola pointing along the second axis, with the point $(0, 1/2)$ closest to the origin, and intersecting the line $x = 1$ at $(1, 1)$ and $(1, -1)$. The area bounded by this parabola and the line $x = 1$ is the desired set of points.

4. $|x + iy|^2 = x^2 + y^2 = x^2 + (-y)^2 = |x - iy|^2$.
$(z + \bar{z})/2 = [(x + iy) + (x - iy)]/2 = x$.
$(z - \bar{z})/2 = [(x + iy) - (x - iy)]/2i = y$.

5. $|z + w|^2 + |z - w|^2 = (z + w)(\bar{z} + \bar{w}) + (z - w)(\bar{z} - \bar{w}) = 2z\bar{z} + 2w\bar{w} = 2(|z|^2 + |w|^2)$. Geometrically, this says that the sum of the squares of the diagonals of a parallelogram equal the sum of the squares of the sides.

CHAPTER 27

1. (i) Converges absolutely, since $|(1+i)^n/n!| = (\sqrt{2})^n/n!$, and $\sum_{n=1} (\sqrt{2})^n/n!$ converges.

(iii) Converges, but not absolutely, since the real terms form the series

$$-\frac{1}{2} + \frac{1}{4} - \frac{1}{6} + \frac{1}{8} - \cdots$$

and the imaginary terms form the series

$$i\left(\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

(v) Diverges, since the real terms form the series

$$\frac{\log 3}{3} + 2\frac{\log 4}{4} + \frac{\log 5}{5} + \frac{\log 7}{7} + 2\frac{\log 8}{8} + \frac{\log 9}{9} + \cdots.$$

2. (i) The limit

$$\lim_n \frac{|z|^{n+1}/(n+1)^2}{|z|^n/n^2} = \lim_n \left(\frac{n+1}{n}\right)^2 |z| = |z|$$

is < 1 for $|z| < 1$, but > 1 for $|z| > 1$.

(iii) The limit

$$\lim_n \frac{|z|^{n+1}}{|z|^n} = |z|$$

is < 1 for $|z| < 1$ but > 1 for $|z| > 1$.

(v) The limit

$$\lim_n \frac{2^{n+1}|z|^{(n+1)!}}{2^n|z|^{n!}} = \lim_n 2|z|^{(n+1)!-n!}$$

is 0 for $|z| < 1$, but for $|z| > 1$.

3. (i) The limits

$$\lim_n \sqrt[2n]{\frac{|z|^{2n}}{3^n}} = \frac{|z|}{3} \quad \text{and} \quad \lim_n \sqrt[2n+1]{\frac{|z|^{2n+1}}{2^{n+1}}} = \frac{|z|}{2}$$

are < 1 for $|z| < 2$, so the series converges absolutely for $|z| < 2$. But the series does not converge absolutely for $|z| > 2$, so the radius of convergence is 2.

(iii) Since

$$\lim_n \sqrt[n]{\frac{n!|z|^n}{n^n}} = \lim_n \frac{|z|\sqrt[n]{n!}}{n} \lim_n \frac{|z|}{n} = 0,$$

the series converges absolutely for all z, so the radius of convergence is ∞.

(v) The limit

$$\lim_n \sqrt[n]{2^n z^{n!}} = 2 \lim_n z^{(n-1)!}$$

is 0 for $|z| < 1$, but for $|z| > 1$, so the radius of convergence is 1.

GLOSSARY OF SYMBOLS

P 9
$|a|$ 11
$\sqrt{x}$ 12
$\max(x, y)$ 16
$\min(x, y)$ 16
ε ("epsilon") 18
$\mathbf{N}$ 21
$\emptyset$ 23
$n!$ 23
$\sum_{i=1}^{n} a_i$ 24
$\mathbf{Z}$ 25
$\mathbf{Q}$ 25
$\mathbf{R}$ 25, 579
$\binom{n}{k}$ 27
$f(x)$ 40, 47, 591
I 43
$f + g$ 43, 242
$A \cap B$ 43
$f \cdot g$ 43
f/g 43
$c \cdot g$ 43
$\{x : \dots\}$ 43
$\{a, \dots, z\}$ 44
$f + g + h$ 44
$f \cdot g \cdot h$ 44
$f \circ g$ 44
$f \circ g \circ h$ 45
$x \to f(x)$ 45
$\prod_{i=1}^{n} a_i$ 49
a^{b^c} 49
C_A 50
$A \cup B$ 50
$\mathbf{R} - A$ 50
$|f|$ 51
$\max(f, g)$ 51
$\min(f, g)$ 51
$f < g$ 53
the pair (a, b) 54
the open interval (a, b) 56
$[a, b]$ 57
(a, ∞) 57
$[a, \infty)$ 57
$(-\infty, a)$ 57
$(-\infty, a]$ 57
$(-\infty, \infty)$ 57
$[x]$ 72
$\{x\}$ 72
$v + w$ 75
$v \cdot w$ 78
$\|v\|$ 78
$\det(v, w)$ 79
δ ("delta") 96
$\lim_{x \to a} f(x)$ 99
$\lim_{a} f$ 99
$\lim_{x \to a^+} f(x)$ 104
$\lim_{x \downarrow a} f(x)$ 104
$\lim_{x \to a^-} f(x)$ 104
$\lim_{x \uparrow a} f(x)$ 104
$\lim_{x \to \infty} f(x)$ 105
$\lim_{x \to -\infty} f(x)$ 111
$\lim_{x \to a} f(x) = \infty$ 111
$\lim_{x \to \infty} f(x) = \infty$ 111
$\sup A$ 132
$\text{lub}\, A$ 132
$\inf A$ 132
$\text{glb}\, A$ 132
$\overline{\lim}\, A$ 141
$\limsup A$ 141
$\underline{\lim}\, A$ 141
$f'(a)$ 149
f' 149
$\frac{df(x)}{dx}$ 152
$\left.\frac{df(x)}{dx}\right|_{x=a}$ 153
f'' 159
f''' 159
$f^{(k)}$ 159
$\frac{d^2 f(x)}{dx^2}$ 160
f^{-1} 228
$\mathbf{e}$ 241, 328
$c + d$ 242
$\alpha \cdot c$ 242
$c \cdot d$ 243
$\det(c, d)$ 243
c' 243
$R(f, a, b)$ 250
$L(f, P)$ 251
$U(f, P)$ 251
$\int_a^b f$ 255
$\int_a^b f(x)\,dx$ 261
$\ell(f, P)$ 274
$\mathcal{L}(x)$ 275
$\mathbf{L}\int_a^b f$ 292
$\mathbf{U}\int_a^b f$ 292
$\int_a^\infty f$ 298
$\int_a^\infty f(x)\,dx$ 298
$\int_{-\infty}^a f$ 298
$\int_{-\infty}^\infty f$ 298
$\sin^\circ$ 301
$\sin^r$ 301
π 302
$A(x)$ 303
cos 303, 305, 554
sin 303, 305, 554
sec 397
tan 307
csc 307
cot 307
arcsin 307
arccos 308
arctan 308
$\mathbf{e}$ 328
log 338
exp 340, 554
e 340

e^x 341
a^x 342
$\log_a$ 343
sinh 349
cosh 349
tanh 349
arg cosh 350
arg sinh 350
arg tanh 350
Nap log 354
$F(x)\Big|_a^b$ 359
$\int f$ 360
$\int f(x)\,dx$ 360
$\Gamma(x)$ 390
$P_{n,a}$ 406
$P_{n,a,f}$ 406
$R_{n,a}$ 415
$\binom{\alpha}{n}$ 429
$\{a_n\}$ 445
$\lim_{n\to\infty} a_n$ 446
$\lim_{n\to\infty} a_n = \infty$ 449
γ 456
$\overline{\lim}_{n\to\infty} x_n$ 460
$\limsup_{n\to\infty} x_n$ 460
$\underline{\lim}_{n\to\infty} x_n$ 461
$\liminf_{n\to\infty} x_n$ 461
$N(n; a, b)$ 462
$\sum_{i=1}^{\infty} a_n$ 465
i 517, 523
$\mathbf{C}$ 522
$\bar{z}$ 525
$|z|$ 525
Re 532
Im 532
θ 533
$\lim_{z\to a} f(z)$ 533
$f'(a)$ 542
sin 554
cos 554
exp 554
b_n 563
B_n 563
D 564
D^k 564
e^D 564
Δ 564
φ_n 566
ψ_n 567
$\mathbf{+}$ 571, 581
$\mathbf{\cdot}$ 571, 584
$\mathbf{0}$ 571, 581
$\mathbf{1}$ 572,586
$\mathbf{-}a$ 572,582
$a^{\mathbf{-1}}$ 572, 586
$\mathbf{P}$ 573
$\mathbf{>}$ 574, 580
$\mathbf{<}$ 574, 580
$\mathbf{\geq}$ 574, 580
$\mathbf{\leq}$ 574, 580
$\mathbf{|}\alpha\mathbf{|}$ 585

INDEX

AalbmcndoE, 273
Abel, Niels Henrik, 404, 513
Abel summable, 514
Abel's formula for summation by parts, 388
Abel's Lemma, 389
Abel's test, 488
Abel's Theorem, 513
Absolute value, 11
  of a complex number, 525
Absolutely convergent, 473, 547
Absolutely summable, 473
Acceleration, 159
Acta Eruditorum, 146
Addition, 3
  associative law for, 9
  commutative law for, 9
  of complex numbers, 522
    geometric interpretation of, 526
  of vector-valued functions, 242
  of vectors, 75
Addition formula
  for arcsin, 314
  for arctan, 314
  for cos, 311
  for sin, 310, 311
  for tan, 314
Additive identity
  existence of, 9
  for vectors, 76
Additive inverses
  existence of, 9
Algebra, Fundamental Theorem of, 373, 529, 539, 558
Algebraic functions, 359
Algebraic number, 435
Algebraist's real numbers, 588
Almost lower bound, 140
Almost upper bound, 140
Analyst's real numbers, 588
Angle, 300
  directed, 300
Antidiagonal, 239
Arabic numerals, multiplication of, 8
Arc length, 275, 281
Arccos, 308
  derivative of, 308
Archimedes, 136, 139, 260
Archimedian property
  for the rational numbers, 574
  for the real numbers, 136
Archimedian spiral, 85, 246
Arcsec, 317, 379
Arcsin, 307
  addition formula for, 314
  derivative of, 308
  Taylor series for, 509
Arctan, 308
  addition formula for, 314
  derivative of, 308
  Taylor polynomial for, 407, 414
    remainder term for, 422
Area, 250, 255
Arg cosh, 350
Arg sinh, 350
Arg tanh, 350
Argument, 527
Argument function, 533
  discontinuities of, 537
Argument of the hyperbolic functions, 350
Arithmetic mean, 33
Arrow, 75, 76
  "x arrow $\sin(x^2)$", 45
Associative law
  for addition, 9
    of vectors, 76
  for multiplication, 9
Average velocity, 150
Axis
  horizontal, 57
  imaginary, 525
  real, 524
  vertical, 57

Bacon, Francis, vi
Basic properties of numbers, 3
"Bent graphs", 147
Bernoulli, 146, 565
Bernoulli numbers, 563
Bernoulli polynomials, 566
Bernoulli's inequality, 32
Big game hunting, mathematical theory of, 543
Binary operation, 571

Binomial coefficient, 27, 429
Binomial series, 487, 510
Binomial theorem, 28
Bisection argument, 140, 543
Bohr, Harold, 390
Bolzano-Weierstrass Theorem, 451, 461, 543
Bound
  almost lower, 140
  almost upper, 140
  greatest lower, 132
  least upper, 131, 574
  lower, 132
  upper, 131
Bounded above, 120, 131, 450, 574
Bounded below, 132, 450
Bourbaki, Nicholas, 146

Cardioid, 89, 247
Cartesian coordinates, 84
Cauchy, 278
Cauchy Condensation Theorem, 488
Cauchy criterion, 466
Cauchy form of the remainder, 417, 419
Cauchy Mean Value Theorem, 201
Cauchy product, 486, 505
Cauchy sequence, 452, 562
  equivalence of, 589
Cauchy-Hadamard formula, 560
Cauchy-Schwarz inequality, 278
Cavalieri, 272
Cesaro summable, 486
Chain Rule, 172 ff.
  proof of, 176
Change, rate of, 150
Characteristic (of a field), 576
Circle, 65
  "f circle g", 44
  unit, 66
Circle of convergence, 550
Classical notation
  for derivatives, 152–154, 160, 165, 184, 238
  for integrals, 262
Cleio, 183
Closed interval, 57
Closed rectangle, 538
Closure under addition, 9
Closure under multiplication, 9
Commutative law
  for addition, 9
    of vectors, 76
  for multiplication, 9
Comparison test, 467, 468
Comparison Theorem, Sturm, 320
Complete induction, 23
Complete ordered field, 574, 593
Completing the square, 17, 375
Complex analysis, 556
Complex function
  continuous, 536
  differentiable, 541
  graph of, 533
  limit of, 533
  nondifferentiable, 542
  Taylor series for, 554
Complex nth root, 527
Complex numbers, 517, 522
  absolute value of, 525
  addition of, 522
    geometric interpretation of, 526
  geometric interpretation of, 525
  imaginary part of, 522
  infinite sequence of, 546
  infinite series of, 546–548
  logarithm of, 561
  modulus of, 525
  multiplication of, 522
    geometric interpretation of, 526–527
  real part of, 522
Complex plane, 524
Complex power series, 548
  circle of convergence of, 550
  radius of convergence of, 550
Complex-valued functions, 532
Composition of functions, 44
Concave function, 217
Conditionally convergent series, 474
Cone, 80
  generating line of, 80
  surface area of, 399
Conic sections, 80; *see also* Ellipse, Hyperbola, Parabola
Conjugate, 525, 530

Conjugate function, 532
Constant function, 43
Construction of the real numbers, 578 ff.
Continued fraction, 455
Continuous, uniformly, 142
Continuous at a, 113, 536
Continuous function, 113, 116, 537
  nowhere differentiable, 157, 501
Continuous on (a, b), 116
Continuous on $[a, b]$, 116
Contraction, 459
Contraction lemma, 459
Converge
  pointwise, 494
  uniformly, 494, 498
Convergent sequence, 446, 546
Convergent series, 465, 547
  absolutely, 473, 547
  conditionally, 474
Convex function, 216
  strictly, 226
  weakly, 226
Convex subset of the plane, 226, 544
Cooling, Newton's law of, 352
Coordinate
  first, 57
  second, 57
Coordinate system, 57
  cartesian, 84
  origin of, 57
Coordinates
  polar, 84
"Corner", 60
Cos, 300, 303, 318–319, 554
  addition formula for, 311
  derivative of, 170, 304
  inverse of, *see* Arccos
  Taylor polynomials for, 407
    remainder term for, 420
Cosh, 349
Cosine, hyperbolic, 349
Cot, 307
  derivative of, 307
Countable, 442
Counting numbers, 21
Critical point, 187
Critical value, 187
Csc, 307
  derivative of, 307
Cubic equation, general solution, 519–520
Curve
  parameterized, tangent line of, 243
  parametric representation of, 241
  reparameterization of, 244
Cycloid, 247

Darboux's Theorem, 211
De Moivre's Theorem, 527
Decimal expansion, 73, 485
Decreasing function, 192
Decreasing sequence, 450
Dedekind, Richard, 38
Defined implicitly, 238
Definite integral, 361
DEFINITION, 47
Definition, recursive, 23
Degree (of a polynomial), 42
Degree measurement, 63, 301–302
Delicate ratio test, 486
Delicate root test, 486
Dense, 138
Derivative, 147 ff., 149
  classical notation for, 152–154, 160, 165, 184, 238
  higher-order, 159
  "infinite", 156
  left-hand, 154
  Leibnizian notation for, *see* Derivative, classical notation for
  logarithmic, 348
  "negative infinity", 156
  of f, 149
  of f at a, 149
  of vector-valued function, 243
  right-hand, 154
  Schwarzian, 182
  second, 159
    Schwarz, 431
Derivative of quotient, incantation for, 169
Descartes, René, 84
Determinant, 79
  of vector-valued functions, 243
Diagonal, 230

Difference operator, 564
Differentiable, 149, 541
Differential equation, 289, 297, 318, 320, 352, 357, 432
  initial conditions for, 433
Differentiation, 166 ff.
  implicit, 238
  logarithmic, 348
Differentiation operator, 564
Dini's Theorem, 515
Directed angle, 300
Dirichlet's test, 488
Disc method, 397
Discontinuities of a nondecreasing function, 443
Discontinuity, removable, 119
Disraeli, Benjamin, 2
Distance, 58, 525
  shortest between two points, 275
Distributive law, 9
Diverge, 446, 547
Division, 6
Division by zero, 6
Domain, 40, 41, 47, 591
Dot product
  of vectors, 78
  of vector-valued functions, 243
Double intersection, 163
Double root, 183
Durège, 38

e, 340
  irrationality of, 425
  relation with π, 441, 555
  transcendentality of, 437
  value of, 341, 422
Eccentricity of ellipse, 87
Elementary function, 359
Ellipse, 66, 82
  axes of, 87
  eccentricity of, 87
  equation in polar coordinates, 86
  focus point of, 66, 86
  major axis of, 87
  minor axis of, 87
Ellipsoid of revolution, 400
Empty collection, 23
Entire function, 558
Epsilon, 18
Equal up to order n, 412
Equality, order of, 412
Equations, differential, *see* Differential equations
Equivalent Cauchy sequences, 589
Etymology lesson, 82
Euler, 565
Euler's number, 456
Euler-Maclaurin Summation Formula, 566
Even function, 51, 196
Even number, 25
Eventually inside, 546
Exhaustion, method of, 139
Exp, 340 ff., 554
  classical approach to, 354
  elementary definition of, 461
  Taylor polynomials for, 407
    remainder term for, 422
Expansion, decimal, 73, 485
Extension of a function, 113–114

Factorial, 23
Factorials, table of, 428
Factorization into primes, 31
Fibonacci, 32
Fibonacci Association, 32
Fibonacci Quarterly, The, 32
Fibonacci sequence, 32, 512, 563
Field, 571
  characteristic of, 576
  complete ordered, 574, 593
  ordered, 573
First coordinate, 57
First Fundamental Theorem of Calculus, 282
Fixed point of a function, 458
Focus point, 66, 86
Force, as vector, 76
Four leaf clover, 88
Fourier series, 315, 317, 320
Fraction, continued, 455
Function, 39, 47
  absolute value, 532

Function (*continued*)
argument, 533
discontinuities of, 537
complex valued, 532
composition of, 44
concave, 217
conjugate, 532
constant, 43
continuous, 113 ff.
convex, 216
critical point of, 187
critical value of, 187
decreasing, 192
derivative of, 147 ff.
differentiable, 149, 541
elementary, 359
entire, 558
even, 51, 196
exponential, 340–341
extension of, 113–114
fixed point of, 458
from A to B, 591
from real numbers to the plane, 241
graphs of, 57–65, 195, 533
hyperbolic, 349
identity, 43
imaginary part, 532
increasing, 192
integrable, 255
integral of, 255
inverse, 228 ff.
linear, 58
local maximum point of, 186
local minimum point of, 186
local strict maximum point of, 215
logarithm, 338, 343
maximum point of, 185
maximum value of, 185
minimum value of, 185
most general definition of, 591
negative part of, 51
nondecreasing, 240
nonincreasing, 240
nonnegative, 51
notation for, 40, 45
odd, 51, 196
periodic, 71, 162, 296
polynomial, 42
positive part of, 51
power, 60
product of, 43
quotient of, 43
rational, 42
real part, 532
real-valued, 532
"reasonable", 68, 116, 178
regulated, 515
step, 275
strict maximum point of, 215
sum of, 43
trigonometric, 300 ff.
value at x, 40
vector-valued, 241
Fundamental Theorem of Algebra, 373, 529, 539, 558
Fundamental Theorem of Calculus
First, 282
Second, 286

Gabriel, 402
Galileo, 146, 162
Gamma function, 390, 437
Generating line, of a cone, 80
Geometric mean, 33
Geometric series, 466
Global property, 121
Goes to, "x goes to $\sin(x^2)$", 45
Graph of polynomial function, 194
Graph sketching, 193–198
Graphs, 57–65, 85 ff., 90–91, 196, 533
Gravitation, 327
Greatest lower bound, 132
Grin and bear it, 381–382
Gronwall's inequality, 353
Grow
at the same rate as, 358
faster than, 358

Hadamard, 560
Half-life (of radioactive substance), 352
Hermite, 436
High-school student's real numbers, 589
Higher-order derivatives, 159
Hilbert, 436
Horizontal axis, 57
Hyperbola, 67, 82
  equation in polar coordinates, 88
Hyperbolic cosine, 349
Hyperbolic functions, 349
Hyperbolic sine, 349
Hyperbolic spiral, 313

Identity
  additive, 9
  multiplicative, 9
Identity function, 43
Identity operator, 564
Imaginary axis, 525
Imaginary part function, 532
Imaginary part of a complex number, 522
Implicit differentiation, 238
Implicitly defined, 238
Improper integral, 298–299, 391–393
Incantation for derivative of quotient, 169
Increasing at a, 214
Increasing function, 192
Increasing sequence, 450
Indefinite integral, 361
Indefinite integrals, short table of, 361–362
Induction, mathematical, 21
  complete, 23
Inductive set of real numbers, 34
Inequalities, 9
  in an ordered field, 574
Inequality
  Bernoulli's, 32
  Cauchy-Schwarz, 278
  geometric-arithmetic mean, 33
  Gronwall's, 353
  Jensen's, 225
  Liouville's, 441
  Schwarz, 17, 32, 278
  triangle, 71
  Young's, 273
Infimum, 132
"Infinite" derivative, 156
Infinite intervals, 57
Infinite product, 326, 391
Infinite products, 489
Infinite sequence, 445, 546
Infinite series, 465
  multiplication of, 479–481
Infinite sum, 426, 464
Infinite trumpet, 402
Infinitely many primes, 32
"Infinitely small", 153, 261
Infinity, 57
  minus, 57
Inflection point, 222
Initial conditions for differential equations, 433
Instantaneous speed, 150
Instantaneous velocity, 150
Integer, 25
Integrable, 255
Integral, 255
  classical notation for, 262
  definite, 361
  improper, 298–299, 391–393
  indefinite, 361
    short table of, 361–362
  Leibnizian notation for, *see* Integral, classical notation for
  lower, 292
  Mean Value Theorem for, 274
  Second Mean Value Theorem for, 387
  upper, 292
Integral form of the remainder, 417, 418
Integral sign, 255
Integral test, 471
Integration
  by parts, 362 ff.
  by substitution, 365 ff.
  limits of, 255
  of rational functions, 373 ff.
Interest (finance), 351

Intermediate Value Theorem, 122, 129, 133, 296
Interpolation, Lagrange, 49
Intersection of sets, 43
Interval, 56
  closed, 57
  infinite, 57
  open, 56; *see also* Nested Intervals Theorem
Inverse
  additive, 9
  multiplicative, 9
Inverse of a function, 228 ff.
Inverse square law, 327
Inverses of trigonometric functions, *see* Trigonometric functions
Irrational numbers, 25
Isomorphic fields, 592
Isomorphism, 592

Jensen's inequality, 225
Johnson, Samuel, 597
Jump, 60

Kepler, 327
Kepler's laws of planetary motion, 327

Lagrange form of the remainder, 417, 418
Lagrange interpolation formula, 49
Large negative, 64
Least upper bound, 131 ff., 574
Least upper bound property, 133
Lebesgue, *see* Riemann-Lebesgue Lemma
Left-hand derivative, 154
Leibniz, 153, 261
Leibniz's formula, 182
Leibniz's Theorem, 474
Leibnizian notation for derivatives, 153–154, 165, 184, 238
  for higher order derivatives, 160
Lemma, 100
Lemniscate, 89
Length, 275, 281
L'Hôpital, Marquis de, 146
L'Hôpital's Rule, 201, 210–211
Limit, 90 ff., 96, 533
  at infinity, 105
  "does not exist", 99
  from above, 104
  from below, 104
  of a sequence, 446
  of vector-valued function, 243, 249
  uniqueness of, 98
Limit of integration, 255
Limit point, 462, 543
Limit superior, 141, 460
Lindemann, 440
Line, real, 56
Line, tangent, *see* Tangent line
Linear functions, 58
Liouville, 441
Liouville's inequality, 441
Liouville's Theorem, 558
Lipschitz of order α, 207
Local maximum point of function, 186
  higher-order derivative test for, 411
  second derivative test for, 198
Local minimum point of function, 186; *see also* Local maximum point
Local property, 107, 121, 164
Local strict maximum point, 215
Log, 338, 343
  Taylor polynomials for, 407
    remainder term for, 423
Logarithm
  classical approach to, 354
  Napierian, 354
  of a complex number, 561
  to the base 10, 336
Logarithmic derivative, 348
Lower bound, 132
  almost, 140
  greatest, 132
Lower integral, 292

Lower limit of integration, 255
Lower sum, 251
Lowest terms, 73

Maclaurin, 566
Major axis of ellipse, 87
Mass, rate of change of, 150
Mathematical induction, 21
Maximum of two numbers, 16
Maximum point of a function, 185
  local, 186; *see also* Local maximum point
  local strict, 215
  strict, 215
Maximum value of function, 185
Mean
  arithmetic, 33
  geometric, 33
Mean Value Theorem, 190, 191
  Cauchy, 201
  for integrals, 274
    Second, 387
Method of exhaustion, 139
Minimum of function, 185
Minimum of two numbers, 16
Minimum point of a function, local, 186; *see also* Local minimum point
Minor axis of ellipse, 87
Minus infinity, 57
Mirifici logarithmonum canonis description, 355
Modulus of a complex number, 525
Mollerup, Johannes, 390
Multiplication, 5
  associative law for, 9
  closure under, 9
  commutative law for, 9
  of arabic numerals , 8
  of complex numbers, 522
    geometric interpretation, 526–527
  of function and vector-valued function, 242
  of infinite series, 479–481
  of number and vector, 77
  of vectors , 77
Multiplicative identity, existence of, 9
Multiplicative inverses, existence of, 9
Multiplicity (of a root), 128

Napier, 355
Napierian logarithm, 354
Natural numbers, 21, 34
Negative, large, 64
"Negative infinity", derivative, 156
Negative number, 9
Negative numbers, product of two, 7
Negative part of a function, 51
Nested Interval Theorem, 140
Newton, 153, 273, 327
Newton's law of cooling, 352
Newton's laws of motion, 159
Newton's method, 457
Nondecreasing function, 240
Nondecreasing sequence, 450
Nondifferentiable complex functions, 542
Nonincreasing function, 240
Nonincreasing sequence, 450
Nonnegative function, 51
Nonnegative sequence, 467
Norm, 78, 249
Notational nonsense, 564
Nowhere differentiable continuous function, 501
*n*th root, 71, 527
  existence of, 123, 527, 544
  primitive, 531
Null set, 23
Number
  algebraic, 435
  complex, 517, 522
  counting, 21
  even, 25
  imaginary, 517
  irrational, 25
  natural, 21, 34
  odd, 25
  prime, 31
  rational, 25
  real, 25, 525, 579
  transcendental, 435
Numbers, basic properties of, 3

Odd function, 51, 196
Odd number, 25
One-one function, 227
Open interval, 56
"Or", 6
Order of equality, 412
Ordered field, 573
  complete, 574
Ordered pair, 47 (footnote), 54
Origin (of a coordinate system), 57

Pair, 46
  ordered, 47 (footnote), 54
Parabola, 60, 82
  area under, 260
  equation in polar coordinates, 88
Parallelogram, 76
Parameterized curve, tangent line of, 243
Parametric representation of a curve, 241
Partial fraction decomposition, 374
Partial sums, 464
Partition, 251
Parts
  Abel's formula for summation by, 388
  integration by, 362 ff.
Pascal's triangle, 27
"Peak", 61
Peak point, 451
Period of a function, 71, 162, 296
Periodic function, 71, 162, 296
Perpendicularity of lines, 70
Petard, H, 543
Pig, yellow, v, 371
Pigheaded, 183
Plane, 58
  complex, 524
Planetary motion, Kepler's laws of, 327
Point, 56
Point of contact, 217
Point-slope form of equation of a line, 59, 70
Polar coordinates, 84 ff.
Polynomial function, 42
  graph of, 61, 194
  multiplicity of roots, 128, 183
Polynomials, Bernoulli, 566
Pope, Alexander, 327
Position, rate of change of, 150
Positive element of **R**, 583
Positive elements of an ordered field, 573
Positive number, 9
Positive part of a function, 51
Power functions, 60
Power series, complex, 548
Power series centered at a, 502, 555
Power series representation, uniqueness of, 512
Powers of 2, table of, 428
Prime number, 31
  characteristic of a field, 576
  infinitely many of, 32
  unique factorization into, 31
Primitive, 359
Primitive nth root, 531
Principia, 273
Product, 5
  Cauchy, 486, 505
  infinite, 326, 391, 489
  of function and vector-valued function, 242
  of functions, 43
  of number and vector, 77
  of two negative numbers, 7
  of vectors, 77
Pyramid
  surface area of, 398
  volume of, 402
Pythagorean theorem, 25, 58
π, 302
  Archimedes' approximation of, 139
  irrationality of, 323
  relation to e, 441, 555
  transcendentality of, 440
  value of, 429
  Viète's product for $2/\pi$, 326
  Wallis' product for $\pi/2$, 391

Quaternions, 577
Quotient of functions, 43

Rabbits
  growth of population, 32
Radian measure, 63, 301–302
Radioactive decay, 352
Radius of convergence of complex power series, 550
Rate of change of mass, 150
Rate of change of position, 150
Ratio test, 469
  delicate, 486
Rational functions, 42
  integration of, 373 ff.
Rational numbers, 25
Real axis, 524
Real line, 56
Real number (formal definition), 579
Real numbers, 25
  algebraist's, 588
  analyst's, 588
  Archimedian property of, 136
  construction of, 578 ff.
  high-school student's, 589
  inductive set of, 34
Real part function, 532
Real part of a complex number, 522
Real-valued function, 532
Rearrangement of a sequence, 476
"Reasonable" function, 68, 116, 147, 178
Rectangle, closed, 538
Recursive definition, 23
Reduction formulas, 373
Regulated function, 515
Remainder term for Taylor polynomials, 415
Removable discontinuity, 119
Reparameterization, 244
Revolution
  ellipsoid of, 400
  solid of, 397
Riemann sum, 279
Riemann-Lebesgue Lemma, 317, 387
Right-hand derivative, 154
Rising Sun Lemma, 141
Rolle, 183
Rolle's Theorem, 190
Root
  multiplicity of, 128
Root of a polynomial function, 50
  double, 183; *see also* *n*th roots
Root test, 485
  delicate, 486

Same sign, 12
Scalar, 78
Scalar product of vectors, 78
Schwarz, H. A., 17, 215
Schwarz inequality, 17, 32, 278
Schwarz second derivative, 431
Schwarzian derivative, 182
Sec, 307
  derivative of, 307
  inverse of, *see* Arcsec
Secant line, 148
Second coordinate, 57
Second derivative, 159
  Schwarz, 431
Second derivative test for maxima and minima, 198
Second Fundamental Theorem of Calculus, 286
Second Mean Value Theorem for Integrals, 387
Sequence
  absolutely summable, 473
  Cauchy, 452
    complex, 562
    equivalence of, 589
  complex numbers, 546
  convergent, 446
    pointwise, 494
    uniformly, 494
  decreasing, 450
  divergent, 446
  Fibonacci, 32, 512, 563
  increasing, 450
  infinite, 445
  limit of, 446
  nondecreasing, 450
  nonincreasing, 450
  nonnegative, 467
  rearrangement of, 476
  summable, 465
Series
  absolutely convergent, 473
  conditionally convergent, 474

Series (*continued*)
convergent, 465
Fourier, 315, 317, 320
geometric, 466
power, 502, 548
Taylor, 503
Set, 22
empty, 23
Sets
intersection of, 43
notation for, 43–44
Shadow point, 141
Shell method, 398
Sigma, 24
Sign, 12
Simpson's rule, 396
Sin, 43, 300, 303, 318–319, 554
addition formula for, 310, 311
derivative of, 170, 304
inverse of, *see* Arcsin
Taylor polynomials for, 406
remainder term for, 420
Sine, hyperbolic, 349
Sine function, 43
Sinh, 349
Sketching graphs, 193–198
Skew field, 577
Slope of a straight line, 58
Solid of revolution, 397
Speed, instantaneous, 150
Spiral
Archimedian, 85
hyperbolic, 313
Square root, 12, 518
existence of, 122
Square root function, 537–538
Square root in a field, 576
Squaring the circle, 440
Step function, 275
Stirling's Formula, 568
Straight line
analytic definition, 58
shortest distance between two points, 275
slope of, 58
Strict maximum point, 215
Strictly convex, 226
Sturm Comparison Theorem, 320
Subsequence, 451
Substitution
integration by, 365 ff.
world's sneakiest, 382
Substitution formula, 365
Subtraction, 5
Sum
finite, 3–4
infinite, 426, 464
lower, 251
of an infinite sequence, 465
of an infinite sequence of complex numbers, 546
of functions, 43
of vector-valued functions, 242
of vectors, 75
partial, 464
sigma notation for, 24
upper, 251
Sum of squares, 543
Summable, 465, 547
Abel, 514
absolutely, 473
Cesaro, 486
uniformly, 498
Summation by parts, Abel's formula for, 388
Supremum, 132
Surface area
of cone, 399
of pyramid, 398
of solid of revolution, 397
Swift, Jonathan, 570
Symmetry in graphs, 196

Tan, 307
derivative of, 307
inverse of, *see* Arctan
Taylor series for, 564
Tangent, hyperbolic, 349
Tangent line, 147, 149
of parameterized curve, 243
point of contact of, 217
"Tangent line", vertical, 156
Tanh, 349
Taylor polynomial, 406 ff.
remainder term of, 415, 417, 418; *see also specific functions*

Taylor series, 503, 554
Taylor's Theorem, 417
Torus, 400
Transcendental number, 435
Trapezoid rule, 394
Triangle inequality, 71
Trichotomy law, 9
Trigonometric functions, 300, *see also* cos, cot, csc, sec, sin, tan
  integration of, 372–373
  inverses of, 307; *see also* Arccos, arcsec, arcsin, arctan
Trumpet
  infinite, 402
Two-time differentiable, 159

Uniform limit, 494
Uniformly continuous function, 142
Uniformly convergent sequence, 494
Uniformly convergent series, 498
Uniformly distributed sequence, 462
Uniformly summable, 498
Uniqueness
  of factorization into primes, 31
  of limits, 98
  of power series representations, 512
Unit circle, 66
Upper bound, 131, 574
  almost, 140
  least, 131
Upper integral, 292
Upper limit of integration, 255
Upper sum, 251

"Valley", 61
Value
  absolute, *see* Absolute value
Value of f at x
Vanishing condition, 466
Vector-valued functions, 241
Vector-valued functions
  determinant of, 243
  derivative of, 243
  dot product of, 243
  limit of, 243, 249
  multiplication of function by, 242
  sum of, 242
Vectors, 75
  addition of, 75
  as forces, 76
  dot product of, 78
  multiplication by numbers, 77
  multiplication of, 77
  scalar product of, 78
Velocity
  average, 150
  instantaneous, 150
Vertical axis, 57
Viète, François, 326
Volume, 397–398

Wallis' product, 391
Weakly convex, 226
Weierstrass, *see* Bolzano-Weierstrass Theorem
Weierstrass M-test, 499
Well-ordering principle, 23
Wright, 383

Young's inequality, 273

Zahl, 25
Zero, division by, 6